Germanic Lexeme Reports: Lexical Derivation Volume

Alpha 01 lexical volume. This volume assembles the current lexeme-report corpus in editorial order without revising entry-level prose, citations, locators, or transducer logic.

Introduction

The catalogue groups words by the kind of historical explanation they require. Regular derivations establish the reach of the sound laws. The remaining classes mark, without concealment, the places where attestation, morphology, analogy, or an unresolved mismatch intervenes.

The catalogue proceeds from words; the sound-change account proceeds from rules. Together they allow each proposed sound law to be tested against its lexical witnesses and each etymology against the complete ordered history. SC-numbered rules cited in these entries are discussed in the companion sound-change volume, or in the sound-change chapters of the combined book.

Data and sources

This volume assembles the lexical corpus from the aligned Germanic dataset and the compact derivation traces that accompany each entry. Comparative dictionaries, Old English dictionaries, and historical grammars are cited in the prose where they bear on particular lexical arguments.

The result is a lexical catalogue rather than a separate report on citation method or trace machinery.

Transducer and derivation method

Four objects must be distinguished in every derivation: the citation reconstruction, the selected input, the transducer outcome, and the Old English target. The summary identifies them where they differ; the boxed trace then divides the changes into Earlier Germanic and Old English stages.

Derivation classes

The lexical catalogue is ordered by seven derivation classes and, within each class, alphabetically by headword. Counts in this alpha are:

- Regular derivations: 91
- Attested variants: 5
- Early analogy: 35
- Late analogy: 29
- Reconstructed Old English comparators: 3
- Known but unmodelled developments: 2
- Unexplained or deliberately unmodelled exceptions: 5

Part I. Regular derivations

In these entries the selected Germanic input yields the Old English reflex by regular sound change. They establish the standard against which the following analogical and exceptional histories must be judged.

adder — OE *nǣdre*

Derivation: **nédrōn* > *nǣdre* (regular).

Derivation trace Proto input: **nédrōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**nǣdræ*
PNWGmc Long E Lowering	**nādrōn*	OE Unstressed AE Merger	**nǣdre*
PNWGmc N Stem N Loss	**nādrǭ*		
EAF Long A Fronting	**nǣdrǭ*		
Early Anglo-Frisian			
[no change]			

Old English form: *nǣdre*

Reconstruction and comparative evidence Kroonen distinguishes the masculine snake word **nadra-* from a feminine ablauting formation **nēdrōn-*, and gives Old English *nǣdre* ‘adder’, *nædre* ‘adder’ under the latter (Kroonen 2013, 381, 386). Orel’s feminine entry gives **nēdrōn* ‘adder’ ~ **nadrōn* ‘adder’ and relates them to the masculine formation. These comparison spellings use plain *d* for Orel’s printed barred *đ*; the quantity and the two alternatives are retained (Orel 2003, 286).

The derivational input retains the long-*e* feminine alternative. The lexical comparison supports that formation without erasing Orel’s alternative or the distinction between his notation and CAPR’s derivational input.

Reconstruction alternatives and their limits Orel’s short-*a* feminine is an admitted alternative within his own account, not proof that he and Kroonen advocate exclusive histories. The unresolved premise is the reflex distribution and formation history licensing that alternative beside the selected long-*ē* feminine (Orel 2003, 286; Kroonen 2013, 381, 386). Kroonen’s uncertain sewing or ‘winder’ connection and masculine comparanda cannot substitute for that feminine argument. CAPR’s acute marks its input stress; it is not independently quoted in these headwords.

Fulk’s probable shortening accompanying pre-*r* gemination concerns the later Old English variant. He calls the motivation obscure and considers analogical extension; this does not explain inherited *ē* versus *a* or supply another complete ancestral adder word (Fulk 2018, 132). Regular production of *nǣdre* therefore supports the selected path without settling the alternative formation.

Old English evidence The Old English word is securely represented by *nǣdre* ‘adder’, with *næddre* ‘adder’ as a secondary variant. Clark Hall cross-references *næddre* to *nædre* ‘adder’, and Fulk treats *næddre* as the later geminated form beside the older base (Clark Hall 1960, 225; Fulk 2018, 132).

Development to Old English From **nédrōn* ‘adder’, the stressed long mid vowel develops to Old English *nǣdre* ‘adder’, and the weak feminine ending remains as final *-e*, giving *nǣdre*. The doubled consonant of *næddre* ‘adder’ is secondary and does not alter the inherited base form.

bake — OE *bacan*

Derivation: **bákaną* > *bacan* (regular).

Derivation trace Proto input: **bákaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**bækaną*
[no change]	OE A Restoration	**bakaną*
	OE Heavy Syllable Nasal Apocope	**bakan*
Early Anglo-Frisian	OE Secondary Nasalization	**bakąn*
[no change]	OE Weak Tail Reduction	**bakan*

Old English form: *bacan*

Reconstruction and comparative evidence Orel reconstructs the verb as **bakanan* ‘bake’ and cites Old English *bacan* ‘bake’ beside Old High German *backan, bahhan* (Orel 2003). Campbell gives *bacan* as one of the standard examples of Old English A-restoration before a single consonant, and Ringe and Taylor state the same development from **bakan* ‘bake’ to Old English *bacan* (Campbell 1959, 61; Ringe and Taylor 2014).

Singleton and geminate present alternatives The selected infinitive keeps singleton *k*. Kroonen explicitly distinguishes **bakan-* beside **bakkan-*, considering contamination with, or back-formation from, the iterative **bak(k)ōn-*; a present *n*-suffix explanation is less likely (Kroonen 2013, 49). Orel prints singleton **bakanan* and separately the long-*ō* preterite (Orel 2003, 33). The inspected passages do not establish Orel’s reason for excluding the geminate from this citation. This is a qualified present-formation question, not proof of a mutually exclusive reconstruction debate.

Later finite-cell front vowels and the inherited preterite grade are not rival infinitive vowels. Ringe and Taylor distinguish West Saxon infinitive *bacan* ‘bake’ from second singular *becst* ‘you bake’, preserving the umlauted vowel in syncopated forms while *a* spread elsewhere (Ringe and Taylor 2014, 232–33). Regular infinitive production does not resolve the earlier singleton/geminate distribution.

Old English evidence Bosworth-Toller and Clark Hall both record *bacan* ‘bake’ as the ordinary Old English verb (Bosworth and Toller 1898, 72; Clark Hall 1960). The target in this entry is therefore the attested infinitive headword itself, not a selected oblique or finite paradigm cell.

Development to Old English From **bákaną* ‘bake’, Anglo-Frisian brightening first gives **bækaną* ‘bake’. A-restoration then returns the stem vowel to *a* before single *k* plus the back-vocalic infinitive suffix, and later apocope and weak-tail reduction yield *bacan* ‘bake’ (Campbell 1959, 61; Ringe and Taylor 2014). The development is therefore straightforward: **bákaną* > *bacan*.

beech — OE *bōc*

Derivation: **bōkō* > *bōc* (regular).

Derivation trace Proto input: **bōkō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**bōk*
PNWGmc Final Long O Raising	**bōku*		
Early Anglo-Frisian			
[no change]			

Old English form: *bōc*

Reconstruction and comparative evidence Kroonen gives the beech noun as **bōk(j)ō-* and cites Old English *boc* ‘beech’, *bēce* ‘beech’ among its reflexes (Kroonen 2013). The derivational input **bōkō* ‘beech’ is the nominative-singular shape of that family, which is the relevant comparison form here.

Old English evidence Kroonen’s Old English evidence already separates the paradigm material: *boc* ‘beech’ as the nominative form and *bēce* ‘beech’ as an oblique form (Kroonen 2013). The relevant comparator is therefore *bōc* ‘beech’; *bēċe* ‘beech’ remains related paradigm evidence rather than the form chosen for this comparison.

Development to Old English With nominative input **bōkō* ‘beech’, the development is compact. Northwest Germanic final long *ō* raises to *u*, and later high-vowel apocope leaves *bōc* ‘beech’. The regular comparison is therefore **bōkō* > *bōc*.

begin — OE *beġinnan*

Derivation: **bigínnaną* > *beġinnan* (regular).

Derivation trace Proto input: **bigínnaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**bigínnan*
[no change]	OE Secondary Nasalization	**bigínnąn*
	OE Velar Palatalization	**bijínnąn*
Early Anglo-Frisian	OE Prefix I Reduction	**beįínnąn*
[no change]	OE Weak Tail Reduction	**beįínnan*
	OE Palatal Fricative Merger	**bejínnan*

Old English form: *beġinnan*

Reconstruction and comparative evidence The verb is modeled here as inherited **bigínnaną* ‘begin’. Ringe and Taylor state that intervocalic **g* is palatalized between front vowels in Old English (Ringe and Taylor 2014), and Campbell lists *ginnan* ‘begin’ among familiar examples of palatal *g* in this verb family (Campbell 1959, 174).

Old English evidence Bosworth-Toller and Clark Hall lemmatize the verb as *be-ginnan* ‘begin’ / *beginnan* ‘begin’ (Bosworth and Toller 1898, 84; Clark Hall 1960). Those plain-*g* dictionary spellings support the same verb that appears here in normalized form as *beġinnan* ‘begin’.

Development note The prefix deserves separate notice. Ringe and Taylor explicitly cite *bi-* > *be-* as an Old English unstressed-prefix development (Ringe and Taylor 2014).

Development to Old English From **bigínnaną* ‘begin’, heavy-syllable nasal apocope yields **bigínnan* ‘begin’. Intervocalic **g* between front vowels then palatalizes to *ġ*, and the unstressed prefix reduces *bi-* to *be-*, giving *beġinnan* ‘begin’.

bier — OE *bǣr*

Derivation: **bḗrō* > *bǣr* (regular).

Derivation trace Proto input: **bḗrō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**bǣr*
PNWGmc Long E Lowering	**bārō*		
PNWGmc Final Long O Raising	**bāru*		
EAF Long A Fronting	**bǣru*		
Early Anglo-Frisian			
[no change]			

Old English form: *bǣr*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **bērō-* f. ‘bier’ and cites Old English *bar* ‘bier’, *bær* ‘bier’ among the reflexes (Kroonen 2013, 717). The derivational input **bḗrō* ‘bier’ is the same lexeme in the accent notation used here.

Old English evidence Clark Hall and Bosworth-Toller lemmatize the noun as *bær* ‘bier’, and Kroonen also records *bar* ‘bier’ beside it (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717). The target *bǣr* ‘bier’ is therefore a normalized long-vowel spelling of the same noun.

Source note Lexicographic spellings vary between *bær* ‘bier’ and *bar* ‘bier’. The normalized target *bǣr* ‘bier’ represents the same long vowel (Clark Hall 1960; Bosworth and Toller 1898, 73; Kroonen 2013, 717).

Development to Old English From **bḗrō* ‘bier’, Northwest Germanic final long *ō* raises to *u*, long *ē* lowers to *ǣ*, and high-vowel apocope yields *bǣr* ‘bier’. The resulting noun matches the normalized Old English target.

birth — OE *byrd*

Derivation: **búrdiz* > *byrd* (regular).

Derivation trace Proto input: **búrdiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**byrdi*
EAF Final Z Deletion	**búrdi*	OE High Vowel Apocope	**byrd*
Early Anglo-Frisian			
[no change]			

Old English form: *byrd*

Reconstruction and comparative evidence Kroonen cites the noun under stem-level **burdi-* and gives Old English *(ge-)byrd* among the reflexes (Kroonen 2013). The derivational input **búrdiz* ‘birth’ is the nominative-style form that stands behind that stem label.

Old English evidence Clark Hall and Bosworth-Toller both attest simplex *byrd* ‘birth’ as an Old English noun meaning ‘birth’ (Clark Hall 1960; Bosworth and Toller 1898, 125). The prefixed form *gebyrd* ‘birth’ is also well established in the tradition: Kroonen lists *(ge-)byrd*, Bosworth-Toller has a separate *ge-byrd* ‘birth’ entry, and Campbell cites *gebyrd* and *gebyrdu* in his grammatical discussion (Kroonen 2013; Bosworth and Toller 1898, 125; Campbell 1959).

Form note The relevant comparator here is the simplex noun *byrd*. The prefixed forms remain related attested material within the same lexical family, and Kastovsky's discussion of deverbal feminines provides the broader derivational setting (Kastovsky 1992, 358).

Development to Old English From **búrdiz* 'birth', loss of final *z* gives **búrdi* 'birth'. I-umlaut fronts *u* to *y*, and high-vowel apocope then yields *byrd*. The result is the ordinary simplex Old English noun.

bone — OE *bān*

Derivation: **báiną* > *bān* (regular).

Derivation trace Proto input: **báiną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**bān*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**bāną*		

Old English form: *bān*

Reconstruction and comparative evidence Kroonen cites the noun as **baina-*, and Orel gives the same lexeme under **bainan* 'bone' (Kroonen 2013; Orel 2003). Both are comparative headword conventions for the same neuter noun whose Old English reflex is *bān* 'bone'.

Old English evidence Clark Hall and Bosworth-Toller record *bān* 'bone' as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898). Bright's glossary also distinguishes citation-form *bān* 'bone' from oblique *bāne* 'bone' (Bright 1971).

Source note The comparative headwords **baina-* and **bainan* 'bone' provide lexeme background. The relevant comparison form here is the nominative-accusative singular **báiną* 'bone'.

Development to Old English West Germanic monophthongization turns stressed **ai* into *ā*, giving **bāną* 'bone'; heavy-syllable nasal apocope then yields *bān* 'bone'. The resulting form matches the attested Old English citation noun.

book — OE *bōc*

Derivation: **bōkz* > *bōc* (regular).

Derivation trace Proto input: **bōkz*

Earlier Germanic changes	Old English changes
Northwest and West Germanic	[no change]
[no change]	
Early Anglo-Frisian	
[no change]	

Old English form: *bōc*

Reconstruction and comparative evidence The noun is an athematic consonant stem (root noun), and the sources print its nominative singular in three different notations that encode the same morphology. Orel's citation form is **bōkz* 'book', with the explicit nominative marker **-z* that he writes across the whole root-noun class (Orel 2003, 52). Kroonen cites the bare stem **bōk-* 'book, beech' and treats Gothic *bōka* 'letter' as a separate ō-stem formation (Kroonen 2013, 71–72). Kluge/Seebold print a third variant with voiceless sibilant, **bōks* 'book' (Kluge 2011, 158). These are citation conventions, not competing facts: Ringe gives the consonant-stem nominative ending as zero, **-z*, or possibly **-s*, and states that the distribution is unrecoverable for monosyllabic stems (Ringe 2017, 306). No scholar is simply "wrong" here; the notations differ in whether they make the inflectional marker visible.

The derivational input **bōkz* 'book' follows Orel's morphologically explicit convention: it records the nominative marker whose historical elimination the derivation then models. The word is distinct from the ō-stem 'beech' word, whose derivational input here is the separate nominative **bōkō* 'beech'.

Old English evidence Old English *bōc* 'book' is the endingless nominative-accusative singular of a root noun, with the mutated plural *bēċ* 'books' preserving the class's characteristic i-umlaut alternation (Fulk 2018, 165–66). Comparative root-noun evidence shows the branch-by-branch fate of the ending: Gothic keeps its sibilant in this class (*baúrgs* 'city', *nahts* 'night'), while Old Norse *bók* 'book' keeps a reflex of **-z* as *-r* only in the masculine root nouns and drops it in feminines (Fulk 2018, 165–67). West Germanic alone is uniform: none of its daughters shows any ending in the root-noun nominative (Ringe and Taylor 2014, 118).

Development to Old English The nominative marker of the consonant stems was eliminated before Proto-West Germanic: endingless nominatives arose by Szemerényi's law after sonorant-final stems and spread analogically through the class (Fulk 2018, 143), so that no West Germanic daughter inherits an ending here (Ringe and Taylor 2014, 118). From the derivational input **bōkz* 'book' the loss of the marker gives **bōk* 'book', which is already the Old English form: *bōc* 'book' differs only in spelling. This early morphological loss is a different development from the later West Germanic loss of final **z* in unstressed syllables (as in **bárdaz* 'beard') and from the still later northern loss in stressed monosyllables (as in the pronouns); the root-noun ending was gone before either of those changes applied.

both — OE *bū*

Derivation: **bō* > *bū* (regular).

Derivation trace Proto input: **bō*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Stressed Monosyllable O Raising	**bū*	
Early Anglo-Frisian		
[no change]		

Old English form: *bū*

Reconstruction and comparative evidence Kroonen treats the Germanic numeral under **ba-* and gives the inherited paradigm **bai* 'both', **bans* 'both (pl.)', **bōz* 'both'/ **bōns* 'both', **bō* 'both', with Old English *bēġen* 'both', *bā* 'both', and neuter *bū* 'both' (Kroonen 2013, 47). For the present entry, the relevant inherited form is the unextended neuter dual **bō* 'both'.

The older explanation of *bēġen* 'both' derives it from **bō-jen-*, and Orel still gives OE *bezen* 'both' (< **bō-jenō* 'both') beside ON *báðir* 'both', OFris *bēthe* 'both', OS *be-thia* 'both', and OHG *bēde* 'both' (Orel 2003, 65). Fulk reports that explanation cautiously and notes Seebold's preference for a **bō-þ-* analysis instead (Fulk 2018, sect. 10.1). That debate concerns *bēġen* and the extended

forms behind Modern English *both* ‘both’, German *beide* ‘both’, and Dutch *beide* ‘both’; it does not displace the inherited neuter **bō* ‘both’ > *bū* ‘both’ treated here.

Old English evidence The Old English dual paradigm is well established. Brunner gives masculine *bēġen* ‘both’, feminine *bā* ‘both’, and neuter *bū* ‘both’ beside *bā*, with compounds such as *bā twā* ‘both (two)’, *bū tū* ‘both (two, neut.)’, and *bām* ‘both’ *twām* ‘two’ (Brunner 1965, sect. 324 Anm. 2). Campbell and Fulk present the same basic pattern: masculine *bēġen* ‘both’, feminine *bā*, neuter *bā*, *bū*, genitive *bēġra* ‘both’, *bēġ(e)a*, and dative *bǣm* ‘both’ (Campbell 1959, sect. 683; Fulk 2018, sect. 10.1).

bū ‘both’ is therefore an attested neuter dual form, not a reconstruction. It is the cleanest target for this entry because *bēġen* ‘both’ belongs to the historically more contested **bō-jen-* / analogical zone, while *bā* ‘both’ remains a partner form within the dual paradigm rather than the most straightforward monosyllabic comparison.

Development to Old English **bō* ‘both’ is a stressed monosyllabic form. Campbell cites *cū* ‘cow’, *hū* ‘how’, *tū* ‘two’, and *bū* ‘both’ as examples of final accented *ō* > *ū* in the West Germanic stage leading to Old English (Campbell 1959, sect. 122). Brunner states the same development more directly: Auslautendes *ō* erscheint als û in *bū* … cu … *hū*, *tū* (Brunner 1965, sect. 69).

The development is therefore straightforward: **bō* ‘both’ > *bū* ‘both’.

Form comparison The comparison below sets the relevant forms side by side. It separates the inherited OE target from the other forms that belong to the same broader lexical history.

Form	Source / stage	Status	Relevance to this entry
**bō* > *bū*	PGmc neuter dual > OE neuter dual	selected regular comparison	main line of the entry
bēġen	OE masculine dual	attested, but historically contested and at least partly analogical in Kroonen	real OE evidence, not the Old English form here
bā	OE feminine dual; also neuter variant	attested partner form	part of the OE paradigm, but not the chosen monosyllabic comparator
báðir, German *beide*, Dutch *beide*, Modern English *both*	Norse, continental West Germanic, Modern English extended forms	related but different formation	useful background, not the direct continuation of OE *bū*

bow — OE *bīeġan*

Derivation: **báugijanǫ* > *bīeġan* (regular).

Derivation trace Proto input: **báugijanǫ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**báeugijaną*
[no change]		OE Diphthong Leveling	**bēagijaną*
		OE Heavy Syllable Nasal Apocope	**bēagijan*
Early Anglo-Frisian		OE Secondary Nasalization	**bēagijąn*
[no change]		Sievers Law Syncope	**bēagjąn*
		OE Velar Palatalization	**bēąjjąn*
		OE I Umlaut	**bīejjąn*
		OE Weak Tail Reduction	**bīejjan*
		OE J Loss After Heavy	**bīejan*
		OE Palatal Fricative Merger	**bīejan*

Old English form: *bīeġan*

Reconstruction and comparative evidence Kroonen reconstructs the weak verb as **baugjan-* 'to (make) bend' and cites Old English *biegan* 'bow, bend' among its reflexes (Kroonen 2013). Ringe and Taylor give the northwest Germanic preform **baugijana* 'bow', with later **béagjan* 'bow' behind West Saxon Old English *biegan* 'bow, bend' (Ringe and Taylor 2014). The entry therefore concerns the weak causative member of the bend-family, alongside the related strong verb and noun.

Old English evidence Clark Hall lemmatizes *biegan* 'bow, bend' as an Old English weak verb, and Bosworth-Toller records *bigan* 'bow, bend' as an attested form (Clark Hall 1960; Bosworth and Toller 1898, 102). The form *bīeġan* 'bow, bend' used here is a normalized spelling of that attested Old English weak verb.

Development to Old English From **báugijaną* 'bow', the stem reaches pre-Old-English **bēagjan* 'bow', after which palatalization of **gj* and i-umlaut yield West Saxon *biegan* 'bow'; Campbell lists *biegan* 'bow' among the regular *ie* outcomes of **éa* under i-umlaut (Ringe and Taylor 2014; Campbell 1959, 80). The development is therefore straightforward: **báugijaną* > *bīeġan* 'bow'.

breeches — OE *brēċ*

Derivation: **brōkiz* > *brēċ* (regular).

Derivation trace Proto input: **brōkiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Palatalization	**brōʧi*
EAF Final Z Deletion	**brōki*	OE I Umlaut	**brēʧi*
Early Anglo-Frisian		OE High Vowel Apocope	**brēʧ*
[no change]			

Old English form: *brēċ*

Reconstruction and comparative evidence Kroonen cites the noun under **brōk-*, with Old English *brōc* 'breeches' and plural 'breeches' among its reflexes (Kroonen 2013). Ringe and Taylor give the plural development directly as PNWGmc **brōkiz* 'breeches' > **breeci* 'breeches' > OE *bréc* 'breeches' (Ringe and Taylor 2014). The deeper verbal base belongs to the noun's etymological background, while the derivational input here is the plural noun form **brōkiz* 'breeches'.

Old English evidence Bright notes *brōc* 'breeches' with plural *brēc* 'breeches', and Clark Hall gives *brēc* fp. breeches while also listing *broc* 'breeches' as a feminine noun probably represented chiefly in the plural (Bright 1971; Clark Hall 1960, 64). I write *brēċ* 'breeches' for the long vowel and palatal consonant; the attested plural is *brēc*.

Development to Old English After loss of final *-z*, the stem ends in *-ki*, so the velar palatalizes and *ō* ‘breeches’ undergoes i-umlaut to *ē* ‘breeches’; final high-vowel apocope then yields *brēċ* ‘breeches’ (Ringe and Taylor 2014). The development is therefore regular: **brōkiz* ‘breeches’ > *brēċ* ‘breeches’.

calf — OE *ċealf*

Derivation: **kálbaz* > *ċealf* (regular).

Derivation trace Proto input: **kálbaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**kálb*
EAF Final Z Deletion	**kálba*	EAF Brightening	**kælb*
Early Anglo-Frisian		OE Breaking	**kealb*
		PGmc B Allophony	**kealβ*
[no change]		OE Velar Palatalization	**ʧealβ*

Old English form: *ċealf*

Reconstruction and comparative evidence Kroonen treats the noun under **kalbiz-* and notes an older s-stem **kalbaz* ‘calf’ with plural **kalbizō* ‘calves’, while Orel cites **kalbaz* ‘calf’ as the citation form and Ringe and Taylor derive West Saxon *Cealf* from **kalbaz*, **kalbiz-* (Kroonen 2013; Orel 2003, 248; Ringe and Taylor 2014, 220). The derivational input here is the singular **kálbaz* ‘calf’, since the entry concerns the citation-form noun.

Old English evidence Clark Hall gives *cealf* ‘calf’ I. (æ, e) nm. (nap. *cealfru* ‘calves’), and Bosworth-Toller likewise records *Caelf* / *Cealf* ‘calf’ beside plural forms such as *calfur* and *cealfru* ‘calves’ (Clark Hall 1960; Bosworth and Toller 1898, 131). Campbell and Fulk show the same singular-plus-*-r-* plural pattern (Campbell 1959; Fulk 2018, 193). I write *ċealf* ‘calf’ for the palatalized initial; the attested dictionary headword is *cealf* ‘calf’.

Development to Old English After loss of final *-z* and bare *-a*, Anglo-Frisian brightening gives **kælb* ‘calf’, and breaking before *l* plus consonant yields **kealb* ‘calf’. Ringe and Taylor’s account of the lexeme and their rule for initial *k* in front-vocalic environments support the West Saxon palatalized onset represented here as *ċ-*, so **kálbaz* ‘calf’ develops regularly to *ċealf* ‘calf’ (Ringe and Taylor 2014, 220).

coat — OE *rocc*

Derivation: **rúkkaz* > *rocc* (regular).

Derivation trace Proto input: **rúkkaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**rókk*
PNWGmc U Lowering	**rókkaz*		
EAF Final Z Deletion	**rókka*		
Early Anglo-Frisian			
[no change]			

Old English form: *rocc*

Reconstruction and comparative evidence Orel cites a masculine **rukkaz* ‘coat’ for the garment word, while Kroonen gives **hrukka-*. Both treat this as the garment lexeme and not as the separate stone word (Orel 2003; Kroonen 2013, 290).

Old English evidence Clark Hall and Bosworth-Toller record *rocc* ‘coat’ as an over-garment or tunic and preserve compounds such as *bisceoprocc* and *breóstrocc* ‘breast-coat’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English With **rúkkaz* ‘coat’ as the derivational input, Northwest Germanic u-lowering and later loss of final *-a* yield *rocc* as a regular outcome.

Source note This entry concerns the garment noun only. The stone word seen in *stānrocc* ‘stone-coat’ belongs to a different lexical history (Clark Hall 1960; Bosworth and Toller 1898).

corn — OE *corn*

Derivation: **kúrną* > *corn* (regular).

Derivation trace Proto input: **kúrną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**kórn*
PNWGmc U Lowering	**kórną*		
Early Anglo-Frisian			
[no change]			

Old English form: *corn*

Reconstruction and comparative evidence Kroonen cites the noun as **kurna-*, and Orel gives the citation form **kurnan* ‘corn’, both with Old English *corn* ‘corn, grain’ among the reflexes (Kroonen 2013; Orel 2003, 264). The singular form **kúrną* ‘corn’ is the nominative-accusative singular appropriate to the citation noun.

Old English evidence Clark Hall gives *corn n. ‘corn,’ grain*, Bright’s glossary lists *corn, n.* with genitive singular *cornes*, and Bosworth-Toller treats *corn* ‘corn, grain’ as an ordinary noun headword (Clark Hall 1960; Bright 1971, 347; Bosworth and Toller 1898, 144). The target is therefore an attested citation form, while forms such as *cornes* simply provide paradigm background.

Development to Old English With northwest Germanic lowering, **kúrną* ‘corn’ becomes **kórną* ‘corn’, and later loss of final nasal after a heavy syllable yields **kórn* ‘corn’, whence *corn* ‘corn, grain’. The oblique form **kurnăn* ‘corn’ belongs to comparative background rather than to the derivational input of this entry.

deed — OE *dǣd*

Derivation: **dǽdiz* > *dǣd* (regular).

Derivation trace Proto input: **dǽdiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**dǣd*
PNWGmc Long E Lowering	**dādiz*		
EAF Final Z Deletion	**dādi*		
EAF Long A Fronting	**dǣdi*		
Early Anglo-Frisian			
[no change]			

Old English form: *dǣd*

Reconstruction and comparative evidence Orel reconstructs the noun as **dēdiz* ‘deed’, and Ringe and Taylor derive the same inherited i-stem from Proto-Germanic **dédiz* ‘deed’ through northwest Germanic **dadiz* ‘deed’ (Orel 2003; Ringe and Taylor 2014). The acute accent in **dédiz* ‘deed’ marks stress on the same reconstructed long vowel.

Old English evidence Campbell states that Primitive Germanic *ē* appears as West Saxon *ǣ* but in other Old English dialects mostly as *ē*, and Brunner gives the contrast explicitly as West Saxon *dǣd* ‘deed’ beside non-West-Saxon *dēd* ‘deed’ (Campbell 1959; Brunner 1965). Clark Hall likewise lists *dæd* ‘deed’ and cross-refers Anglian *dēd* to it (Clark Hall 1960). West Saxon *dǣd* is therefore the relevant Old English form here, with Anglian *dēd* as a dialectal doublet.

Development to Old English From inherited **dēdiz* ‘deed’, loss of final *-z* and the West Saxon lowering of stressed long *ē* yield *dǣd* ‘deed’; Anglian *dēd* ‘deed’ preserves the non-West-Saxon outcome (Campbell 1959; Brunner 1965). The development treated here is therefore the regular West Saxon line.

dew — OE *dēaw*

Derivation: **dáwwō* > *dēaw* (regular).

Derivation trace Proto input: **dáwwō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**dáeuwu*
OE WW Simplification	**dáuwō*	OE Diphthong Leveling	**dēawu*
PNWGmc Final Long O Raising	**dáuwu*	OE High Vowel Apocope	**dēaw*
Early Anglo-Frisian			
[no change]			

Old English form: *dēaw*

Reconstruction and development The selected PGmc **dáwwō* gives OE *dēaw* ‘dew’. Its inherited geminate is reanalyzed early: short **a* plus **ww* becomes **au* with retained consonantal **w*. The nucleus subsequently undergoes English fronting and offglide realization. Early reanalysis must therefore be distinguished from the later long English surface diphthong (Ringe and Taylor 2014, 65–66, 171–75; Campbell 1959, 45–47).

The current path is SC031 OEWWSimplification, then SC030 OEAuBrightening, then SC032 OEDiphthongLeveling. The unchanged reconstruction and final form no longer depend on a late unconditional ww deletion followed by singleton-aw lengthening. The selected evaluation uses the explicitly declared strong-final citation context; lexical accent does not infer sentence stress (Ringe and Taylor 2014, 55, 57–58).

door — OE *dor*

Derivation: **dúrą* > *dor* (regular).

Derivation trace Proto input: **dúrą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**dór*
PNWGmc U Lowering	**dórą*		
Early Anglo-Frisian			
[no change]			

Old English form: *dor*

Reconstruction and comparative evidence Kroonen reconstructs a neuter **dura-* ‘gate, (single) door’ and cites Old English *dor* ‘door’ among its reflexes. In the same entry he separates Old English duru ‘door’, Old Frisian *dore*, and Old High German *tura* ‘door’ as reflexes of **durō-* instead (Kroonen 2013).

Old English evidence Clark Hall records *dor* ‘door’ as a neuter noun and separately records feminine duru ‘door’ with its own inflection (Clark Hall 1960). Ringe and Taylor likewise treat duru ‘door’ as an early Old English u-stem, originally a root noun shifted into that class (Ringe and Taylor 2014). The Old English form here is therefore the attested neuter *dor* ‘door’, while duru ‘door’ remains a parallel Old English reflex from another stem history.

Development to Old English From **dúrą* ‘door’, Northwest Germanic u-lowering gives **dórą* ‘door’, and heavy-syllable nasal apocope then yields *dor* ‘door’. The regular development treated in this entry is therefore **dúrą* > *dor*; the feminine duru ‘door’ belongs to the separate line identified by Kroonen and Ringe-Taylor (Kroonen 2013; Ringe and Taylor 2014).

fare — OE *faran*

Derivation: **fáraną* > *faran* (regular).

Derivation trace Proto input: **fáraną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**færaną*
[no change]	OE A Restoration	**faraną*
	OE Heavy Syllable Nasal Apocope	**faran*
Early Anglo-Frisian	OE Secondary Nasalization	**farąn*
[no change]	OE Weak Tail Reduction	**faran*

Old English form: *faran*

Reconstruction and comparative evidence Kroonen gives the inherited strong verb as **faran-*, and Orel gives the same lexeme as **faranan* ‘fare’, both with Old English *faran* ‘fare’ among the reflexes (Kroonen 2013; Orel 2003, 132). Campbell also uses *faran* ‘fare’ as a standard example of Old English A-restoration (Campbell 1959, 61).

Old English evidence Clark Hall lemmatizes the strong verb as *faran* ‘fare’ and separately records weak *færan* ‘to frighten’; *fære* ‘fare’, *færst* ‘fare’, and *færð* ‘fare’ belong to present-tense forms of *faran* rather than to the infinitive itself (Clark Hall 1960). Bosworth-Toller preserves the same distinction (Bosworth and Toller 1898, 108). The Old English form here is therefore the attested citation infinitive *faran* ‘fare’.

Development to Old English From **fáraną* ‘fare’, Anglo-Frisian brightening first gives **færaną* ‘fare’, but A-restoration before single *r* returns **faraną* ‘fare’; later apocope and weak-tail reduction yield *faran* ‘fare’ (Campbell 1959, 61). Fulk’s contrast with participial faren- < **farœn-* < **faran-* shows why fronting elsewhere in the paradigm does not alter the infinitive headword (Fulk 2018).

fell — OE *fell*

Derivation: **féllą* > *fell* (regular).

Derivation trace Proto input: **félląˀ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**féll*
[no change]			
Early Anglo-Frisian			
[no change]			

Old English form: *fell*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **fella-* 'membrane, skin, hide' and cites Old English *fell* 'skin, fell' beside Dutch *vel* 'skin, fell' and German *Fell* (Kroonen 2013). The derivational input **félląˀ* 'fell' is the derivable singular form of that same inherited noun.

Old English evidence Clark Hall records *fell* 'skin, fell' as the noun 'fell, skin, hide', and Bright's glossary likewise gives *fell* 'skin, fell' with inflected forms such as accusative singular *fel* and dative plural *fellum* (Clark Hall 1960; Bright 1971). The target is therefore the attested noun *fell* 'skin, fell', not the verb *fellan* or the preterite *feoll*.

Development to Old English With **félląˀ* 'fell', no special earlier reshaping is needed: heavy-syllable nasal apocope yields **féll* 'fell', surfacing as *fell* 'skin, fell'. The regular development treated here is therefore **félląˀ* > *fell*.

fern — OE *fearn*

Derivation: **fárnaz* > *fearn* (regular).

Derivation trace Proto input: **fárnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fárn*
EAF Final Z Deletion	**fárna*	EAF Brightening	**fœrn*
Early Anglo-Frisian		OE Breaking	**fearn*
[no change]			

Old English form: *fearn*

Reconstruction and comparative evidence Kroonen cites the noun as masculine **farna-* and gives Old English *fearn, fern*, while Orel gives the same lexeme as neuter **farnan* 'fern' with Old English *fearn* 'fern' (Kroonen 2013; Orel 2003, 133). Those are comparative headword conventions rather than competing Old English outcomes; the modeled input here is the nominative-style **fárnaz* 'fern'.

Old English evidence Clark Hall gives *fearn* 'fern' as an Old English noun, and Bosworth-Toller records *fearn* 'fern' with inflected forms such as *fearnes*, *fearna*, and *fearne* (Clark Hall 1960; Bosworth and Toller 1898, 219). Kroonen also records *fern* 'fern', but the local lexical sources give stronger support to *fearn* 'fern' as the citation target (Kroonen 2013).

Development to Old English From **fárnaz* 'fern', loss of final *-z* and final *-a* gives **fárn* 'fern'; Anglo-Frisian brightening then yields **fœrn* 'fern', and breaking before *r* plus consonant gives *fearn* 'fern' (Campbell 1959; Ringe and Taylor 2014). The development treated here is therefore the regular *rC*-breaking line.

field — OE *feld*

Derivation: **félθuz* > *feld* (regular).

Derivation trace Proto input: **félθuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**féld*
EAFL Th Voicing	**félduz*		
EAF Final Z Deletion	**féldu*		
Early Anglo-Frisian			
[no change]			

Old English form: *feld*

Reconstruction and comparative evidence Ringe and Taylor treat Old English *feld* ‘field’ as one of the cases where earlier **felþu- ~ *feldu-* may reflect either inherited *þ ~ *d* alternation or the regular West Germanic development **lþ* > *ld* (Ringe and Taylor 2014, 170). The broader proto label **félθuz* ‘field’ can remain as comparative background, while that narrower historical ambiguity does not affect the regular classification.

Old English evidence Clark Hall records *feld* ‘field’ with oblique forms such as *felda* and *felde*, and Campbell notes early place-name spellings in *-felth* beside the later standard form (Clark Hall 1960, 114; Campbell 1959, 169). The Old English form here is therefore the attested citation noun *feld* ‘field’, with the older *-felth* spellings as historical support rather than as rival targets.

Development to Old English In the modeled pathway, medial **lþ* becomes *ld*, final *-z* is lost, and high-vowel apocope then yields *feld* ‘field’. Whether the voiced dental ultimately reflects inherited alternation or the regular **lþ* > *ld* development, both accounts converge on the same Old English form (Ringe and Taylor 2014, 170).

filth — OE *fȳlþ*

Derivation: **fūliþō* > *fȳlþ* (regular).

Derivation trace Proto input: **fūliθō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**fȳliθu*
PNWGmc Final Long O Raising	**fūliθu*	OE High Vowel Apocope	**fȳliθ*
Early Anglo-Frisian		OE Medial Syncope	**fȳlθ*
[no change]			

Old English form: *fȳlþ*

Old English evidence Old English fȳlþ ‘filth, foulness, impurity’ is a feminine abstract noun on the adjective *fūl* ‘foul’. The Toller Supplement lemmatizes fylþ ‘filth’ without marking vowel quantity, in both physical and moral senses: it glosses Latin *putor* in the Aldhelm glosses (An. Ox. 3323) and *putredo* in the *Liber scintillarum* (Scint. 38, 19), and the entry also records the inflected form fylþu ‘filth’ (An. Ox. 1790) (Toller 1921, 276). Clark Hall lemmatizes fýlð ‘filth’ as feminine and marks the long vowel (Clark Hall 1960, 125). The Old English form here *fȳlþ* is the endingless nominative singular, normalized with marked length after Clark Hall and with thorn for the dental fricative; the substitution of thorn for eth changes no segment. The inflected *fylþu* beside endingless *fylþ* is the familiar alternation of the *-iþō* abstracts, seen also in *strengþu* ‘strength’ beside *hǣlþ* ‘health’, *þīefþ* ‘theft’ and *wiermþu* ‘warmth’ (Campbell 1959, 236, §589.6).

Reconstruction and comparative evidence Kroonen reconstructs the adjective base **fūla-* 'foul, rotten' and cites among its reflexes an exact West Germanic abstract, Old Saxon fulitha 'filth' (Kroonen 2013, 158). Orel likewise prints the base adjective **fūlaz*, with OE *fūl* 'foul, dirty, rotten' among its reflexes, but lists no derived abstract (Orel 2003, 121). The Toller entry itself cites the Old Saxon noun together with Old High German fülida 'filth' (Toller 1921, 276). Bammesberger documents the suffix type: the feminine abstracts in **-iþō* show the vowel **-i-* before the dental throughout, as in **mēriþō* > *mǣrþu* 'fame' and **hauhiþō* > *hīehþu* 'height' (Bammesberger 1990, 117–18, §4.5).

The derivational input **fūliþō* 'filth' is therefore an assembled morphological reconstruction: the attested base **fūla-* composed with the attested suffix **-iþō*. No single authority prints the composite verbatim; its two members are each independently supported, and the Old Saxon and Old High German nouns attest the derivative itself on the continent. Those cognates retain the medial vowel, so the medial loss seen in Old English is an English development, not a common West Germanic one.

Development to Old English The adopted derivation is regular throughout. SC019, final long-*ō* raising gives **fūliþō* > **fūliþu*. SC055, i-umlaut fronts the root vowel over the medial syllable, **fūliþu* > **fȳliþu*. SC063, high-vowel apocope removes the final high vowel after the heavy sequence, **fȳliþu* > **fȳliþ*, and SC065, medial syncope deletes the now-medial **i* before the dental, **fȳliþ* > *fȳlþ*.

The word is a live chronological witness. Had medial syncope preceded i-umlaut, the **i* that conditions the umlaut would have been deleted first, and the outcome would have been unattested **fūlþ* rather than attested *fȳlþ*; umlaut must therefore precede this syncope. The same ordering is visible across the *-iþō* class, whose members uniformly surface with umlauted root vowels beside syncopated suffixes (Campbell 1959, 236, §589.6; Bammesberger 1990, 117–18, §4.5).

Campbell derives the endingless nominative singular of these abstracts analogically, after the *lār* type, with the medial syncope taken as given for the class (Campbell 1959, 236, §589.6). The adopted derivation reaches the same surface form by the regular phonological route, through raising, apocope and syncope, without an analogical step. Both accounts converge on the attested form, and the corpus output does not decide between them; the alternative is recorded rather than suppressed. The inflected *fylþu* beside *fylþ* shows that the paradigm retained vocalic case forms either way.

flea — OE *flēah*

Derivation: **fláuxz* > *flēah* (regular).

Derivation trace Proto input: **fláuxz*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**fláeux*
[no change]	OE Diphthong Leveling	**flēax*
Early Anglo-Frisian		
[no change]		

Old English form: *flēah*

Reconstruction and comparative evidence This word carries a genuine stem-class dispute, not merely a notational one. Orel reconstructs a root noun with his class-wide explicit nominative marker, **flauxz* 'flea' (Orel 2003, 105). Kroonen instead reconstructs an ō-stem **flauhō-* 'flea', with no *-z* at any layer of the analysis (Kroonen 2013, 145). Kluge and Seebold offer yet a third analysis, a masculine a-stem **flauha-* 'flea' (Kluge 2011, 304), whose nominative singular **flauhaz* would have carried its *-z* in an unstressed syllable and lost it under the West Germanic unstressed loss rather than with the root nouns. The attested forms do not decide the question: Old English *flēah* 'flea' and Old Norse *fló* 'flea' are compatible with all three reconstructions. This is a case

where reputable sources disagree about the morphological class itself, and the disagreement is documented here rather than silently normalized away.

The derivational input **fláuxz* ‘flea’ adopts Orel’s root-noun analysis with its morphologically explicit nominative marker. Under Kroonen’s ō-stem analysis the nominative would instead have ended in **-ō* and developed like the other ō-stems; the choice of Orel’s form keeps the word in the root-noun class alongside ‘book’, ‘goose’, and ‘louse’, whose endingless West Germanic nominatives it shares (Ringe and Taylor 2014, 118).

Old English evidence Old English *flēah* ‘flea’ is the endingless nominative-accusative singular. The root-noun class to which Orel assigns it shows no nominative ending anywhere in West Germanic (Ringe and Taylor 2014, 118); on Kroonen’s ō-stem analysis the endingless form would instead reflect the regular fate of the ō-stem nominative in this shape. Either way the attested Old English form is the expected one, which is precisely why the class question remains open.

Development to Old English On the selected root-noun analysis, the nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118), giving **fláux* ‘flea’ from the derivational input **fláuxz* ‘flea’. The diphthong is then fronted and levelled in Old English, yielding **flēax* ‘flea’, spelled *flēah* ‘flea’ with final *h* for the fricative. The early loss of the root-noun marker is historically distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; the derivational input is a stressed monosyllable, but its **-z* stands after a consonant and was gone long before the northern change, whose domain is vowel-final monosyllables.

fly — OE *flēogan*

Derivation: **fléuganą* > *flēogan* (regular).

Derivation trace Proto input: **fléuganą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**flēoganą*
[no change]	OE Heavy Syllable Nasal Apocope	**flēogan*
	OE Secondary Nasalization	**flēogą̄n*
Early Anglo-Frisian	OE Weak Tail Reduction	**flēogan*
[no change]		

Old English form: *flēogan*

Reconstruction and comparative evidence Ringe and Taylor derive the verb as **fleugana* ‘fly’ > OE *fléogan* ‘fly’ and elsewhere contrast West Saxon *fléogan* ‘fly’ with Anglian *flégan* ‘fly’, alongside related forms noted in the form note below (Ringe and Taylor 2014). The derivational input **fléuganą* ‘fly’ represents that inherited strong verb in the notation used here.

Old English evidence Clark Hall and Bosworth-Toller record *flēogan* ‘fly’ as the ordinary Old English strong verb, and Bright gives the familiar paradigm *flēag* ‘flew’, *flugon* ‘flew’, *flogen* ‘flown’ with present *fleogeð* ‘flies’ (Clark Hall 1960; Bosworth and Toller 1898; Bright 1971). The target in this entry is therefore the attested verbal infinitive itself.

Form note Ringe and Taylor also list related *fléoge* ‘fly’ / *flége* ‘fly’ and Anglian *flégan* ‘fly’, which belong to the same family but do not replace the infinitive *flēogan* ‘fly’ treated here (Ringe and Taylor 2014).

Development to Old English From **fléuganą* ‘fly’, Old English diphthong leveling gives **flēoganą* ‘fly’; heavy-syllable nasal apocope and weak-tail reduction then yield *flēogan* ‘fly’ (Ringe and Taylor 2014). The development is therefore regular: **fléuganą* > *flēogan* ‘fly’.

forlorn — OE *lēosan*

Derivation: **léusanǫ* > *lēosan* (regular).

Derivation trace Proto input: **léusanǫ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Diphthong Leveling	**lēosanǫ*
[no change]		OE Heavy Syllable Nasal Apocope	**lēosan*
		OE Secondary Nasalization	**lēosǫn*
Early Anglo-Frisian		OE Weak Tail Reduction	**lēosan*
[no change]			

Old English form: *lēosan*

Reconstruction and comparative evidence Kroonen reconstructs the verb under **leusan-* and cites prefixed daughters such as Gothic *fra-liusan* ‘lose’ and Old English *for-lēosan* ‘lose, forlorn’; Orel likewise gives Old English *for-leósan* ‘lose’ (Kroonen 2013; Orel 2003). The inherited verbal base is therefore clear, though the daughter set often appears with the prefix.

Old English evidence The direct Old English evidence behind English *forlorn* lies in the prefixed verb *forlēosan* ‘lose, forlorn’ and especially in the participle *forloren* ‘forlorn’, recorded by Ringe and Taylor and in the dictionaries (Ringe and Taylor 2014; Clark Hall 1960; Bosworth and Toller 1898). The simplex infinitive *lēosan* ‘lose’ represents the verbal base itself.

Form note As a base-form comparison, the simplex infinitive is *lēosan* ‘lose’, while the English adjective continues the prefixed Old English family *forlēosan* ‘lose’ / *forloren* ‘forlorn’ (Ringe and Taylor 2014).

Development to Old English From **léusanǫ* ‘lose’, Old English diphthong leveling gives **lēosanǫ* ‘lose’, and later nasal apocope and weak-tail reduction yield *lēosan* ‘lose’ (Ringe and Taylor 2014). The prefixed forms follow the same verbal base with added *for-*.

four — OE *fēower*

Derivation: **fédwōr* > *fēower* (regular).

Derivation trace Proto input: **fédwōr*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Diphthong Leveling	**fēowar*
PWGmc Final Or Lowering	**fédwar*	EAF Brightening Unstressed	**fēowær*
PWGmc Coronal W Assimilation	**féwwar*	OE Unstressed AE Merger	**fēower*
OE WW Simplification	**féuwar*		
Early Anglo-Frisian			
[no change]			

Old English form: *fēower*

Reconstruction and development The selected PGmc **fédwōr* gives OE *fēower* ‘four’. Coronal-w assimilation creates the geminate; short **e* plus **ww* is then reanalyzed as an **eu* nucleus with retained consonantal **w*. Ringe and Taylor explicitly use this history alongside the oblique plural pronoun *ēow* ‘you’ (Ringe and Taylor 2014, 41–42).

Thus SC008 PWGmcCoronalWAssimilation feeds SC031 OEWWSimplification. The earlier product reaches English long-diphthong realization through SC032 OEDiphthongLeveling,

rather than receiving an English surface vowel in the West Germanic reanalysis (Ringe and Taylor 2014, 65–66, 171–75). The selected context is the ordinary strong-final citation convention, not the weak-final selection of the pronoun. No reconstruction or final target has changed.

gang — OE *gang*

Derivation: **gángaz* > *gang* (regular).

Derivation trace Proto input: **gángaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**gáng*
EAF Final Z Deletion	**gánga*		
Early Anglo-Frisian			
[no change]			

Old English form: *gang*

Reconstruction and comparative evidence Orel reconstructs the noun as **gangaz* ‘gang’ and cites Old English *gang* ‘going, way’ beside Old Norse *gangr*, Old Frisian *gang* / *gong*, Old Saxon *gang* ‘going, way’, and Old High German *gang* ‘going, way’ (Orel 2003). The derivational input **gángaz* ‘gang’ is the same lexeme in the accent notation used here.

Old English evidence Clark Hall and Bosworth-Toller both record *gang* ‘going, way’ as the noun ‘going, journey, way’, and Bright’s glossary gives *gong (gang), m., path, course* (Clark Hall 1960; Bosworth and Toller 1898, 159; Bright 1971, 392). The target is therefore the attested noun headword itself.

Form note This entry concerns the noun *gang* ‘going, way’, not the separate verb *gangan* (Clark Hall 1960; Bosworth and Toller 1898, 159).

Development to Old English From **gángaz* ‘gang’, loss of final *-z* gives **gánga* ‘gang’, and later loss of final bare *-a* yields *gang* ‘going, way’. The development is therefore regular: **gángaz* > *gang*.

gift — OE *ġift*

Derivation: **gíftiz* > *ġift* (regular).

Derivation trace Proto input: **gíftiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Palatalization	**jífti*
EAF Final Z Deletion	**gífti*	OE High Vowel Apocope	**jíft*
Early Anglo-Frisian		OE Palatal Fricative Merger	**jíft*
[no change]			

Old English form: *ġift*

A. The selected reconstruction and the question it answers Old English *ġift* ‘gift’ is a small word with unusually important chronological consequences. A simple high vowel follows an initial palatal consonant, whereas *ġiefan* ‘give’ shows the diphthongal treatment associated with that consonant. To explain the difference we must ask which vowel was present when ordinary palatal diphthongization operated, not simply which order of rules can reproduce the final

spellings. Campbell lists the short-i gift headword among his initial-palatal examples (Campbell 1959, 173–74, §427).

We select PGmc **giftiz* ‘gift’ following Ringe’s explicit reconstruction (Ringe 2017, 135). Our acute root-stress convention does not add a different vowel or claim that the source prints an accent. Both the comparative reconstruction and the selected computational input use the raised vowel. This is a regular inherited derivation, not a late analogical substitute input. The decision is nevertheless not a claim that every etymological source prints the same reconstruction. A substantial e-form tradition must be explained and preserved rather than suppressed.

B. Orel: the full e-vowel noun reconstruction Orel’s noun entry gives an e-vowel reconstruction, normalized here as **geftiz* ‘gift’, and connects it with the verb ‘give’ (Orel 2003, 130). It has the full iz ending, not merely a bare root or a stem ending in a hyphen. His evidence includes the Old English gift headword and the Old Frisian counterpart *jeft* ‘gift, privilege’, alongside the wider inherited formation. The e is plainly printed in the original page: it is not an OCR accident or a vowel introduced by our transcription. Only the initial velar sign is normalized to g here; no argument about stop versus fricative history is drawn from that typographic normalization.

Orel thus provides genuine published support for the older e-input used in our model. However, the entry does not explain why e is reconstructed despite the high-vowel reflex, identify a specific later raising date, or argue against the early raising account. Its derivation from ‘give’ supplies an etymological relationship; that alone does not demonstrate that the noun’s realized root vowel remained identical to the verb’s at every historical stage. The cited bibliography points to morphology and verbal etymology treatments, including Bammesberger and Seebold, considered below (Orel 2003, 130). Those references are evidence of the scholarly tradition behind the entry, not a license to attribute an unstated phonological argument to Orel.

C. Kluge–Seebold: an e-vowel feminine stem Kluge–Seebold gives **gefti-* ‘gift’, explicitly a Germanic feminine stem, and relates the noun to ‘give’ (Kluge 2011, 359). This is independent confirmation that an e-vowel etymological presentation is established in major reference works. It is not the same printed object as Orel’s full nominative reconstruction. We must not silently append z and then present the expanded form as a quotation from Kluge–Seebold.

The entry cites the Old English gift headword and the Old Frisian form *jeft(e)* ‘gift’ within the inherited family (Kluge 2011, 359). Its principal discussion concerns the word’s semantic history; it does not supply a worked chronology of root-vowel raising versus palatal diphthongization. Consequently the entry establishes the stem and its etymological affiliation more directly than it establishes the vowel quality at a narrowly dated phonological node. This does not prove that the stem notation is purely abstract. It means that such an interpretation, if adopted, would require an argument beyond the entry.

D. Bammesberger: the e-grade derivational analysis Bammesberger places the formation among ti-stems built on an e-grade verbal root and prints a segmented e-vowel form (Bammesberger 1990, 142, §5.4.2). His discussion relates the noun to the Germanic root of ‘give’ and treats the consonantal sequence ft as regular in the ti-abstract formed from a labial-final root. On the same page the formation also participates in his explanation of the structural pattern of other abstracts. The continuation lists further e-grade formations with Old English i-vowel reflexes (Bammesberger 1990, 142–43).

This is a substantive reason for retaining e in the derivational analysis: the noun belongs to an e-grade formation, not an arbitrarily invented high-vowel root. But derivational grade and realized surface quality after a regular conditioned sound change are different questions. An e-grade formation can have a later i-reflex without becoming a zero-grade formation. The passage supports the morphological analysis; it does not itself decide that the inherited e survived until Old English mutation or state a competing date for raising. We therefore retain its e-grade evidence in the deeper explanation without using it alone to override a separately argued phonological reconstruction (Bammesberger 1990, 142–43).

E. Seebold: segmented forms and the limits of notation Seebold prints a segmented e-vowel formation with the full nominal ending and supplies more differentiated Old English evidence than a single normalized citation (Seebold 1970, 218). He distinguishes *giftu* 'wedding', explicitly plural, *gyft* 'marriage', and *feoh-gift* 'gift of money'. A separate homonymous noun associated with 'food' belongs to a different formation and must not be recruited as attestation for the noun treated here. These distinctions matter: the formation's identity is well supported, but a plural or compound citation is not automatically evidence for a particular singular manuscript token.

His methodological introduction also makes clear that the reconstructed notation is not uniformly a direct transcription of one phonological moment. Earlier states are reconstructed for selected developments, including the development of ei to long i (Seebold 1970, 35). The symbols for stem classes are modelled on common-Germanic forms without claiming exact phonetic reproduction (Seebold 1970, 35–36). These explicit cautions make it particularly important not to read every segment mechanically as an input to a later sound-change cascade. They do not, however, specifically explain the root e in this noun: the warning about stem-class endings cannot be generalized into proof that all reconstructed root vowels are abstract morphological labels. We leave that interpretive limit visible rather than declaring the e/i discrepancy merely notational.

F. Ringe: the realized PGmc i-form and its phonological rationale Ringe explicitly gives PGmc **giftiz* 'gift' as the nominal formation associated with the verb 'give' (Ringe 2017, 135). His discussion places it in the broader treatment of productive nominal formation and consonantal developments. The Old English comparison is plural *ġifta* 'wedding', with a change of stem class. We do not turn that plural into a claim that Ringe attests our normalized singular target.

The vowel has a separate phonological explanation. Ringe states that e was raised to i when a high front vocalic followed in the same or immediately succeeding syllable (Ringe 2017, 151–52). The noun's following i supplies precisely such an environment. This explains how an etymological e-grade formation can be represented with i in a realized phonological reconstruction. It is not a lexical exception, a grammatical condition on nouns, or a repair licensed only because this one word otherwise produces an unwanted diphthong.

Ringe is more cautious about the exact ancestral dating than a bare reconstruction might suggest. The merger of stressed e and i in Gothic prevents direct demonstration of the raising there. His indirect argument depends on raising preceding j-loss and subsequent contraction, particularly the history of the numeral 'three'. He calls a PGmc date the most probable hypothesis, not completely certain (Ringe 2017, 152–53). That qualification belongs in our account. It concerns the placement of the change in the deeper ancestry; it must not be misrepresented as doubt that his noun entry actually prints an i-vowel reconstruction.

Ringe and Taylor independently insist on separating this inherited raising from Old English front mutation: the older change occurred hundreds of years earlier, and alleged later repetitions are rare and doubtful (Ringe and Taylor 2014, 220). Their discussion therefore supplies an important chronological check against explaining this noun's inherited high vowel by the general Old English mutation rule.

G. Why we choose i, and what changes in the derivation For the input to a PGmc-to-OE cascade we prefer the explicitly staged phonological reconstruction, rather than carrying an earlier etymological e-grade representation forward until a convenient later rule can correct it. Ringe's noun and regular raising account together supply that reconstruction (Ringe 2017, 135, pp. 151–153). The competing reference forms remain part of the evidence: we neither count dictionary spellings as votes nor conclude that their authors were wrong merely because they did not give a worked chronology.

With earlier i already present, the noun undergoes the inherited consonantal and ending developments, then initial palatalization. The singleton initial is a palatal fricative, distinct from gg/ng stops; the adopted model keeps it distinct from inherited j during mutation (Fulk 2018, 130–32; Ringe and Taylor 2014, 204). Ordinary palatal diphthongization does not change its high vowel, and Old English mutation need not raise an e that is no longer there. Loss of the final high vowel and surface spelling yield *ġift* 'gift'. Our earlier implementation instead allowed a selected

e-input to reach late mutation and used that correction to justify putting all palatal diphthongization afterward. The right final spelling concealed the wrongly conflated histories.

The independent ordinary chronology is breaking, then palatal diphthongization, then mutation (Fulk 2018, 74; Ringe and Taylor 2014, 215–17, 222). The worked contrast is *ġiest* 'guest': its low-front vowel first becomes ea after the palatal, and ea then becomes ie under mutation. Gift's earlier i is a negative control on ordinary diphthongization, not an independent proof of that relative order. The later treatment of *sċēaþ* 'sheath' is separately retained after mutation; its mutation-derived long vowel does not require ordinary diphthongization as a whole to be late (Campbell 1959, 68–69; Ringe and Taylor 2014, 235). The complete conditioning of that later treatment remains unresolved.

H. Confidence and the remaining alternatives Our confidence is strongest in the distinction between inherited raising and later OE mutation, and in the independently supported ordinary diphthongization chronology. The choice of Ringe's i-input is well motivated as an explicit working phonological reconstruction, but is not advertised as unanimous: Orel's e-form, Kluge–Seebold's e-stem and the e-grade derivational presentations remain genuine alternatives (Orel 2003, 130; Kluge 2011, 359; Bammesberger 1990, 142–43; Seebold 1970, 218). We have not established that every one of those forms denotes exactly the same historical phonological checkpoint as Ringe's input.

Confidence in the precise PGmc date is qualified by Ringe's own indirect evidence (Ringe 2017, 152–53). Confidence in our normalized target as a headword is distinct from certification of a diplomatic paradigm cell (Campbell 1959, 173–74). Variant spellings alone do not establish a different exceptionless sound law, lexical diffusion, or an analogical history; those explanations would need their own evidence. We make no such new claim here.

The computational comparison is deliberately secondary. The adopted input and ordinary chronology preserve every previously generated final form, while realizing the source sequence for guest and leaving sheath's derivation unchanged. That demonstrates compatibility of the selected account with the present corpus, not its historical truth by output score. If stronger evidence eventually requires an e-vowel at the input node, the necessary earlier regular raising must be specified and tested as such; it must not be silently delegated again to late OE mutation.

give — OE *ġiefan*

Derivation: **gébanǫ* > *ġiefan* (regular).

Derivation trace Proto input: **gébanǫ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**géban*
[no change]	OE Secondary Nasalization	**gébąn*
	PGmc B Allophony	**géβąn*
Early Anglo-Frisian	OE Velar Palatalization	**ǰéβąn*
[no change]	OE Ws Palatal Diphthongization	**ǰíeβąn*
	OE Weak Tail Reduction	**ǰíeβan*
	OE Palatal Fricative Merger	**jíeβan*

Old English form: *ġiefan*

Reconstruction and comparative evidence Kroonen reconstructs the strong verb as **geban-* and cites Old English *giefan* 'give' among its reflexes (Kroonen 2013). Ringe and Taylor contrast West Saxon *giefan* 'give' with Mercian *for-geofan* 'give' and Northumbrian *geafa* 'give', showing that the inherited verb takes different later dialectal shapes (Ringe and Taylor 2014).

Old English evidence Campbell gives *gefan* (W-S *giefan* ‘give’) among examples of initial palatalization, and Clark Hall records the verb under plain *giefan* ‘give’ with forms such as *geaf* and *giefen* (Campbell 1959; Clark Hall 1960). I write *ġiefan* ‘give’ for the palatal initial.

Dialect note West Saxon *ie* here reflects palatal diphthongization after initial palatalization; non-West-Saxon forms such as *geafa* ‘give’ or *for-geofan* ‘give’ continue the same verb without the West Saxon vocalism (Ringe and Taylor 2014).

Development to Old English The singleton initial belongs to the fricative class, not the gg/ng stop class. The adopted consonantal account distinguishes its palatal j̨ from inherited j during mutation and merges them afterwards (Fulk 2018, 130–32; Ringe and Taylor 2014, 204).

From **gébaną* ‘give’, initial *g* palatalizes before *e*; West Saxon palatal diphthongization then yields *ie*, and later tail reduction gives *giefan* ‘give’ (Campbell 1959; Ringe and Taylor 2014). The result is therefore the regular West Saxon infinitive.

gold — OE *gold*

Derivation: **gúlθą* > *gold* (regular).

Derivation trace Proto input: **gúlθą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**góld*
EAFL Th Voicing	**gúldą*		
PNWGmc U Lowering	**góldą*		
Early Anglo-Frisian			
[no change]			

Old English form: *gold*

Reconstruction and comparative evidence Ringe and Taylor cite the noun as **gulþa-* / **gulda-*, and Kroonen gives the same pair (Ringe and Taylor 2014, 42; Kroonen 2013). The derivational input **gúlθą* ‘gold’ preserves the older consonantal form while leaving open whether the medial stop reflects inherited alternation or regular West Germanic development.

Old English evidence Bosworth-Toller and Clark Hall both record *gold* as the ordinary Old English neuter noun (Bosworth and Toller 1898, 121; Clark Hall 1960, 152). The target is therefore the attested citation form itself.

Development note Ringe and Taylor note that the medial stop can be understood either as alternation **gulþa-* / **gulda-* or as the ordinary West Germanic change **lþ* > *ld*; both routes lead to the same Old English consonantism (Ringe and Taylor 2014, 42).

Development to Old English From **gúlθą* ‘gold’, the regular consonant development gives **gúldą* ‘gold’; Northwest Germanic / Old English lowering then yields **góldą* ‘gold’, and apocope gives *gold* (Campbell 1959; Ringe and Taylor 2014, 42).

goose — OE *gōs*

Derivation: **gánsz* > *gōs* (regular).

Derivation trace Proto input: **gánsz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
EAF Nasal Spirant Lengthening	**gãns*	
EAF Nasal Spirant Loss	**gãs*	
EAF Nasalized Low Rounding	**gōs*	
Early Anglo-Frisian		
[no change]		

Old English form: *gōs*

Reconstruction and comparative evidence The ‘goose’ word is a root noun whose nominative singular has an unusually layered history. Orel’s citation form is **gansz* ‘goose’, with his class-wide explicit nominative marker **-z* stacked on the root-final *s* (Orel 2003, 126). Kroonen reconstructs an endingless nominative **gans* ‘goose’ with Verner-voiced *z* only in the oblique cells (genitive **gunzaz* ‘goose (gen.)’) (Kroonen 2013, 168–69), and Kluge/Seebold treat the German cognate under the same root noun (Kluge 2011, 331). Bammesberger supplies the deeper layer: by Szemerényi’s law the Proto-Indo-European nominative **ǵʰanss* yielded **ǵʰān*, after which the Proto-Germanic nominative was rebuilt with a final voiced sibilant, **ganz*, reanalyzed within the paradigm (Bammesberger 1990, 196). A mainstream specialist thus reconstructs a Proto-Germanic ‘goose’ nominative with a final voiced sibilant; Orel’s notation is not idiosyncratic, and Kroonen’s endingless citation reflects a different analytical layer of the same paradigm rather than a contradictory fact.

The derivational input **gánsz* ‘goose’ follows Orel’s morphologically explicit convention, keeping the nominative marker visible so that its historical elimination is modelled rather than presupposed.

Old English evidence Old English *gōs* ‘goose’ is the endingless nominative-accusative singular, with mutated plural *ġēs* ‘geese’ (Fulk 2018, 165–66). The comparative picture matches the rest of the root-noun class: Gothic replaced the word (its ‘goose’ is the u-stem remodeling behind Spanish *ganso* ‘goose’ (Orel 2003, 126)), and no West Germanic daughter shows any nominative ending (Ringe and Taylor 2014, 118).

Development to Old English The root-noun nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118), giving **gáns* ‘goose’ from the derivational input **gánsz* ‘goose’. The remaining development is the familiar Ingvaeonic nasal-spirant sequence: the vowel lengthens and rounds before the nasal + voiceless fricative cluster and the nasal is then lost, giving **gōs* ‘goose’, which is the attested Old English form. The early loss of the nominative marker is historically distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; had the marker survived to the later monosyllabic change, a lengthened vowel before *s* would not be the expected outcome of this shape, since the root-noun ending was already gone.

grave — OE *grafan*

Derivation: **grábaną* > *grafan* (regular).

Derivation trace Proto input: **grábaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**græbanǫ*
[no change]	OE A Restoration	**grabanǫ*
	OE Heavy Syllable Nasal Apocope	**graban*
Early Anglo-Frisian	OE Secondary Nasalization	**grabǫn*
[no change]	PGmc B Allophony	**graβǫn*
	OE Weak Tail Reduction	**graβan*

Old English form: *grafan*

Reconstruction and comparative evidence Campbell gives *grafan* ‘dig, grave’ among the standard examples of Old English a-restoration before a single consonant and a following back vowel, and Ringe and Taylor describe the same development for Class VI infinitives (Campbell 1959, 61; Ringe and Taylor 2014).

Old English evidence Clark Hall records *grafan* ‘dig, grave’ as the verb ‘to dig, grave’ and separately records noun *græf* ‘grave, trench’ (Clark Hall 1960). The target here is the attested infinitive headword of the verb.

Development to Old English From **grábanǫ* ‘grave’, Anglo-Frisian brightening first gives a fronted stem vowel. A-restoration then returns *a* before single *b* plus the back-vocalic infinitive ending, and later apocope and weak-tail reduction yield *grafan* ‘dig, grave’ (Campbell 1959, 61; Ringe and Taylor 2014).

Form note Noun *græf* ‘grave’ and verbal forms such as *grǽfð* ‘graves’ or past participial *grǽfen* ‘graven’ belong to other lexical or paradigm positions and do not replace the infinitive *grafan* ‘grave’ as the target here (Clark Hall 1960).

guest — OE *ġiest*

Derivation: **gástiz* > *ġiest* (regular).

Derivation trace Proto input: **gástiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**gæsti*
EAF Final Z Deletion	**gásti*	OE Velar Palatalization	**ǰæsti*
Early Anglo-Frisian		OE Ws Palatal Diphthongization	**ǰeasti*
		OE I Umlaut	**ǰiesti*
[no change]		OE High Vowel Apocope	**ǰiest*
		OE Palatal Fricative Merger	**jiest*

Old English form: *ġiest*

Reconstruction and comparative evidence Campbell and Ringe-Taylor treat the noun as an ordinary i-stem whose West Saxon development shows palatal diphthongization, while non-West-Saxon evidence preserves forms of the *gest* ‘guest’ type (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller and Clark Hall record the word under forms such as *gist* ‘guest’, *gest* ‘guest’, *giest* ‘guest’, and *gyst* ‘guest’ (Bosworth and Toller 1898; Clark Hall 1960). The Old English form here, *ġiest*, ‘guest’ is the normalized West Saxon form within that attested family.

Development to Old English From **gástiz* ‘guest’, ordinary English fronting supplies the low-front vowel. Initial singleton g palatalizes as a fricative, not an early affricate (Fulk 2018, 130–32). Ordinary West Saxon palatal diphthongization changes æ to ea before mutation changes ea to ie. The high-vocalic ending is lost afterwards; the regular outcome is *ġiest* ‘guest’ (Ringe and Taylor 2014, 215–17, 222, 287).

Dialect note West Saxon *ġiest* ‘guest’ is the Old English form here. Anglian *gest* ‘guest’ and related spellings remain real Old English comparators rather than corrections to that choice (Ringe and Taylor 2014; Bosworth and Toller 1898).

hair — OE *hǣr*

Derivation: **xḗrą* > *hǣr* (regular).

Derivation trace Proto input: **xḗrą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çǣrą*
PNWGmc Long E Lowering	**xārą*	OE Heavy Syllable Nasal Apocope	**çǣr*
EAF Long A Fronting	**xǣrą*		
Early Anglo-Frisian			
[no change]			

Old English form: *hǣr*

Reconstruction and comparative evidence Kroonen cites the ordinary Proto-Germanic hair word as **hēra-* (Kroonen 2013). The derivational input **xḗrą* ‘hair’ represents that same long-ē stem in the present derivation.

Old English evidence Clark Hall and Bosworth-Toller record *hær* ‘hair’ / *hǣr* ‘hair’ as the ordinary Old English noun (Clark Hall 1960, 158; Bosworth and Toller 1898, 510). The target is therefore the attested headword itself.

Development to Old English From **xḗrą* ‘hair’, Northwest Germanic lowering gives a long front vowel, and later loss of the final nasal leaves the Old English form *hǣr* ‘hair’. The development treated here is straightforward and does not require any special paradigm choice.

Form note Older references to **xazwăz* ‘hair’ belong to a different lexeme, and the separate *haddr* / *heordan* / *hād-* material does not displace the ordinary simplex *hǣr* ‘hair’ treated here (Kroonen 2013; Clark Hall 1960, 158).

harvest — OE *hierfest*

Derivation: **xárbistuz* > *hierfest* (regular).

Derivation trace Proto input: **xárbistuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xærbistu*
EAF Final Z Deletion	**xárbistu*	OE Breaking	**xearbistu*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çearbistu*
		PGmc B Allophony	**çearβistu*
[no change]		OE I Umlaut	**çierβistu*
		OE High Vowel Apocope	**çierβist*
		OE Med Unstressed I Lowering1	**çierβest*

Old English form: *hierfest*

Reconstruction and comparative evidence Bammesberger and Ringe-Taylor treat **harbist-* as the inherited base and explain that the regular native West Saxon development would be of the *hierfest* / *hyrfest* type (Bammesberger 1997; Ringe and Taylor 2014).

Old English evidence Bosworth-Toller and Clark Hall record *hærfest* 'harvest', with *herfest* 'harvest' as a variant in the lexical tradition (Bosworth and Toller 1898; Clark Hall 1960). Those attested forms remain the main dictionary evidence for the noun.

Development to Old English From **xárbistuz* 'harvest', Anglo-Frisian brightening, breaking, and i-mutation produce a *hierbist-* stage, and later lowering of unstressed medial *i* to *e* gives *hierfest* 'harvest'. That is the regular West Saxon development treated here (Ringe and Taylor 2014; Campbell 1959).

Source note The Old English form here, *hierfest*, represents the regular native West Saxon outcome discussed by Bammesberger and Ringe-Taylor. The attested Old English lexical tradition, however, is chiefly *hærfest* 'harvest' / *herfest* 'harvest', commonly treated as non-West-Saxon or Anglian material in West Saxon transmission (Bammesberger 1997; Ringe and Taylor 2014).

hedge — OE *heġġ*

Derivation: **xágjaz* > *heġġ* (regular).

Derivation trace Proto input: **xágjaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xággj*
PWGmc J Gemination	**xággjaz*	EAF Brightening	**xæggj*
EAF Final Z Deletion	**xággja*	OE Velar Fricative Palatalization	**çæggj*
Early Anglo-Frisian		OE Velar Palatalization	**çæʤʤj*
[no change]		OE I Umlaut	**çeʤʤj*
		OE J Loss After Heavy	**çeʤʤ*

Old English form: *heġġ*

Reconstruction and comparative evidence The derivational input models a palatal **-gj-** 'hedge' noun whose Old English development includes gemination, palatalization, and i-mutation. The current derivation therefore reaches a palatal-geminate outcome of the *heġġ* 'hedge' type (Campbell 1959).

Old English evidence Bosworth-Toller and Clark Hall record the noun under standard spellings *hecg* 'hedge' / *heċġ* 'hedge' (Bosworth and Toller 1898; Clark Hall 1960). The lexical item itself is therefore well attested even though the form compared here is normalized.

Development to Old English This is the geminate-stop class, distinct from the singleton fricative in *dæġ* 'day'. The model's palatal stop-reflex proxy does not assert that affrication had already occurred at the articulation checkpoint (Ringe and Taylor 2014, 203–4; Fulk 2018, 131–32).

From **xágjaz* 'hedge', West Germanic j-gemination first yields a geminate stop, and later Old English palatalization and loss of final *j* produce *heġġ* 'hedge'. The development is treated as regular rather than exceptional.

Form note Standard dictionary spelling is *hecġ* ‘hedge’ or *hecg* ‘hedge’. Normalized *hegġ* ‘hedge’ is the Old English form here, while the ordinary lexicographic forms remain the main Old English citation evidence (Bosworth and Toller 1898; Clark Hall 1960).

helm — OE *helm*

Derivation: **xélmaz* > *helm* (regular).

Derivation trace Proto input: **xélmaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xélm*
EAF Final Z Deletion	**xélma*	OE Velar Fricative Palatalization	**çélm*
Early Anglo-Frisian			
[no change]			

Old English form: *helm*

Reconstruction and comparative evidence Kroonen cites the helmet noun as **helma-* and separately distinguishes a different lexeme **helman-* ‘rudder’ (Kroonen 2013). The derivational input **xélmaz* ‘helm’ is the nominative-style form used for the helmet noun itself.

Old English evidence Clark Hall and Bosworth-Toller record *helm* ‘helm, helmet’ as the ordinary Old English noun for ‘helmet’, while *helma* ‘helm’ belongs to a separate rudder lexeme (Clark Hall 1960; Bosworth and Toller 1898, 542).

Development to Old English From **xélmaz* ‘helm’, loss of final *z* and later loss of the short final vowel yield *helm* ‘helm, helmet’. The development is therefore a straightforward citation-form match.

Form note Comparative **helma-* is headword notation for the helmet cognate set. It should not be confused with Old English *helma* ‘helm’, which is a different noun meaning ‘helm, rudder’ (Kroonen 2013; Clark Hall 1960).

help — OE *helpan*

Derivation: **xélpaną* > *helpan* (regular).

Derivation trace Proto input: **xélpaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Velar Fricative Palatalization	**çélpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**çélpan*
	OE Secondary Nasalization	**çélpąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**çélpan*
[no change]		

Old English form: *helpan*

Reconstruction and comparative evidence The entry treats the strong verb itself rather than the separate noun *help* ‘help’. Bright’s principal parts *helpan, healp, hulpon, holpen* show the ordinary Old English strong-verb family continued by this input (Bright 1971).

Old English evidence Clark Hall and Bosworth-Toller record *helpan* ‘help’ as the verbal headword ‘to help’ (Clark Hall 1960; Bosworth and Toller 1898, 542). The target is therefore the attested infinitive citation form.

Development to Old English From **xélpanǫ* ‘help’, no special repair is needed beyond the ordinary reduction of the infinitive ending. The derivation therefore reaches *helpan* ‘help’ directly.

Form note Noun *help* ‘help’ belongs to a separate lexical line and should not replace verbal *helpan* ‘help’ as the target here (Clark Hall 1960; Bosworth and Toller 1898, 542).

hew — OE *hēawan*

Derivation: **xáwwanǫ* > *hēawan* (regular).

Derivation trace Proto input: **xáwwanǫ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**xáeuwanǫ*
OE WW Simplification	**xáuwanǫ*	OE Diphthong Leveling	**xēawanǫ*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çēawanǫ*
		OE Heavy Syllable Nasal Apocope	**çēawan*
[no change]		OE Secondary Nasalization	**çēawǫn*
		OE Weak Tail Reduction	**çēawan*

Old English form: *hēawan*

Reconstruction and development PGmc **xáwwanǫ* gives OE *hēawan* ‘hew’. Its inherited short-vowel-plus-geminate sequence yields **au* plus retained consonantal **w* in the earlier West Germanic reanalysis. English fronting and realization then produce the long diphthong. The geminate is not simply deleted at the later English checkpoint (Ringe and Taylor 2014, 65–66, 171–75; Campbell 1959, 45–47).

Together with dew, this word follows SC031 OEWWSimplification, SC030 OEAuBrightening, and SC032 OEDiphthongLeveling. It contrasts with the j-created geminates of hay and reconstructed West Saxon strew, which retain their independently governed resolution. The earlier and later components are separate even though the final spelling is unchanged (Campbell 1959, 45–47; Ringe and Taylor 2014, 53, 173).

hind — OE *hind*

Derivation: **xéndjō* > *hind* (regular).

Derivation trace Proto input: **xéndjō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Velar Fricative Palatalization	**çéndju*
PNWGmc Final Long O Raising	**xéndju*	OE I Umlaut	**çindju*
Early Anglo-Frisian		OE High Vowel Apocope	**çindj*
		OE J Loss After Heavy	**çind*
[no change]			

Old English form: *hind*

Reconstruction and comparative evidence Kroonen cites the animal name as **hindō-* f. ‘hind’ (Kroonen 2013). The derivational input **xéndjō* ‘hind’ represents that same noun in the present derivation.

Old English evidence Clark Hall and Bosworth-Toller record *hind* ‘hind, doe’ as the noun ‘hind, female deer’ (Clark Hall 1960; Bosworth and Toller 1898, 554). The target is therefore the attested lexical item itself.

Development to Old English From **xéndjō* ‘hind’, i-mutation produces the front-vocalic Old English stem, and later apocope plus loss of final *j* yield *hind*. The outcome is therefore a regular citation-form derivation.

Form note *hindan* ‘from behind, behind’ is a different Old English lexeme and does not belong to the noun history of *hind* (Clark Hall 1960; Bosworth and Toller 1898, 554).

hold — OE *healdan*

Derivation: **xáldaną* > *healdan* (regular).

Derivation trace Proto input: **xáldaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**xældaną*
[no change]	OE Breaking	**xealdaną*
	OE Velar Fricative Palatalization	**çealdaną*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**çealdan*
[no change]	OE Secondary Nasalization	**çealdą̊n*
	OE Weak Tail Reduction	**çealdan*

Old English form: *healdan*

Reconstruction and comparative evidence Campbell and Ringe-Taylor treat the verb as a regular *a* + *lC* breaking case, with West Saxon *healdan* ‘hold’ opposed to Anglian and Mercian *haldan* ‘hold’ (Campbell 1959; Ringe and Taylor 2014).

Old English evidence Bright gives the ordinary strong-verb citation form and principal parts *healdan, heold, heoldon, healden* (Bright 1971). The target is therefore the attested infinitive headword itself.

Development to Old English From **xáldaną* ‘hold’, Anglo-Frisian brightening first yields a fronted vowel, and West Saxon breaking then produces *ea* before *ld*. Later reduction of the infinitive ending gives *healdan* ‘hold’ (Campbell 1959; Ringe and Taylor 2014).

Dialect note West Saxon *healdan* ‘hold’ is the Old English form here. Anglian and Mercian *haldan* ‘hold’ are genuine non-West-Saxon doublets rather than corrections to that choice (Campbell 1959; Ringe and Taylor 2014).

horn — OE *horn*

Derivation: **xúrną* > *horn* (regular).

Derivation trace Proto input: **xúrną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**xórn*
PNWGmc U Lowering	**xórną*		
Early Anglo-Frisian			
[no change]			

Old English form: *horn*

Reconstruction and comparative evidence Kroonen and Orel cite lemma-style Proto-Germanic headwords of the **hurna-* / **xurnan* ‘horn’ type for this noun (Kroonen 2013; Orel 2003, 234). The derivational input **xúrną* ‘horn’ is the nominative-style form used in the derivation here.

Old English evidence Clark Hall, Bosworth-Toller, and Bright all record *horn* as the ordinary Old English noun (Clark Hall 1960; Bosworth and Toller 1898, 108; Bright 1971). The target is therefore the attested citation form.

Development to Old English From **xúrną* ‘horn’, Northwest Germanic u-lowering gives **xórną* ‘horn’, and later loss of the final nasal leaves *horn*. The development treated here is fully regular.

Form note The note’s oblique **xurnăn* ‘horn’ belongs to comparative stem background only. It does not replace the derivational input **xúrną* ‘horn’ as the derivational form used here (Kroonen 2013; Orel 2003, 234).

hue — OE *hīew*

Derivation: **xíwją* > *hīew* (regular).

Derivation trace Proto input: **xíwją*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Ew Long Diphthong	**xíuwją*
PWGmc J Gemination	**xíwwją*	OE J Glide IO	**xīowją*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**çīowją*
		OE Heavy Syllable Nasal Apocope	**çīowj*
[no change]		OE I Umlaut	**çīewj*
		OE J Loss After Heavy	**çīew*

Old English form: *hīew*

Reconstruction and comparative evidence The selected reconstruction is realized Proto-Germanic **xíwją* ‘form, appearance’, a neuter *ja*-stem. It follows Ringe and Taylor’s explicit PGmc **hiwją* ‘form, appearance’, normalized with CAPR’s dorsal *x* and lexical acute (Ringe and Taylor 2014, 250). Both the citation reconstruction and the derivational input use i. This is an adopted working reconstruction, not a claim that all dictionaries print it or that their e forms have been shown to belong to a different historical stage.

The published e reconstructions Orel gives **xewjan* ‘form, appearance’, neuter, with Gothic *hiwi* ‘surface appearance’, Old Norse *hý* ‘down on plants’ and Old English *hīw* ‘form’. He probably derives the noun from **xawwanan* and lists older etymological alternatives (Orel 2003, 171–72). The e vowel is therefore not an invented CAPR spelling. The dictionary entry supplies a real competing reconstruction; it does not license us to call its e an abstract underlying grade or an automatically pre-PGmc form. His final *-an* must also be distinguished from the other authors’ stem citation or nasal-vowel notation.

Kroonen gives **heuja-*, a neuter stem ‘visible layer, appearance’, on the comparative set Gothic hiwi ‘surface appearance, complexion’, Old Norse hý ‘down on plants’, Norwegian *hy*, and Old English *hīw*, *hēow* ‘shape, form, hue, colour’. He considers an early derivative of **kieh₁-u-*, with **heuja-* probably arising from **kēu-ió-* by Dybo’s law (Kroonen 2013, 224). This is not the same derivational argument as Orel’s. Their agreement on e cannot be treated as two identical histories, and their different endings cannot be silently collapsed into one quoted whole-word form. Writing prevocalic u as w and h as x explains a normalization; it does not by itself explain the disagreement between e and i.

The realized-i account and its early conditioning Ringe and Taylor explicitly begin with PGmc **hiwją*. They distinguish a nongeminated **hiwi* from the stem with palatalized geminate w, and propose its development through **iuw* to pre-OE long **īow*, with a still-palatalizing semivowel, then early West Saxon long *īew*. They call the development surprising and its phonemic analysis unclear (Ringe and Taylor 2014, 250). The i choice is consequently not inferred merely from the final Old English spelling: an author explicitly assigns it to Proto-Germanic and offers a complete subsequent history for this noun.

Ringe separately describes early e-to-i raising before a high front vocalic in the same or following syllable. His example **néwios* to PGmc **niwjaz* 'new' supplies the relevant w+j class; he qualifies the evidence for the precise early date (Ringe 2017, 151–53). This passage explains the inherited-i preference but is not itself a quotation of a whole-word hue reconstruction. Fulk likewise treats PGmc e-to-i, and eu-to-iu, before following i/j separately from later front umlaut (Fulk 2018, 59). Under this working account, the raising precedes the supplied PGmc input. It must not be replaced by a later one-word e-to-i operation simply because that would conceal an input disagreement.

The glide objection and the strength of the choice Campbell distinguishes the traditional awj and iwj developments and prints the hue reflex in the iwj class (Campbell 1959, 46, §120.2). Ringe and Taylor's independent gemination comparison uses new, sew and mirth, including nongeminated beside geminated paradigm stems (Ringe and Taylor 2014, 53). These support distinguishing the consonant classes and successive operations; they do not settle every phonemic detail of the offglide.

Fulk presents a substantive objection to the traditional dismantling account. He questions how w retained the presumed consonantal status in these inputs and how the resulting geminate was dismantled. His discussion considers a different strew input and paradigm regularization for new, rather than simply endorsing every conventional arrow (Fulk 2018, 71–72). His general statement of j-gemination after short vowels, not diphthongs, belongs to that issue (Fulk 2018, 126, §6.15). He does not supply a replacement complete hue history in these passages. Neither his objection nor Ringe and Taylor's uncertainty is removed by our choosing a computable working account.

We prefer realized PGmc i because the explicit reconstruction, independently motivated early raising and complete hue chain form a coherent account of the selected West Saxon reflex. This is not an author majority, an automatic preference for the most recent book, or an argument from successful final spelling. Confidence is stronger in the inherited high-front conditioning and the selected dialectal reflex than in a uniquely recoverable phonemic analysis of palatalized w. The e reconstructions remain alternatives with positive published support; they have not been disproved or silently reclassified.

These differences therefore warrant extended discussion despite a regular selected result: the authors' formation premises, citation endpoints and later phonemic uncertainties are distinct questions. No majority vote or normalization of final spellings resolves them. The existing source-led *i* choice and full glide analysis are retained, not replaced by a summary of dictionary vowels.

Old English evidence Campbell prints hīew 'form' beside *hīow* 'form' in the section that derives this stem type, alongside *nīowe*, *nīewe* 'new' and *glīow*, *glīw* 'mirth' (Campbell 1959, sect. 120.2, p. 46). He distinguishes non-West-Saxon io from West Saxon ie. CAPR selects *hīew* as the source-supported West Saxon counterpart, not as the only attested spelling. Orel gives *hīw*, and Kroonen gives *hīw, hēow* (Orel 2003, 171–72; Kroonen 2013, 224).

Development to Old English Campbell's decisive class contrast is that awj and iwj share gemination but thereafter lose different glides: the low-vowel type yields the palatal consonant, whereas the high-front type retains w and yields io/ie (Campbell 1959, 46, §120.2). For hue, the adopted cascade now follows the independently specified Ringe-Taylor chain rather than using long ēo as an output-compatible proxy for their long īo (Ringe and Taylor 2014, 250).

Step	Selected reconstruction / represented checkpoint	Interpretation
PGmc input	**xíwją*	Inherited i, with early raising already reflected in the reconstruction
J-gemination	**xíwwją*	Palatalized geminate carried as ww+j
J-created reanalysis	**xíuwją*	Offglide formation, retaining palatalized w as w+j
English realization	**xīowją*	Long īo, distinct from ordinary ēo
Palatal articulation and ending loss	**çīowj*	Independent dorsal history and reduction in the existing cascade
Mutation	**çīewj*	Existing long-diphthong mutation conditioned by the retained palatal carrier
Later glide loss and spelling	*hīew*	Retained w and the selected native orthography

These are CAPR's checkpoints, not diplomatic quotations of all the authors' intermediate forms. In particular, superscript j on a source w denotes palatalization. Our separate j symbol carries that phonetic conditioning until the relevant consumer; it does not decide that a second j phoneme existed alongside every palatalized w. No new phonetic marker or lexical exception is needed.

Knee now selects the regular dative *cneowe* 'knee (dat.sg.)', whose short diphthong is produced by SC044 OEBreaking; it no longer shares this long-output component. The source distinguishes its short pre-ending stem from long endingless *cnēo* 'knee' and restored *w* in *cnēow* (Hogg 2011, 21–22, 86; Ringe and Taylor 2014, 387).

The j-created geminate is distinct from the earlier inherited-Vww operation. Its offglide checkpoint belongs to SC033 OEEwLongDiphthong; its long-io realization is separately represented by SC110 OEJGlideIO, before unchanged ordinary SC032 OEDiphthongLeveling. The composed SC106 OEJWWSimplification compatibility operation no longer acts on hue, which already has singleton w. Its absence of a live application is not universal proof of redundancy. The earlier inherited-reanalysis placement remains a representative serialization, not a newly proved strict chronology (Campbell 1959, 45–47; Ringe and Taylor 2014, 53, 65–66, 171–75, 250).

The contrast with *hīeġ* 'hay', from **xáwją*, remains exact: the selected inputs differ in one vowel and both geminate, but the low-vowel word loses w and retains the palatal consonant while hue retains w. The low-vowel resolution and subsequent fronting do not apply here (Campbell 1959, 46; Ringe and Taylor 2014, 53, 250).

Why this row is in the corpus The corpus originally lacked a high-front witness separating gemination of w from the low-vowel resolution illustrated by hay and strew. Hue remains a positive witness for SC010 PWGmcJGemination and a negative control for SC029 OEAwwjResolution. The corrected input and distinct intermediate history now make that contrast scientifically informative rather than merely spelling-compatible. No additional lexeme or analogical mechanism is introduced.

lead — OE *lǣdan*

Derivation: **láidijaną* > *lǣdan* (regular).

Derivation trace Proto input: **láidijaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**lādijan*
[no change]		OE Secondary Nasalization	**lādijąn*
		Sievers Law Syncope	**lādjąn*
Early Anglo-Frisian		OE I Umlaut	**lædjąn*
EAF Ai Monophthongization	**lādijaną*	OE Weak Tail Reduction	**lædjan*
		OE J Loss After Heavy	**lædan*

Old English form: *lædan*

Reconstruction and comparative evidence Ringe and Taylor derive Old English *lædan* ‘lead’ from Proto-Germanic **laidijaną* ‘lead’, and Kroonen likewise cites a weak verb of the **laidjan*-type for ‘lead’ (Ringe and Taylor 2014; Kroonen 2013, 363).

Old English evidence Clark Hall and Bosworth-Toller both record *lædan* ‘lead’ / *lædan* ‘lead’ as the ordinary Old English verb ‘to lead, guide, conduct’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English From **láidijaną* ‘lead’, monophthongization of **ai* first gives a **lād-* stage. Later syncope, i-mutation, weak-tail reduction, and loss of *j* after a heavy stem yield *lædan* ‘lead’, so the development represented here is fully regular (Ringe and Taylor 2014).

learn — OE *leornian*

Derivation: **líznōjaną* > *leornian* (regular).

Derivation trace Proto input: **líznōjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**léornōjaną*
PNWGmc I Lowering	**léznōjaną*	OE Heavy Syllable Nasal Apocope	**léornōjan*
EAF Rhotacism	**lérnōjaną*	OE Secondary Nasalization	**léornōjąn*
Early Anglo-Frisian		OE I Umlaut	**léornējąn*
		OE Unstressed Long Vowel Shortening	**léornejąn*
[no change]		OE Weak Tail Reduction	**léornejan*
		OE Intervocalic J Vocalization	**léorneian*
		OE Unstressed EI Contraction	**léornian*

Old English form: *leornian*

Comparative reconstruction and the inherited formation Old English *leornian* ‘learn’ belongs with Old High German *lernēn* ‘learn’ beside *lirnēn* ‘learn’, and Old Saxon *līnon* ‘learn’. Kroonen reconstructs the weak verb **liznōn-*; Orel likewise preserves a high-vowel comparative formation. The reconstructed **z* is important both to the nasal formation and to the later rhotic of English. Neither the dictionary spelling of the English lemma nor its modern pronunciation establishes an inherited **e* (Kroonen 2013, 340; Orel 2003, 247).

The selected infinitive **líznōjaną* ‘learn’ retains that inherited root. Its extended weak-verb shape should not be confused with a verbatim quotation of Kroonen’s citation stem, or with the distinct imperative and third-singular cells. The family underwent conjugational remodelling; an English class II paradigm does not by itself recover every Proto-Germanic ending. Ringe and Taylor discuss the earlier formation separately from its later English continuation (Ringe and Taylor 2014, 38, 84).

Attested vowels and selected cells The infinitive selected here is independently attested. In Ælfric’s Treatise the clause beginning *Gif þu nelt witan* continues *þu scealt leornian*: the modal requires an infinitive, so the grammatical identification does not depend on a dictionary headword.

The edition prints this on p.71. The same volume gives the singular imperative *leorna* ‘learn!’ in Deuteronomy14:23, followed by a clause with the singular addressee *þu*. These are textual occurrences, not cells generated from an instructional paradigm (Crawford 1922, 71, 351).

For the companion present indicative third singular, Miller’s Bede prints *leornað* ‘learns’ after *swa hwylc swa* ‘whoever’, coordinated with *rædeð* ‘reads’. The singular subject disambiguates the ending, which in isolation could resemble a plural. CAPR represents final dental-fricative spelling with thorn and therefore selects *leornaþ* ‘learns’, without changing the root vowel, ending vowel or grammatical cell (Miller 1891, 380, ll. 26–27).

The alternative *liornian* ‘learn’ and finite *liorna* ‘learn!’ remain important. Campbell’s traditional account relates *io/eo* to breaking, later vowel developments and levelling. It must not be compressed into a universal equation of *liorn- with conservative Northumbrian and* leorn- with secondary West Saxon: early West Saxon also has the high-vowel spelling. A conservative spelling is not necessarily an unchanged inherited vowel (Campbell 1959, 48, 59, 125).

Lowering, mutation and the weak paradigm The account adopted in SC015 PNWGmcILowering lowers **i* before preconsonantal **z*, independently of the following vowel. For the selected infinitive, early **lizn-* > *lezn-* is followed by rhotacism and English breaking before **rn*. The resulting *leorn- is not obtained by selecting an inherited mid-vowel reconstruction. Hill explains the competing high-vowel forms through later i-mutation in extended stems and subsequent paradigmatic history (Hill 2017, 153, n. 25).

That explanation is plausible but not a recovered complete paradigm. Jasanoff had already connected the family with the history of weak class III. He considers a lost third-singular **lizniþ* as a possible source of the high vowel of OHG *lirnēn*, rather than requiring a disputed first-singular reconstruction. The proposal matters because it supplies a particular morphological source for mutation or levelling, not a general appeal to analogy wherever the desired vowel is missing (Jasanoff 1973, 865, 868).

The original distribution of suffixes, their extension and the direction of later levelling remain uncertain. Later class II morphology cannot be projected backwards merely to exclude every earlier high-front trigger; conversely, one possible trigger does not establish that all three selected cells once contained it. The modern discussion of the class III family preserves these limitations (Ringe and Taylor 2014, 38, 84; Tarsi and Laker 2026, 42–43).

Interpretation of the selected derivation The three entries represent one etymological family: an infinitive, an imperative singular and a present third singular, not three independent proofs of a sound law. Their inherited inputs remain distinct, while the selected *leorn- vocalism receives a regular consonant-conditioned history. Selection of these textually supported vowels does not disprove the* liorn- tradition or resolve the whole weak-verb paradigm (Crawford 1922, 71, 351; Miller 1891, 380; Jasanoff 1973, 865, 868).

The finite entries retain an analogical family classification for the conjugational remodelling, not because finite-cell selection is itself analogy. No additional analogical replacement of the selected root vowel is required. The distinction between regular phonology of the form compared here and uncertain earlier paradigm history is essential to interpreting the derivations (Jasanoff 1973, 865, 868; Tarsi and Laker 2026, 42–43).

lid — OE *hlid*

Derivation: **xlídą* > *hlid* (regular).

Derivation trace Proto input: **xlídą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**xlíd*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *hlid*

Reconstruction and comparative evidence Orel cites a neuter lexeme of the **xliđ-* type with Old English *hlid* ‘lid, cover’, and Lloyd includes OE *hlid* ‘lid’ beside ON *hliþó* ‘lid’ and OHG *(h)lit* ‘lid’ among forms that retain *i* (Orel 2003; Lloyd 1966).

Old English evidence Clark Hall and Bosworth-Toller record *hlid* ‘lid, cover’ as the noun ‘lid, cover, door, gate’ (Clark Hall 1960; Bosworth and Toller 1898, 563).

Development to Old English The derivational input already represents the later Germanic *hliđ-* stage used for the derivation here. From **xlídą* ‘lid’, heavy-syllable apocope yields *hlid* ‘lid’, and the form belongs to the retained-*i* set noted by Lloyd rather than to the lowered *e* type (Lloyd 1966).

Form note An earlier etymological stage **liþuz* ‘lid’ belongs to comparative background only. The form represented here is the later **xlídą* ‘lid’ > *hlid* ‘lid’ line that matches the attested Old English noun (Orel 2003; Lloyd 1966).

light — OE *līehtan*

Derivation: **léuxtijaną* > *līehtan* (regular).

Derivation trace Proto input: **léuxtijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**lēoxtijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**lēoxtijan*
	OE Secondary Nasalization	**lēoxtijąn*
Early Anglo-Frisian	Sievers Law Syncope	**lēoxtjąn*
[no change]	OE I Umlaut	**līextjąn*
	OE Weak Tail Reduction	**līextjan*
	OE J Loss After Heavy	**līextan*

Old English form: *līehtan*

Reconstruction and comparative evidence Fulk gives Proto-Germanic **liuxtijanan* ‘light’ with Old English *līehtan* ‘illuminate’, and Ringe and Taylor likewise derive West Saxon *liehtan* ‘light, illuminate’ from the same weak-verb formation (Fulk 2018; Ringe and Taylor 2014).

Old English evidence Clark Hall and Bosworth-Toller preserve the verb family under spellings such as *liehtan* ‘illuminate’, *lihtan* ‘light, illuminate’, and *līhtan* ‘illuminate’, distinct from the related noun *lēoht* ‘light’ and adjective *leoht* / *liht* ‘light’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English From **léuxtijaną* ‘light’, the regular verbal line preserves **xt*, passes through a West Saxon *liehtan* ‘illuminate’ stage, and is represented here by normalized *līehtan* ‘illuminate’. The word treated in this entry is therefore the verb ‘to light, illuminate’, not the related noun from **leuxtą* ‘light’ (Fulk 2018; Ringe and Taylor 2014).

Dialect note Ringe and Taylor and Campbell distinguish West Saxon *liehtan* ‘light, illuminate’ from Anglian *lihtan* ‘light, illuminate’, while later West Saxon also shows *lyhtan* ‘light, illuminate’ (Ringe and Taylor 2014; Campbell 1959).

linden — OE *lind*

Derivation: **líndō* > *lind* (regular).

Derivation trace Proto input: **líndō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**línd*
PNWGmc Final Long O Raising	**líndu*		
Early Anglo-Frisian			
[no change]			

Old English form: *lind*

Reconstruction and comparative evidence Kroonen cites **lindō-* ‘lime tree’ and gives Old English *lind* ‘linden, lime tree’ as the relevant reflex (Kroonen 2013).

Old English evidence Clark Hall and Bosworth-Toller both record *lind* ‘linden, lime tree’ as the noun ‘lime-tree, linden’ (Clark Hall 1960; Bosworth and Toller 1898, 630).

Development to Old English From **líndō* ‘linden’, Northwest Germanic final **ō* raising first gives a **líndu* ‘linden’ stage, and later high-vowel apocope yields *lind* ‘linden, lime tree’. The development is therefore straightforward and regular.

Form note The Old English noun represented here is *lind* ‘linden, lime tree’. Clark Hall also has a separate adjectival *linden* ‘made of linden-wood’, but that is not the noun counterpart for this entry (Clark Hall 1960).

louse — OE *lūs*

Derivation: **lūsz* > *lūs* (regular).

Derivation trace Proto input: **lūsz*

Earlier Germanic changes	Old English changes
Northwest and West Germanic	[no change]
[no change]	
Early Anglo-Frisian	
[no change]	

Old English form: *lūs*

Reconstruction and comparative evidence Orel’s citation form is **lusz* ‘louse’, with the explicit nominative marker **-z* he writes across the root-noun class; his entry points to Bammesberger and Griepentrog for the morphology (Orel 2003, 252). Kroonen has no ‘louse’ entry, so his dictionary cannot be cited for this word (Kroonen 2013, 345). Kluge/Seebold carry the German cognate as the same root noun (Kluge 2011, 563). Bammesberger shows that the root-final *s* is itself secondary: the root **luw-* ‘louse’ was extended with *s* on the model of the ‘mouse’ word (Bammesberger 1990, 195). The ‘mouse’ parallel also shows why no extra sibilant is audible in

the nominative: Ringe reconstructs endingless **mūs* ‘mouse’, with the expected double sibilant of the nominative removed by degemination (Ringe 2017, 149).

The derivational input **lūsz* ‘louse’ follows Orel’s morphologically explicit convention. The final **-z* is an inflectional marker made visible in the citation form, not a claim that a phonetic cluster *[sz]* was ever pronounced; the notation records the morphology whose elimination the derivation models.

Old English evidence Old English *lūs* ‘louse’ is the endingless nominative-accusative singular of a root noun, with mutated plural *lȳs* ‘lice’ (Fulk 2018, 165–66). Old Norse *lús* ‘louse’ is likewise endingless, in keeping with the North Germanic redistribution that drops the reflex of **-z* in feminine root nouns (Fulk 2018, 167). No West Germanic daughter shows any ending in this class (Ringe and Taylor 2014, 118).

Development to Old English The root-noun nominative marker was eliminated before Proto-West Germanic (Ringe and Taylor 2014, 118). From the derivational input **lūsz* ‘louse’ the loss of the marker gives **lūs* ‘louse’, which is already the attested Old English form. Because the root itself ends in *s*, the marker’s elimination here is best read together with the degemination that Ringe describes for ‘mouse’: at no stage does the reconstruction require an audible *-sz* sequence. This early morphological development is distinct from the later West Germanic loss of final **z* in unstressed syllables and from the northern monosyllabic loss with compensatory lengthening; the root-noun ending was gone before either applied, and the root-final *s* of *lūs* ‘louse’ was never eligible for either change.

milk — OE *meoloc*

Derivation: **mélukz* > *meoloc* (regular).

Derivation trace Proto input: **mélukz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Back Mutation	**méoluk*
EAF Final Z Deletion	**méluk*	OE Med Unstressed U Lowering	**méolok*
Early Anglo-Frisian			
[no change]			

Old English form: *meoloc*

Reconstruction and comparative evidence Kroonen and Orel reconstruct the noun with stem notation **meluk-* and nominative **melukz* ‘milk’, and the nominative-style input used here is **mélukz* ‘milk’ (Kroonen 2013; Orel 2003, 306).

Old English evidence Old English preserves a mixed dossier for this noun. Ringe and Taylor describe West Saxon *meolc* ‘milk’ < *meoluc* ‘milk’ < **mélukz* ‘milk’, Campbell likewise discusses *meoluc* ‘milk’ and *meoloc* ‘milk’, and Anglian shows *milc* ‘milk’ (Ringe and Taylor 2014; Campbell 1959).

Development to Old English The unsyncopated line from **mélukz* ‘milk’ loses final **z*, lowers unstressed *u* to *o*, and with back mutation yields *meoloc* ‘milk’. That fuller unsyncopated outcome is the form represented here (Ringe and Taylor 2014).

Form comparison Syncopated *meolc* and Anglian *milc* belong to the competing leveled tradition associated with oblique forms, whereas *meoloc* / *meoluc* preserves the fuller nominal shape (Campbell 1959; Ringe and Taylor 2014).

mother — OE *mōder*

Derivation: **mōdēr* > *mōder* (regular).

Derivation trace Proto input: **mōdēr*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**mōder*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *mōder*

Reconstruction and comparative evidence Kroonen and Orel cite the Proto-Germanic r-stem kinship noun as **mōder-* with citation **mōdēr* ‘mother’ (Kroonen 2013; Orel 2003).

Old English evidence The transmitted Old English headword tradition is *mōdor* ‘mother’ / *modor* ‘mother’, with oblique *mēder* ‘mother’ in the paradigm. Clark Hall, Campbell, and Ringe and Taylor all preserve that contrast (Clark Hall 1960; Campbell 1959; Ringe and Taylor 2014).

Development to Old English From **mōdēr* ‘mother’, the regular suffixal development yields *mōder* ‘mother’. That regular nominative reflex is the form represented here, while the more familiar citation form *mōdor* ‘mother’ reflects later levelling within the r-stem paradigm (Campbell 1959; Ringe and Taylor 2014).

Form comparison The note therefore concerns inherited vocalism rather than a different lexeme: *mōder* ‘mother’ is the regularized nominative represented here, but dictionaries usually print *mōdor* / *modor* ‘mother’, and the oblique evidence survives in *mēder* ‘mother’ (Clark Hall 1960; Ringe and Taylor 2014).

nest — OE *nest*

Derivation: **nístą* > *nest* (regular).

Derivation trace Proto input: **nístą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**nést*
PNWGmc I Lowering **néstą*		
Early Anglo-Frisian		
[no change]		

Old English form: *nest*

Comparative formation and lowering Old English nest ‘nest’ continues an inherited neuter formation. Kroonen reconstructs **nista-*, Orel the full-word **nistan*. Their comparanda include Latin *nīdus* ‘nest’ and Sanskrit *nīḍá-* ‘nest’. The traditional analysis combines an element meaning ‘down’ with a zero-grade formation of the root ‘sit’. The English word must not be confused with the separate food noun also spelled *nest* (Kroonen 2013, 388, 391; Orel 2003, 287).

The selected **nístą* ‘nest’ retains the inherited high vowel. SC015 PNWGmcILowering lowers it immediately before **st*, rather than because of the next vowel. This is the principal independent lexical support for Hill’s **st* condition. The contrast with **sk* in the fish family is useful, but it does not turn one inherited nest etymon into numerous independent witnesses (Hill 2017, 153–55).

Later mutation in a derivative with a high-front trigger can conceal an earlier mid vowel. Such derivatives test chronological compatibility; they do not establish every detail of a lost paradigm. The general productivity of lowering before **st* is less securely demonstrated than lowering before inherited **r*, and the adopted unified law retains that evidential qualification (Hill 2017, 153–56).

net — OE *nett*

Derivation: **nátją* > *nett* (regular).

Derivation trace Proto input: **nátją*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**nættją*
PWGmc J Gemination	**náttją*	OE Heavy Syllable Nasal Apocope	**nættj*
Early Anglo-Frisian		OE I Umlaut	**nettj*
		OE J Loss After Heavy	**nett*
[no change]			

Old English form: *nett*

Reconstruction and comparative evidence Orel gives **natjan* ‘net’ with Old English *nett* ‘net’, and Fulk’s account of West Germanic gemination before *j* explains the geminate outcome after a short vowel (Orel 2003; Fulk 2018).

Old English evidence Clark Hall and Bosworth-Toller record *nett* ‘net’ as the noun, and Campbell notes that final geminates are often graphically simplified in Old English spelling (Clark Hall 1960; Bosworth and Toller 1898, 29; Campbell 1959).

Development to Old English From **nátją* ‘net’, West Germanic j-gemination first gives **náttją* ‘net’. Later brightening, loss of the weak ending, and loss of final *j* after a heavy stem yield *nett* ‘net’, so the development represented here is regular (Fulk 2018).

Form note Spellings in *net* can therefore be graphic simplifications, but the lexical target supported by the dictionary evidence is *nett* ‘net’ (Campbell 1959; Orel 2003).

nightmare — OE *mare*

Derivation: **márōn* > *mare* (regular).

Derivation trace Proto input: **márōn*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**mærǭ*
PNWGmc N Stem N Loss	**márǭ*	OE A Restoration	**marǭ*
Early Anglo-Frisian		OE Unstressed Long Vowel Shortening	**marœ*
		OE Unstressed AE Merger	**mare*
[no change]			

Old English form: *mare*

Reconstruction and comparative evidence Ringe and Taylor treat the lexeme as Proto-Germanic / Proto-Northwest-Germanic **marōn-*, with Old English *mare, maran*, and variant *mere*; Orel preserves the same comparative lemma though with a different Old English headword tradition (Ringe and Taylor 2014; Orel 2003).

Old English evidence Clark Hall records *mare* ‘nightmare, monster’ and also preserves related variant forms *mera* / *mere* (Clark Hall 1960, 213).

Development to Old English The selected simplex input **márōn* ‘nightmare’ regularly gives *mare* ‘nightmare’ after brightening, A-restoration before the n-stem ending, and later reduction of the final vowel. The word represented here is the attested simplex noun, not an attested compound (Ringe and Taylor 2014).

Form note The concept corresponds to an unattested compound **nihtmare* ‘nightmare’, but the Old English lexical evidence is for simplex *mare* ‘nightmare’, with oblique *maran* and variant *mere* / *mera* (Ringe and Taylor 2014; Clark Hall 1960, 213).

rainbow — OE *reġnboga*

Derivation: **régna-bùgô* > *reġnboga* (regular).

Derivation trace Proto input: **régnabùgô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Compound Linking Syncope	**régnbògô*
PNWGmc U Lowering	**régnabògô*	OE Strip Secondary Stress	**régnbogô*
Early Anglo-Frisian		OE Velar Palatalization	**réjnbogô*
		OE Unstressed Long Vowel Shortening	**réjnboga*
[no change]		OE Palatal Fricative Merger	**réjnboga*

Old English form: *reġnboga*

Reconstruction and compound formation The compound means ‘rain-bow’. Orel reconstructs **reʒna-buʒōn*, explicitly joining the rain noun **reʒnan* **reʒnaz* and the bow noun **buʒōn*. The latter belongs to the family of ‘bend’. Kroonen independently gives masculine **bugan-* ‘bow’, an n-stem derivative of that verb; his rain entry supplies the first constituent, not an independently printed reconstruction of the complete accented compound (Orel 2003, 61, 300; Kroonen 2013, 61–62, 82, 408). The bow noun, the bend verb and the different noun for an arm-ring must not be treated as interchangeable second members.

The selected **régna-bùgô* is a complete nominative in CAPR’s quantity and accent notation, not a quotation of Kroonen’s bare stem or a mechanical transcription of Orel’s citation ending. Old English masculine weak *boga* ‘bow’ and oblique *bogan* reflect the n-stem history. The compound takes the same inflection, with nominative *-boga* and oblique *-bogan*. The accentuation assigns primary prominence to the first root and subordinate prominence to the second. Its exact prehistoric realization is reconstructed; neither the etymological hyphen nor a dictionary’s normalized spelling independently establishes that accent.

The early vowel of the second root The ordinary short *o* of *boga* has an early explanation: inherited short *u* lowered before a non-high vowel in the next syllable. Campbell, Hogg, Fulk, and Ringe and Taylor describe the change and its nasal and intervening-*j* restrictions. Colman expressly applies this explanation to the noun *boga* (Campbell 1959, 43, §§115–116; Hogg 2011, 54–55, §§3.10–3.11; Fulk 2018, 56–57, §4.3; Ringe and Taylor 2014, 27–29; Colman 2011, 95–96).

CAPR infers that the same assimilation applied to the secondarily stressed root in the compound: **régna-bùgô* > **régna-bògô*. The following non-high vowel was still present, and neither a nasal immediately after the target nor an intervening *j* blocks this particular sequence. The compound extension is a phonological inference, not a derivation printed with these accents by the cited authorities. It does not extend early lowering to ordinary unstressed medial *u*. A regular account is preferable to replacement from the independent noun unless positive evidence requires that analogy; none has been established for this selected history.

The name *Buga* beside *Boga* remains relevant. Colman's coins preserve both vowel spellings, and transmission between moneyer and die-cutter complicates dialect attribution. The name's retained *u* cannot be explained merely by saying that names escape sound laws. Nor does this distribution erase the ordinary noun's independently supported lowering. It remains a qualification of the broader retention problem, not a reason to implement an optional change (Colman 2011, 95–98).

Linking syncope and the regular development Early lowering retains the subordinate accent. Consequently **régna-bògô* still supplies the prosodic environment for linking-vowel deletion, giving **régnbògô*. Only then does removal of the earlier accent annotation give **régnbogô*. Campbell's discussion of weakened compound junctures supplies the broader historical context; the exact ordering of these serialized annotations is CAPR's explicit representation (Campbell 1959, 148–49, §§356–357, p. 153, §367).

The first velar undergoes palatalization, the final long weak vowel shortens, and the palatal consonant is represented in the selected spelling *reġnboga*. The second velar remains before a back vowel. Thus the adopted chain is **régna-bùgô* > **régna-bògô* > **régnbògô* > **régnbogô* > **réjnbogô* > **réjnboga* > **réjnboga*, then *reġnboga*. The intermediate accented and palatal symbols are analytical representations, not alternative manuscript spellings. The principal account is SC017 PNWGmcULowering, with linking deletion at SC037 OECompoundLinkingSyncope.

The word already contains *o* when SC040 OEMedUnstressedULowering operates. It therefore supplies no evidence that compound secondary stress had disappeared before that late lowering. Its genuine dependency is survival of the earlier accent information until linking syncope. This does not alter the separate weak-vowel chronology of *weorold* 'world' or establish a universal date for compound deaccentuation.

Old English documentary evidence and spelling Bosworth–Toller gives *regn-boga* 'rainbow' as masculine, with oblique *-an*. Its quotation "Hwí wæs se rénboga tó wedde gesette mancynne?" has nominative singular *rénboga*, the subject of the passive question. The following statement has *ðone rénbogan*, accusative singular. The online entry assigns this passage to *Boutr. Scrd.* 21, 19–26. Its separate *Genesis* 9:13 quotation, "Ic sette mínne rénbogan … on wolcnum", likewise has accusative singular *rénbogan*, not a competing nominative target. The two vocabulary citations, *Wrt. Voc.* i. 52, 42 and 76, 33, pair the noun with Latin *iris/yris* and *arcus* (Bosworth and Toller 1898, s.v. regn-boga).

These are dictionary-mediated documentary witnesses, checked in the online entry, not a new manuscript collation. The dictionary's *rén-* represents the first member without written *g* and marks vowel length; its accent is not CAPR's reconstructed primary-stress mark. The selected *reġn-* retains the palatal consonant in modern scholarly notation. Hyphenation and the palatal dot likewise belong to editorial representation. Neither the contracted first member nor oblique *-bogan* changes the evidence for second-root *o*. The quotations do not justify attaching an exact manuscript date or a narrower dialect label to every normalized spelling. The selected regular span is retained without claiming to model every attested orthographic or inflectional variant.

sheep — OE *sċēap*

Derivation: **skép̄ą* > *sċēap* (regular).

Derivation trace Proto input: **skép̄ą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**skǣp*
PNWGmc Long E Lowering	**skāpą*	OE Sk Palatalization	**ʃǣp*
EAF Long A Fronting	**skǣpą*	OE Ws Palatal Diphthongization	**ʃēap*
Early Anglo-Frisian			
[no change]			

Old English form: *sċēap*

Reconstruction and comparative evidence Ringe and Taylor cite a later West Germanic **skap* ‘sheep’ > WS *scéap* ‘sheep’, while Orel preserves a Proto-Germanic noun of the **skēp-* type for the same lexeme (Ringe and Taylor 2014; Orel 2003).

Old English evidence Clark Hall records *scēap* ‘sheep’ with spelling variation, and Campbell likewise lists West Saxon *scéap* ‘sheep’ among the palatal-diphthongized forms (Clark Hall 1960; Campbell 1959).

Development to Old English From **sképą* ‘sheep’, Northwest Germanic lowering gives **skǣpą* ‘sheep’; after apocope and palatalization the West Saxon branch diphthongizes to *sċēap* ‘sheep’. The development represented here is therefore fully regular.

Dialect note Ringe and Taylor contrast West Saxon *scéap* ‘sheep’ with Mercian and Kentish *scép* ‘sheep’, and Campbell also notes Northumbrian *scip* ‘sheep’. The form represented here is the West Saxon headword (Ringe and Taylor 2014; Campbell 1959).

shilling — OE *sċilling*

Derivation: **skíllingaz* > *sċilling* (regular).

Derivation trace Proto input: **skíllingaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**skílling*
EAF Final Z Deletion	**skíllinga*	OE Sk Palatalization	**fílling*
Early Anglo-Frisian		OE Med Unstressed I Lowering1	**fílleng*
		OE Med Unstressed I Lowering	**fílling*
[no change]			

Old English form: *sċilling*

Reconstruction and comparative evidence Kroonen treats the cognate set under **skellinga-* ~ **skillinga-* and connects it with **skeld-linga-*, while Orel likewise gives the coin word with OE *scilling* ‘shilling’ among the reflexes (Kroonen 2013; Orel 2003). The derivational input **skíllingaz* ‘shilling’ is the nominative-style form used here to represent that inherited **-ing-* derivative.

Old English evidence Clark Hall records *scilling* ‘shilling’, and Campbell cites it among nouns with unstressed *i* in derivational *-ing* (Clark Hall 1960; Campbell 1959). The target is the ordinary OE citation form, normalized as *sċilling* ‘shilling’.

Development to Old English From **skíllingaz* ‘shilling’, loss of final *-az* yields **skílling* ‘shilling’. Old English palatalization of initial *sk* before front vocalism then gives *sċilling* ‘shilling’. The *i* of derivational *-ing-* remains, so the regular outcome is *sċilling* ‘shilling’, not **sċilleng* ‘shilling’ (Campbell 1959; Hogg 1992a).

Form note Kroonen’s **skellinga-* ~ **skillinga-* and his internal analysis **skeld-linga-* belong to the etymological background of the cognate set. The derivational input **skíllingaz* ‘shilling’ is the specific form used for the derivation represented here (Kroonen 2013).

shovel — OE *sċofl*

Derivation: **skúflō* > *sċofl* (regular).

Derivation trace Proto input: **skúflō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Sk Palatalization	**fóflu*
PNWGmc U Lowering	**skóflō*	OE High Vowel Apocope	**fófl*
PNWGmc Final Long O Raising	**skóflu*		
Early Anglo-Frisian			
[no change]			

Old English form: *sċofl*

Reconstruction and development The selected short-root-vowel **skúflō* ‘shovel’ follows the feminine formation discussed by Orel (Orel 2003, 347). With lowering before final oral raising, its root becomes **o* before the ending becomes **u*; subsequent apocope and consonant development give *sċofl* ‘shovel’. No later root-vowel levelling is needed for the form compared here.

This is a conditional witness, not independent proof of the order. Earlier raising instead predicts a high-root nominative beside a lowered oblique. Luick’s inferred **scufel* beside *scofel* ‘shovel’ makes alternation credible but does not directly establish o-generalization in the early English paradigm (Luick 1914--1940, 105–6, §78). The assessment of that additional analogy, alongside wool and the other ō-stems, belongs to SC019 PNWGmcFinalLongORaising.

show — OE *sċēawian*

Derivation: **skáwōjaną* > *sċēawian* (regular).

Derivation trace Proto input: **skáwōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Aw Long Diphthong	**skɛ́awōjaną*
[no change]	OE Heavy Syllable Nasal Apocope	**skɛ́awōjan*
Early Anglo-Frisian	OE Secondary Nasalization	**skɛ́awōjąn*
[no change]	OE Sk Palatalization	**ʃɛ́awōjąn*
	OE I Umlaut	**ʃɛ́awējąn*
	OE Unstressed Long Vowel Shortening	**ʃɛ́awejąn*
	OE Weak Tail Reduction	**ʃɛ́awejan*
	OE Intervocalic J Vocalization	**ʃɛ́aweian*
	OE Unstressed EI Contraction	**ʃɛ́awian*

Old English form: *sċēawian*

Reconstruction and comparative evidence Orel and Kroonen cite a Class II verb of the type **skawōjan-*, with OE *scēawian* ‘show’ among the reflexes (Orel 2003, 337; Kroonen 2013, 442). The dictionary singleton stem is not silently identified with Campbell and Brunner’s inherited-geminate analysis (Campbell 1959, 45; Brunner 1965, 49). Brunner also records the later West Saxon variant *scāwian* ‘show’ in his discussion of diphthong accent shift; it is not evidence for the same prehistoric singleton quantity (Brunner 1965, 106–7).

Old English evidence The held fourth edition of Bright’s reader contains *scēawian* ‘show’ and the related imperative *scēawa* ‘show’ in the Blickling homily (Bright 1917, 71). This is not a newly verified passage in the separate Cassidy/Ringler edition. The sources use *scēawian* ‘show’, while *sċēawian* ‘show’ supplies CAPR’s normalized spelling.

Development to Old English From **skáwōjaną* ‘show’, the transducer retains SC034 OE-AwLongDiphthong as a computational proxy mapping to long *ea*+*w* before the class-II suffix. Its **ō* separates root **w* from **j*, excluding direct *aw*+*j* gemination. Neither that exclusion nor the successful *sċēawian* ‘show’ output establishes an exceptionless singleton lengthening law. The derivational input and *regular* computational classification remain unchanged, conditional

on this retained proxy; their historical basis is provisional, not a resolution of the root-history premise. Campbell and Brunner's inherited-geminate account differs from the dictionary singleton stem (Campbell 1959, 45–46; Brunner 1965, 49; Orel 2003, 337; Kroonen 2013, 442).

The finite companions select an imperative singular and a present third singular, not alternative dictionary infinitives. Their class-II endings are independently supported even though successful root outputs cannot decide the disputed prehistoric quantity (Ringe and Taylor 2014, 80, 314; Campbell 1959, 148).

Form note The difference between *scēawian* 'show' and *sċēawian* 'show' is CAPR's editorial normalization of initial <*sc*>, not a different lexeme or paradigm cell. It does not itself settle the consonant's prehistoric articulation.

sleep — OE *slǣpan*

Derivation: **slépaną* > *slǣpan* (regular).

Derivation trace Proto input: **slépaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**slǣpan*
PNWGmc Long E Lowering	**slāpaną*	OE Secondary Nasalization	**slǣpąn*
EAF Long A Fronting	**slǣpaną*	OE Weak Tail Reduction	**slǣpan*
Early Anglo-Frisian			
[no change]			

Old English form: *slǣpan*

Reconstruction and comparative evidence Kroonen preserves the comparative verb as **slēpan-*, and Fulk cites the same family under root **slēb-* (Kroonen 2013; Fulk 2018, 120). The derivational input **slépaną* 'sleep' is the infinitive-style form used here for that inherited sleep-verb.

Old English evidence Clark Hall gives *slæpan* 'sleep' with preterite *slēp* 'slept', *slēap* 'slept', and Bright likewise lists *slæpan* 'sleep' (*slāpan* 'sleep'), *slēp* 'slept', *slēpon* 'slept', *slēpen* 'slept' (Clark Hall 1960; Bright 1971, 435). The target represented here is therefore the normalized infinitive *slǣpan* 'sleep', not the preterite forms and not the separate noun *slǣp* 'sleep'.

Development to Old English From **slépaną* 'sleep', Northwest Germanic lowering gives **slǣpaną* 'sleep'. The later OE tail developments then yield *slǣpan* 'sleep' regularly. Brunner and Bülbring show that the OE tradition also has variant spellings such as West Saxon *slāpan* 'sleep' / *slæpan* 'sleep' and Anglian or Kentish *slēpan* 'sleep', but those do not displace the infinitive chosen here (Brunner 1965; Bülbring 1902).

Form note The note concerns lemma type rather than a special derivational problem: this row represents the verb *slǣpan* 'sleep', whereas *slǣp* 'sleep' belongs to noun or lookup background and *slēp* 'slept' / *slēap* 'slept' are preterite forms (Clark Hall 1960; Bright 1971, 435).

smear — OE *smierwan*

Derivation: **smérwijaną* > *smierwan* (regular).

Derivation trace Proto input: **smérwijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**sméorwijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**sméorwijan*
	OE Secondary Nasalization	**sméorwijąn*
Early Anglo-Frisian	Sievers Law Syncope	**sméorwjąn*
[no change]	OE I Umlaut	**smíerwjąn*
	OE Weak Tail Reduction	**smíerwjan*
	OE J Loss After Heavy	**smíerwan*

Old English form: *smierwan*

Reconstruction and comparative evidence Kroonen gives the comparative headword as **smerwjan-* (Kroonen 2013). Ringe and Taylor instead cite a later-stage **smirwijana* ‘smear’, from which they derive West Saxon *smierwan* ‘smear’, Mercian *smirwan* ‘smear’, and Northumbrian *smiriga* (Ringe and Taylor 2014). The derivational input **smérwijaną* ‘smear’ therefore represents the Kroonen-aligned PGmc layer, while the later-stage dialect split belongs to a different chronological level.

Old English evidence The target represented here is the West Saxon citation form *smierwan* ‘smear’. Campbell’s Anglian discussion explains the contrasting *smirwan* ‘smear’, and Clark Hall, Brunner, and Bright show that the same lexical family later also includes forms such as *smirian*, *smyrian*, and preterite *smyrode* (Campbell 1959; Clark Hall 1960; Brunner 1965; Bright 1971).

Development to Old English From **smérwijaną* ‘smear’, breaking before *r* + *consonant* yields *eo*, and later i-umlaut produces *ie*. The result is West Saxon *smierwan* ‘smear’. Anglian *smirwan* ‘smear’ reflects the well-known failure of breaking in this environment, not a different lexeme (Campbell 1959; Ringe and Taylor 2014).

Dialect note The entry therefore represents the West Saxon member of a broader OE family: *smierwan* ‘smear’ in West Saxon, *smirwan* ‘smear’ in Anglian or Mercian, and related later class-II forms such as *smirian* or *smyrian* in the same lexical field (Ringe and Taylor 2014; Clark Hall 1960).

sorrow — OE *sorg*

Derivation: **súrgō* > *sorg* (regular).

Derivation trace Proto input: **súrgō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**sórg*
PNWGmc U Lowering	**sórgō*		
PNWGmc Final Long O Raising	**sórgu*		
Early Anglo-Frisian			
[no change]			

Old English form: *sorg*

Reconstruction and development The selected **súrgō* ‘sorrow’ belongs to the feminine formation discussed by Kroonen (Kroonen 2013, 493). Lowering before final oral raising gives a low root before the ending becomes **u*; later apocope yields *sorg* ‘sorrow’. This accounts for the selected form without an additional root-vowel generalization.

Earlier raising would instead protect the nominative root while leaving the nasal accusative’s non-high conditioner available for lowering. Generalization of that oblique root could recover

the low-root paradigm, but has not been independently established for this particular early English noun. Generic evidence for such levelling is not specific attestation of this operation (Ringe and Taylor 2014, 32–34; Stiles 2012, 47–48). The comparative economy argument and its limits are developed in SC019 PNWGmcFinalLongORaising.

span — OE *spannan*

Derivation: **spánnaną* > *spannan* (regular).

Derivation trace Proto input: **spánnaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**spánnan*
[no change]	OE Secondary Nasalization	**spánnąn*
	OE Weak Tail Reduction	**spánnan*
Early Anglo-Frisian		
[no change]		

Old English form: *spannan*

Reconstruction and comparative evidence Kroonen cites the inherited verb as **spannan-*, with OE *spannan* ‘span, fasten’ among the reflexes (Kroonen 2013). The derivational input **spánnaną* ‘span’ is the infinitive-style form used here for that same verbal lexeme.

Old English evidence Clark Hall lists noun *spann* and verb *spannan* ‘span, fasten’ as separate headwords, and Brunner likewise records *sponnan, spannan stv.* (Clark Hall 1960; Brunner 1965). The target is the strong-verb infinitive, not the noun.

Development to Old English From **spánnaną* ‘span’, the final nasal ending is lost and the regular OE weak-tail steps surface *spannan* ‘span, fasten’. No paradigm-cell substitution is needed: the current derivation already lands on the infinitive directly.

Form note English *span* can also reach the noun *spann* in local lookup material. The form represented here is the verb *spannan* ‘span, fasten’, with the noun treated elsewhere (Clark Hall 1960).

spar — OE *spearra*

Derivation: **spárrô* > *spearra* (regular).

Derivation trace Proto input: **spárrô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**spærrô*
[no change]	OE Breaking	**spearrô*
	OE Unstressed Long Vowel Shortening	**spearra*
Early Anglo-Frisian		
[no change]		

Old English form: *spearra*

Reconstruction and comparative evidence Kroonen and Orel place this noun in the beam or rafter set **spar(r)an-*, with cognates such as Old Saxon and Old High German *sparro* ‘spar, rafter’ (Kroonen 2013; Orel 2003). The derivational input **spárrô* ‘spar’ is the OE-facing nominal form used here for that same lexeme.

Old English evidence The noun is *spearra* ‘spar, rafter’. English gloss overlap also reaches the unrelated verb *sperran* ‘to bar’, which does not belong to this row.

Development to Old English From **spárrô* ‘spar’, Anglo-Frisian brightening gives **spærrô* ‘spar’, and OE breaking before geminate *rr* yields **spearrô* ‘spar’, later *spearra* ‘spar, rafter’. The development is therefore regular for a breaking-conditioned noun of this type (Luick 1914--1940).

Form note This entry concerns the noun *spearra* ‘spar, rafter’ only. It should be kept separate from verb *sperran*, even though the Modern English glosses overlap (Kroonen 2013; Orel 2003).

still — OE *stillan*

Derivation: **stéllijaną* > *stillan* (regular).

Derivation trace Proto input: **stéllijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**stéllijan*
[no change]	OE Secondary Nasalization	**stéllijąn*
	Sievers Law Syncope	**stélljąn*
Early Anglo-Frisian	OE I Umlaut	**stilljąn*
[no change]	OE Weak Tail Reduction	**stilljan*
	OE J Loss After Heavy	**stillan*

Old English form: *stillan*

Reconstruction and comparative evidence The wider West Germanic family includes adjective *still* ‘still, quiet’ and verb *stillen* (Kluge 2011). The derivational input **stéllijaną* ‘still’ represents the verbal j-formation used for the OE row.

Old English evidence Clark Hall gives *stillan* as the verb and separately *stille* ‘still, quiet’ as the adjective (Clark Hall 1960). Bosworth-Toller likewise preserves a substantial prefixed verbal family under *ge-stillan* ‘still, stop’ and related forms (Bosworth and Toller 1898, 724). The Old English form here is the verb *stillan*, not the adjective.

Development to Old English As a heavy-stem Class I weak verb, **stéllijaną* ‘still’ undergoes the expected syncope and i-umlaut, and later loss of *j* after a heavy stem yields *stillan*. The development represented here is regular.

Form note The note concerns lexical framing rather than sound law: *stillan* is the verb represented here, while *stille* ‘still, quiet’ belongs to the related adjectival branch of the family (Clark Hall 1960; Kluge 2011).

summer — OE *sumer*

Derivation: **súmaraz* > *sumer* (regular).

Derivation trace Proto input: **súmaraz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**súmar*
EAF Final Z Deletion	**súmara*	EAF Brightening Unstressed	**súmær*
Early Anglo-Frisian		OE Unstressed AE Merger	**súmer*
[no change]			

Old English form: *sumer*

Reconstruction and comparative evidence Kroonen gives the lexeme as **sumara-*, and Ringe and Taylor likewise use **sumaraz* ‘summer’, while Orel preserves an alternate **sumeraz* ‘summer’ (Kroonen 2013; Ringe and Taylor 2014; Orel 2003, 425). The derivational input **súmaraz* ‘summer’ follows the **a* vocalism that underlies the regular development represented here.

Old English evidence Clark Hall gives *sumor* ‘summer’ m., gs. *sumeres* ‘summer’, ds. *sumera* ‘summer’ / *sumere* ‘summer’, and Bright likewise lists *sumor (sumer)* with genitive *sumeres* ‘summer (gen.)’ (Clark Hall 1960; Bright 1971, 440). The tradition therefore preserves both *sumor* ‘summer’ and *sumer* ‘summer’, with the oblique forms strongly supporting second-syllable *e*.

Development to Old English From **súmaraz* ‘summer’, loss of final *-az* is followed by fronting and merger in the unstressed second syllable, yielding *sumer* ‘summer’. The form compared here is the regularized *e*-form, while the common citation form *sumor* ‘summer’ remains part of the attested OE tradition (Ringe and Taylor 2014).

Form note The entry does not deny *sumor* ‘summer’. It represents *sumer* ‘summer’ as the regular outcome chosen here, while *sumor* ‘summer’ remains a common headword spelling and *sumeres* ‘summer (gen.)’ / *sumere* ‘summer (dat.)’ show that the *e*-vocalism was also real in Old English (Clark Hall 1960; Bright 1971, 440).

sunder — OE *sundrian*

Derivation: **súndrōjaną* > *sundrian* (regular).

Derivation trace Proto input: **súndrōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**súndrōjan*
[no change]	OE Secondary Nasalization	**súndrōjąn*
	OE I Umlaut	**súndrējąn*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**súndrejąn*
[no change]	OE Weak Tail Reduction	**súndrejan*
	OE Intervocalic J Vocalization	**súndreian*
	OE Unstressed EI Contraction	**súndrian*

Old English form: *sundrian*

Reconstruction and comparative evidence Orel distinguishes three related formations: adverbial **sunþraz* ‘asunder’ > *sundor* ‘apart, asunder’, Class I verbal **sunþrjanan* ‘sunder’ > *syndrian* ‘separate, sunder’, and Class II verbal **sunþrōjanan* ‘sunder’ > *sundrian* ‘sunder’ (Orel 2003).

Kluge-Seebold aligns the cognate set with German *sondern* ‘separate’ and OE *gesundrian* ‘separate’, so this entry belongs with the Class II verb, not the adverb (Kluge 2011).

Old English evidence Clark Hall and Bosworth-Toller list *sundrian* ‘sunder’ and *syndrian* ‘sunder’ separately from adverbial *sundor* ‘asunder’; both also record the prefixed verbal family *ā-sundrian* ‘sunder apart’ (Clark Hall 1960, 296; Bosworth and Toller 1898). The target is therefore the weak verb *sundrian* ‘sunder’.

Development to Old English From **súndrōjaną* ‘sunder’, the Class II weak-verb suffix yields regular OE *-ian*, producing *sundrian* ‘sunder’. Because this is the **-ōjan-* verb and not the Class I **-jan-* formation, the form represented here does not belong to the umlauted *syndrian* ‘separate’ branch.

Form note The earlier confusion was lexical, not phonological: *sundor* ‘apart’ is the separate adverb, and *syndrian* ‘separate’ is a related but different verb. The verb treated here is the Class II verb *sundrian* (Orel 2003; Clark Hall 1960, 296).

swallow — OE *swealwe*

Derivation: **swálwōn* > *swealwe* (regular).

Derivation trace Proto input: **swálwōn*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**swælwǭ*
PNWGmc N Stem N Loss **swálwǭ*	OE Breaking	**swealwǭ*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**swealwæ*
	OE Unstressed AE Merger	**swealwe*
[no change]		

Old English form: *swealwe*

Reconstruction and comparative evidence Kroonen gives the bird name as **swalwōn-*, and Ringe and Taylor cite the later West Germanic stage **swalwa* ‘swallow’, from which West Saxon *swealwe* ‘swallow’ and Mercian *swalwe* ‘swallow’ develop (Kroonen 2013, 535; Ringe and Taylor 2014, 200). The selected etymological comparison belongs to the swallow-bird family, not to the verb *swelgan*.

Old English evidence Clark Hall records *swealwe (a, o)* as the noun headword (Clark Hall 1960). Campbell and Brunner also preserve later or oblique-family forms such as *swaluwe*, *swalewan*, and *swealuwe*, but those belong to wider variation around the noun rather than to the citation form represented here (Campbell 1959; Brunner 1965).

Development to Old English From **swálwōn* ‘swallow’, brightening yields **swælw-*, and breaking before *lw* gives **swealw-*. The later noun ending develops regularly to *swealwe*. The relevant point is that the bird name has no inherited **g*: that consonant belongs to the separate verb *swelgan* (Ringe and Taylor 2014, 200; Kroonen 2013, 535).

swine — OE *swīn*

Derivation: citation reconstruction **swínᵠ*; form followed here **swínᵠ* > *swīn* (regular).

Derivation trace Proto input: **swínᵠ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**swín*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *swīn*

Reconstruction and comparative evidence Kroonen cites the noun as **swina-* ‘pig’ (Kroonen 2013). The selected comparative form here is the bare neuter citation cell **swínᵠ* ‘swine’, which matches the singular noun aimed at in Old English rather than an oblique stem form.

Old English evidence Clark Hall records *swin (y)* as the ordinary noun headword (Clark Hall 1960). The target here is that singular citation form *swīn* ‘swine’, not a plural glossed in Modern English as *swine*.

Development to Old English From derivational input **swínᵠ* ‘swine’, loss of the final nasal vowel yields *swīn* ‘swine, pig’. The outcome is therefore the regular monosyllabic noun with preserved long root *ī*.

Source note The derivational input writes stressed long *ī* as **í*, so comparative **swínᵠ* ‘swine’ and derivational **swínᵠ* ‘swine’ represent the same lexical form.

there — OE *þǣr*

Derivation: **θár* > *þǣr* (regular).

Derivation trace Proto input: **θár*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Monosyllabic Final R Lengthening	**θār*	
EAF Long A Fronting	**θǣr*	
Early Anglo-Frisian		
[no change]		

Old English form: *þǣr*

Old English evidence Old English þǣr ‘there’ is the normalized West Saxon adverb. Clark Hall lemmatizes ðær ‘there’; the substitution of thorn for eth does not change the vowel or the identity of the word (Clark Hall 1960, 305). Hill contrasts West Saxon *ðǣr* with Anglian ðēr ‘there’: both have long vowels, with different dialectal qualities (Hill 2017, 142–43). These are normalized lexical and dialect forms, rather than diplomatic transcriptions of individual manuscripts.

Reconstruction and comparative evidence Kroonen reconstructs **þar* ‘there’, formed from the demonstrative stem **þa-* with locative **-r* (Kroonen 2013, 534). Orel likewise identifies the demonstrative formation, but prints the alternatives **þar* and **þēr* (Orel 2003, 416). His long-vowel citation must be kept distinct from the earlier short-vowel antecedent. The selected **θár* ‘there’ represents the short antecedent with an explicit root accent and a phonological dental symbol; neither addition is a literal quotation of the dictionaries.

The comparative distribution leaves the date of lengthening open. Gothic *þar* ‘there’ has a spelling that does not distinguish short *a* from long *ā*. More informative for conditioning is the contrast between Old High German *dār* ‘there’ and the short vowels of *dăra* ‘thither’ and *dărōt* ‘thither’. Hill uses this contrast to support lengthening in monosyllables ending in inherited *r* (Hill 2017, 142–48).

Development to Old English The adopted derivation applies SC112, stressed monosyllabic lengthening before inherited final *r*, giving **þar* > **þār*. Subsequent SC101, long-*ā* fronting supplies **þǣr*, yielding West Saxon *þǣr* without a change of formation or an analogical replacement. The vowel becomes long while it is still back; this is distinct from lengthening an already fronted short vowel.

Hill argues for a Proto-Germanic date, whereas Ringe and Taylor place the lengthening in West Germanic and discuss deictic stress and reanalysis as possible explanations (Hill 2017, 146–48; Ringe and Taylor 2014, 65). The regular short-*a*-to-*ǣ* development is conditional on the chosen antecedent and early lengthening. Its successful outcome does not establish that absolute date, and a word without inherited **i* cannot independently order SC015 relative to lengthening.

think — OE *þencan*

Derivation: **θánkijaną* > *þencan* (regular).

Derivation trace Proto input: **θánkijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**θánkijan*
[no change]	OE Secondary Nasalization	**θánkijąn*
	Sievers Law Syncope	**θánkjąn*
Early Anglo-Frisian	OE Velar Palatalization	**θántʃjąn*
[no change]	OE I Umlaut	**θentʃjąn*
	OE Weak Tail Reduction	**θentʃjan*
	OE J Loss After Heavy	**θentʃan*

Old English form: *þencan*

Reconstruction and comparative evidence Kroonen gives the verb as **þankjan-* ‘to think’, and Ringe and Taylor cite fully inflected **þankijaną* ‘think’ beside OE *þencan* ‘think’ (Kroonen 2013; Ringe and Taylor 2014). The noun **þankaz* ‘think’ belongs only to the wider derivational background.

Old English evidence Bosworth-Toller preserves the verb under *þencan* ‘think’ / *geþencan* ‘think’, and the citation form here is the ordinary infinitive *þencan* ‘think’ (Bosworth and Toller 1898).

Development to Old English From **θánkijaną* ‘think’, palatalization before **j* and i-umlaut produce *þencan* ‘think’. The infinitive is therefore a straightforward weak-verb outcome.

Lexical note Campbell’s assibilation discussion uses the same verb *þencan* ‘think’; the class-III relic *hycgan* is a different lexeme (Campbell 1959; Hogg 1992a).

thorn — OE *þorn*

Derivation: **θúrnaz* > *þorn* (regular).

Derivation trace Proto input: **θúrnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**θórn*
PNWGmc U Lowering	**θórnaz*		
EAF Final Z Deletion	**θórna*		
Early Anglo-Frisian			
[no change]			

Old English form: *þorn*

Reconstruction and comparative evidence Kroonen gives **þurna-* ‘thorn, briar’, while Orel preserves the masculine pair **þurnuz* ‘thorn’ and **þurnaz* ‘thorn’ (Kroonen 2013; Orel 2003). The derivational input **θúrnaz* ‘thorn’ belongs to that same comparative family.

Old English evidence Bright lists *þorn* ‘thorn’ (m.) and Clark Hall likewise treats *þorn* ‘thorn’ as the ordinary noun headword (Bright 1971; Clark Hall 1960).

Development to Old English The inherited stem shows regular lowering of *u* to *o* before *r*, and final loss yields *þorn* ‘thorn’. The noun is therefore a regular Old English continuation of the Proto-Germanic thorn-family.

Source note The comparative sources preserve more than one stem formation, but the Old English target itself is simply the citation form *þorn* ‘thorn’.

thought — OE *þōhte*

Derivation: **θánxtē* > *þōhte* (regular).

Derivation trace Proto input: **θánxtē*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**θōxte*
EAF Nasalized Low Rounding	**θōxtē*		
Early Anglo-Frisian			
[no change]			

Old English form: *þōhte*

Reconstruction and comparative evidence The derivational input is the class I weak preterite third singular of the verb ‘think’, PGmc **θánxtē* ‘thought’ (traditionally *þanhtē). Ringe gives the principal parts in full as *þankijaną, *þanhtē, *þanhtaz, with the comparative set Gothic þagkjan ‘think’, þāhta ‘thought’, ON þekkja ‘think’, þátti ‘thought’, OE *þencan*, *þōhte*, *þōht*, OHG denken ‘think’, dāhta ‘thought’ (Ringe 2017, 281). The preterite stem is reconstructed a second time, independently of the paradigm list, among the past participles that show pre-*t devoicing: post-PIE *tong- gives PGmc *þank- in *þankijaną beside *þanhtaz ‘thought’ (Ringe 2017, 136). In both passages the Proto-Germanic form carries the nasal before the dorsal fricative. The nasal is therefore present in the derivational input and its loss is derived by the cascade instead of being built into the reconstruction.

The corpus already holds the present stem of the same verb as *think*, PGmc **θánkijaną* ‘think’. The preterite is entered as a separate row because the two stems have entirely different vowel histories, and only the preterite passes through the nasal-loss complex.

The connectorless formation Ringe’s full principal parts establish an inherited finite word, not a CAPR assembly from a present stem and a general ending. His set contains exactly five reconstructable Proto-Germanic class-I verbs with connectorless pasts; Ringe and Taylor distinguish that inherited set from later extension (Ringe 2017, 281; Ringe and Taylor 2014, 71–75, 97–99). The selected third singular is the verb ‘thought’, not the noun.

Its deeper formation is a different question. Fulk discusses **þaŋxta* and the cluster **þaŋk-t-*, reviewing retention of a present-only suffix, loss of an earlier connector, and analogical remodelling without finding a consensus (Fulk 2018, 299–301). Those morphological illustrations are not additional complete selected words. In particular, his counterfactual buy form **bu3-dē* predicts the wrong Gothic result; it is neither an endorsed input nor evidence for thought’s finite cell (Fulk 2018, 300). Successful nasal-loss and rounding phonology does not explain the connector’s absence. The inherited reconstruction is secure within this account while that formation-level premise remains qualified.

Old English evidence West Saxon þōhte ‘thought’ is the standard preterite singular of *þencan* (Clark Hall 1960; Bosworth and Toller 1898). Fulk names this exact form as the Anglo-Frisian witness for the rounding of the lengthened nasalized low vowel: the lengthened vowels “may have remained nasalized for a considerable time, well past the close of the NWGmc. period, since *ą produced this way developed to ō in Anglo-Frisian (as in OE pret. sg. þōhte, OFris. thochte

‘thought’) and did not fall together with OE ā < ai or OFris. ā < ai, au” (Fulk 2018, sect. 4.1, p. 55). The Old Frisian counterpart thochte ‘thought’ confirms that the development is Anglo-Frisian and not an English innovation.

Development to Old English The diagnostic root-vowel history consists of nasal loss and rounding. Loss of the nasal before the dorsal fricative nasalizes and lengthens the preceding vowel, so *θánxtē becomes *θãxtē; the Anglo-Frisian rounding of the nasalized low vowel then gives *θōxtē, which surfaces as *þōhte* with the fricative written *h* (Ringe 2017, 281; Fulk 2018, sect. 4.1, p. 55). The trace also records ordinary final-vowel shortening; that endpoint adjustment is separate from the diagnostic root-vowel history.

Three non-firings matter as much as the firings. The Ingvaeonic nasal-spirant law does not apply, correctly, because the dorsal fricative is not one of the *f, *þ, *s spirants that law governs. Preconsonantal loss of the dorsal fricative does not apply either, so the fricative survives to the surface; this is what makes the preterite a cleaner diagnostic than *fist* (Ringe 2017, 281). Finally the rounding does not touch ordinary long low vowels of other origin, as *stone* and *home* show.

Why this row is in the corpus The nasal-loss rule has a high-vowel branch and a low-vowel branch. Before this row only the high-vowel branch had a corpus witness, in *fist* *fúnxstiz, whose input vowel is *u and which never reaches the rounding rule at all. The low-vowel pathway *aNx > *ą̃x > *ōx rested on the handbooks alone. This row supplies an attested instance of it, and with it the feeding relation between the nasal loss and the rounding, which was thereby promoted from stage entailment to an independently demonstrated ordering. See SC103 PGmcNasalLossBeforeX and SC104 EAFNasalizedLowRounding.

tide — OE *tīd*

Derivation: citation reconstruction **tī́diz*; form followed here **tī́diz* > *tīd* (regular).

Derivation trace Proto input: **tī́diz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**tī́d*
EAF Final Z Deletion	**tī́di*		
Early Anglo-Frisian			
[no change]			

Old English form: *tīd*

Reconstruction and comparative evidence Kroonen gives **tīdi-* ‘time’, and Orel’s **tīðiz* ‘tide’ points to the same feminine noun (Kroonen 2013; Orel 2003). The related verb *tīdan* ‘happen’ is separate.

Old English evidence Bright records *tīd* ‘tide’ with singular *tīde* ‘tide’ and plural *tīda* ‘tides’, and Clark Hall treats *tīd* ‘tide’ as the ordinary noun ‘time, period, season’ (Bright 1971; Clark Hall 1960, 309).

Development to Old English From **tī́diz* ‘tide’, final *z* is lost and the high final vowel drops, leaving *tīd* ‘tide’. The development is straightforward for a feminine i-stem.

Lexical note English *tide* can also lead to the separate weak verb *tīdan* ‘happen’; the noun is *tīd* ‘tide’.

token — OE *tācn*

Derivation: **táiknq* > *tācn* (regular).

Derivation trace Proto input: **táiknq*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**tākn*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**tāknq*		

Old English form: *tācn*

Reconstruction and comparative evidence Kroonen cites **taikna-* and Orel **taiknan* 'token' for the noun 'sign, token' (Kroonen 2013; Orel 2003, 438). The derivational input **táiknq* 'token' is the simple citation-form noun used for the derivation.

Old English evidence Campbell and Sievers-Brunner preserve both unbroken *tācn* 'token' and broken *tācen* 'token', with oblique *tācnes* 'token' remaining unbroken (Campbell 1959; Brunner 1965).

Development to Old English Monophthongization of *ai* yields *ā*, and loss of the final nasal vowel leaves *tācn* 'token'. The unbroken form is therefore a regular Old English outcome.

Form note *tācn* 'token' is the attested unbroken citation form selected here. Later West Saxon prose often prefers *tācen* 'token', but that does not displace the older unbroken form (Campbell 1959; Brunner 1965).

town — OE *tūn*

Derivation: **tūnq* > *tūn* (regular).

Derivation trace Proto input: **tūnq*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**tūn*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *tūn*

Reconstruction and comparative evidence Kroonen cites **tūna-* 'fenced area', while Orel gives **tūnan* 'town' and **tūnaz* 'town' (Kroonen 2013; Orel 2003, 452). The derivational input **tūnq* 'town' is the simple citation-form noun used in the derivation.

Old English evidence Clark Hall records *tūn* 'town' as the ordinary headword 'enclosure, yard, village, town' (Clark Hall 1960).

Development to Old English The inherited long *ū* is preserved, and loss of the final nasal vowel yields *tūn* 'town' regularly (Brunner 1965).

Source note The comparative headwords vary, but the Old English target here is the direct citation form *tūn* ‘town’, not an oblique **tūnăn* ‘town’.

wade — OE *wadan*

Derivation: **wádaną* > *wadan* (regular).

Derivation trace Proto input: **wádaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wædaną*
[no change]	OE A Restoration	**wadaną*
	OE Heavy Syllable Nasal Apocope	**wadan*
Early Anglo-Frisian	OE Secondary Nasalization	**wadąn*
[no change]	OE Weak Tail Reduction	**wadan*

Old English form: *wadan*

Reconstruction and comparative evidence Campbell and Ringe and Taylor describe A-restoration before a following back vowel, and Luick explicitly includes *wadan* ‘wade, go through’ among the standard open-syllable examples (Campbell 1959; Ringe and Taylor 2014; Luick 1914–1940, 239).

Old English evidence Clark Hall gives *wadan* ‘wade, go through’ as the verb ‘to go, move, stride, advance’, and Bright lists the same infinitive in the strong-verb paradigm (Clark Hall 1960; Bright 1971).

Development to Old English From **wádaną* ‘wade’, Anglo-Frisian brightening first gives **wædaną* ‘wade’. A-restoration before the back-vocalic infinitive ending then returns *a*, and later reduction yields *wadan* ‘wade, go through’ (Campbell 1959; Ringe and Taylor 2014).

Development note The infinitive belongs to the A-restoration class. The citation form is therefore *wadan* ‘wade’, not a fronted *wæden* ‘waded’-type output.

warp — OE *weorpan*

Derivation: **wérpaną* > *weorpan* (regular).

Derivation trace Proto input: **wérpaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**wéorpaną*
[no change]	OE Heavy Syllable Nasal Apocope	**wéorpan*
	OE Secondary Nasalization	**wéorpąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**wéorpan*
[no change]		

Old English form: *weorpan*

Reconstruction and comparative evidence Ringe and Taylor distinguish preterite **warp* ‘warp’ from infinitive **werpana* ‘warp’, and the derivational input here is the verbal form **wérpaną* ‘warp’ (Ringe and Taylor 2014).

Old English evidence Clark Hall records *weorpan* as the strong verb headword and separately lists *wearp* as both noun and preterite. Bright gives the paradigm *weorpan, wearp, wurpon, worpen* (Clark Hall 1960; Bright 1971).

Development to Old English Breaking before *r* + *C* yields *weor-*, and the infinitive develops regularly to *weorpan* (Campbell 1959; Hogg 1992a).

Lexical note English *warp* also points to related *wearp* material. The target is the infinitive *weorpan*.

wash — OE *wascan*

Derivation: **wáskaną* > *wascan* (regular).

Derivation trace Proto input: **wáskaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wæskaną*
[no change]	OE A Restoration	**waskaną*
	OE Heavy Syllable Nasal Apocope	**waskan*
Early Anglo-Frisian	OE Secondary Nasalization	**waskąn*
[no change]	OE Weak Tail Reduction	**waskan*

Old English form: *wascan*

Reconstruction and comparative evidence Kroonen cites **waskan-*, Orel **waskanan* ‘wash’, and Ringe and Taylor likewise derive Old English *wascan* from the same verb family (Kroonen 2013; Orel 2003, 489; Ringe and Taylor 2014, 142).

Old English evidence Clark Hall heads the verb as *wascan* ‘wash’, while Sievers-Brunner also notes the variant *wæscan* ‘wash’ (Clark Hall 1960; Brunner 1965).

Development to Old English From **wáskaną* ‘wash’, brightening gives **wæskaną* ‘wash’. A-restoration before the *sC* cluster restores *a*, and medial *sc* remains unpalatalized before the following back vowel, yielding *wascan* (Campbell 1959; Ringe and Taylor 2014, 142).

Form note The conservative citation form *wascan* ‘wash’ is selected here. Spellings such as *wæscan* ‘wash’ or *wasċan* ‘wash’ belong to variant or normalized background rather than to the target of this entry.

wax — OE *weaxan*

Derivation: **wáxsaną* > *weaxan* (regular).

Derivation trace Proto input: **wáxsaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wæxsaną*
[no change]	OE Breaking	**weaxsaną*
	OE Heavy Syllable Nasal Apocope	**weaxsan*
Early Anglo-Frisian	OE Secondary Nasalization	**weaxsąn*
[no change]	OE Weak Tail Reduction	**weaxsan*
	OE Xs Merge	**weaXSan*

Old English form: *weaxan*

Reconstruction and comparative evidence Kroonen cites the verb as **wahs(j)an-*, Orel as **waxsanan* ‘wax’, and Ringe and Taylor discuss the prehistory of Old English *weaxan* within the same verbal family (Kroonen 2013; Orel 2003, 478; Ringe and Taylor 2014).

Old English evidence Clark Hall gives *weaxan* ‘wax’ as the verb headword and separately records bare *wax* ‘wax’ as a preterite form; Bright likewise treats *weaxan* ‘wax’ as the infinitive (Clark Hall 1960; Bright 1971).

Development to Old English From **wáxsanǫ* ‘wax’, brightening and breaking yield *weax-*, and the infinitive develops regularly to *weaxan*. The cluster is preserved here, since *xs* > *s* belongs to forms where another consonant follows, such as *wæstm* ‘growth’, not to the infinitive itself (Campbell 1959; Brunner 1965).

Lexical note The target here is the infinitive *weaxan* ‘wax’. Noun *weax* ‘wax’ and preterite *wax* / *wēox* ‘waxed’ belong to different lexical or paradigm slots.

way — OE *weġ*

Derivation: **wégaz* > *weġ* (regular).

Derivation trace Proto input: **wégaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wég*
EAF Final Z Deletion	**wéga*	OE Velar Palatalization	**wéj*
Early Anglo-Frisian		OE Palatal Fricative Merger	**wéj*
[no change]			

Old English form: *weġ*

Reconstruction and comparative evidence Kroonen cites the noun as **wega-* ‘way, road’, while the selected derivational form here is nominative-singular **wégaz* ‘way’ (Kroonen 2013). Campbell, Hogg, and Ringe and Taylor use the same word as the standard contrast between singular palatal *weġ* ‘way’ and inflected *wegas* / *wegum* (Campbell 1959; Hogg 1992a; Ringe and Taylor 2014, 341).

Old English evidence The Old English singular is the ordinary noun *weg* ‘way’, here normalized as *weġ* ‘way’ to show the palatal final. The contrasting plural and oblique forms *wegas* ‘ways’, *wegum* ‘ways’ keep a velar stop before the following back vowel (Campbell 1959; Ringe and Taylor 2014, 341).

Development to Old English From **wégaz* ‘way’, final **z* is lost and the weak tail apocopates, leaving word-final **g* after a front vowel. In that environment Old English palatalization yields *weġ* ‘way’, whereas *wegas* ‘ways’ remains velar because the following *a* blocks the same outcome (Campbell 1959; Hogg 1992a; Ringe and Taylor 2014, 341).

Form note Normalized *weġ* ‘way’ and dictionary *weg* ‘way’ represent the same noun. No Old English form *wē* ‘we’ is supported by the evidence for ‘way’.

weapon — OE *wǣpn*

Derivation: **wépnǫ* > *wǣpn* (regular).

Derivation trace Proto input: **wépną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**wǣpn*
PNWGmc Long E Lowering	**wāpną*		
EAF Long A Fronting	**wǣpną*		
Early Anglo-Frisian			
[no change]			

Old English form: *wǣpn*

Reconstruction and comparative evidence Kroonen reconstructs a double-stem noun **wēbna- ~ *wēpna-* and cites OE *wæpn* ‘weapon’ among its reflexes (Kroonen 2013, 617). The derivational input **wépną* ‘weapon’ represents the unbroken citation-form noun rather than the later broken simplex.

Old English evidence Campbell’s cluster-noun discussion preserves unbroken *wépn* ‘weapon’ beside broken *wépen* ‘weapon’-type forms (Campbell 1959, 150; 1959, 226–27). Bright contrasts broken nominative *wǣpen* ‘weapon’ / *wapen* ‘weapon’ with unbroken oblique *wǣpnes* ‘weapon’, while Clark Hall lemmatizes the noun under *wapen* ‘weapon’ and also preserves unbroken forms in compounds and related spellings (Bright 1971, 29; Clark Hall 1960, 355).

Development to Old English Northwest Germanic lowering gives *wǣpn* ‘weapon’, and loss of the final nasal vowel leaves the unbroken cluster word-finally. The Old English form here is the attested unbroken form *wǣpn* ‘weapon’.

Form note The ordinary late West Saxon simplex headword is *wǣpen* ‘weapon’, but Campbell’s noun-class discussion also preserves unbroken *wépn* ‘weapon’ beside broken *wépen* ‘weapon’-type forms (Campbell 1959, 150; 1959, 226–27). *wǣpnes* ‘weapon’ remains the regular unbroken oblique comparator (Bright 1971, 29; Clark Hall 1960, 355).

wether — OE *weþer*

Derivation: **wéθruz* > *weþer* (regular).

Derivation trace Proto input: **wéθruz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**wéθr*
EAF Final Z Deletion	**wéθru*	OE Epenthetic Vowel	**wéθer*
Early Anglo-Frisian			
[no change]			

Old English form: *weþer*

Comparative reconstruction and development Old English *weþer* ‘wether, ram’ belongs with Gothic *wiþrus* ‘lamb, young ram’, Old Norse *veðr* ‘ram’, Old Saxon *withar* ‘wether’ and Old High German *widar* ‘ram’. These are masculine animal names, not the neuter weather noun, despite the similar spellings. Kroonen reconstructs **weþru-*; Kluge–Seebold likewise gives a masculine Germanic **weþru-*, while allowing a similar stem formation (Kroonen 2013, 584; Kluge 2011, 985).

The inherited root has short **e*. Gothic *i* does not require Proto-Germanic **i*: inherited short **e* and **i* merged as *i* in Gothic outside its breaking environments (Fulk 2018, 59). The continental *i*-forms require equal care. Raising before a following *u* is independently documented in

Old Saxon and Old High German, but it is not universal there; that general development is not a complete, independently dated history of every cell of this noun (Fulk 2018, 69). The comparison consequently supports inherited **e* without requiring every daughter-language vowel to preserve it unchanged.

The principal formation question concerns the inflection, not the root vowel or masculine gender. Orel records both **weþruz* and **weþraz*, respectively masculine *u*- and *a*-stem alternatives, and notes traces of *u*-inflection in Norse (Orel 2003, 459–60). Gothic nominative *wiþrus* supplies particularly useful positive evidence for the older *u*-stem. CAPR therefore adopts the masculine nominative singular **weþruz*, represented as **wéθruz*, while retaining **weþraz* as a legitimate alternative. The latter may represent early inflectional remodelling; the dictionaries do not establish its precise date.

The Old English masculine paradigm, including genitive *weþeres*, does not decide the original Germanic stem class. Bosworth–Toller records *weþer, es; m.* and the senses ‘a wether, a ram’ (Bosworth and Toller 1898, 1212). Masculine *u*-stems underwent reclassification towards other declensions in West Germanic, and heavy-stem nominatives lost their final short **u* (Fulk 2018, 163). In the selected nominative, loss of final **-z* and **-u* leaves the consonantal ending resolved as unstressed *-er* in *weþer*, consistent with the vocalization of final postconsonantal sonorants (Fulk 2018, 91–92). The root vowel is already **e*; no early lowering of **i* is needed for this derivation.

The further etymology is less secure. Orel connects the noun with the Indo-European word for ‘year’, comparing Hittite, Greek and Albanian year-words and names for young cattle (Orel 2003, 459–60). Kroonen makes that connection conditional: if correct, the original designation was ‘yearling, one-year-old animal’, with Sanskrit *vatsá-* ‘calf’ among the comparanda (Kroonen 2013, 584). Kluge–Seebold notes phonological difficulties with the traditional comparison involving Latin *vitulus* ‘calf’, and also considers derivation from a root meaning ‘sheep’ (Kluge 2011, 985–86). A young-animal designation could develop into ‘ram’ or ‘wether’, but this attractive semantic account is not as secure as the Germanic masculine *u*-stem reconstruction.

where — OE *hwǣr*

Derivation: **xwár* > *hwǣr* (regular).

Derivation trace Proto input: **xwár*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc Monosyllabic Final R Lengthening	**xwār*	
EAF Long A Fronting	**xwǣr*	
Early Anglo-Frisian		
[no change]		

Old English form: *hwǣr*

Old English evidence Clark Hall records hwǣr ‘where’, the West Saxon comparison form used here (Clark Hall 1960, 170). Hill sets it beside Anglian hwēr ‘where’ (Hill 2017, 142–43). The contrast concerns the quality of a long vowel, not lengthening in West Saxon alone. The macrons belong to normalized scholarly forms; they do not imply that every manuscript token explicitly marks quantity.

Reconstruction and comparative evidence Kroonen gives **hwar* ‘where’, an interrogative formation with locative **-r*. He explicitly derives the long-vowel West Germanic forms by monosyllabic lengthening, rejecting inherited **hwēr* as their source (Kroonen 2013, 263). Orel instead preserves both **xwar* ‘where’ and **xwēr* as citation alternatives, connecting the formation with pronominal **xwa-* (Orel 2003, 198). The selected **xwár* ‘where’ follows the short-vowel line. Its

acute marks the accented root, while **xw-* represents the Germanic dorsal-plus-glide sequence corresponding to the dictionaries' **hw-* or **xw-*.

The extended interrogative supplies an important morphological comparison. Gothic *ƕarjis* 'which' and Old Norse *hverr* 'who, which' support an inherited disyllabic **hwarja-* with short root vowel. Hill argues that such a formation could precede lengthening of the monosyllabic adverb; its short vowel therefore need not date lengthening to West Germanic. Gothic adverbial *ƕar* 'where' itself leaves quantity ambiguous in writing (Hill 2017, 142, 147–48).

Development to Old English From short **hwar*, SC112, stressed monosyllabic lengthening before inherited final *r* gives **hwār*; then SC101, long-*ā* fronting gives **hwǣr*. The initial consonant sequence is represented orthographically as *hw-*. The derivational input with short **a* thus reaches *hwǣr* through lengthening followed by fronting, without inflectional retargeting. The parallel *þǣr* 'there' makes the same vowel development visible in a different pronominal formation.

The historical interpretation remains narrower than a uniquely dated reconstruction. Hill's Proto-Germanic account depends in part on his explanation of the Scandinavian short-vowel forms, while Ringe and Taylor retain a later West Germanic lengthening (Hill 2017, 146–48; Ringe and Taylor 2014, 65). The early placement adopted here preserves inherited-*r* conditioning before rhotacism could obscure consonant ancestry. That analytical choice does not establish an independently dated ordering against SC015 or the lowering of inherited **ē₁* in SC024.

who — OE *hwā*

Derivation: **xwáz* > *hwā* (regular).

Derivation trace Proto input: **xwáz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
Monosyllabic Final Z Loss	**xwā*	
Early Anglo-Frisian		
[no change]		

Old English form: *hwā*

Reconstruction and comparative evidence The derivational input is the nominative singular masculine of the interrogative pronoun, PGmc **xwáz* 'who' (traditionally *hwaz; Gothic ƕas 'who'). Ringe and Taylor cite exactly this form among the witnesses for the northern West Germanic loss of word-final *-z in stressed monosyllables: "*hwaz 'who' > OE hwā" (Ringe and Taylor 2014, 86). The nominative is the natural citation cell of the interrogative and the direct ancestor of the Old English headword form, so no paradigm-cell retargeting is involved.

Old English evidence West Saxon hwā 'who' is the standard headword form (Campbell 1959, sect. 125, p. 49). The vowel is decisive for the derivation: Campbell states that the back vowel of *hwā* never underwent Anglo-Frisian brightening — "*hwǣ does not exist" — and Sievers-Brunner records the same restriction (Campbell 1959, sect. 125, p. 49; Brunner 1965, sect. 137 Anm. 1, p. 129).

Development to Old English The derivation is two historical steps. First, the later northern West Germanic loss of word-final *-z in stressed monosyllables removed the sibilant with compensatory lengthening of the short nucleus: **xwáz* 'who' > **hwā* 'who' (Ringe and Taylor 2014, 86). Old High German instead rhotacized the sibilant (cf. OHG wer 'who' type inflection), which is why this loss counts among the diagnostic Ingvaeonic features (Fulk 2018, 18). Second, the resulting long back *ā stayed back: Anglo-Frisian brightening of long final *ā requires a preceding

nucleus in the word, and in this monosyllable there is none, so the form surfaces directly as OE *hwā* (Campbell 1959, sect. 125, p. 49).

An alternative account derives the endingless pronoun from a generalized unaccented sentence variant *hwa with later lengthening (Campbell 1959, 166; Brunner 1965, sect. 182, p. 160). Ringe and Taylor reject the unaccented-variant analysis for this witness set because *mā* 'more' and *cū* 'cow' are not plausibly unaccented words (Ringe and Taylor 2014, 86). The project follows Ringe and Taylor; the alternative is recorded, not adopted.

Why this row is in the corpus This row makes SC097, the monosyllabic final *-z loss, corpus-witnessed: before its addition the change was validated only by synthetic controls. It also witnesses the ordering SC097 before SC003 rhotacism — a rhotacized* †hwar *could never yield* hwā* — and exercises the narrowed long-final clause of Anglo-Frisian brightening. See `docs/sound_changes/audits/corpus-maturation-01-candidate-adjudication.md` §1.

will — OE *willa*

Derivation: **wéljô* > *willa* (regular).

Derivation trace Proto input: **wéljô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**willjô*
PWGmc J Gemination	**wélljô*	OE Unstressed Long Vowel Shortening	**willja*
Early Anglo-Frisian		OE J Loss After Heavy	**willa*
[no change]			

Old English form: *willa*

Reconstruction and comparative evidence Kroonen separates noun **weljan* 'will, wish' from verb **weljan* 'to want', while Orel and Kluge represent the noun as **weljōn* 'will' (Kroonen 2013; Orel 2003; Kluge 2011). The selected derivational form **wéljô* 'will' is the noun-side input used for this row.

Old English evidence Clark Hall lemmatizes noun *willa m.* separately from verb *willan* 'will, want' (Clark Hall 1960, 368). The Old English form here is the noun citation form, not the related verb.

Development to Old English From **wéljô* 'will', j-gemination yields a heavy stem, i-umlaut gives *will-*, and later shortening plus j-loss produce *willa* 'will, desire'. The noun is therefore a regular weak masculine outcome.

Lexical note The target here is the noun *willa* 'will, wish'. Related verb *willan* 'will, want' belongs to a separate lexeme and should not be substituted for the noun row (Kroonen 2013; Clark Hall 1960, 368).

wind — OE *windan*

Derivation: **wíndaną* > *windan* (regular).

Derivation trace Proto input: **wíndanᾱ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**wíndan*
[no change]		OE Secondary Nasalization	**wíndąn*
Early Anglo-Frisian		OE Weak Tail Reduction	**wíndan*
[no change]			

Old English form: *windan*

Reconstruction and comparative evidence Kroonen distinguishes noun **winda-* from verb **windan-*, and the present row belongs to the verb (Kroonen 2013). Fulk and Ringe and Taylor derive the dental directly from PIE **wendh-*, not from a Verner alternant (Fulk 2018; Ringe and Taylor 2014).

Old English evidence Clark Hall and Bosworth-Toller record *windan* ‘wind, coil’ as the verb headword (Clark Hall 1960; Bosworth and Toller 1898, 101). The Old English form here is the ordinary infinitive of the strong verb.

Development to Old English The derivational input **wíndanᾱ* ‘wind’ yields the regular infinitive *windan* by ordinary heavy-syllable apocope and weak-tail reduction. The form is therefore a straightforward strong-verb outcome.

Lexical note English *wind* also names the noun. This row represents the class-III verb, not the noun (Kroonen 2013; Clark Hall 1960).

wold — OE *weald*

Derivation: **wálθuz* > *weald* (regular).

Derivation trace Proto input: **wálθuz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**wældu*
EAFL Th Voicing	**wálduz*	OE Breaking	**wealdu*
EAF Final Z Deletion	**wáldu*	OE High Vowel Apocope	**weald*
Early Anglo-Frisian			
[no change]			

Old English form: *weald*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **walþu-* and gives OE *weald* ‘forest, woodland’ beside other West Germanic *wald* ‘forest’ forms (Kroonen 2013). The derivational input **wálθuz* ‘wold’ is the nominative singular used for the derivation.

Old English evidence Clark Hall makes *weald* ‘forest, woodland’ the main noun headword and cross-refers *wald* ‘forest’ and *wold* ‘plain, open country’ to it (Clark Hall 1960). Clark Hall glosses *wald* as *(N,VPs)* — Northumbrian and Vespasian Psalter (Mercian) — making this an explicitly dialect-labelled Anglian form.

Development to Old English **lþ* voices to *ld*, Anglo-Frisian brightening yields *wæld-*, and breaking before the cluster gives *weald-*; apocope then yields *weald* ‘forest, woodland’. The noun is therefore a regular breaking outcome.

Dialect note *wald* ‘forest’ survives as an Anglian variant in the same family (Clark Hall *wald (N,VPs)*: Northumbrian and Vespasian Psalter). The Old English form here is normalized *weald* ‘forest, woodland’, not the variant form (Clark Hall 1960; Ringe and Taylor 2014).

world — OE *weorold*

Derivation: citation reconstruction **wíra-àldiz*; form followed here **wír-àldiz* > *weorold* (regular).

Derivation trace Proto input: **wíràldiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Strip Secondary Stress	**wérald*
PWGmc Early I Apocope	**wíràldz*	OE Inter Stress Raising	**wéruld*
PNWGmc I Lowering	**wéràldz*	OE Back Mutation	**wéoruld*
EAF Final Z Deletion	**wéràld*	OE Med Unstressed U Lowering	**wéorold*
Early Anglo-Frisian			
[no change]			

Old English form: *weorold*

Reconstruction and comparative evidence The word is the old compound ‘age of men’. Orel reconstructs **wira-alđiz*, with the older first-element vowel, while Ringe and Taylor discuss the form **weraldiz* ‘world’ and its pre-Old-English chain **weraldu* ‘world’ > **weruld* ‘world’ (Orel 2003, 462; Ringe and Taylor 2014, 326). Kluge-Seebold gives the compound **wira-aldō* ‘world’ and explicitly includes Old English *weorold* ‘world’ (Kluge 2011, 981).

Kroonen’s separate entries for **wira-* ‘man’ and **aldi-* ‘age’ support the components, not independently the complete compound or its accentuation. The first vowel is not beyond dispute: Ringe and Taylor allow **weraz* beside **wiraz* for the simplex, since Gothic pre-*r* developments and Norse lowering obscure the comparison. Initial *i* is therefore a supported reconstruction, not something established by its usefulness in the derivation (Kroonen 2013, 20, 588; Ringe and Taylor 2014, 34).

The derivational input **wír-àldiz* ‘world’ adopts the inherited compact i-stem nominative. No early ō-stem intervention is required in its selected phonological history. The compound’s later class mixing remains relevant, rather than being silently denied because the conservative nominative suffices (Orel 2003, 462; Ringe and Taylor 2014, 202, 326).

The class contrast is real: Ringe and Taylor oppose i-stem Old Saxon *werold* ‘world’ and Old High German *weralt* ‘world’ to ō-stem Old Frisian *warld* ‘world’ and Old Norse *verǫld* ‘world’. Their separate qualification that the late Norse word is probably borrowed prevents counting it as independent evidence for a common inherited ō-stem innovation (Ringe and Taylor 2014, 202, 326). Nor did every Old English cell necessarily abandon the older class. Campbell explicitly records frequent endingless accusative singular *weorold*, characteristic of feminine i-stems. This is evidence for a conservative ending, not by itself proof of an entirely unchanged stem history (Campbell 1959, 244, §610(1) n.2).

The selected inherited nominative is sufficient; a complete reconstruction of its oblique paradigm is not assumed. Different nominal endings must still be distinguished, rather than replacing them indiscriminately with *-u* (Fulk 2018, 154–56, 161–62).

No mixed-stage early analogical transponent is now necessary. The compact stem nevertheless presupposes resolution or reduction of the compound junction. The etymological spelling **wira-aldi-* does not independently establish a literal hiatus at the chosen stage, and consonant-separated linking-vowel syncope is not a demonstration of how adjacent *a*+*a* would have resolved. An already **wer-* compound requires no lowering in its own derivation. Accordingly the derivational input cannot prove that SC015 PNWGmcILowering historically preceded SC036

OEInterStressRaising. The adopted consonantal lowering has no following-vowel trigger for interstress raising to remove. The two processes remain compatible in this derivation without that supposed ordering argument (Orel 2003, 462; Ringe and Taylor 2014, 326; Hill 2017, 153–56).

Old English evidence Old English does not preserve a single isolated form. Ringe and Taylor give West Saxon *weorold* 'world' ~ *worold* 'world', Mercian *weoruld* 'world', Northumbrian *woruld* 'world', and Kentish *wiarald* 'world' (Ringe and Taylor 2014, 326). Sievers-Brunner and Bright present the same wider set, including the syncopated *world* 'world' and later rounded *wurold* 'world' (Brunner 1965, sect. 113; Bright 1971, 465).

The Old English form used here is the West Saxon form *weorold* 'world'. It is an attested Old English form within that broader variant cluster, not the only form the lexeme ever shows.

The two sets of vowels carry different evidence. Initial *eo/o* concerns back mutation and subsequent rounding after *w*; medial *a/o/u* concerns weak vocalism. Kentish medial *a* does not show the breaking expected under stronger prominence. Later *world* demonstrates loss of that weak vowel, not another raising event. The variants do not require every medial *o* to have passed through *u* (Campbell 1959, 142, §338; Hogg 2011, 217–18, §§6.6–6.7; Ringe and Taylor 2014, 326).

Čermák places *world* in his class of obscured compounds affected by loss of an unstressed vowel, comparing *woruld/worold*. This is evidence for later medial syncope, not another instance of the earlier raising. His possible reinterpretation of the second element is explicitly a suggestion; the entry supplies no new primary-text locator (Čermák 2005, 36, 39). Sauer contrasts Old English feminine genitive *worulde* with later Middle English genitive-*s* and manuscript variation. These developments help distinguish inherited inflection from subsequent restructuring, but cannot date a prehistoric shift into the ō-stems (Sauer 1992, 89–90, 354–56).

Compound prominence and early Old English footing Goering argues that early Old English *weruld* 'world' belonged to a single initial, trimoraic foot, rather than an initial light foot followed by a secondary-stressed heavy foot. Apocope would occur under either analysis and cannot decide between them. The ordinary reduction and eventual loss of the medial vowel, together with resolution of inflected *worulde* 'world' in *Beowulf* 3068a and 2343a, support his analysis (Goering 2023, 79–82, §4.5.1.2).

This does not rule out earlier compound secondary accent. Kaluza's older theory allowed subsidiary stress in *weorold* and other disyllables; Stanley's presentation is evidence for that theory, not direct written accent evidence (Stanley 2022, 34). The adopted relative history loses independent second-element prominence before weak rounding. Goering's early Old English system, perhaps around 600 and compatible with the seventh-/eighth-century poetic horizon, supplies an approximate upper horizon, not a precise date for this word's deaccentuation (Goering 2023, 84–85, §4.6). Foot structure, lexical prominence and metrical ictus remain distinct. Hogg's discussion warns against making "tertiary stress" an unexplained name for the desired outcome; Minkova likewise requires independent linguistic controls on metrical reconstruction (Hogg 1996, 3, 9–11; Minkova 1996, 20–29). The grave in the selected reconstruction marks the earlier second-root accent, not an unstressed linking vowel. It survives the linking-syncope operation and is then removed before weak raising, not asserted as actual independent prominence at the latter change. True compound secondary stress can protect a vowel, while obscured compounds reduce; the history of transparency cannot be inferred from a hyphen alone (Fulk 2018, 36–38, §2.2).

Competing histories of the weak vowel The starting point is weak vocalism, not an obligatory analogy. Campbell describes retraction before *l/r* plus consonant, rounding to *o*, and further *u* in some forms. Hogg similarly distinguishes retraction of a weak vowel from the breaking of a more prominent one. Luick particularly associates rounding with labials or following *l*, while expressly leaving the connection between the *o* and *u* developments uncertain (Campbell 1959, 142, §338; Hogg 2011, 217–18, §§6.6–6.7; Luick 1914--1940, 125–26, §113). Bülbring's assimilation before following *u* supplies one account of *world*, but Luick's criticism and the wider weak-vowel evidence prevent treating it as the only possible environment (Bülbring 1902, 147–49, §§366–367; Luick 1914--1940, 126, §113; Ringe and Taylor 2014, 202).

CAPR adopts weak short *a* > *u* before *l* plus consonant in a light-initial primary-stressed domain: one intervening consonant separates the initial short nucleus from the medial weak vowel. This bounded interpretation joins Luick's environment to Goering's LH analysis; the exact formal restriction is CAPR's qualified reconstruction, not either author's quoted law. Heavy-initial retained-prominence compounds such as *ānfeald* 'onefold' argue against unrestricted raising after removal of accent annotations. Weak *twifold* 'twofold' is a relevant rounded comparison; primary medial stress is not rewritten (Bülbring 1902, 148–49, §367; Campbell 1959, 142, §338; Hogg 2011, 217–18; Goering 2023, 79–82).

The selected operation precedes early unstressed fronting and breaking. A post-fronting compression of weak *æ* retraction, rounding and raising also succeeds and resembles Hogg's sequence. The chosen ordering does not require that intermediate vowel, without claiming an independently dated universal law. Versloot's alternative conditional fronting and later breaking challenge compulsory fronting before *lC*; his principally stressed-domain analysis does not itself establish weak *a* > *u* in *world* (Hogg 2011, 217–18; Versloot 2026, 111–14, 122–23).

Selected regular projection to Old English From selected **wír-àldiz*, early short-*i* apocope, SC015 PNWGmcILowering, and final-*z* loss give **wér-àld*. After linking-syncope has read the earlier accent information, grave removal gives **wérald*. SC036 OEInterStressRaising supplies **wéruld*; SC059 OEBackMutation gives **wéoruld*; and SC040 OEMedUnstressedULowering gives *weorold*. No final high vowel survives to undergo later apocope.

This is the longest defensible regular sequence from the selected compact inherited nominative, not a derivation of the unresolved earlier compound junction. The *regular* classification concerns that selected span. The inherited accent, original *i/e*, and exact weak-vowel chronology retain their qualifications. One raising rewrite may compress gradual rounding and raising; it does not assert universal *a* > *u* > *o* across the dialect variants (Luick 1914–-1940, 125–26, §113; Ringe and Taylor 2014, 202, 326).

An early rounded *o* alone fails in the existing model at SC100 OEFinalUnstressedOLowering, not because historical rounding is intrinsically impossible. Furthermore, the present SC059 OEBackMutation also admits *o* as a trigger for initial *e*; its executable behaviour cannot prove that only *u* feeds the diphthong. The adopted inherited high-vowel path also succeeds under explicitly restricted *u*-triggered back mutation. No back-mutation or later merger rule is changed to rescue it.

Stage comparison The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing stage chosen for the derivation.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
comparative compound with older first-element vowel	**wíra-àldiz*	citation reconstruction / lexeme label	preserves the older **wir-* tradition of the compound
literature-stage lowered compound after early stem-class shift	**weraldiz* > **weraldu* > **weruld*	Ringe-Taylor background chain to OE *weorold* 'world' ~ *worold* 'world'	explains the older comparative literature cited for the word
selected inherited compact nominative	**wír-àldiz*	regular output: *weorold* 'world'	exact West Saxon match; junction pre-encoded, no necessary early analogy
broader OE variant cluster	—	*worold* 'world', *weoruld* 'world', *woruld* 'world', *wiarald* 'world', *world* 'world'	real attested comparanda that remain outside that West Saxon line

yarn — OE *ġearn*

Derivation: **gárną* > *ġearn* (regular).

Derivation trace Proto input: **gárną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**gærną*
[no change]	OE Breaking	**gearną*
	OE Heavy Syllable Nasal Apocope	**gearn*
Early Anglo-Frisian	OE Velar Palatalization	**jearn*
[no change]	OE Palatal Fricative Merger	**jearn*

Old English form: *ġearn*

Reconstruction and comparative evidence Kroonen cites the noun as **garna-*, and Ringe and Taylor give the early chain **garna* ‘yarn’ > **geern* ‘yarn’ > **gearn* ‘yarn’ > OE *gearn* ‘yarn’ (Kroonen 2013; Ringe and Taylor 2014, 220). The derivational input **gárną* ‘yarn’ is the nominal citation form used here, while oblique **garnăn* ‘yarn’ belongs only to comparative background.

Old English evidence Clark Hall records *gearn (e) n.* ‘yarn, spun wool’, and Bosworth-Toller glosses *gearn* ‘yarn’ with Latin *filatum* ‘spun thread’ (Clark Hall 1960; Bosworth and Toller 1898).

Development to Old English From **gárną* ‘yarn’, brightening and breaking before *rn* yield *gearn* ‘yarn’; palatalization of initial *g* before the resulting front-vocalic sequence gives normalized *ġearn* ‘yarn’. The derivation is regular.

Form note Dictionary *gearn* ‘yarn’ and normalized *ġearn* ‘yarn’ refer to the same noun. The comparative stem **garna-* and oblique **garnăn* ‘yarn’ do not replace the derivational input **gárną* ‘yarn’.

you — OE *ēow*

Derivation: **ízwiz* > *ēow* (regular).

Derivation trace Proto input: **ízwiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Diphthong Leveling	**ēow*
PWGmc Coronal W Assimilation	**íwwiz*		
OE WW Simplification	**íuwiz*		
EAF Final Z Deletion	**íuwi*		
PWGmc Unstressed Word Final I Apocope	**íuw*		
Early Anglo-Frisian			
[no change]			

Old English form: *ēow*

Reconstruction and comparative evidence The derivational input is the dative(-accusative) plural of the second-person plural pronoun, PGmc **ízwiz* ‘you (dat. pl.)’ (Gothic izwis ‘you’). Ringe and Taylor print the development in full: “PGmc *izwiz ‘you (dat. pl.)’ (Goth. izwis) > *iwwi > PWGmc *iuwi ~ *iuw (see 3.1.4) > OE īow, OF iū, OS, OHG iu” (Ringe and Taylor 2014, 41–42), following Stiles’s demonstration of the coronal-w assimilation (Stiles 1985, 89–94). Fulk’s discussion of inherited geminate glides after short vowels supplies the wider reanalysis context, not a replacement citation for that assimilation (Fulk 2018, 117). The oblique plural is the cell that survives as the Old English pronoun *ēow*.

Competing origins of the inherited oblique Kroonen derives **izwiz* by metathesis from **iwsiz*, with a suffix abstracted from the first-person plural, and separately distinguishes older **iwiz* and the dual formations (Kroonen 2013, 275–76). Fulk regards the reported Krahe–Meid reduplication **wiz-wiz* and initial-*w* dissimilation as probably most plausible, while explicitly stating that there is no consensus (Fulk 2018, 187–88). This is his assessment of a reported theory, not a newly verified original argument or a named rebuttal of Kroonen.

Ringe's Katz-based account instead combines case formation and sandhi history from Indo-European obliques including **uswé*, with later prothetic *i*; he also records Weiss's dissimilatory alternative (Ringe 2017, 233–35). The competing proposals concern the genealogy and segmentation of the same relevant plural oblique. Agreement on a later inherited input does not choose between metathesis and suffix abstraction, reduplication and dissimilation, or sandhi and case formation. The regular English path and its independently selected weak-final context cannot settle that earlier morphology.

Old English evidence West Saxon ēow 'you' beside early West Saxon and Northumbrian īow 'you' shows the normal West Saxon treatment of the diphthong (Campbell 1959, sect. 702, p. 283). The corpus targets the West Saxon citation form.

Development to Old English Coronal-w assimilation first supplies the geminate in **ízwiz* > **íwwiz*. Early inherited-glide reanalysis then gives an **iu* nucleus with retained consonantal **w*, rather than an already completed English long diphthong. Final-z loss exposes the last high vowel, and weak-final apocope selects the apocopated member of Ringe and Taylor's PWGmc **iuwi* / **iuw* doublet. The later English realization gives *ēow* without an i-mutation trigger (Ringe and Taylor 2014, 41–42, 44–45, 57–58, 171–75).

The selected sentence context is weak and phonologically final at early apocope. It is recorded independently of the unchanged PGmc reconstruction and independently of the root accent: the acute is not a sentence-stress mark. Under a retained-i context the present model produces predicted †*īei*, because w-loss before i intervenes before mutation. That is a computational negative control, not an attested strong Old English spelling. Proclitic *ymbe* 'around', weak-final *and* 'and', and stressed *ġiest* 'guest' / *fȳr* 'fire' test the boundary and stress contrasts; no pronoun-specific or lexeme-ID sound-law condition is used (Ringe and Taylor 2014, 55, 57–58).

Why this row is in the corpus The word supplies the assimilation input to SC031 OEWWSimplification and the selected weak-final context for SC098 PWGmcUnstressedWordFinalIApocope. It demonstrates loss of the mutation trigger, not a late English vocalization-before-degemination order. That older interpretation belonged to the superseded ww proxy. The independently justified assimilation feeder and the prosodic condition now remain separate (Ringe and Taylor 2014, 41–42, 55, 57–58).

Part II. Attested variants and comparison forms

Here the Old English comparator belongs to a documented set of variants. The variation is part of the evidence and not an inconvenience to be normalized away.

cud — OE *cwedu*

Derivation: citation reconstruction **kwíθuz*; form followed here **kwéðuz* > *cwedu* (attested variant).

Derivation trace Proto input: **kwéðuz*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PWGmc Dental Hardening	**kwéduz*	
EAF Final Z Deletion	**kwédu*	
Early Anglo-Frisian		
[no change]		

Old English form: *cwedu*

Reconstruction and comparative evidence Kroonen gives the resin word under homonym 2 of **kwedu-* 'cud' and supplies Old English variants *cwidu* 'cud', *cweodu* 'cud', and *c(w)udu* 'cud' (Kroonen 2013, 315). The homonym number is not part of the reconstructed phonological stem. Orel also lists *cwidu*, but his feminine headword is a different formation, not another quotation of Kroonen's masculine u-stem (Orel 2003, 227). The derivational input **kwéðuz* 'cud' supplies an e-vowel and voiced dental for the chosen comparison *cwedu* 'cud'. It is a CAPR whole-word selection, not a diplomatic quotation of either dictionary's complete form. Its relationship to the different citation reconstruction above remains under comparative review.

Old English evidence The Old English word survives in a wider variant set than one dictionary headword suggests. Ringe and Taylor discuss *cwidu* 'cud' > *cwudu* 'cud' > *cudu* 'cud' and also note late West Saxon *cweodu* 'cud'. Their starting reconstruction is explicitly West Germanic with i, not an explicitly printed Proto-Germanic form in this passage (Ringe and Taylor 2014, 323). Clark Hall's headword *cwudu* 'cud' admits eo and i variants, but does not itself attest the selected bare-e spelling (Clark Hall 1960, 69). The claim that CAPR's comparison is the conservative attested variant needs further source verification; these passages do not establish it.

Development to Old English The current modeled chain takes **kwéðuz* 'cud' to *cwedu* 'cud', with the voiced dental hardened along the way. This records the executable result, not independent proof that the derivational input is the best source-backed history. Ringe and Taylor support back umlaut and further reduction within the i-starting variant family; their passage does not establish the older account's claim that the i-form is leveled from the bare-e comparison (Ringe and Taylor 2014, 323). Corpus fields and classification remain unchanged pending the complete source audit.

Variant comparison

Variant type	Old English form	Comment
selected comparison	*cwedu*	current target; its attestation and conservative status require verification

Variant type	Old English form	Comment
i-grade comparison	*cwidu*	source-backed variant; leveling from the Old English form here is not established by the cited passage
back-umlauted forms	*cweodu*, *cwudu*	later developments within the same OE tradition
reduced form	*cudu*	further reduced member of the same variant set

earthly — OE *eorþcunde*

Derivation: citation reconstruction **érθa-kùnðaz*; form followed here **érθa-kùnðǭ* > *eorþcunde* (attested variant).

Derivation trace Proto input: **érθakùnðǭ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Compound Linking Syncope	**érθkùndǭ*
PWGmc Dental Hardening	**érθakùndǭ*	OE Strip Secondary Stress	**érθkundǭ*
Early Anglo-Frisian		OE Breaking	**éorθkundǭ*
[no change]		OE Unstressed Long Vowel Shortening	**éorθkundæ*
		OE Unstressed AE Merger	**éorθkunde*

Old English form: *eorþcunde*

Selected form and documentary status

Item	Value
comparative citation in CAPR notation	**érθa-kùnðaz*
selected inherited weak neuter accusative singular	**érθa-kùnðǭ*
Old English target	*eorþcunde* ‘earthly’
documentary classification	attested_variant
adopted regular span under medial-foot protection	**érθa-kùnðǭ* → *eorþcunde*

The Old English form here is an attested inflected adjective, not the dictionary headword *eorþcund* ‘earthly’. Its documentary classification does not assert an irregular sound change, a dialectal exception or analogical levelling. The citation nominative and the selected weak neuter accusative are different cells reconstructed at the same comparative stage.

Comparative formation and the limits of the citation Orel gives the whole formation **erþakunđaz*, with Old English *eorþ-cund* and Gothic *airþa-kunds* ‘earthly’ (Orel 2003, 85). This is comparative evidence for the compound, not a printed reconstruction of its accented weak-neuter cell. CAPR’s *θ* and *ð* serialize the source’s dental signs; its acute, grave and selected ending are analytical additions, not diplomatic transcription. The Gothic comparison supports a formation extending beyond Old English, without independently proving the date or uninterrupted inheritance of every Old English inflection.

Kroonen independently gives **kunda-* ‘born’, with Gothic *-kunds*, Norse *-kundr* and Old English *-cund*, as the old participle of the root ‘be born’. His **erþō-* ‘earth’ supplies the lexical first constituent, not the compound’s linking vowel. The separate **kunþa-* ‘known’, associated with ‘know’, is

not the same formation. The bounded consultation recovered component evidence, not an authored whole earthly compound or a weak-neuter ending (Kroonen 2013, 118–19, 310–12).

The second constituent's subsequent suffixal behavior matters. Hogg includes *-cund* among elements historically derived from compound constituents, distinguishing older compound stress from later subordinate stress in disyllabic formations or inflected forms (Hogg 2011, 49–50, §2.89). Neither a dictionary hyphen nor the spelling *-cund* independently establishes prehistoric prominence. No unsourced derivation from the verb 'know' is needed here.

Three located passages, not a complete six-token concordance Ciszek-Kiliszewska's published DOEC survey reports six occurrences of *eorþcund,* against 761 of *godcund* 'divine', and includes inflections in its wildcard searches. The lexeme-level table is not a six-item concordance (Ciszek-Kiliszewska 2015, 489–90). The bounded documentary record locates three distinct passages: Genesis 1627, An. Ox. 691 and R. Ben. 139, 27–28. It does not identify the remaining occurrences or explain the discrepancy by counting unrelated forms. Two dictionary quotations of the Benedict sentence represent one occurrence, not two.

The Genesis object and its weak neuter agreement Bosworth–Toller quotes "Se rinc ageaf eorþ-cunde eád", with locators Cd. 79, Th. 98, 8 and Gen. 1627. Its separate *eád* 'wealth' entry explicitly marks the noun as neuter. In context *eád* is the object of *ageaf* 'gave up'; *eorþcunde* agrees with it as a weak neuter accusative singular (Bosworth and Toller 1898, s.vv. eorþ-cund, eád).

The passage has also been located in the OCR of Thorpe's 1832 edition on the page numbered 98. That edition-OCR consultation supports the sentence context and absence of a determiner. It is not a collation of Junius 11, Thorpe's page image or Krapp's critical edition. Apparent eth/thorn differences and hyphenation in OCR must not be described as verified printed or manuscript spellings. The target *eorþcunde* is a normalized scholarly representation of the dictionary-located form. No exact manuscript date or narrower dialect is inferred from it.

Weak inflection without a determiner The absence of a demonstrative or possessive does not make the weak ending an anomaly requiring repair. Campbell describes the freer use of weak adjectives in poetry, and Fulk independently recognizes the poetic use without the usual specifier. Their weak-adjective paradigms give neuter nominative–accusative singular *-e*, including the ordinary comparison *blinde* 'blind' (Campbell 1959, 261, 272, §§638, 656; Fulk 2018, 216–17, §§9.7–9.8).

Applying those general grammatical statements to this sentence is the report's syntactic inference. The adjective need not be reinterpreted as a masculine accusative with an inexplicably lost *n*. Nor is weak inflection here evidence for an otherwise unrecorded levelling event.

The inherited weak-neuter ending Fulk explicitly treats weak adjectives as n-stems. His n-stem discussion and paradigms supply the relevant neuter singular ending, **-ō^{n}*, for the West Germanic and Norse comparison (Fulk 2018, 168 n. 1, pp. 169–170, §§7.29–7.31). The macron denotes a bimoric long vowel; superscript ^{n} marks nasalization, not a consonant to be added to the string. CAPR therefore represents this ending as **-ǭ*, not trimoric **-ô^{n}* or literal consonantal **-ōn*.

The ending belongs on the unaugmented compound stem. Fulk's neuter *augō* 'eye' paradigm contrasts singular forms without the oblique *n* formative with plural forms containing it. The class name "n-stem" is not permission to insert **-an-* before the selected singular ending. CAPR consequently assembles **erþa-kunđ-ō^{n}*, serialized **érθa-kùnðǭ*. This whole weak adjective is a comparative inference from Fulk's class statement and noun paradigm, not a whole word printed by him. Campbell independently supports the Old English ending and syntax; he has not been used as an independent check of that entire Proto-Germanic assembly.

Inheritance and the reported origin alternative Fulk connects the neuter ending with inherited PIE **-ōn* < **-on-h₂*, citing Jasanoff. He also reports Boutkan's alternative: analogical extension from the neuter plural (Fulk 2018, 170, citing Jasanoff 2002, p. 33, and Boutkan 1995b, pp. 285–286). These are alternative accounts of the ending's prehistory, not different readings of the OE-relevant macron. Neither cited work has been independently consulted for this entry.

The report prefers Fulk's regular inherited account, without claiming that successful execution decides between it and the reported analogy. Fulk's separate discussion of a perplexing masculine trimoric origin must not be transferred to this bimoric neuter cell. No trimoric-to-bimoric repair or additional ending law is adopted.

Medial prominence after an initial heavy foot The adopted protection concerns phonological weight and domain, not membership in *-cund* or the weak-adjective paradigm. In *eorþ.cun.de*, the initial syllable is heavy and the following nonfinal closed syllable can head a medial foot. Goering distinguishes this configuration from an initial light syllable accommodated together with a following heavy syllable in the initial foot (Goering 2023, 79–84, §§4.5.1.1–4.5.2).

McCully and Hogg's revised quantity-sensitive conditions likewise make initial-foot accommodation essential, rather than stressing every medial heavy-looking string (McCully and Hogg 1990, 323–33, especially pp. 330–333). These analyses motivate CAPR's bounded projection; neither author prints a full accented derivation of earthly. The exact syllabification and prosodic domain remain explicit modelling assumptions, not a complete Old English cyclic stress grammar.

World, rainbow and noncircular controls *Weorold* 'world' has the contrasting initial light-plus-heavy configuration. At the late-lowering checkpoint the inflected diagnostic **weo.rul.des* contains one short diphthongal *eo* nucleus in its initial light syllable. The analytical accented **éo* is neither two separately represented vowels nor a long vowel. Initial L+H is accommodated together in the initial foot rather than giving the second nucleus an independent medial head. By contrast, *eorþ.cun.de* has initial H followed by nonfinal H, permitting a medial head. Its selected weak-vowel history therefore remains compatible with late lowering (Goering 2023, 79–82, §4.5.1.2). *Reġnboga* 'rainbow' already has second-root *o* from SC017 PNWGmcULowering; it cannot establish that subordinate prominence disappeared before late lowering. Neither adopted history is reopened.

Hogg warns that "tertiary stress" can merely rename a desired retention. Ordinary *-un* > *-on* also prevents a blanket dental-nasal exemption; labial-nasal preservation is a different environment (Hogg 1996, 8–11, especially p. 11 n. 7; Hogg 2011, 239–41, §§6.55–6.58). Dictionary-located *godcundne* 'divine' and *godcundre* 'divine' are real retention comparisons, but do not by themselves establish every ending's history. The earlier record's *godcundum* in *in godcundum mægne* is dative singular in context; its syncretic *-um* must not be counted as an independently demonstrated plural. Final uninflected *godcund* remains a qualification of an ordinary nonfinal-foot account (Toller 1921, s.v. god-cund). Metrical prominence is not automatically attested speech stress (Minkova and Zhou 2023, 86–90).

The selected regular span and its executable representation The supported chain is **érθa-kùnðǭ* > **érθa-kùndǭ* > **érθkùndǭ* > **érθkundǭ* > **éorθkundǭ* > **éorθkundæ* > **éorθkunde*, followed by *eorþcunde*. Dental hardening, linking-vowel deletion, breaking and existing final shortening and merger account for the changes other than medial protection. The selected bimoric nasal ending is already within the input notation.

SC037 OECompoundLinkingSyncope reads the earlier compound accent before SC038 OESstripSecondaryStress removes it at its inherited position. At SC040 OEMedUnstressedULowering, the independently computed medial prominence excludes this *u* from unstressed lowering; without that protection the experimental comparison gives **eorþconde*.

The adopted calculation retains temporary grave marking for all five short vowels *a e i o u* in the computed heads and immediately decodes the marks after lowering. It accommodates an existing pretonic prefix in the sentence domain and permits consonants preceding the first syllable

boundary. Such consonants do not add a nucleus or change its weight. No new initial boundary is inserted, and the computation is not wrapped in the separate pretonic-first-syllable context used by the lowering operation itself.

Lowering retains its canonical consonant context. Its only additional preceding-vowel exclusion is temporarily headed *ù*: a headed *u* must not license lowering of a subsequent *u* merely because its technical annotation changes. This restores the existing *u*-exclusion after marking rather than introducing a new historical condition. Complete-relation checks establish a single-valued calculation, while complete-corpus controls preserve the existing derivations. Both the full weak-neuter and genitive probes succeed. Those results support the selected regular span without establishing an entire adjective paradigm or removing its documentary qualifications.

The temporary marks are not survival of the original grave, a second historical deaccentuation, or a new chronology edge. The projection addresses the short-*u* lowering question; marking five short-vowel classes does not supply a full stress representation for all nuclei, boundaries, final-overheavy syllables or transparent compound domains.

Other cells and the limits of adoption The Oxford gloss has *eorþcundre* ‘earthly’, feminine genitive singular, in *ealre eorþcundre gesceafte*, translating Latin *totius terrestris (i. terrene) creaturae* (Toller 1921, s.v. eorþ-cund, An. Ox. 691). The exact edition, compiler and manuscript behind that locator have not been independently established. Fulk’s strong paradigm allows *-*(a)izōz*; the experimental **eorþcundere* > **eorþcundre* requires additional pre-*r* syncope research, not an adopted ending repair (Fulk 2018, 209–11; Campbell 1959, 157–59, §§388–393; Ringe and Taylor 2014, 264–66).

The Benedict sentence has *eorþcundum* ‘earthly’ agreeing with masculine dative plural *lustum* ‘desires’, governed by *filigaþ* ‘follow’. The dictionary cites it at 139, 27 under *grædiglice* and 139, 28 under *sceamleáslice*. The underlying edition and manuscript have not been independently checked (Toller 1921, s.vv. grædiglice, sceamleáslice). The conventional strong dative plural *-*aimiz* currently gives experimental **-em*, not the attested *-um*. Secure synchronic inflection does not resolve that whole inherited derivation (Fulk 2018, 209–11).

The strong genitive **eorþcundes*, from the supported **-as(a)* ending, is an experimentally regular but unattested fallback, not a documentary substitute for the Genesis cell. Bounded negative searches are not proof that it never existed. *Eardcundes* ‘native’ has a different first constituent; *eorþcundlic* ‘earthly’ is a separately suffixed lexeme. Their dictionary grouping does not make either an occurrence of the selected adjective. The chosen weak-neuter span needs neither Oxford r-syncope, a dative ending repair, final-overheavy protection, nor an additional analogy.

ten — OE *tēon*

Derivation: **téxun* > *tēon* (attested variant).

Derivation trace Proto input: **téxun*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Breaking	**téoxun*
[no change]	OE Med Unstressed U Lowering	**téoxon*
	OE H Loss	**téoon*
Early Anglo-Frisian	OE Contraction	**tḗon*
[no change]		

Old English form: *tēon*

Reconstruction and comparative evidence **tēon* ‘ten’ is the bare cardinal form. Fulk states that Old English *tien* ‘ten’ shows umlaut from the inflected forms, whereas the uninflected form without umlaut is reflected in *tēon* ‘ten’ (Fulk 2018, sect. 10.2). Brunner gives the same contrast more broadly: **tēon* ‘ten’ develops from **tëhun* ‘ten’, while West Saxon *tien* ‘ten’, *tȳn* ‘ten’ belong

to a different, umlauted branch of the numeral history (Brunner 1965, sects 129.2, 129 Anm. 6, 234).

The comparison form *tēon* ‘ten’ therefore represents the bare cardinal’s un-umlauted line, not the later umlauted simplex tradition.

Old English evidence The attested simplex forms are varied. Campbell gives *tien* ‘ten’, northwestern West Saxon *tēn* ‘ten’, and late Northumbrian *tēo* ‘ten’, *tēa* ‘ten’ (Campbell 1959, sect. 682). Brunner likewise lists West Saxon *tien* ‘ten’, *tȳn* ‘ten’ beside *tēn* ‘ten’, *tēo* ‘ten’, *tēa* ‘ten’ in other dialects (Brunner 1965, sect. 325).

Exact simplex *tēon* ‘ten’ is weaker as a directly cited headword than those spellings. The un-umlauted stem is, however, explicit in *tēoða* ‘tenth’ and *tēontig* ‘ten’ (Brunner 1965, sect. 129.2; Fulk 2018, sect. 10.2). The comparison form *tēon* ‘ten’ is therefore a normalized spelling of that un-umlauted base.

Development to Old English From **téxun* ‘ten’, lowering of medial unstressed *u* gives **téxon* ‘ten’, breaking gives **téoxon* ‘ten’, loss of intervocalic *h*/*x* yields **téoon* ‘ten’, and contraction produces **tḗon* ‘ten’, written *tēon* ‘ten’. This is the regular bare-cardinal path.

The umlauted forms *tien* / *tīen* ‘ten’ belong to a different branch, created when the numeral was levelled from inflected forms with a front-vocalic trigger (Fulk 2018, sect. 10.2; Brunner 1965, sect. 129 Anm. 6).

Variant comparison The comparison below sets the relevant forms side by side. It distinguishes the normalized un-umlauted base from the attested simplex variants.

Form or branch	Status	Relevance to this entry
tēon	normalized un-umlauted comparison form; trace-supported	Old English form here
tien / *tīen*	attested West Saxon umlauted simplex forms	genuine OE variants, but not the bare-cardinal line modeled here
tēn / *tēo* / *tēa*	attested un-umlauted simplex variants in other dialects	support the same branch as the comparison form

three — OE *þrīe*

Derivation: **θréjez* > *þrīe* (attested variant).

Derivation trace Proto input: **θréjez*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**θrije*
EAF Final Z Deletion	**θréje*	OE Intervocalic J Vocalization	**θriie*
Early Anglo-Frisian		OE Contraction	**θrīe*
[no change]			

Old English form: *þrīe*

Reconstruction and comparative evidence Kroonen cites the numeral under a broader stem-style reconstruction rather than under one Old English-ready paradigm cell (Kroonen 2013, 586). The input **θréjez* ‘three’ is therefore best understood as the inherited masculine nominative-accusative singular.

The Old English numeral has no uniform citation form across the paradigm. The masculine singular line must be kept apart from feminine-neuter *þrēo* ‘three’ and from later reduced spellings of the masculine form.

Old English evidence Campbell gives masculine nominative-accusative *þrīe* ‘three’, feminine and neuter nominative-accusative *þrēo* ‘three’, genitive *þrēora* ‘three’, and dative *þrim* ‘three’, adding that late West Saxon has *þry* ‘three’ / *þri* ‘three’ for masculine nominative-accusative beside *þrīe* ‘three’ (Campbell 1959, sect. 683). Fulk presents the same masculine *þrīe* ‘three’ beside the wider numeral paradigm (Fulk 2018, sect. 10.1).

The target is therefore an attested Old English paradigm cell. *þrī* ‘three’ / *þry* ‘three’ belongs to later reduction or headword-style citation, whereas *þrīe* ‘three’ is the conservative masculine nominative-accusative form.

Development to Old English From **θréjez* ‘three’, loss of final *-z* leaves **θréje* ‘three’ as an intermediate form. The following **θrije* ‘three’ fronts the stem vowel, then vocalizes between vowels, and contraction yields **þrīe* ‘three’. The compact trace records the same sequence as θrije > θriie > *þrīe*.

Variant comparison The comparison below sets the relevant forms side by side. It separates the selected masculine cell from the later reduced form and from the rest of the numeral paradigm.

Form	Status	Relevance to this entry
þrīe	attested masculine nom./acc.; regular output	Old English form here
þrī / *þry*	later reduced masculine variant	genuine OE variant, but not the conservative comparison form
þrēo	attested feminine-neuter nom./acc.	same numeral, different paradigm cell
þrēora, *þrim*	attested genitive and dative forms	confirm the wider paradigm, not the cell compared here

wasp — OE *wæfs*

Derivation: **wábsaz* > *wæfs* (attested variant).

Derivation trace Proto input: **wábsaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wábs*
EAF Final Z Deletion	**wábsa*	EAF Brightening	**wæbs*
Early Anglo-Frisian		PGmc B Allophony	**wæβs*
[no change]			

Old English form: *wæfs*

Reconstruction and comparative evidence The Proto-Germanic form **wábsaz* ‘wasp’ reaches Old English without any special change of stem or paradigm cell. The question in this entry is instead which attested Old English member of the variant set should serve as the comparison form.

Fulk presents the Old English forms together as *wæfs* ‘wasp’ with variants *wæsp* ‘wasp’ and *wæps* ‘wasp’ (Fulk 2018, sect. 6.5). Bülbring and Brunner then make the chronology more explicit by

deriving later *wæps* and late West Saxon *wasp* ‘wasp’ from earlier *waefs* / *wæfs* through restricted metatheses (Bülbring 1902, sect. 484 Anm. 3; Brunner 1965, sects 193, 204).

Old English evidence The earliest directly cited Old English form is *wæfs* ‘wasp’, written *waefs* ‘wasp’ in the Épinal-Corpus material discussed by Bülbring and Brunner (Bülbring 1902, sect. 484 Anm. 3; Brunner 1965, sect. 193). Later Old English also shows *wæps* ‘wasp’ and *wæsp* ‘wasp’ / *wasp* ‘wasp’, and dictionary practice often favors *wæps* or later spellings as headwords (Clark Hall 1960, 341).

This entry therefore distinguishes chronological priority from headword habit. *wæfs* ‘wasp’ is not a convenient reconstruction: it is an attested Old English form and also the one that matches the regular development most closely.

Development to Old English From **wábsaz* ‘wasp’, the regular Old English path passes through loss of final *z*, Anglo-Frisian fronting, and the allophonic development of *b* to a fricative before *s*, yielding *wæfs* ‘wasp’.

The later forms *wæps* ‘wasp’ and *wæsp* ‘wasp’ / *wasp* ‘wasp’ belong to subsequent, lexically restricted metatheses. They are genuine Old English forms, but they are later within the variant history.

Variant comparison The comparison below sets the relevant forms side by side. It separates the earliest attested and regular form from the later metathesized doublets.

Form	Status	Relevance to this entry
wæfs	earliest attested OE form; regular output	Old English form here
wæps	later attested metathesized variant	genuine OE doublet, but secondary
wæsp / *wasp*	later West Saxon metathesized variant	genuine OE doublet, but not the form compared here

Part III. Early analogy and pre-Old-English input selection

Here an analogical change already separates the transducer input from the lexeme's citation reconstruction before the specifically Old English changes apply. Each entry must therefore justify the remodeled input independently of the successful output.

bottom — OE *botm*

Derivation: citation reconstruction **búdmaz*; form followed here **búttmaz* > *botm* (early analogy).

Derivation trace Proto input: **búttmaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**bóttm*
PNWGmc U Lowering	**bóttmaz*	OE Preconsonantal Degemination	**bótm*
EAF Final Z Deletion	**bóttma*		
Early Anglo-Frisian			
[no change]			

Old English form: *botm*

Reconstruction and comparative evidence Kroonen reconstructs the word as a stem complex **budmō* 'bottom' with genitive **buttaz* 'bottom', summarized as **budman-* ~ **buttman-*, and gives Old English *botm* 'bottom' as the reflex (Kroonen 2013, 120). The comparative label **búdmaz* 'bottom' names the lexeme-level stem complex, while the derivational input **búttmaz* 'bottom' represents the pre-Old-English form with oblique **butt-* generalized into the nominative formation.

Orel likewise preserves both sides of the comparison under **budmaz* 'bottom' and **butmaz* 'bottom' (Orel 2003, 100). The derivational input is thus a historical stem choice, not an arbitrary respelling.

Old English evidence The Old English noun itself is secure. Clark Hall gives *botm* 'bottom' (Clark Hall 1960, 63). Bosworth-Toller cross-references *bodan* 'bottom, floor' to *botm* 'bottom', showing the wider reflex family without weakening the attested lemma (Bosworth and Toller 1898, 112).

Development to Old English Once the oblique **butt-* stem has been generalized, the derivational input **búttmaz* 'bottom' develops regularly to *botm* 'bottom'. The analogical step is therefore early: it belongs to pre-Old-English stem formation rather than to a later choice among Old English paradigm cells.

brand — OE *brandes*

Derivation: citation reconstruction **brándaz*; form followed here **brándas* > *brandes* (early analogy).

Derivation trace Proto input: **brándas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**brándæs*
[no change]	OE Unstressed AE Merger	**brándes*
Early Anglo-Frisian		
[no change]		

Old English form: *brandes*

Reconstruction and comparative evidence The inherited noun is the masculine a-stem **brándaz* ‘brand’, continued by Old English *brand* ‘brand’ and its continental cognates (Orel 2003, 53). The selected input **brándas* ‘brand’ is not a different lexeme but the genitive singular of that same a-stem noun.

Both forms belong to the same root and stem class but occupy different inherited inflectional cells. The derivational input is the oblique cell.

Old English evidence Old English dictionaries lemmatize the noun as *brand* ‘brand’ (Clark Hall 1960, 49; Bosworth and Toller 1898, 116). Bosworth-Toller also records inflectional forms such as *brandas* ‘brand’, *branda* ‘brand’, and *brandum* ‘brand’ under the same entry (Bosworth and Toller 1898, 116).

The specific comparison form in this entry, *brandes* ‘brand’, is the expected genitive singular of that a-stem noun. It is therefore an inferred Old English paradigm form rather than the ordinary dictionary headword.

Development to Old English From **brándas* ‘brand’, the regular Old English development passes through the usual unstressed-vowel weakening of the inflectional ending, yielding *brandes* ‘brand’. Nothing in the stem itself requires a special repair. The root consonants and the stressed vowel are the same as in the citation lemma *brand* ‘brand’.

The analytical weight of the entry lies in the ending. By choosing the oblique singular rather than the nominative citation form, the entry presents the same lexeme in a different inherited cell.

Form comparison The comparison below sets the relevant forms side by side. It separates the citation lemma from the selected oblique singular.

Form / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**brándaz*	expected OE lemma *brand*	*brand*	regular headword-level outcome
genitive singular	**brándas*	regular output: *brandes*	*brandes*	exact match for the oblique cell

The noun itself is straightforwardly inherited. The main point of the entry is that *brandes* ‘brand (gen.)’ belongs to the same regular a-stem paradigm as *brand* ‘brand’, even though the citation lemma remains the nominative singular.

breast — OE *brēost*

Derivation: citation reconstruction **brústz*; form followed here **bréustą* > *brēost* (early analogy).

Derivation trace Proto input: **bréustą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Diphthong Leveling	**brēostą*
[no change]	OE Heavy Syllable Nasal Apocope	**brēost*
Early Anglo-Frisian		
[no change]		

Old English form: *brēost*

Reconstruction and comparative evidence The word family shows two related but distinct Proto-Germanic formations. The root noun **brust-* lies behind forms such as Gothic *brusts* ‘breast’, whereas Old English *brēost* ‘breast’ belongs to a thematic formation **breusta-*, alongside Old Norse *brjóst* ‘breast’ and Old Saxon *briost* ‘breast’ (Kroonen 2013, 114; Orel 2003, 95; Ringe and Taylor 2014, 43).

The derivational input **bréustą* ‘breast’ therefore differs from the citation label **brústz* ‘breast’ because Old English reflects the thematic branch rather than the root noun. The morphological choice comes before the Old English sound changes themselves.

Old English evidence Clark Hall records the noun as *brēost* ‘breast’ / *breóst* ‘breast’ (Clark Hall 1960, 65). The form is an established Old English lexeme, not a reconstructed target assembled from comparative evidence alone.

What requires explanation is not the Old English attestation but the relation between that attested noun and the broader Germanic word family. The relevant comparison form is therefore the thematic Old English noun *brēost* ‘breast’.

Development to Old English From **bréustą* ‘breast’, the regular Old English development gives *brēost* ‘breast’, with the expected *eu* > *ēo* vowel history (Campbell 1959, sect. 115). No special repair is needed once the correct thematic formation is chosen.

The earlier mismatch arose only if the word was forced into the root-noun line. The Old English noun itself continues the thematic branch cleanly and directly.

Formation comparison The comparison below sets the relevant forms side by side. It separates the broader root-noun family label from the thematic formation actually continued in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader root-noun family	**brústz*	root-noun type outcomes outside OE	non-OE comparanda	useful family label, but not the direct source of *brēost*
selected thematic formation	**bréustą*	regular output: *brēost*	*brēost*	exact match between formation and attested OE noun

The relevant point is the formation split. *brēost* ‘breast’ is the regular Old English outcome of the thematic **breusta-* branch, not of the root noun **brust-*.

craft — OE *crœft*

Derivation: citation reconstruction **kráftiz*; form followed here **kráftaz* > *crœft* (early analogy).

Derivation trace Proto input: **kráftaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**kráft*
EAF Final Z Deletion	**kráfta*	EAF Brightening	**kræft*
Early Anglo-Frisian			
[no change]			

Old English form: *cræft*

Reconstruction and comparative evidence Comparative sources disagree about the older stem class. Kroonen gives a u-stem **kraftu-*, while Orel prints **kraftiz* ‘craft’ and **kraftuz* ‘craft’ (Kroonen 2013, 340; Orel 2003, 259). The comparative label **kráftiz* ‘craft’ remains in view as a lexeme-level shorthand, while **kráftaz* ‘craft’ is the pre-Old-English form used for the Old English derivation.

Old English evidence The Old English noun itself is secure. Clark Hall and Bosworth-Toller both give *cræft* ‘craft’ as the headword (Clark Hall 1960, 19; Bosworth and Toller 1898, 145).

Development to Old English The comparison is between possible pre-Old-English inputs. The i-stem comparator **kráftiz* ‘craft’ gives *creft*, while the u-stem comparator **kráftuz* ‘craft’ gives *craft*. The a-stem-shaped input **kráftaz* ‘craft’ yields *cræft* ‘craft’ and is therefore the form used for the Old English derivation. This does not require treating the comparative dictionaries as identical; it shows the narrower point that the Old English derivation needs a pre-Old-English form without the i-umlaut trigger of **-iz* ‘craft’ and without the back-vowel outcome associated with the u-stem comparator.

Form comparison

Candidate input	OE output	Result
*kráftiz	*creft*	non-match; i-stem comparator
*kráftuz	*craft*	non-match; u-stem comparator
*kráftaz	*cræft*	exact match; selected pre-OE input

dill — OE *dile*

Derivation: citation reconstruction **déljaz*; form followed here **déliz* > *dile* (early analogy).

Derivation trace Proto input: **déliz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**dili*
EAF Final Z Deletion	**déli*	OE Med Unstressed I Lowering1	**dile*
Early Anglo-Frisian			
[no change]			

Old English form: *dile*

Reconstruction and comparative evidence Comparative evidence preserves both an i-stem and a ja-stem formation, with Old English *dile* ‘dill’ on one side and continental forms such as Old Saxon *dilli* ‘dill’ and Old High German *tilli* ‘dill’ on the other (Fulk 2018, 170). The derivational input **déliz* ‘dill’ therefore represents the i-stem side of the paradigm, whereas the citation label **déljaz* ‘dill’ is a broader comparative headword.

The stem class determines the Old English consonant shape. A ja-stem with **-lj-* would be expected to produce gemination, but the Old English noun shows a single *l*. Fulk's discussion of ja-stems transferred to the i-stems provides the relevant morphological background for the OE side (Fulk 2018, 170).

Old English evidence Old English dictionaries record the plant name as *dile* 'dill', alongside the variant *dili* 'dill' (Bosworth and Toller 1898, 164; Clark Hall 1960, 95). The form discussed here is therefore an attested Old English noun with single *l*.

The Old English evidence is the relevant point. Whatever broader comparative headword is chosen for the family, the inherited form reflected in OE is the i-stem type *dile* 'dill', not a geminated *dill* 'dill' outcome.

Development to Old English From **déliz* 'dill', regular loss of final *z* and the later lowering of unstressed *i* yield *dile* 'dill'. The stem itself remains ungeminated throughout that path.

The contrast is morphological rather than phonological. If the word were forced through a ja-stem **-lj-* pathway, the expected result would show *ll*. The attested Old English noun instead matches the i-stem development.

Formation comparison The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the stem class actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative ja-stem label	**déljaz*	ja-stem type outcome with gemination	*dill*-type comparison	useful comparative label, but not the OE form
selected i-stem formation	**déliz*	regular output: *dile*	*dile*	exact match between formation and attested OE noun

The single *l* is the decisive diagnostic. It identifies *dile* 'dill' with the i-stem formation rather than with the continental ja-stem branch.

fast — OE *festan*

Derivation: citation reconstruction **fastēną*; form followed here **fástijaną* > *festan* (early analogy).

Derivation trace Proto input: **fástijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**fæstijaną*
[no change]	OE Heavy Syllable Nasal Apocope	**fæstijan*
	OE Secondary Nasalization	**fæstijąn*
Early Anglo-Frisian	Sievers Law Syncope	**fæstjąn*
[no change]	OE I Umlaut	**festjąn*
	OE Weak Tail Reduction	**festjan*
	OE J Loss After Heavy	**festan*

Old English form: *festan*

Reconstruction and comparative evidence Kroonen places the verb within the wider **fastu-* adjective family and its derived **fasten-* verbal line, the comparative background behind Old English ‘to fast’ (Kroonen 2013, 171). Ringe and Taylor, however, distinguish the Old English verb more closely: they treat OE ‘to fast’ as originally a class-I weak verb that later acquired the stative meaning through lexical confusion (Ringe and Taylor 2014, 110).

The derivational input **fástijaną* ‘fast’ therefore represents the inherited class-I formation reflected in Old English, whereas the citation label **fastēną* ‘fast’ belongs to the broader comparative presentation of the lexeme.

Old English evidence Old English dictionaries record forms such as *festan* ‘fast’, alongside related *fæstan* ‘fast’ / *fǣstan* ‘fast’ spellings and meanings (Bosworth and Toller 1898, 213). The form selected here is *festan*, which fits the regular class-I phonological development.

The *æ*-forms remain relevant, but they do not control the entry. In the present analysis they belong to a later analogical reshaping under the adjective *fæst* ‘fast’, whereas *festan* ‘fast’ is the regular inherited class-I comparison form.

Development to Old English From **fástijaną* ‘fast’, Anglo-Frisian brightening and subsequent i-umlaut produce the fronted vowel seen in *festan* ‘fast’. The later weak-tail reductions and loss of *j* after a heavy syllable complete the regular Old English outcome.

What makes the entry non-regular is not the phonology of *festan* ‘fast’ itself, but the choice of formation. Old English continues the class-I verb, even though the comparative headword is often given under the parallel **fastēn-* family.

Class comparison The comparison below sets the relevant forms side by side. It distinguishes the comparative class-III headword from the class-I formation actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative class-III headword	**fastēną*	class-III type outcome, not *festan*	wider family context	useful family label, but not the direct source of the target
selected class-I weak verb	**fástijaną*	regular output: *festan*	*festan*	exact match between formation and attested OE verb
later analogical reshaping	adjective-driven *fæst* influence	*fæstan* / *fǣstan* type spellings	fæstan-type evidence	genuine later OE reshaping, but secondary to the Old English form here

The relevant point is the class split. *festan* ‘fast’ is the regular Old English outcome of the class-I formation, while the better-known *æ*-forms belong to a later analogical layer.

flask — OE *flasce*

Derivation: citation reconstruction **flaskō*; form followed here **fláskōn* > *flasce* (early analogy).

Derivation trace Proto input: **fláskōn*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**flæskǭ*
PNWGmc N Stem N Loss **fláskǭ*	OE A Restoration	**flaskǭ*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**flaskæ*
	OE Unstressed AE Merger	**flaske*
[no change]		

Old English form: *flasce*

Reconstruction and comparative evidence The wider Germanic family is often cited under a form such as **flaskō* ‘flask’, but the evidence relevant for Old English points instead to a weak feminine formation **fláskōn* ‘flask’ / **flaskǭ* ‘flask’ (Orel 2003, 104). That distinction is crucial for the suffixal history of the noun.

The derivational input therefore differs from the citation label in stem class. Old English *flasce* ‘flask’ belongs with the weak feminine line, and the plural or oblique forms *flascan* ‘flask’ support that analysis (Ringe and Taylor 2014, 192).

Old English evidence Old English dictionaries record the noun as *flasce* ‘flask’, with inflectional support from forms such as *flascan* ‘flask’; a later West Saxon *flaxe* ‘flask’ is also noted as a secondary variant (Bosworth and Toller 1898, 235; Clark Hall 1960, 121).

The relevant comparison form is therefore the weak feminine noun *flasce* ‘flask’. The plural and oblique evidence helps explain why the vowel and ending are preserved as they are in the singular.

Development to Old English From **fláskōn* ‘flask’, the weak feminine passes through the expected loss of *n* and the later Old English development of the unstressed ending, reaching *flasce* ‘flask’. Campbell cites restored *a* in exactly this environment, including *flasce* ‘flask’ after inflected *flascan* ‘flask’ (Campbell 1959, sect. 158). Once the weak feminine formation is chosen, the noun follows a regular path to its Old English shape.

The decisive issue is morphological rather than phonological. A simple strong feminine citation form does not capture the OE weak noun as cleanly as the selected **fláskōn* ‘flask’ does.

Formation comparison The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak feminine formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader comparative headword	**flaskō*	broader family label	wider family context	useful lexeme label, but not the cleanest OE-facing derivation
selected weak feminine formation	**fláskōn*	regular output: *flasce*	*flasce*	exact match between formation and attested OE noun

The weak feminine suffix is the relevant point. It aligns the inherited form with attested *flasce* ‘flask’ and its supporting paradigm forms.

follow — OE *fylġan*

Derivation: citation reconstruction **fulgēną*; form followed here **fúlgijaną* > *fylġan* (early analogy).

Derivation trace Proto input: **fúlgijaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**fúlgijan*
[no change]	OE Secondary Nasalization	**fúlgijąn*
	Sievers Law Syncope	**fúlgjąn*
Early Anglo-Frisian	OE Velar Palatalization	**fúljjąn*
[no change]	OE I Umlaut	**fyljjąn*
	OE Weak Tail Reduction	**fyljjan*
	OE J Loss After Heavy	**fyljan*
	OE Palatal Fricative Merger	**fyljan*

Old English form: *fylġan*

Reconstruction and comparative evidence Kroonen reconstructs the verb as **fulgen-* and gives Old English *fylgan* ‘follow’, *folgian* ‘follow’, adding that Old Norse *fylgja* ‘follow’ and Old English *fylg(e)an* continue a formation **fulgjan-* (Kroonen 2013, 159). The comparative headword and the class-I formation are therefore related but not identical.

Ringe and Taylor distinguish PNWGmc **fulgija-* ‘follow’ ~ **fulgai-* ‘follow’ > OE *fylgan* ‘follow’ ~ *folgian* ‘follow’ and describe it as a dual formation that probably reflects an older alternation between j-present and e-stative (Ringe and Taylor 2014, 293–94). This is a stem-class choice, not a spelling choice. The derivational input **fúlgijaną* ‘follow’ belongs to the class-I **fulgija-* / **fulgjan-* branch; the citation form **fulgēną* ‘follow’ belongs to the parallel class-II history behind *folgian* ‘follow’.

Old English evidence The Old English evidence preserves both formations. Clark Hall lists *fylgan* ‘follow’ with variant spellings *fylgian* ‘follow’ and *fyligan* ‘follow’ (Clark Hall 1960, 125). Bosworth-Toller likewise has a separate *fylgean* ‘follow’ entry (Bosworth and Toller 1898, 275).

Bright notes traces of the older conjugation in *fylg(e)an* ‘follow’ (Bright 1971, 77) and lists *folgian* ‘follow’ (*fylgean* ‘follow’) in the glossary (Bright 1971, 364). The relevant comparison form in this entry is therefore the class-I verb *fylgan* ‘follow’ / *fylgean*, here normalized as *fylġan* ‘follow’. The spelling with ġ represents the palatalized velar before a front-vocalic environment.

Development to Old English **fúlgijaną* ‘follow’ is a class-I weak-verb formation. In the class-I branch the **j* blocks NWGmc lowering of *u* to *o*, since Ringe and Taylor formulate that lowering for environments in which no **j* intervened (Ringe and Taylor 2014, 96). The same front-vocalic environment then triggers i-umlaut, so *u* becomes *y* (Ringe and Taylor 2014, sect. 6.6.2).

The subsequent Old English developments are palatalization of the velar, weak-tail reduction, and loss of *j* after a heavy syllable, yielding *fylġan* ‘follow’. This is the regular outcome of the class-I formation. The class-II form *folgian* ‘follow’ belongs to the parallel **-ē-* / **-ai-* branch and is not the form modeled here.

Class comparison A class comparison identifies which inherited formation corresponds to the established Old English form under discussion. The comparison below is manual; no full automatic class probe is presented here.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation class-II formation	**fulgēną*	probe output: *folgon*	*folgian* ‘follow’	mismatch: the regular output is not the remodeled infinitive *folgian*
parallel class-II branch	PNWGmc **fulgai-*	Ringe-Taylor: OE *folgian* ‘follow’	*folgian* ‘follow’	documents the separate class-II branch, but not the target of this entry
selected class-I formation	**fúlgijaną*	regular output: *fylġan* ‘follow’	*fylġan* ‘follow’ / *fylgan* ‘follow’	exact match between input, output, and class

The relevant point is the class split. *fylġan* ‘follow’ is the regular Old English outcome of the class-I **fulgija-* / **fulgjan-* formation, whereas *folgian* ‘follow’ belongs to the parallel class-II branch.

gall — OE *ġealla*

Derivation: citation reconstruction **gállą*; form followed here **gállô* > *ġealla* (early analogy).

Derivation trace Proto input: **gállô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**gœllô*
[no change]	OE Breaking	**geallô*
	OE Velar Palatalization	**jeallô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**jealla*
[no change]	OE Palatal Fricative Merger	**jealla*

Old English form: *ġealla*

Reconstruction and comparative evidence The wider cognate family can be presented under a form such as **gállą* ‘gall’, but the Old English noun itself belongs with a weak noun **gallōn-* ‘gall’, cited here as **gállô* ‘gall’ (Kroonen 2013, 165). The derivational input therefore differs from the broader comparative headword in stem class.

The stem class determines the Old English shape. The weak masculine pathway preserves the ending needed for *ġealla* ‘gall’, whereas a simple strong-noun headword does not align as closely with the attested OE noun.

Old English evidence Old English dictionaries record the noun as *gealla* ‘gall’, and Bright also gives the dative *geallan* ‘gall’, confirming a weak-noun paradigm (Bosworth and Toller 1898, 297; Clark Hall 1960, 145; Bright 1971, 372). The normalized spelling *ġealla* ‘gall’ uses ġ for the palatal consonant.

Campbell also notes dialectal variation, contrasting West Saxon or Kentish *gealla* ‘gall’ with Anglian *galla* ‘gall’ (Campbell 1959, sect. 486). The target of this entry is the West Saxon type *ġealla* ‘gall’.

Development to Old English From **gállô* ‘gall’, the weak noun develops through the expected Old English history of the suffix and the regular breaking environment before *ll*, yielding *ġealla*

‘gall’ (Campbell 1959, sect. 486). Once the weak masculine input is chosen, the noun follows a regular path to its attested Old English form.

The decisive issue is therefore morphological. Old English reflects the weak noun, while the broader family label belongs to a different way of presenting the cognate set.

Stem comparison The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the weak noun formation actually reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
broader family label	**gállą*	broader cognate-set headword	wider family context	useful lexeme label, but not the direct source of *ġealla*
selected weak noun	**gállô*	regular output: *ġealla* ‘gall’	*ġealla* ‘gall’	exact match between formation and attested OE noun
dialectal Anglian continuation	weak noun branch	Anglian *galla* ‘gall’ type	*galla* ‘gall’	genuine OE variant, but not the West Saxon form used here

The weak-noun stem class is the relevant point. It gives a direct route to attested *ġealla* ‘gall’, while the broader comparative label serves only as a family heading.

heaven — OE *heofon*

Derivation: citation reconstruction **xémenaz*; form followed here **xébun* > *heofon* (early analogy).

Derivation trace Proto input: **xébun*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Velar Fricative Palatalization	**çébun*
[no change]	PGmc B Allophony	**çéβun*
	OE Back Mutation	**çéoβun*
Early Anglo-Frisian	OE Med Unstressed U Lowering	**çéoβon*
[no change]		

Old English form: *heofon*

A. Selected CAPR derivation

Item	Value
lexical item	heaven
citation reconstruction / lexeme label	*xémenaz
form followed here	*xébun (northern WGmc / pre-OE* hebun)
Old English target	heofon
classification	early_analogy
derivational result	*xébun > heofon (multiplicity 1, exact match)

Compact live trace (corrected cascade, literal adjacent-*mn* SC022):

step	form
proto input	**xébun*
OE Med Unstressed U Lowering	**xébon*
OE Velar Fricative Palatalization	**çébon*
PGmc B Allophony	**çéβon*
OE Back Mutation	**çéoβon*
Old English Orthography	*heofon*

No rule in this span creates the labial: it enters with the input *b* (→ *β* → *f*). Only vocalic and surface changes are modelled.

B. Why *xébun is not arbitrary Ringe and Taylor reconstruct northern West Germanic **hebun* 'sky, heaven', gen. **hebunas* 'heaven', whence West Saxon and Northumbrian *heofon* 'heaven', Mercian *heofen* 'heaven' (Ringe and Taylor 2014, 272, 324). R&T's *hebun* is the lexeme CAPR writes **xébun* (canonical *x* for source *h*), dated to that northern West Germanic stage, not Proto-Germanic; §D explains its labial.

The word is an inherited mn-stem (Kroonen 2013, 220): PIE *h₂ék-mōn, gen.* h₂k-mn-ós, remodelled in Germanic to nom. **hemō* 'heaven', gen. **hemnaz* 'heaven'. In the zero-grade oblique cluster cells the sequence *-mn- is adjacent, and the change* mn > βn/bn/fn* is regular there (Fulk §6.14 (Fulk 2018, 121)). Old Norse preserves exactly this paradigmatic distribution — *himinn* 'heaven' (nominative, with *m*) beside dative *hifni* 'heaven' (oblique, with *f* from the syncopated cluster) — while Old English and Old Saxon (OS *heban* 'heaven', *hebenes* 'heaven' (Fulk 2018)) generalized the labial into the vowel-bearing stem. The *b* of *xébun is therefore analogical* within this nominative input*, but its historical source is regular and recoverable — back to the inherited genitive (see §E).

C. The unresolved *u* The medial *u* is the less secure element and must not be treated as certain merely because the cascade needs a back vowel to trigger back mutation. It is the n-stem suffixal vowel, the genuine conditioning environment for West Saxon back umlaut of *e* (parallel to *seven*, §D), and it is supported by Ringe–Taylor's reconstruction and by the Old English back-mutation pattern (Ringe and Taylor 2014, 324). But its exact deeper ablaut grade — *u* here, against the *i* of the cognate vowel-stem **hemina-* (*himinn* 'heaven', Go. *himins* 'heaven') — is not fully recoverable. It is the irreducible residue of the reconstruction; the *b* is not.

D. Regular modelled span The northern-WGmc / pre-OE → Old English development **xébun* > *heofon* is regular and is the exact structural parallel of the independently regular numeral *seven*, **sébun* > *seofon* (row 2174, *DERIVATION_CLASS* regular). Both have a medial *u* and a labial, and both undergo West Saxon back umlaut of *e* before *u* across that labial (Ringe and Taylor 2014, 324; Hogg 1992a). The live traces run in lock-step (medial *u*-lowering, labial spirantization, back mutation, orthography). Ringe and Taylor explicitly treat *heaven* and *seven* as parallel back-umlaut examples, which is why *seven* is retained here as the control.

E. Deep oblique controls The obliques go deeper than the selected nominative and derive the labial themselves. Live outputs under the corrected literal adjacent-*mn* cascade, with evidential status marked:

cell / form	type	corrected-cascade output	status	note
nom. citation **xémenaz*	reconstructed (citation)	*hemen*	counterfactual (no labial)	citation a-stem; no cluster, so no regular *f*
gen.sg cluster **xémnas*	reconstructed (PIE *h₂k-mn-ós)	*hefnes*	reconstructed, **not attested in OE**	labial **regular** here (mn > βn)

cell / form	type	corrected-cascade output	status	note
dat.sg cluster **xémni*	reconstructed	*hifn*	attested only as ON *hifni* ‘heaven’	labial + i-umlaut regular
dat.pl cluster **xémnum*	reconstructed	*hefnum*	reconstructed, **not attested in OE**	labial regular
remodelled stem **xébun*	reconstructed (R&T *hebun)	*heofon*	**attested** (selected row)	labial pre-supplied
R&T genitive **hebunas*	reconstructed	*heofones*	**attested**	labial pre-supplied
OE oblique *heofnes* / *heofnum*	**attested**	—	actual OE forms	carry *eo*; from the re-vowelled stem + late syncope
ON *himinn* : *hifni* ‘heaven’	**attested** (North Gmc)	—	attested cluster line	keeps the *m* : *f* alternation
OS *heban* / *hebenes*	**attested** (Old Saxon)	—	labial generalized	vowel-bearing throughout

Distinctions kept explicit: attested OE forms (*heofon*, *heofones*, *heofnes*, *heofnum*) descend from the re-vowelled stem; the cluster outputs *hefnes*/*hifn*/*hefnum* are reconstructed regular outputs, and for Old English they are counterfactual — the actual attested obliques carry *eo* from back mutation on the retained/restored medial vowel, then late syncope (*heofonum* > *heofnum*), not from a bare *-fn-* cluster. The genuine attested witness of the pure cluster line is Old Norse *hifni* ‘heaven’.

F. Why the oblique is not the selected row The oblique lines are historically essential — they show that the labial has a regular origin, all the way back to the inherited genitive *h_2k-mn-ós* — *but they are inferior as the single CAPR input→target datum. In Old English every oblique cluster form encounters morphological re-vowelling (restoration of a medial vowel into the labialized* -βn- *stem) before it reaches an attested Old English shape. That re-vowelling is a non-regular event; it interrupts any single deterministic path from a cluster oblique to an attested OE form. The nominative input* xébun begins after that restructuring and then proceeds by regular sound change to a secure attested target — the longest clean modellable span.

G. Attestation warning Bare Old English *hefn* is not securely attested and must not be reintroduced as an attested comparator. The earliest witnesses (Cædmon’s Hymn) are *hefun* ‘heaven’, *hefen* ‘heaven’, and (Moore Bede) *heben* ‘heaven’ — all with a medial vowel. The reconstructed cluster output *hefn* (§E) is a reconstruction, never a manuscript form.

H. Classification rationale The row is *early_analogy*, not *late_analogy* and not *known_unmodelled*.

- The selected row models exactly one form, **xébun → heofon*. The necessary paradigm restructuring (labial levelling + suffix-vowel generalization) precedes the derivational input; once that input is chosen, the modelled development is regular. This is the definition of *early_analogy* (analogy separates the input from the citation reconstruction before the specifically Old English changes apply).
- It is not *late_analogy*: the analogy is in the input, not a later paradigm-cell selection; the later history of *other* paradigm cells (oblique syncope, *heo-* levelling) does not touch the modelled nominative derivation.
- It is not *known_unmodelled*: that label (a legitimate, conservative alternative discussed in `dossier-heaven-paradigm-history-2026.md`) would discard a real, defensible regular span. **xébun* > *heofon is* a clean regular derivation to a secure attested target, so CAPR models it.

The old cross-syllable *mV…n* proxy in SC022 — which formerly fabricated the labial from an intervocalic *m* so that the previous input **xémonų* could reach *heofon* — has been retired in favour of the literal historical adjacent *mn* > *βn*. See `dossier-heaven-paradigm-history-2026.md` §§13–15 and the implementation audit `audits/heaven-sc022-implementation-2026.md`.

knight — OE *cniht*

Derivation: citation reconstruction **knixtaz*; form followed here **knéxtaz* > *cniht* (early analogy).

Derivation trace Proto input: **knéxtaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**knéxt*
EAF Final Z Deletion	**knéxta*	OE Breaking	**knéoxt*
Early Anglo-Frisian		OE Ws Palatal Umlaut	**knixt*
[no change]			

Old English form: *cniht*

Reconstruction and comparative evidence The comparative sources align on an *e*-grade reconstruction for this noun. Ringe and Taylor cite **kneht* ‘knight’, and Orel gives **knextaz* ‘knight’ (Ringe and Taylor 2014, 142; Orel 2003, 256). Kluge-Seebold likewise points to **knehta-* (Kluge 2011, 506). The derivational input **knéxtaz* ‘knight’ follows that comparative evidence.

A competing citation reconstruction **knixtaz* ‘knight’ remains possible as a label for the word family, but it is not the reconstruction followed here. The Old English development discussed below is based on **knéxtaz* ‘knight’.

Old English evidence Old English dictionaries record the noun as *cniht* ‘knight’ (Clark Hall 1960, 63; Bosworth and Toller 1898, 71). Campbell cites plural *cneohtas* ‘knights’ among the broken forms, showing the same vowel environment from another point in the paradigm (Campbell 1959, sect. 146).

The target is therefore an ordinary attested Old English noun. No reconstructed OE comparator is needed here.

Development to Old English From **knéxtaz* ‘knight’, the relevant Old English changes include breaking before the velar cluster and then the later reduction that yields *cniht* ‘knight’. Campbell later notes the early West-Saxon alternation *cniht* ‘knight’ beside plural *cneohtas* ‘knights’ (Campbell 1959, sect. 305). Sievers-Brunner gives the same contrast as *cniht … cneohtas* (Brunner 1965, sect. 122). With that corrected input, the derivation is straightforward.

Stem comparison The comparison below sets the relevant forms side by side. It separates the handbook-supported *e*-grade input from a competing citation reconstruction.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing citation reconstruction	**knixtaz*	not the reconstruction followed here	broader citation tradition	useful as a competing label, but not the source-based choice used for the OE derivation

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
handbook-supported reconstruction	**knéxtaz*	regular output: *cniht* ‘knight’	*cniht* ‘knight’	exact match between comparative reconstruction and attested OE noun
related plural evidence	same stem family	plural *cneohtas* ‘knights’ type background	*cneohtas* ‘knights’	supports the vowel environment, but not the Old English form here cell

lade — OE *hladan*

Derivation: citation reconstruction **laθōjanǫ*; form followed here **xláðanǫ* > *hladan* (early analogy).

Derivation trace Proto input: **xláðanǫ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**xlædanǫ*
PWGmc Dental Hardening **xládanǫ*	OE A Restoration	**xladanǫ*
Early Anglo-Frisian	OE Heavy Syllable Nasal Apocope	**xladan*
	OE Secondary Nasalization	**xladǫn*
[no change]	OE Weak Tail Reduction	**xladan*

Old English form: *hladan*

Reconstruction and comparative evidence Ringe and Taylor cite the strong verb *hladan* ‘lade, load’ directly (Ringe and Taylor 2014, 248). The citation label **laθōjanǫ* ‘lade’ is used here only as a broader family heading, not as the direct source of the OE derivation.

The derivational input therefore marks an early stem choice. The entry follows the strong Verner-grade form that reaches Old English *hladan* directly.

Old English evidence Bosworth-Toller records the verb as *hladan* and preserves the expected strong-verb paradigm material around it (Bosworth and Toller 1898, 559). Clark Hall likewise records *hladan* (Clark Hall 1960, 159). The target is an attested infinitive rather than a reconstructed paradigm cell.

For this entry the relevant comparison form is the infinitive *hladan* itself. The question is how that attested strong verb relates to the broader comparative family.

Development to Old English From **xláðanǫ* ‘lade’, the verb passes through the expected early voiced stop stage, Anglo-Frisian brightening, and the later A-restoration that returns the root vowel to *a* before the full infinitival ending. Campbell treats verbs such as *hladan* as showing restored *a* in the present system (Campbell 1959, sect. 744). Ringe and Taylor likewise derive *hladan* from the voiced strong-verb line (Ringe and Taylor 2014, 248). The resulting Old English infinitive is *hladan*.

Class comparison The comparison below sets the relevant forms side by side. It separates the wider weak-verb family label from the strong verb actually reflected in Old English.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative weak-verb label	*laθōjaną	wider family background	broader family context	useful family label, but not the direct source of *hladan*
selected strong Verner-grade input	*xláðaną	regular output: *hladan*	*hladan*	exact match between formation and attested OE infinitive

lap — OE *lappa*

Derivation: citation reconstruction **lábbaz*; form followed here **láppô* > *lappa* (early analogy).

Derivation trace Proto input: **láppô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**læppô*
[no change]	OE A Restoration	**lappô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**lappa*
[no change]		

Old English form: *lappa*

Reconstruction and comparative evidence The comparative sources point to a weak noun with *pp*: Orel gives **lappōn* ‘lap’, and the Old English dictionary tradition preserves variant *læppa* ‘lap’ (Orel 2003, 236; Clark Hall 1960, 180). The derivational input **láppô* ‘lap’ follows that evidence.

A competing comparative label **lábbaz* ‘lap’ has also circulated for the word family, but the cited handbooks do not make it the direct source of the Old English weak noun. The form relevant to the OE development is the weak masculine input **láppô* ‘lap’.

Old English evidence Campbell cites *lappa* ‘lap’ as a case of restored *a* (Campbell 1959, sect. 158). Sievers-Brunner records *lappa* ‘lap’ beside variant *læppa* ‘lap’ (Brunner 1965, sect. 10). The dictionary tradition also preserves *læppa* ‘lap’ (Clark Hall 1960, 180; Bosworth and Toller 1898, 613).

The target of this entry is the restored singular *lappa* ‘lap’. The variant *læppa* ‘lap’ and the oblique or plural *leappan* remain part of the Old English record and help frame the noun’s vowel history.

Development to Old English Campbell explicitly lists *lappa* ‘lap’ among the forms with restored *a* (Campbell 1959, sect. 158). Sievers-Brunner records *lappa* ‘lap’ beside variant *læppa* ‘lap’ at the same Old English stage (Brunner 1965, sect. 10). With the weak masculine input chosen, the selected *lappa* ‘lap’ outcome is therefore the regular Old English comparison.

Stem comparison The comparison below sets the relevant forms side by side. It separates the weak masculine formation from a competing voiced comparative label.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing voiced comparative label	*lábbaz	not the form followed for the OE weak-noun derivation	broader comparative background	useful as a competing label, but not the source-based choice used here
selected weak masculine noun	*láppô	regular output: *lappa*	*lappa*	exact match between formation and attested OE noun
attested OE variant line	same noun family	*læppa*, *leappan*	*læppa* / *leappan*	useful control forms within the same OE tradition

laugh — OE *hliehhan*

Derivation: citation reconstruction **lákaną*; form followed here **xláxjaną* > *hliehhan* (early analogy).

Derivation trace Proto input: **xláxjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xlæxxjaną*
PWGmc J Gemination	**xláxxjaną*	OE Breaking	**xleaxxjaną*
Early Anglo-Frisian		OE Velar Fricative Palatalization	**xleaxçjaną*
		OE Heavy Syllable Nasal Apocope	**xleaxçjan*
[no change]		OE Secondary Nasalization	**xleaxçjąn*
		OE I Umlaut	**xliexçjąn*
		OE Weak Tail Reduction	**xliexçjan*
		OE J Loss After Heavy	**xliexçan*

Old English form: *hliehhan*

Reconstruction and comparative evidence The wider Germanic family includes a non-j branch represented here by the citation label **lákaną* ‘laugh’, while the derivational input **xláxjaną* ‘laugh’ reflects the j-present line behind Old English *hliehhan* ‘laugh’.

This branch supplies the geminate fricative and the vowel development characteristic of the Old English verb. The comparative family label and the OE-facing input are therefore related but not identical.

Old English evidence Bosworth-Toller records *hlihhan* ‘laugh’ as the verb ‘to laugh’ (Bosworth and Toller 1898, 551). Clark Hall cross-references *hlæhan* ‘laugh’, *hlehhan* ‘laugh’, and *hlihhan* ‘laugh’ to *hliehhan* ‘laugh’ (Clark Hall 1960, 160–61). Bright’s glossary likewise gives *hlihhan (hliehhan, hlyhhan)* (Bright 1971, 315). The target of this entry is the West Saxon *hliehhan* ‘laugh’.

The variants belong to the same background, but the attested lemma *hliehhan* ‘laugh’ supplies the evidence followed here.

Development to Old English From **xláxjaną* ‘laugh’, West Germanic j-gemination yields the doubled consonant (Campbell 1959, sect. 407). Ringe and Taylor derive Old English *hliehhan* ‘laugh’ from the j-present branch via breaking before the palatalized geminate (Ringe and Taylor 2014, 240). They separately compare the related noun *hleahtor* as the outcome of **hlahtraz* ‘laugh’ (Ringe and Taylor 2014, 328).

Branch comparison The comparison below sets the relevant forms side by side. It separates the wider non-j family label from the j-present branch actually reflected in Old English.

Formation / branch	Candidate input	Expected or documented OE outcome	OE comparison form	Result
wider non-j family	*lákaną	comparative background outside the selected OE line	wider family context	useful family label, but not the direct source of *hliehhan*
selected j-present branch	*xláxjaną	regular output: *hliehhan*	*hliehhan*	exact match between branch and attested OE lemma
attested OE variants	same OE verb line	*hlæhhan*, *hlehhan*	*hlæhhan* / *hlehhan*	genuine variant evidence, but secondary to the form compared here

loam — OE *lām*

Derivation: citation reconstruction **laimōn*; form followed here **láimą* > *lām* (early analogy).

Derivation trace Proto input: **láimą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Heavy Syllable Nasal Apocope	**lām*
[no change]			
Early Anglo-Frisian			
EAF Ai Monophthongization	**lāmą*		

Old English form: *lām*

Reconstruction and comparative evidence The inherited comparative noun is given as **laimōn* ‘loam’ or **laiman-*, and both Orel and Kroonen identify Old English *lām* ‘loam’ as a neuter reflex of that family (Orel 2003, 272; Kroonen 2013, 363). The form followed here, **láimą* ‘loam’, differs from the comparative headword because it represents the stem class that matches the Old English noun most directly.

This is therefore a class shift within the history of the English branch rather than a dispute about the OE target itself.

Old English evidence Old English dictionaries record the noun as *lām* ‘loam’, a neuter word for ‘loam, clay, mud’ (Bosworth and Toller 1898, 604; Clark Hall 1960, 196). The target is an attested citation form rather than a reconstructed comparator.

The relevant question is not whether *lām* ‘loam’ is Old English, but which inherited formation best accounts for that attested neuter noun.

Development to Old English From **láimą* ‘loam’, regular monophthongization of *ai* and the later loss of the final nasal syllable yield *lām* ‘loam’. With that OE-facing input, the phonological development is straightforward.

Class comparison The comparison below sets the inherited comparative n-stem label beside the OE-facing stem class used to derive the attested noun.

Formation / class	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited comparative noun	*laimōn	comparative family background	wider family context	useful headword, but not the direct OE-facing input
OE-facing stem class followed here	*láimą	regular output: *lām*	*lām*	exact match between the form followed here and the attested OE noun

lung — OE *lungen*

Derivation: citation reconstruction **lungō*; form followed here **lúnganjō* > *lungen* (early analogy).

Derivation trace Proto input: **lúnganjō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE I Umlaut	**lúngennju*
PWGmc J Gemination	**lúngannjō*	OE High Vowel Apocope	**lúngennj*
PNWGmc Final Long O Raising	**lúngannju*	OE J Loss After Heavy	**lúngenn*
Early Anglo-Frisian		OE Final Geminate Simplification	**lúngen*
[no change]			

Old English form: *lungen*

Reconstruction and comparative evidence Kroonen treats the basic noun as **lungōn-* and also cites an OE-facing derivative **lungunjō-*, continued by Old English *lungen* ‘lung’ and close West Germanic cognates (Kroonen 2013, 384). The derivational input **lúnganjō* ‘lung’ models that derived feminine formation rather than the base noun. The notation differs slightly from Kroonen’s **lungunjō-*, but both point to the same derived feminine line.

The difference between the citation label and the derivational input is therefore derivational. Old English *lungen* ‘lung’ is not a direct reflex of the bare base noun **lungō* ‘lung’; it belongs to an expanded feminine formation.

Old English evidence Old English dictionaries record the noun as *lungen* ‘lung’, with inflected forms such as *lungenne* and *lungena* (Bosworth and Toller 1898, 634). Clark Hall also preserves a small family of compounds such as *lungenādl* ‘lung-disease’, *lungensealf*, and *lungenwyrt* (Clark Hall 1960, 191).

The target is an attested Old English lexeme with its own paradigm, not a rescued inflectional cell.

Development to Old English From the selected derived input, the expected derivational consonant and vowel adjustments lead to *lungen* ‘lung’. Once the expanded feminine formation is chosen, the Old English outcome is regular.

Formation comparison The comparison below sets the relevant forms side by side. It separates the base noun from the derived feminine formation reflected in Old English.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
base noun	*lungō	base-noun outcome without the OE derivative suffix	broader family context	useful headword, but not the direct source of *lungen*
derived OE-facing formation	*lúnganjō	regular output: *lungen*	*lungen*	exact match between the derived formation and the attested OE noun
Kroonen's cited derivative	**lungunjō-*	comparative support for the same OE-facing formation	*lungen* and cognate set	supports the derived feminine formation, with notation differing from the normalized input form used here

navel — OE *nafola*

Derivation: citation reconstruction **nablô*; form followed here **nábulô* > *nafola* (early analogy).

Derivation trace Proto input: **nábulô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**nœbulô*
[no change]	OE A Restoration	**nabulô*
	PGmc B Allophony	**naβulô*
Early Anglo-Frisian	OE Unstressed Long Vowel Shortening	**naβula*
[no change]	OE Med Unstressed U Lowering	**naβola*

Old English form: *nafola*

Reconstruction and comparative evidence Kroonen instead gives a nasal-suffix navel formation with Old English *nafela* 'navel' among its reflexes (Kroonen 2013, 420), while Ringe and Taylor give the derivational pathway **nabulō* 'navel' > **nœbula* 'navel' > *nafola* 'navel' (Ringe and Taylor 2014, 270). The difference is one of stage and notation rather than of lexeme identity: the derivational input **nábulô* 'navel' is the pre-syncope form needed for the Old English development.

For the Old English comparison, the crucial point is simply that the pre-OE form still contains a medial vowel.

Old English evidence Ringe and Taylor note the early West Saxon shift *nafola* 'navel' > *nafela* 'navel' (Ringe and Taylor 2014, 336). Campbell likewise records *nafela* 'navel' beside Corpus *nabula* 'navel' (Campbell 1959, sect. 159). The target of this entry is the nominative singular *nafola* 'navel', the form that matches the selected derivational pathway most directly.

nafela ‘navel’ is the better-known later West Saxon spelling, while *nabula* ‘navel’ preserves a less reduced medial vowel. These forms belong to the same lexical history, but this entry is centered on *nafola* ‘navel’.

Development to Old English Ringe and Taylor give the pre-OE line **nabulō* ‘navel’ > **næbula* ‘navel’ > OE *nafola* ‘navel’ (Ringe and Taylor 2014, 270). The trace represents that same development with stress-marked notation and explicit intermediate weakening. Intervocalic *b* then surfaces as *f*, and final weak-tail shortening gives *nafola* ‘navel’.

The medial vowel is still present when A-restoration applies. That is why the selected pre-syncope input differs from the syncopated comparative headword.

Stage comparison The comparison below sets the relevant forms side by side. It separates the comparative citation form from the pre-syncope input and from the later OE spellings.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
syncopated comparative headword	**nablô*	reduced *næfla*-type outcome rather than *nafola* ‘navel’	not the Old English form here	useful citation form, but too reduced for the pathway modeled here
selected pre-syncope input	**nábulô*	regular output: *nafola* ‘navel’	*nafola* ‘navel’	exact match between derivational input and target
later OE reduction stages	same lexical history	attested *nafela* ‘navel’; Corpus *nabula* ‘navel’	*nafela* ‘navel’ / *nabula* ‘navel’	related OE spellings, but not the chosen comparator

neck — OE *hnecca*

Derivation: citation reconstruction **xnákkaz*; form followed here **xnékkô* > *hnecca* (early analogy).

Derivation trace Proto input: **xnékkô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**xnékka*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *hnecca*

Reconstruction and comparative evidence The noun belongs to an ablauting n-stem family. Kroonen reconstructs a paradigm with nominative **hnekkō* ‘neck’, genitive **hnukkaz* ‘neck’, and accusative plural **hnakkuns* ‘neck’, and he places Old English *hnecca* ‘neck, nape’ among the e-grade descendants (Kroonen 2011, 167). Kluge-Seebold likewise identifies *ae. hnecca* as an ablaut partner of the a-grade *Nacken* family (Kluge 2011, 347).

A competing comparative label **xnákkaz* ‘neck’ belongs to the wider family, and Orel also gives an a-grade headword line (Orel 2003, 218). The derivational input **xnékkô* ‘neck’, however, is the form that matches the Old English branch.

Old English evidence Clark Hall records the weak masculine noun *hnecca* ‘neck, nape’ (Clark Hall 1960, 162). Bosworth-Toller likewise records *hnecca* ‘neck, nape’ (Bosworth and Toller 1898, 567). The target is therefore an attested citation form, not an oblique cell or a reconstructed lemma.

The phonological question is upstream of the Old English evidence. The attested noun already shows that the branch continued an e-grade form rather than the a-grade seen in much of the continental material.

Development to Old English From **xnékkô* ‘neck’, the derivation is straightforward. The trace shortens the final long vowel to **xnékka* ‘neck’, and Old English orthography gives *hnecca* ‘neck’.

The derivation depends on the earlier selection of the e-grade weak-noun form continued by Old English.

Stem comparison The comparison below sets the relevant forms side by side. It separates the wider a-grade family from the selected e-grade Old English branch.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
competing comparative label	**xnákkaz*	broader a-grade family rather than the selected OE source	continental *Nacken* line	useful family label, but not the input followed for the Old English derivation
weak noun with a-grade	**xnakkô*	expected *hnacca* type outcome	*hnacca*	fixes the class, but not the vowel grade
selected e-grade nominative	**xnékkô*	regular output: *hnecca*	*hnecca*	exact match between derivational input and attested OE noun
oblique paradigm background	**hnukkaz*, **hnakkuns*	ON/OHG/German a-grade continuation	*hnakki* / *Nacken*	shows the wider ablaut family, but not the chosen OE branch

needle — OE *nǣdl*

Derivation: citation reconstruction **nḗθlō*; form followed here **nḗðlō* > *nǣdl* (early analogy).

Derivation trace Proto input: **nḗðlō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**nǣdl*
PNWGmc Long E Lowering	**nāðlō*		
PWGmc Dental Hardening	**nādlō*		
PNWGmc Final Long O Raising	**nādlu*		
EAF Long A Fronting	**nǣdlu*		
Early Anglo-Frisian			
[no change]			

Old English form: *nǣdl*

Reconstruction and comparative evidence Ringe and Taylor treat the word as a voiced/voiceless alternant, citing **nēþlō* 'needle' ~ **nēdlō* 'needle' > OE *nédl* 'needle' (Ringe and Taylor 2014, 329). The form followed here, **néðlō* 'needle', is the voiced Verner-grade form used for the Old English comparison, while the citation form **néθlō* 'needle' remains the broader lexeme label.

The development discussed here follows the Ringe-Taylor alternant framework.

Old English evidence Clark Hall records the attested citation form *nǣdl* 'needle' (Clark Hall 1960, 210). Campbell lists *nédl* 'needle' among the expected unbroken forms after *t* and *d* (Campbell 1959, sect. 367). Hogg also includes *nidi* / *nǣdl* 'needle' in the same broader cluster history (Hogg 1992b).

The target is therefore an attested citation form. No oblique-cell substitution is involved in this entry.

Development to Old English Ringe and Taylor give the historical line **nēþlō* 'needle' ~ **nēdlō* 'needle' > OE *nédl* 'needle' (Ringe and Taylor 2014, 329). Campbell likewise lists *nédl* 'needle' among the expected unbroken forms after *t* and *d* (Campbell 1959, sect. 367). The trace expresses the same pathway with the voiced alternant followed here for the Old English comparison.

The essential choice lies in which Proto-Germanic alternant is taken as the starting point. Once the voiced form is chosen, the rest of the pathway is regular.

Alternant comparison The comparison below sets the relevant forms side by side. It separates the broader citation headword from the voiced alternant used for Old English.

Formation / stage	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative voiceless headword	**néθlō*	broader word-family label rather than the OE-facing alternant	**nēþlō* line	useful citation form, but not the derivational input for the Old English comparison
voiced Verner alternant followed here	**néðlō*	regular output: *nǣdl*	*nǣdl*	exact match between the form followed here and the attested OE noun
later hardening stage	*nédlō	intermediate pre-OE stage in the same derivation	*nǣdl*	genuine stage in the pathway, but not the Proto-Germanic form followed here

nose — OE *nosu*

Derivation: citation reconstruction **nasō*; form followed here **núsō* > *nosu* (early analogy).

Derivation trace Proto input: **núsō*

Earlier Germanic changes		Old English changes
Northwest and West Germanic		[no change]
PNWGmc U Lowering	**nósō*	
PNWGmc Final Long O Raising	**nósu*	
Early Anglo-Frisian		
[no change]		

Old English form: *nosu*

Reconstruction and comparative evidence Kroonen reconstructs a Germanic ablaut pair **nasō- ~ *nusō-* and adds that the root **nus-* is likely to have arisen as a secondary zero grade after a remodeling of the older paradigm (Kroonen 2013, 383). Campbell is more specific for Old English, citing *nosu* 'nose' < **nusō* 'nose' (Campbell 1959, 43).

The citation reconstruction **nasō* 'nose' is therefore best treated as the full-grade comparative headword, while the derivational input **núsō* 'nose' represents the remodeled zero-grade line continued by the Old English form discussed here. Orel's **nasō* 'nose' alongside OE *nasu* 'nose' preserves the competing full-grade notation and shows that the two lines should not be collapsed without comment (Orel 2003, 320).

Old English evidence *Nosu* 'nose' is an attested Old English noun. Ringe and Taylor list it among the few surviving early Old English feminine u-stems (Ringe and Taylor 2014, 385). Clark Hall likewise gives *nosu f.* 'nose', with genitive-dative singular *nosa*, and cross-refers *nasu* 'nose' to *nosu* 'nose' (Clark Hall 1960, 211, 217).

The selected OE target is therefore attested *nosu* 'nose', not a reconstructed placeholder. The lexicographical record also gives *nasu* 'nose' for the full-grade side of the tradition.

Development to Old English From **núsō* 'nose', the regular path is the one documented by the current trace: **núsō* 'nose' > **nósō* 'nose' > **nósu* 'nose' > *nosu* 'nose'. The early special step lies in the choice of the zero-grade input, not in any late Old English repair.

With that input chosen, the OE development is straightforward. The full-grade line behind **nasō* 'nose' instead points toward *nasu* 'nose', not to the form treated here.

The low root of *nosu* 'nose' supports the economy of lowering before final oral raising, but only conditional on that selected remodeled stem. Reversing the changes yields a high-root nominative, whose replacement by a lowered oblique root would be an additional, individually unproved levelling. The word is therefore not a pristine inherited nominative proving the chronology. The comparative argument and its limits are developed in SC019 PNWGmcFinalLongORaising (Fulk 2018, 56; Kroonen 2013, 383).

Stem comparison The comparison below sets the relevant forms side by side. It separates the full-grade comparative line from the remodeled zero-grade input that yields the Old English form.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
full-grade comparative line	*nasō	expected full-grade continuation *nasu*	*nasu*	useful comparative background, but not the Old English-facing input

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
remodeled zero-grade line	*núsō	regular output: *nosu*	*nosu*	exact match between derivational input and attested OE noun

sap — OE *sæp*

Derivation: citation reconstruction **sapōn*; form followed here **sápą* > *sæp* (early analogy).

Derivation trace Proto input: **sápą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**sæpą*
[no change]	OE Heavy Syllable Nasal Apocope	**sæp*
Early Anglo-Frisian		
[no change]		

Old English form: *sæp*

Reconstruction and comparative evidence The comparative sources do not give one uniform inherited stem. Kroonen preserves the word family as **saf* 'sap' / **ppan-*, with Old English *sæp* 'sap' (Kroonen 2013, 420). Orel preserves the comparative notation **sapōn* 'sap' and **sapan* 'sap' (Orel 2003, 319).

The derivational input **sápą* 'sap' therefore does not replace those comparative labels. It identifies the OE-facing stem shape that yields the attested noun treated here.

Old English evidence Clark Hall records *sæp* 'sap' (e) n. (Clark Hall 1960, 247). The target is therefore an attested neuter Old English noun. Orel's plain *sap* 'sap, juice' notation belongs to comparative normalization, not to the spelling adopted here for the Old English form (Orel 2003, 319).

Development to Old English From **sápą* 'sap', Anglo-Frisian brightening yields *sæ* 'sap', and heavy-syllable nasal apocope then produces *sæp* 'sap'. That is the regular path documented by the current trace.

The competing comparative lines do not give the same result. The inherited n-stem notation **sapōn* 'sap' yields *sape* 'sap', while an i-stem continuation from the **sapi-* line leads to *sep* 'sap' / *sepe* 'sap' rather than to *sæp* 'sap'. The special step in this entry is therefore the early stem choice, not a late OE paradigm-cell selection.

Stem comparison The comparison below sets the relevant forms side by side. It separates the competing comparative stem lines from the Old English-facing input.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative n-stem line	**sapōn*	local comparator output: *sape* ‘sap’	*sape* ‘sap’	useful comparative background, but not the source of attested *sæp*
inferred i-stem comparator from **sapi-*	**sapiz*	local comparator output: *sepe* ‘sap’	*sepe* ‘sap’	confirms that an i-triggering stem does not reach the target
selected a-stem input	**sápą*	regular output: *sæp* ‘sap’	*sæp* ‘sap’	exact match between derivational input and attested OE noun

sea — OE *sǣ*

Derivation: citation reconstruction **sái*; form followed here **sáiwiz* > *sǣ* (early analogy).

Derivation trace Proto input: **sáiwiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE W Loss Before I	**sāi*
EAF Final Z Deletion	**sáiwi*	OE I Umlaut	**sǣi*
Early Anglo-Frisian		OE High Vowel Apocope	**sǣ*
EAF Ai Monophthongization	**sāwi*		

Old English form: *sǣ*

Reconstruction and comparative evidence Kroonen gives the noun in stem notation as **saiwi-*, an i-stem whose English reflex is cited as OE *sæ* ‘sea’ (Kroonen 2013, 423). Ringe and Taylor write the fuller form **saiwiz* ‘sea’ and derive it through **sawi* ‘sea’ > **sei* ‘sea’ > OE *sǣ* ‘sea’ (Ringe and Taylor 2014, sect. 6.7.1). The comparative headword is therefore shorter than the form required for the English history: **sái* ‘sea’ names the lexeme, but **sáiwiz* ‘sea’ preserves the medial **w* and the final high vowel that control the later development.

Old English evidence The Old English noun is the ordinary word for ‘sea’. Kroonen cites it as *sæ* ‘sea’; the normalized form here is *sǣ* ‘sea’ (Kroonen 2013, 423). Campbell likewise treats *sea* as continuing the same **saiui* ‘sea’ > **sǣi* ‘sea’ history, with loss of *u*/*w* before *i* (Campbell 1959, sect. 406).

Development to Old English Once the fuller i-stem input is chosen, the development is regular. After Proto-West-Germanic monophthongization **sáiwiz* ‘sea’ > **sāwiz* ‘sea’ and final **-z* loss **sāwiz* ‘sea’ > **sāwi* ‘sea’, the non-initial **w* disappears before unstressed **i*, and the following high vowel fronts the root vowel before final apocope. The documented chain is **sáiwiz* ‘sea’ > **sāwiz* ‘sea’ > **sāwi* ‘sea’ > **sāi* ‘sea’ > **sǣi* ‘sea’ > *sǣ* ‘sea’ (Ringe and Taylor 2014, sect. 6.7.1; Campbell 1959, sect. 406).

Stem and stage comparison The comparison below sets the relevant forms side by side. It separates the abbreviated comparative headword from the fuller i-stem input that yields the Old English form.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
abbreviated comparative headword	**sái*	too short to preserve the **w … *i* environment needed for the documented chronology	*sǣ*	useful comparative label, but not the Old English-facing input
selected i-stem input	**sáiwiz*	documented regular output: *sǣ*	*sǣ*	exact match between derivational input and Old English target

sieve — OE *sife*

Derivation: citation reconstruction **síbaz*; form followed here **síbi* > *sife* (early analogy).

Derivation trace Proto input: **síbi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	PGmc B Allophony	**síβi*
[no change]	OE Med Unstressed I Lowering1	**síβe*
Early Anglo-Frisian		
[no change]		

Old English form: *sife*

Reconstruction and comparative evidence Kluge-Seebold gives wg. **sibi-* n. … ae. *sife*, and Campbell groups *sife* ‘sieve’ with short neuter i-stems such as *spere* (Kluge 2011, 847; Campbell 1959, sect. 609). The older morphological background is the s-stem **sib-iz* ‘sieve’, but the derivational input is the normalized i-stem form **síbi* ‘sieve’.

Kroonen’s nearby **sebjō-* entry belongs to the separate kinship lexeme that yields Old English *sibb* ‘sieve’, not to the sieve word. Orel’s **sibaz* ‘sieve’ … OE *sife* ‘sieve’ preserves a broader handbook notation, but that a-stem shape does not fit the Old English form treated here (Orel 2003, 328).

Old English evidence Clark Hall gives *sibi* ‘sieve’ (GL) … = *sife* ‘sieve’ and also *sife* ‘sieve’ n. ‘sieve’ (Clark Hall 1960, 263). Campbell likewise cites Corpus Glossary *sibi* ‘sieve’ and treats *sife* ‘sieve’ as a short neuter i-stem (Campbell 1959, sects 444, 609). The normalized Old English target is therefore *sife* ‘sieve’, while *sibi* ‘sieve’ is an attested earlier spelling rather than a separate lexeme.

Development to Old English From **síbi* ‘sieve’, the regular derivation gives **síβi* ‘sieve’ > **síβe* ‘sieve’ > *sife* ‘sieve’. Medial *b* is realized as a spirant and later written *f*, while the final unstressed *i* lowers to *e*. The older s-stem background **sib-iz* ‘sieve’ explains the morphology, but the derivational input **síbi* is the immediate pre-Old-English form.

Stem comparison The comparison below sets the relevant forms side by side. It distinguishes the accepted i-stem line from its rejected competitors.

Formation / label	Candidate input	Expected or documented OE outcome	OE comparison form	Result
ja-stem kinship line	Kroonen **sebjō-* / comparator **sibja*	OE *sibb* ‘sieve’	*sibb* ‘sieve’	separate lexeme, not the target treated here
a-stem handbook line	**síbaz*	expected *sif*	*sif*	wrong ending for the attested noun
selected i-stem line from older **sib-iz*	**síbi*	documented regular output: *sife* ‘sieve’	*sife* ‘sieve’; early spelling *sibi* ‘sieve’	exact match between derivational input and Old English evidence

spare — OE *sparian*

Derivation: citation reconstruction **sparēną*; form followed here **spárōjaną* > *sparian* (early analogy).

Derivation trace Proto input: **spárōjaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**spærōjaną*
[no change]	OE A Restoration	**sparōjaną*
	OE Heavy Syllable Nasal Apocope	**sparōjan*
Early Anglo-Frisian	OE Secondary Nasalization	**sparōjąn*
[no change]	OE I Umlaut	**sparējąn*
	OE Unstressed Long Vowel Shortening	**sparejąn*
	OE Weak Tail Reduction	**sparejan*
	OE Intervocalic J Vocalization	**spareian*
	OE Unstressed EI Contraction	**sparian*

Old English form: *sparian*

Reconstruction and comparative evidence Kroonen keeps the inherited verb under class-III **sparēn-* (Kroonen 2013, 465). Orel similarly preserves **sparēnan* ‘spare’ (Orel 2003, 362). Ringe and Taylor, however, reconstruct **sparai-* ~ **sparja-* for the English branch and derive the citation verb from a class-II line (Ringe and Taylor 2014, 162, 191). The derivational input **spárōjaną* ‘spare’ therefore represents the refashioned class-II formation behind Old English *sparian* ‘spare’, while the citation reconstruction **sparēną* ‘spare’ remains the inherited comparative headword.

Old English evidence Campbell says that *sparian* ‘spare’ does not show the ordinary class-III characteristics, but the Ritual forms, normalized here as *spæria* ‘spare’, *spær* ‘spare’, and *spærede* ‘spare’, together with Vespasian Psalter *spearad* ‘spare’, point to primitive Old English forms both with and without back vowels (Campbell 1959, sect. 764). Brunner likewise records Northumbrian *spæria* ‘spare’, *spærede* ‘spare’ beside common Old English *sparian* ‘spare’ and Vespasian Psalter *spearad* ‘spare’ (Brunner 1965, sect. 364 Anm. 11). The citation form treated here is *sparian* ‘spare’; the Anglian forms are relics of the older formation, not alternative headwords of equal status.

Development to Old English Once the class-II formation **spárōjaną* ‘spare’ is chosen, the remaining development is regular. The regular derivation shows brightening, restoration of *a* before the back vocalism of the suffix, later i-mutation within the weak ending, weak-tail reduction, and contraction to *sparian* ‘spare’. By contrast, Brunner’s rule against further apocope of final *-e* explains why Ritual *spær* ‘spare’ cannot be the regular continuation of inherited **spárē* ‘spare’ (Brunner 1965, sect. 150).

Formation comparison The comparison below sets the relevant forms side by side. It contrasts the inherited class-III formation with the refashioned class-II one that yields the citation verb.

Formation / comparison	Candidate input	Expected or documented OE outcome	OE comparison form	Result
inherited class-III infinitive	**spárēną*	paradigm comparison / probe output: *sparen*	*sparian* ‘spare’	wrong class and wrong ending for the citation verb
inherited class-III imperative singular	**spárē*	paradigm comparison / probe output: *spære*	Ritual *spær* ‘spare’	loss of final *-e* is not regular, so the relic form cannot control the entry
inherited class-III finite present	**spárēθi*	paradigm comparison / probe output: *spæreþ*	*spearad* ‘spare’	attested form is mixed, not a direct continuation of the inherited cell
selected class-II formation	**spárōjaną*	documented regular output: *sparian* ‘spare’	*sparian* ‘spare’	exact match between derivational input and Old English citation form

staff — OE *stæf*

Derivation: citation reconstruction **stábiz*; form followed here **stábaz* > *stæf* (early analogy).

Derivation trace Proto input: **stábaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**stáb*
EAF Final Z Deletion	**stába*	EAF Brightening	**stæb*
Early Anglo-Frisian		PGmc B Allophony	**stæβ*
[no change]			

Old English form: *stæf*

Reconstruction and comparative evidence The comparative dictionaries do not give one uniform stem class. Kroonen reconstructs an a-stem **staba-* (Kroonen 2013, 471). Orel writes **stábiz* ‘staff’ ~ **stábaz* ‘staff’ (Orel 2003, 368). A direct i-stem input in **-iz* ‘staff’ would predict i-mutation in Old English, whereas the attested noun keeps *æ*.

Old English evidence The Old English noun itself is the ordinary citation form *stæf* ‘staff’. Luick lists *stæf* ‘staff’ among closed monosyllables with *æ* (Luick 1914--1940, 176). Ringe and Taylor pair singular *stæf* ‘staff’ with plural *stafas* ‘staffs’ (Ringe and Taylor 2014, 193). The normalized form here is therefore *stæf*; later English *staff* with *a* belongs to a later stage of the word’s history.

Development to Old English With the selected a-stem input, the development is regular. Final *-*z* disappears, bare final -*a* is lost, Anglo-Frisian brightening gives *æ* in the closed monosyllable, and medial *b* surfaces as a fricative written *f*. The documented chain is **stábaz* 'staff' > **stába* 'staff' > **stáb* 'staff' > *stæb* 'staff' > *stæf* 'staff'. A direct continuation of **stábiz* 'staff', by contrast, would produce i-mutated *stefe* 'stave, staff' rather than the attested singular.

Formation comparison The comparison below sets the relevant forms side by side. It separates the rejected i-stem line from the selected a-stem input.

Formation / label	Candidate input	OE output or comparison	OE comparison form	Result
comparative i-stem line	**stábiz*	expected *stefe* after i-mutation	*stæf*	wrong vowel for the attested singular
mixed comparative notation	Orel **stabiz* ~ **stabaz*; Kluge **stabi-*/a-	source-level stem-class uncertainty	*stæf*	useful comparative background, but not a single OE-facing input
selected a-stem input	**stábaz*	documented regular output: *stæf*	*stæf*	exact match between derivational input and Old English target

stem — OE *stefn*

Derivation: citation reconstruction **stámnaz*; form followed here **stámniz* > *stefn* (early analogy).

Derivation trace Proto input: **stámniz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**stæβni*
EAF Final Z Deletion	**stámni*	OE I Umlaut	**steβni*
PNWGmc Mn Dissimilation	**stáβni*	OE High Vowel Apocope	**steβn*
Early Anglo-Frisian			
[no change]			

Old English form: *stefn*

Reconstruction and comparative evidence Orel reconstructs the Proto-Germanic source as **stámnaz* 'stem, trunk' (also variant **stamniz* 'stem, trunk'), citing Old Norse *stafn* 'stem of a ship', Old Frisian *stevene* 'stem of a ship' (fem.), and Old Saxon *stamn* 'stem' as its main continuants (Orel 2003, 371).

This word is etymologically unrelated to the Old English homonym *stefn* 'voice, sound' / *stemn* 'voice, sound', which descends from a distinct Proto-Germanic **stebnō* 'voice, sound' / **stemnō* 'voice, sound' etymon. The two OE lexemes are distinguished in Clark Hall, Bosworth-Toller, Brunner, and Luick; they collide on the surface forms *stefn* and *stemn* through historically independent developments.

Kroonen relates the 'stem' family to the *stam(m)* 'stem, trunk' group attested in Old High German, Dutch, and German, without reconstructing a separate 'prow' proto-form (Kroonen 2013, 479–80).

Old English evidence Clark Hall records *stefn* ‘stem, trunk, root, prow, foundation’, the weak n-stem *stefna* ‘prow or stern of a ship’, and the related *stofn* ‘trunk, stem, branch, shoot’ (Clark Hall 1960, 276, 341).

Luick §211 explicitly distinguishes OE *stefn* ‘stem, trunk’ / *stemn* ‘stem, trunk’ — noting cognates with Old Saxon *stamn* ‘stem’ and Middle English *stam* ‘stem’ — from the separate OE voice/sound word (*stefn* ‘voice, sound’ / *stemn* ‘voice, sound’) treated in §75 (Luick 1914--1940, sect. 211). Brunner §205 lists *stefn, stemn Stamm* among words showing the *fn*/*mn* alternation (Brunner 1965, sect. 205).

The primary comparison form used here is *stefn* ‘stem, trunk, root, prow’, the form attested most directly in the semantic range relevant to English *stem*. The n-stem *stefna* ‘prow/stern’ is a closely related but more narrowly attested nautical specialization.

Development to Old English The derivation of OE *stefn* ‘stem, trunk’ from the i-stem input **stámniz* ‘stem, trunk’ (Orel’s attested i-stem variant; the citation reconstruction remains **stámnaz* ‘stem, trunk’) is now regular and modelled, with multiplicity 1. Live trace under the corrected literal adjacent-*mn* SC022:

step	form
proto input	**stámniz*
EAF Final Z Deletion	**stámni*
PNWGmc Mn Dissimilation (adjacent mn > βn)	**stáβni*
EAF Brightening (á > æ)	**stæβni*
OE i-Umlaut (æ > e, from the i-stem ending)	**steβni*
OE High Vowel Apocope	**steβn*
Old English Orthography (β > f)	*stefn*

The consonantal change mn → *βn*/*fn* is attested comparatively: Old Norse *stafn* ‘stem of a ship’ and Old Saxon *stamn* ‘stem’ preserve the *fn*/*mn* variants expected from this family. The handbooks treat the change as a pre-Old-English development (Luick §211; Brunner §205), and the FST now encodes it as the historical adjacent *mn* > *βn* (the rule *PNWGmcMnDissimilation* / SC022). The *e* of *stefn* is regular from *á* via brightening (*á* > *æ*) plus i-umlaut triggered by the i-stem ending (*æ* > *e*), so both the vowel and the consonant follow by regular sound change once the i-stem input is selected.

Note the contrast with *heaven*. Here SC022 (*PNWGmcMnDissimilation*) fires directly inside the selected derivation, deriving the labial of *stefn* from the adjacent *-mn-* cluster of **stámniz*. In *heaven*, by contrast, the same historical change *mn* > *βn*/*fn* operated only in the deeper cluster-bearing oblique prehistory; the resulting labial was then generalized into the vowel-bearing **hebun-* stem, so the selected CAPR path **xébun* > *heofon* begins *after* that analogy and does not itself contain SC022 (see `dossier-heaven-paradigm-history-2026.md` §§13–15 and `audits/heaven-sc022-implementation-2026.md`). The earlier cross-syllable *mV...n* proxy — which had formerly fabricated a labial from an intervocalic *m* — has been retired in favour of this literal adjacent *mn* > *βn*. Selecting the i-stem input **stámniz* over the a-stem citation **stámnaz* is a pre-OE input selection; once selected, the Old English development is regular. Hence the classification *early_analogy* (analogy separates the input from the citation reconstruction before the specifically Old English changes apply), not *known_unmodelled*.

Homonym note Old English *stefn*/*stemn* is a surface homonym for two etymologically unrelated words:

1. *stefn*/*stemn* ‘voice, sound’: from PGmc **stebnō* (Ringe-Taylor, Orel); via *bn* → *fn* (b-allophony), giving *stefn* ‘voice, sound’. This is a different row from 2216.
2. *stefn*/*stefna*/*stofn*/*stemn* ‘stem, trunk, prow’: from PGmc **stámnaz* ‘stem/trunk’ (Orel); via the pre-OE cluster change *mn* → *fn*. This is row 2216.

The form followed here, **stébnō*, used in earlier versions of this entry was the wrong homonym’s transponent and must not be used here.

Source comparison

Form or label	Status	OE relation	Result
**stámnaz* ‘stem, trunk’	Orel’s PGmc a-stem citation for the stem/trunk/prow family	lexeme-level citation	retained as PROTO (citation)
**stámniz* ‘stem, trunk’	Orel’s attested i-stem variant	**selected derivational input**	regular OE output *stefn* (mult 1)
stefn III ‘stem, trunk, root, prow’	primary OE target (strong masc.)	stem/trunk sense per Clark Hall	target form
stefna ‘prow/stern’	OE n-stem specialization	nautical sense per Clark Hall	comparison form
stofn ‘trunk’	OE *o*-grade variant or earlier stage	trunk/stem sense	comparison form
stemn ‘trunk’	late West Saxon *fn* → *mn* doublet	secondary form	comparison form
**stébnō* (voice word)	**wrong homonym** — belongs to voice/sound dossier	no relation to stem/trunk sense	**must not be used as row 2216’s derivational input**

swan — OE *swanes*

Derivation: citation reconstruction **swánaz*; form followed here **swánas* > *swanes* (early analogy).

Derivation trace Proto input: **swánas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**swánæs*
[no change]	OE Unstressed AE Merger	**swánes*
Early Anglo-Frisian		
[no change]		

Old English form: *swanes*

Reconstruction and comparative evidence The Germanic noun is ordinarily cited as the masculine a-stem **swánaz* ‘swan’ (Orel 2003, 367). The derivational input **swánas* ‘swan’ is not a competing lexeme reconstruction. It is the genitive singular **swánas* of the same paradigm.

The question here is therefore one of paradigm cell rather than stem history. The citation form remains **swánaz* ‘swan’ > *swan* ‘swan’; the comparison form is the genitive singular **swánas* ‘swan’ > *swanes* ‘swan’.

Old English evidence Bright’s glossary records the ordinary noun as *swan* ‘swan’, m. and also gives the exact inflected form *swanes* ‘swan’, citing the phrase *swanes feðre* (Bright 1971, 441).

The target is therefore an attested Old English genitive singular, not a reconstruction or the ordinary citation lemma.

Development to Old English From **swánas* ‘swan’, Anglo-Frisian brightening gives **swánæs* ‘swan’, and subsequent merger of unstressed *æ* with *e* yields *swanes* ‘swan’. The comparison is straightforward once the genitive singular is chosen as the relevant cell.

Paradigm-cell comparison The comparison below sets the relevant forms side by side. It separates the ordinary citation form from the selected inflected cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**swánaz*	OE headword *swan* ‘swan’	*swan* ‘swan’	ordinary lexeme line
genitive singular	**swánas*	regular output: *swanes* ‘swan’	*swanes* ‘swan’	attested cell

thousand — OE *þūsend*

Derivation: citation reconstruction **θūs-undī*; form followed here **θūs-èndi* > *þūsend* (early analogy).

Derivation trace Proto input: **θūsèndi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Strip Secondary Stress	**θūsend*
PWGmc Early I Apocope **θūsènd*		
Early Anglo-Frisian		
[no change]		

Old English form: *þūsend*

Reconstruction and comparative evidence Kroonen reconstructs the Germanic numeral as **þūsundī-* and cites Old English *þūsend* ‘thousand’ among its continuations (Kroonen 2013, 554). The derivational input **θūs-èndi* ‘thousand’ is not the same claim. It is an OE-oriented transponent with the second-member vowel already resolved to *e* and the final high vowel already shortened for apocope.

The chronology must explain why Old English shows *þūsend* ‘thousand’ while related languages such as Old Saxon and Old High German keep *u* in the second syllable? (Kroonen 2013, 554).

Old English evidence Old English *þūsend* ‘thousand’ is an ordinary citation form, not a selected oblique or paradigm cell. Campbell treats it as a neuter noun with normal case forms (Campbell 1959, sect. 689). The problem lies in the internal history of the word, not in its lexical status.

Development to Old English If the old final *-ī* had remained long enough to trigger ordinary double umlaut, Campbell’s rule would point toward a form of the **þȳsend* ‘thousand’ type rather than attested *þūsend* ‘thousand’ (Campbell 1959, sect. 203). Preserved root *ū* therefore argues that the umlaut-triggering vowel was lost or neutralized before the ordinary OE umlaut could develop.

That early loss, however, does not by itself explain the medial *e*. Luick compares the word with *ærende* ‘errand/message’ and later groups *thousand* with forms reshaped on that pattern (Luick 1914--1940, sects 198, 492). Viredaz is more cautious, arguing that Old English *e* in this weak position may simply write schwa and so need not prove a unique *ærende* ‘message’-type analogy (Viredaz 2025, sect. 2.1.4).

The selected transponent **θūs-èndi* ‘thousand’ captures the OE-side state from which the regular derivation reaches *þūsend* ‘thousand’.

Stage comparison The comparison below sets the relevant forms side by side. It separates the secure chronology from the more interpretive account of the second-syllable vowel.

Stage / interpretation	Candidate form	OE relation	Result
surviving *-ī* with ordinary double umlaut	**þūsundī-* treated as still umlaut-active in OE	would point toward **þȳsend*	excluded by preserved *ū*
early loss of the trigger without further reshaping	**þūsund-* type	explains *ū*, but not why OE alone has medial *e*	incomplete account
selected OE-oriented transponent	**θūs-èndi*	regular output: *þūsend* ‘thousand’	selected modeling input

timber — OE *timber*

Derivation: citation reconstruction **tímrą*; form followed here **tímbrą* > *timber* (early analogy).

Derivation trace Proto input: **tímbrą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**tímbr*
[no change]	OE Epenthetic Vowel	**tímber*
Early Anglo-Frisian		
[no change]		

Old English form: *timber*

Reconstruction and comparative evidence Kroonen reconstructs the noun as **timbra-* and cites Old English *timber* ‘timber’ among its continuations (Kroonen 2013, 517). Ringe and Taylor instead state the history from **timra* ‘timber’ through West Germanic **timbr* ‘timber’ to Old English *timber* ‘timber’ (Ringe and Taylor 2014, 327).

The difference is therefore not over the Old English noun itself. It concerns whether medial *b* belongs in the comparative citation form or appears in an early pre-Old-English stage of the cluster.

Old English evidence Clark Hall lemmatizes the noun as *timber* ‘timber’ and also records *timbor* ‘timber’ as a variant spelling (Clark Hall 1960, 294). The Old English form is thus an ordinary citation noun, not a selected oblique cell or a reconstructed convenience form.

Development to Old English With the consonantal frame **timbr-* ‘timber’ in place, the rest of the development is straightforward. Loss of final *-ą* leaves **tímbr* ‘timber’, and epenthetic *e* in the final cluster yields timber ‘timber’. Ringe and Taylor’s *timra* > *timbr* > OE *timber* and the handbook treatment of this epenthetic vowel point to the same Old English result (Ringe and Taylor 2014, 327; Campbell 1959, sects 463–464).

Formation comparison The comparison below sets the relevant forms side by side. It separates the comparative headword from the OE-facing consonantal input.

Formation or notation	Candidate form	OE relation	Result
Kroonen's comparative citation	**timbra-*	already matches the consonantal frame of OE *timber* 'timber'	closest comparative support for the derivational input
Ringe-Taylor citation line	**timra* > **timbr* 'timber'	reaches the same OE noun through early cluster expansion	compatible comparative background
modeled input	**tímbrą*	regular output: *timber* 'timber'	Old English-facing input

wake — OE *wacan*

Derivation: citation reconstruction **wakēną*; form followed here **wákaną* > *wacan* (early analogy).

Derivation trace Proto input: **wákaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**wækaną*
[no change]	OE A Restoration	**wakaną*
	OE Heavy Syllable Nasal Apocope	**wakan*
Early Anglo-Frisian	OE Secondary Nasalization	**wakąn*
[no change]	OE Weak Tail Reduction	**wakan*

Old English form: *wacan*

Reconstruction and comparative evidence Kroonen gives the strong verb as **wakan-* with Old English *wacan* 'wake' (Kroonen 2013, 568). Ringe and Taylor separately derive Old English *wacian* 'wake' from weak **wakai-* ~ **wakja-* (Ringe and Taylor 2014, sect. 3.3.2).

The difference is therefore lexical and class-based, not graphic. Strong *wacan* 'wake' 'wake up, arise' and weak *wacian* 'wake' 'be awake, watch' belong to related but distinct histories.

Old English evidence Clark Hall lists *wacan* 'wake' and *wacian* 'wake' as separate headwords (Clark Hall 1960, 338). Bosworth-Toller cautions under *wacan* 'wake': the simplex infinitive itself does not occur, its place seeming to be taken by *wæcnan* 'wake' (Bosworth and Toller 1898, 226).

The target *wacan* 'wake' is therefore best understood as a normalized strong headword for the verb family, not as a directly quoted simplex infinitive. It still remains the correct Old English comparison form for the strong branch.

Development to Old English With strong **wákaną* 'wake', Anglo-Frisian brightening first gives a form of the **wækaną* 'wake' type. A-restoration then returns *a*, and the ordinary tail reductions yield *wacan* 'wake'. The weak verb *wacian* 'wake' belongs to a different prehistory and is not the expected outcome of this input.

Class comparison The comparison below sets the relevant forms side by side. It separates the strong and weak verb lines.

Formation / class	Candidate input	OE outcome or comparison	Result
weak class-III / class-II branch	**wakēną*, **wakai-* ~ **wakja-*	OE *wacian* 'wake' and related weak forms	related lexeme, but not the target of this entry

Formation / class	Candidate input	OE outcome or comparison	Result
strong class-VI branch	**wákaną*	regular output: *wacan* ‘wake’	Old English-facing input
strong normalized headword	*wacan* ‘wake’	dictionary comparison form beside attested strong-family forms	correct Old English comparator, though not a directly quoted simplex infinitive

water — OE *wæter*

Derivation: citation reconstruction **wátną*; form followed here **wátōr* > *wæter* (early analogy).

Derivation trace Proto input: **wátōr*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening Unstressed	**wátær*
PWGmc Final Or Lowering	**wátar*	EAF Brightening	**wætær*
Early Anglo-Frisian		OE Unstressed AE Merger	**wæter*
[no change]			

Old English form: *wæter*

Reconstruction and comparative evidence Kroonen reconstructs a heteroclitic noun **watar-* ~ **watan-* and states that the Proto-Germanic material points to **watōr* ‘water’ and **watenaz* ‘water’ (Kroonen 2013, 616). Ringe and Taylor likewise start from singular **wator* ‘water’ before the Old English branch (Ringe and Taylor 2014, sect. 3.1.4).

The generalized comparative label is therefore broader than the singular form that actually corresponds to Old English *wæter* ‘water’. The relevant comparator is the inherited nominative-accusative singular **wátōr* ‘water’.

Old English evidence Bright gives the noun as *wæter* ‘water’ with the regular paradigm *wæteres* ‘water (gen.)’, *wætere* ‘water (dat.)’, *wæter(u)*, *wætera* ‘water (gen.pl.)’, *wæterum* ‘water (dat.pl.)’ (Bright 1971, 29). Ringe and Taylor add the dialectal contrast between West Saxon *weeter* ‘water’ and Mercian *weter* ‘water’ (Ringe and Taylor 2014, sect. 6.5.2).

The target is therefore an attested Old English citation form within a normal paradigm. The complication lies on the comparative side of the lexeme, not in Old English attestation.

Development to Old English From **wátōr* ‘water’, pre-final *ō* becomes *a*, yielding **watar* ‘water’ (Ringe and Taylor 2014, sect. 3.1.4). Anglo-Frisian brightening then gives **wætær* ‘water’, and merger of unstressed *æ*/*e* yields wæter ‘water’.

Stage comparison The comparison below sets the relevant forms side by side. It separates the generalized lexeme label from the singular input that matches the Old English citation form.

Stage or notation	Candidate form	OE relation	Result
generalized comparative label	**wátną*	broader lexeme shorthand, not the singular that corresponds directly to *wæter* ‘water’	useful background only

Stage or notation	Candidate form	OE relation	Result
heteroclitic stem notation	**watar- ~ *watan-*	source-faithful comparative reconstruction	explains why a singular comparator is needed
inherited singular input	**wátōr*	regular output: *wæter* ‘water’	Old English-facing input

whale — OE *hwæl*

Derivation: citation reconstruction **wálaz*; form followed here **xwálaz* > *hwæl* (early analogy).

Derivation trace Proto input: **xwálaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**xwál*
EAF Final Z Deletion	**xwála*	EAF Brightening	**xwæl*
Early Anglo-Frisian			
[no change]			

Old English form: *hwæl*

Reconstruction and comparative evidence The comparative sources are not uniform. Orel gives **xwalaz* ‘whale’ and notes some mixed **xwaliz* ‘whale’ evidence (Orel 2003, 197). Kroonen instead cites **ẖwali-* (Kroonen 2013, 262).

Both notations agree on inherited initial *hw-/xw-*, but they differ in stem label. The a-stem-like input followed here is closer to Orel’s notation than to Kroonen’s citation form.

Old English evidence Clark Hall lemmatizes the noun as *hwal* ‘whale’, and Bosworth-Toller preserves plural *hwalas* ‘whales’ (Clark Hall 1960, 170; Bosworth and Toller 1898, 326). The comparison form is normalized here as *hwæl* ‘whale’ for the singular citation form with Anglo-Frisian fronting.

The plural *hwalas* ‘whales’ supplies control evidence. It shows the same lexeme with *a* in an open syllable beside singular *hwæl* ‘whale’ in a closed monosyllable.

Development to Old English From **xwálaz* ‘whale’, final *-z* disappears and bare final *-a* is lost. Anglo-Frisian fronting then yields *æ* in the closed monosyllable, and Old English orthography writes *hwæl* ‘whale’.

Formation comparison The comparison below sets the relevant forms side by side. It separates the competing comparative notations from the normalized Old English singular.

Comparative line	Candidate form	OE relation	Result
Orel’s citation	**xwalaz*	same stem notation as the modeled singular line	closest comparative support for the derivational input
Kroonen’s citation	**ẖwali-*	same initial cluster, different stem label	important comparative rival, but not the notation followed here
modeled input	**xwálaz*	regular output: *hwæl* ‘whale’	Old English-facing input

Comparative line	Candidate form	OE relation	Result
plural control	*hwalas* ‘whales’	attested open-syllable plural beside singular *hwœl* ‘whale’	confirms that the lexeme also preserves an *a*-vocalism branch

whine — OE *hwīnan*

Derivation: citation reconstruction **wainōjaną*; form followed here **xwḯnaną* > *hwīnan* (early analogy).

Derivation trace Proto input: **xwḯnaną*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Heavy Syllable Nasal Apocope	**xwḯnan*
[no change]	OE Secondary Nasalization	**xwḯnąn*
Early Anglo-Frisian	OE Weak Tail Reduction	**xwḯnan*
[no change]		

Old English form: *hwīnan*

Reconstruction and comparative evidence The citation reconstruction preserved in the header belongs to the lament-family verb seen in German *weinen* ‘whine’ and Old English *wānian* ‘wane’. Kroonen instead separates Old English *hwīnan* ‘whine’ under **hwinan-* (Kroonen 2013, 267). Orel likewise distinguishes strong **xwinanan* ‘whine’ from weak **wainōjanan* ‘whine’ (Orel 2003, 201). Ringe and Taylor make the same split at the Northwest Germanic level, linking Old Norse *hvina* ‘whine’ and Old English *hwinan* ‘whine’ to the same strong verb (Ringe and Taylor 2014, 130).

The two families also differ phonologically and morphologically. The lament family has initial *w-*, diphthongal *ai*, and weak-II morphology, whereas the verb behind Old English *hwīnan* ‘whine’ has initial *hw-/xw-*, long *ī*, and strong-verb inflection. The derivational input **xwḯnaną* ‘whine’ therefore represents a competing comparative identification rather than a hidden cell of **wainōjaną* ‘whine’.

Old English evidence Clark Hall records *hwinan* ‘whine’ with the gloss ‘to hiss, whizz, whistle’ (Clark Hall 1960, 171). Seebold keeps the verb among the strong verbs and notes that only a present-tense attestation is directly preserved (Seebold 1970, 280).

The Old English form is normalized here as *hwīnan* ‘whine’. That normalization adds the usual vowel length marking to the dictionary spelling *hwinan* ‘whine’; it does not turn an unattested verb into a reconstructed one.

Development to Old English Once the strong-verb input **xwḯnaną* ‘whine’ is selected, the path to Old English is straightforward. The compact trace shows heavy-syllable nasal apocope, secondary nasalization, and weak-tail reduction, after which the form surfaces as *hwīnan* ‘whine’.

No special paradigm maneuver is needed for this verb. The comparison is between two different Germanic verb families: the Old English form belongs with the strong verb **hwīnan-*, not with the weak lament verb.

Verb-family comparison The comparison below sets the relevant forms side by side. It separates the competing comparative labels that stand behind the inherited Old English forms.

Verb family / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lament-family weak verb	**wainōjaną*	comparative continuation in OE *wānian* ‘wane’	*wānian* ‘wane’	competing citation reconstruction, but not the source of *hwīnan*
selected strong verb	**xwī́naną*	regular output: *hwīnan* ‘whine’	*hwīnan* ‘whine’	exact match between derivational input and OE verb
comparative North Germanic cognate	Northwest Germanic strong verb behind ON *hvina* ‘whine’ / OE *hwinan* ‘whine’	ON *hvina* ‘whine’ / OE *hwinan* ‘whine’	*hwīnan* ‘whine’	supports the strong-verb identification

withy — OE *wīþiġ*

Derivation: citation reconstruction **wáiθiz*; form followed here **wī́θagą* > *wīþiġ* (early analogy).

Derivation trace Proto input: **wī́θagą*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**wī́θægą*
[no change]	OE Heavy Syllable Nasal Apocope	**wī́θæg*
	OE Velar Palatalization	**wī́θæj*
Early Anglo-Frisian	OE Unstressed AE Merger	**wī́θej*
[no change]	OE Late Unstressed Ag Suffix	**wī́θij*
	OE Palatal Fricative Merger	**wī́θij*

Old English form: *wīþiġ*

Reconstruction and comparative evidence The comparative evidence groups the word with Germanic forms of the **wīþja* ‘withy’ (or **wīþjō-* variant) or **wiþ-* type (Orel 2003, 503). These forms establish the cognate set but do not by themselves explain the Old English suffix of *wīþiġ* ‘withy’.

For Old English, the relevant point is the suffix history. Campbell’s account of OE *-ig*, including forms such as *hunig*, supports an analysis in which the *-iġ* of *wīþiġ* ‘withy’ continues a derivational **-ag-* sequence rather than a heavy ja-stem **-ij-* (Campbell 1959, sects 275, 376). The derivational input **wī́θagą* ‘withy’ is thus a formation choice rather than a mere respelling of the comparative headword.

Old English evidence Clark Hall records the noun as *wiðig* ‘withy’ (Clark Hall 1960, 358). The form used here, *wīþiġ* ‘withy’, is a normalized Old English spelling with macrons and palatal ġ marked explicitly.

The relevant comparison form is therefore not a reconstructed dictionary convenience but an established Old English noun. What requires explanation is why the selected Proto-Germanic input is **wī́θagą* ‘withy’ rather than a comparative headword of the **wīþja* ‘withy’ type.

Development to Old English From **wíθagą* ‘withy’, the retained unstressed-fronting component gives the suffixal front vowel. Later raising recognizes the singleton palatal fricative, distinct from gg/ng stops; this late suffix history is not prehistoric i-mutation (Ringe and Taylor 2014, 334–35; Fulk 2018, 130–32). Postmutation merger and native realization supply the final *ġ*, reaching *wīþiġ* ‘withy’.

This derivation is regular for the form compared here. The central claim of the entry is therefore morphological: Old English *wīþiġ* ‘withy’ belongs with an **-ag-* derivative, whereas the comparative **wīþja* ‘withy’ label belongs to a different way of presenting the cognate family.

Formation comparison The comparison below sets the relevant formations side by side. It distinguishes the comparative headword from the Old English-facing formation that actually yields the attested noun.

Formation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
comparative family label	**wáiθiz*	broader cognate-set headword	OE family context	useful lexeme label, but not the direct source of *wīþiġ*
heavy ja-stem analysis	**wīþja* ‘withy’ type	Campbell/Adamczyk-style heavy ja-stem *-e* / zero outcome	*wīþiġ* ‘withy’	does not account cleanly for the OE suffix
**-ag-* derivative followed here	**wíθagą*	regular output: *wīþiġ* ‘withy’	*wīþiġ* ‘withy’	exact match between formation and target

youth — OE *ġeoguþ*

Derivation: citation reconstruction **júgunθiz*; form followed here **júgunθ* > *ġeoguþ* (early analogy).

Derivation trace Proto input: **júgunθ*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**júguθ*
EAF Nasal Spirant Lengthening	**júgūnθ*		
EAF Nasal Spirant Loss	**júgūθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *ġeoguþ*

Reconstruction and comparative evidence The wider etymological tradition reconstructs an earlier form of the word as **ju(w)unþi-*. The comparative label **júgunθiz* ‘youth’ already stands at a later Germanic stage with *g*, and the derivational input **júgunθ* ‘youth’ is later again: it represents the form after final *-i* has been lost.

Ringe and Taylor explicitly give the sequence **jugunþi* ‘youth’ > **juguþ* ‘youth’ > OE geoguþ ~ iuguþ (Ringe and Taylor 2014, 141). The derivational input therefore differs from the broader comparative headword because the Old English development must begin after early loss of final *-i*.

Old English evidence The Old English noun is attested with varying spellings. Ringe and Taylor cite *geoguþ* ‘youth’ ~ *iuguþ* ‘youth’ (Ringe and Taylor 2014, 141). The form is normalized here as *ġeoguþ* ‘youth’: the initial palatal is written with *ġ*, and the attested spelling variation is treated as orthographic rather than lexical.

Nothing in the source stack suggests that a different paradigm cell should be chosen. The relevant Old English comparison form is the noun *ġeoguþ* ‘youth’ itself.

Development to Old English The decisive early step is the loss of final *-i* before the Old English umlaut stage. If that high vowel remained, the word would develop an over-umlauted *y*-type vowel instead of the attested form (Ringe and Taylor 2014, 141).

From the derivational input **júgunθ* ‘youth’, the later development is regular: nasal-spirant lengthening and loss give **júgūθ* ‘youth’ (Fulk 2018, 109); unstressed long-vowel shortening then produces **júguθ* ‘youth’. The root vowel remains phonological *u* through medial-vowel lowering; late West Saxon spelling, not early palatal fronting, yields *ġeoguþ* ‘youth’ (Ringe and Taylor 2014, 5; Brunner 1965, 64–65, §92.1). Campbell preserves *u* after accented *u* in forms such as *duguþ* ‘troop’ and *munuc* ‘monk’ (Campbell 1959, 156, §373). Brunner likewise cites *iuzuð* ‘youth’ and *munuc* ‘monk’ under the same development (Brunner 1965, sect. 150.3).

Stage comparison The comparison below sets the relevant forms side by side. It separates the broader comparative headword from the later stages relevant to the Old English noun.

Stage / interpretation	Candidate form	Old English outcome or comparison	Relevance to this entry
earlier etymological headword	**ju(w)unþi-*	comparative family background	older comparative reconstruction of the lexeme
later g-bearing comparative label	**júgunθiz*	citation reconstruction / lexeme label	preserves the later Germanic stage behind the selected entry
Old English-facing input	**júgunθ*	regular output: *ġeoguþ*	exact match for the Old English form used here
full *-i* stage retained too long	**jugunþi*	expected over-umlauted *y*-type result	negative control showing why early *-i* loss must precede the OE umlaut stage

Part IV. Late analogy and paradigm-cell selection

These Old English forms continue a particular paradigm cell or a later analogical remodeling, not the unaltered citation form. The phonology may be regular once the proper morphological history has been identified.

ban — OE *bannes*

Derivation: citation reconstruction **bánną*; form followed here **bánnas* > *bannes* (late analogy).

Derivation trace Proto input: **bánnas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**bánnæs*
[no change]	OE Unstressed AE Merger	**bánnes*
Early Anglo-Frisian		
[no change]		

Old English form: *bannes*

Reconstruction and comparative evidence Orel cites a bann-noun under **bannan* ‘ban’, while Seebold distinguishes bann-stems of both masculine and neuter type and gives Old English *gebann* ‘ban’ as the noun reflex (Orel 2003, 35; Seebold 1970, 89). The citation reconstruction **bánną* ‘ban’ names the lexeme, but the comparison here turns on the genitive singular **bánnas* ‘ban’.

The analysis therefore depends on medial, not final, gemination.

Old English evidence Old English lexicographic evidence securely supports the noun itself. Bosworth-Toller records the noun under nominative *ge-bann* ‘ban’, with oblique usage such as *gebanne* ‘ban’ (Bosworth and Toller 1898, 303). The exact unprefixed genitive *bannes* ‘ban’ is less directly cited in the dictionaries, so it is best treated here as the regular genitive form used for comparison rather than as a dictionary headword.

Development to Old English From **bánnas* ‘ban’, the geminate remains medial before the case ending and the unstressed vowel develops regularly to give *bannes* ‘ban’. The paradigm comparison therefore sets the genitive against nominative *ban* ‘ban’, the ordinary nominative form of the same noun, rather than against a directly cited genitive headword.

Paradigm comparison The comparison below sets the relevant forms side by side. It shows why the genitive singular is the conservative cell used for the entry.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bánną*	regular output: *ban*	ban	regular nominative outcome, but not the Old English form here
genitive singular	**bánnas*	regular output: *bannes*	*bannes*	direct match for the conservative genitive

berry — OE *berġes*

Derivation: citation reconstruction **bázją*; form followed here **bázjas* > *berġes* (late analogy).

Derivation trace Proto input: **bázjas*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening Unstressed	**bárjæs*
EAF Rhotacism	**bárjas*	EAF Brightening	**bærjæs*
Early Anglo-Frisian		OE I Umlaut	**berjæs*
		OE Unstressed AE Merger	**berjes*
[no change]			

Old English form: *berġes*

Reconstruction and comparative evidence Kroonen reconstructs the berry noun as **basja-* ~ **bazja-* (Kroonen 2013, 54). The derivational input **bázjas* ‘berry’ is therefore not a rival lexeme headword, but a specific genitive singular cell drawn from that paradigm.

The relevant point is that **rj* did not geminate in Proto-West Germanic. Ringe and Taylor’s *here* ‘army’, *herges* ‘army’s’ comparison shows the same *rj* environment in an Old English paradigm without any hidden gemination repair (Ringe and Taylor 2014, 181).

Old English evidence Campbell cites feminine *berige* ‘berry’ and notes that *-j-* is retained after *r* in this type (Campbell 1959, 250). The reviewed evidence therefore supports the citation form more directly than the exact genitive *berġes* ‘berry’, which is best read here as the regular genitive comparison form rather than as a dictionary headword.

Development to Old English Citation **bázją* ‘berry’ gives *bere* ‘berry (variant)’, not the Old English form here. The genitive singular **bázjas* ‘berry’, however, gives *berġes* ‘berry’, with medial *-rġ-* preserved in the same way that Ringe and Taylor cite *herges* ‘army’s’ beside *here* ‘army’ (Ringe and Taylor 2014, 181). This points to paradigm choice rather than to an extra phonological rule.

Paradigm comparison The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**bázją*	regular output: *bere*	*berige* / *berġe*	useful citation-form background, but not the Old English form here
genitive singular	**bázjas*	regular output: *berġes*	*berġes*	exact match for the conservative cell

bore (3sg) — OE *boraþ*

Derivation: citation reconstruction **burōną*; form followed here **búrōθi* > *boraþ* (late analogy).

Derivation trace Proto input: **búrōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Late O Shortening	**bóroθ*
PWGmc Early I Apocope	**búrōθ*	OE Final Unstressed O Lowering	**bóraθ*
PNWGmc U Lowering	**bórōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *boraþ*

Reconstruction and comparative evidence Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* ‘bore’ among its reflexes (Kroonen 2013, 85). The form compared here isolates the finite 3sg cell **búrōθi* ‘bores’ rather than the infinitive.

Campbell’s class-II pattern lufas, *-aþ* (< *-ōsi*, *-ōþi)* and Ringe and Taylor’s account of stable class-II *a* in finite forms make this 3sg cell the relevant comparison form for the ending (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Old English evidence Clark Hall lemmatizes the verb as *borian* ‘bore’; the imperative singular *bora* and present-third-singular *boraþ* ‘bores’ are the relevant comparison forms (Clark Hall 1960, 48).

Campbell’s dialect survey allows secondary *-e-* and *-o-* spellings in 2sg and 3sg class-II forms, but the basic ending remains *-aþ* (Campbell 1959, sect. 757). The regular comparison form for this non-*j* 3sg cell is *boraþ* ‘bores’.

Development to Old English From **búrōθi* ‘bores’, loss of final *-i* first gives **búrōθ* ‘bores’. Northwest Germanic lowering then produces **bórōθ* ‘bores’, and late shortening of unstressed *ō* yields **bóraθ* ‘bores’. *Boraþ* ‘bores’ is therefore the regular comparison form for this non-*j* 3sg cell.

Campbell’s class-II ending evidence and Ringe and Taylor’s discussion of stable *a* in finite non-*j* cells support this sequence (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the bore-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
3sg present	**búrōθi*	regular output *boraþ* ‘bores’	*boraþ* ‘bores’	exact match
imperative singular companion	**búrô*	related finite form *bora* ‘bore’	*bora* ‘bore’	useful control, but not the target

bore (iptv.2sg) — OE *bora*

Derivation: citation reconstruction **burōną*; form followed here **búrô* > *bora* (late analogy).

Derivation trace Proto input: **búrô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**bóra*
PNWGmc U Lowering **bórô*		
Early Anglo-Frisian		
[no change]		

Old English form: *bora*

Reconstruction and comparative evidence Kroonen reconstructs the bore-family verb as **burojan-* and cites Old English *borian* ‘bore’ among its continuants (Kroonen 2013, 85). Ringe and Taylor give the class-II weak imperative singular ending as -a < **-ō* (Ringe and Taylor 2014, 314).

The derivational input **búrô* ‘bore (iptv.2sg)’ is therefore an imperative cell of the same family, not the citation form of the verb.

Old English evidence Clark Hall lemmatizes the verb as *borian* ‘bore’ (Clark Hall 1960, 48). The comparison form here is the normalized imperative singular *bora* ‘bore (iptv.2sg)’, used beside the headword and the related 3sg form *boraþ* ‘bores’.

The imperative is thus a paradigm form rather than a replacement for the dictionary lemma. It is the most direct Old English comparator for the non-*j* finite cell represented by **búrô* ‘bore (iptv.2sg)’.

Development to Old English Northwest Germanic lowering first gives **bórô* ‘bore (iptv.2sg)’ from **búrô* ‘bore (iptv.2sg)’, and late shortening of the unstressed long vowel then yields **bóra* ‘bore (iptv.2sg)’, whence *bora* ‘bore’.

Ringe and Taylor’s class-II imperative singular -a < **-ō* points to exactly this type of outcome (Ringe and Taylor 2014, 314). The form compared here therefore isolates the regular finite-cell development more cleanly than the remodelled infinitive does.

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the selected imperative cell from the bore-family lemma and from the companion 3sg form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**burōjaną*	dictionary headword *borian*	*borian*	family background, but not the cell compared here
imperative singular	**búrô*	regular output *bora* ‘bore’	*bora* ‘bore’	exact match
3sg present companion	**búrōθi*	related finite form *boraþ* ‘bores’	*boraþ* ‘bores’	useful control, but not the target

bow — OE *bēag*

Derivation: citation reconstruction **béuganą*; form followed here **báug* > *bēag* (late analogy).

Derivation trace Proto input: **báug*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**báeug*
[no change]	OE Diphthong Leveling	**bēag*
Early Anglo-Frisian		
[no change]		

Old English form: *bēag*

Reconstruction and comparative evidence The inherited verb belongs to the class-II strong-verb family **béuganą* ‘bow’ (Ringe and Taylor 2014, 55). Within that paradigm, however, the infinitive and the singular preterite continue different ablaut grades. The derivational input **báug* ‘bow’ is the singular preterite cell, whereas the citation form **béuganą* is the infinitive.

Campbell’s account of Old English class-II strong verbs treats the singular preterite *au* > *ēa* development as regular in this environment (Campbell 1959, 53). That is the phonological path relevant for *bēag* ‘bowed’, whereas the analogical *ū* of the present stem belongs to the separate history behind *būgan* ‘to bow’ (Ringe and Taylor 2014, 55).

Old English evidence Bosworth-Toller and Clark Hall both record *bēag* ‘bowed’ as a preterite form of *būgan* ‘to bow’ (Bosworth and Toller 1898, 122; Clark Hall 1960, 45). The form discussed here is therefore an attested Old English verbal form, not a reconstructed substitute for the infinitive.

The ordinary dictionary headword remains *būgan* ‘to bow’, but the relevant comparison form for this entry is the singular preterite *bēag* ‘bow, ring’ ‘bowed’. That is the paradigm cell in which the inherited **au* grade is preserved most directly.

Development to Old English From **báug* ‘bow’, Anglo-Frisian fronting and the later leveling of the diphthong produce *bēag* ‘bow, ring’ (Campbell 1959, 53). No special analogical repair is needed for that cell. The form is the regular Old English outcome of the singular-preterite grade.

The analogical element in the wider lexeme belongs instead to the present stem seen in *būgan* ‘bow, bend’. The derivational input differs from the citation form because the regular inherited pathway survives more transparently in the preterite than in the infinitive.

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the regular singular preterite from the more familiar infinitival citation form.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**béuganą*	inherited present-stem history behind *būgan*	*būgan*	establishes the lexeme, but not the Old English form here
singular preterite	**báug*	regular output: *bēag*	*bēag*	exact match between input, output, and attested cell
past participial branch	participial **bugan-* type	later participial outcomes	bogen-type evidence	relevant to the paradigm, but not the clearest match for this entry

The singular preterite is the relevant comparison form. It gives a direct lautgesetzlich path to attested *bēag* ‘bow, ring’, while the citation form *būgan* ‘bow, bend’ belongs to a paradigm whose present stem has already undergone later leveling.

cow — OE *cȳ*

Derivation: citation reconstruction **kōz*; form followed here **kūi* > *cȳ* (late analogy).

Derivation trace Proto input: **kūi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE I Umlaut	**kȳi*
[no change]	OE High Vowel Apocope	**kȳ*
Early Anglo-Frisian		
[no change]		

Old English form: *cȳ*

Reconstruction and comparative evidence Kroonen reconstructs a root noun with the inherited alternation **kō- ~ *ku-*, explicitly nom. **kōz* ‘cow’, obl. **kū-* (Kroonen 2013, 299). The citation form therefore belongs to the nominative singular, whereas the derivational input **kūi* ‘cow’ belongs to the oblique stem.

Ringe and Taylor also posit a later PNWGmc nominative **kūaz* ‘cow’ > **kūz* ‘cow’ behind Old English *cū* ‘cow’ (Ringe and Taylor 2014, sect. 3.1.3). That nominative history accounts for the headword, whereas the form cȳ ‘cow’ depends on the oblique **kū-* stem and a following **i*.

Old English evidence Clark Hall lemmatizes the noun under *cū* ‘cow’ and records gen.sg. *cū* ‘cow’ (or *cū(e)*), *cȳ* ‘cow’, *cūs* ‘cow (gen.)’, dat.sg. *cȳ* ‘cow’, nom.-acc.pl. *cȳ* ‘cow’, and dat.pl. *cūm* ‘cow’ (Clark Hall 1960, 67). Ringe and Taylor likewise state that the root-noun *cū* ‘cow’ exhibits dat.sg., nom.-acc.pl. *cȳ* ‘cow’ < **cūi* ‘cow’, dat.pl. *cūm* < **cūm*, and apparently gen.sg. *cā* ‘cow (gen.)’ < **cūiz* ‘cow (gen.)’ (Ringe and Taylor 2014, sect. 6.6.1).

The dictionary headword is therefore *cū* ‘cow’, but *cȳ* ‘cow’ is an established Old English paradigm form rather than a convenient reconstruction. It serves as dative singular and also as nominative-accusative plural. The genitive singular is less stable, with *cā* ‘cow (gen.sg.)’, *cȳ* ‘cow’, *cū(e)*, and *cūs* ‘cow (gen.sg.)’ all appearing in the local source record.

Development to Old English **kūi* ‘cow’ is the dative singular of the oblique **kū-* stem. The following **i* triggers i-umlaut, so *ū* becomes *ȳ*, and loss of the final high vowel leaves *cȳ* ‘cow’. The development is **kūi* ‘cow’ > **kȳi* ‘cow’ > **kȳ* ‘cow’ > cȳ.

This is the regular oblique-cell path recognized by Ringe and Taylor’s paradigm statement (Ringe and Taylor 2014, sect. 6.6.1). It also explains why *cȳ* ‘cow’ is the cleanest comparison form for the present entry, even though the ordinary headword is *cū* ‘cow’.

Paradigm comparison A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	*kōz	OE headword *cū* ‘cow’ belongs to the nominative history of the lexeme	*cū*	useful background, but not the chosen comparison for *cȳ* ‘cow’
later generalized nominative	PNWGmc *kūaz* > kūz	inferred nominative *cū* ‘cow’	*cū*	explains the leveled headword, not the oblique target
dative singular oblique	*kūi	regular output: *cȳ* ‘cow’	*cȳ*	exact match between input, output, and paradigm cell
genitive singular oblique	*kūiz	Ringe-Taylor: apparently *cā* ‘cow (gen.sg.)’; Hall: *cū(e)*, *cȳ*, *cūs*	gen.sg. variable	too unstable to control the entry

The dative singular is the relevant comparison form. It gives a regular path to attested *cȳ* ‘cow’, while the broader Old English paradigm shows how far the oblique **kū-* grade spread beyond that one cell.

find — OE *fundene*

Derivation: citation reconstruction **fínθaną*; form followed here **fúnðanǭ* > *fundene* (late analogy).

Derivation trace Proto input: **fúnðanǭ*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Fronting Early	**fúndœnǭ*
PWGmc Dental Hardening **fúndanǭ*	OE Unstressed Long Vowel Shortening	**fúndœnæ*
Early Anglo-Frisian	OE Unstressed AE Merger	**fúndene*
[no change]		

Old English form: *fundene*

Reconstruction and comparative evidence The inherited verb is the strong verb **fínθaną* ‘find’, continued by Old English *findan* ‘find’ (Ringe and Taylor 2014, 344). The form followed here, **fúnðanǭ* ‘find’, belongs to the past-participial paradigm rather than to the infinitive. It represents an oblique singular form of the participle.

The familiar dictionary form *funden* ‘found’ is not the form compared here. The derivational input instead models an attested participial form directly, rather than treating the infinitive or the ordinary dictionary headword as primary. It therefore reaches *fundene* ‘found (p.p.)’ in the form where the trace and the attested evidence match directly.

Old English evidence Bosworth-Toller records *fundene* ‘found (p.p.)’ ‘found’ under *findan* ‘find’, citing the form in *Beón þá herigeata swa fundene* (Bosworth and Toller 1898, 219). Clark Hall likewise preserves the participial stem in forms such as *funden* ‘found’ and *tō-fundennes* ‘finding’ (Clark Hall 1960, 124).

The ordinary dictionary headword for the participle is *funden* ‘found’, but the relevant comparison form for this entry is the attested oblique *fundene* ‘found (p.p.)’ ‘found’. It is an Old English form in its own right, not a merely convenient probe.

Development to Old English From **fúnðanǭ* ‘find’, the participial oblique develops through regular loss and weakening of the final ending, yielding *fundene* ‘found (p.p.)’. In that cell both the consonantism and the medial vowel history remain regular.

The broader participial paradigm fixes the interpretation. The more familiar nominative *funden* ‘found (p.p.)’ is the ordinary dictionary form, whereas the oblique form *fundene* ‘found (p.p.)’ is the attested form compared here.

Paradigm comparison

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	**fínθaną*	inherited verb *findan*	*findan*	establishes the lexeme, but not the form compared here
nominative participial line	**fúnðanaz*	ordinary dictionary *funden* type	*funden*	important paradigm background, but not the form compared here
oblique participle compared here	**fúnðanǭ*	regular output: *fundene*	*fundene*	exact match between input, output, and attested cell

The oblique participle is the relevant comparison form. It matches the derivational input and Old English form directly, while the nominative participial headword remains a different presentation cell within the same paradigm.

fright — OE *fyrhte*

Derivation: citation reconstruction **furxtīn*; form followed here **fúrxtīnaz* > *fyrhte* (late analogy).

Derivation trace Proto input: **fúrxtīnaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fúrxtīn*
EAF Final Z Deletion	**fúrxtīna*	OE I Umlaut	**fyrxtīn*
Early Anglo-Frisian		NWGmc In Stem N Loss	**fyrxtī*
		OE Unstressed Long Vowel Shortening	**fyrxti*
[no change]		OE Med Unstressed I Lowering1	**fyrxte*

Old English form: *fyrhte*

Reconstruction and comparative evidence The noun belongs to the inherited in-stem abstract **furxtīn* ‘fright’, the same family as Gothic *faurhtei* ‘fear, fright’ (Orel 2003, 120). The derivational input **fúrxtīnaz* ‘fright’ is not a different lexeme but an oblique singular cell within that in-stem paradigm.

Ringe and Taylor treat the later nominative forms with *-u* or *-o* as analogically remodeled (Ringe and Taylor 2014, 395–96). I therefore use the oblique in-stem form, which continues the older formation rather than the remodeled nominative.

Old English evidence Bosworth-Toller records *fyrhte* ‘fright’ with textual attestation, and it also records nominative forms such as *fyrhtu* ‘fright’ and *fyrhto* ‘fright’ (Bosworth and Toller 1898, 160). Clark Hall lists adjective and verb material separately under *fyrht* / *fyrhtan* (Clark Hall 1960, 141).

The relevant comparison form is therefore the attested oblique *fyrhte* ‘fright’. The nominative lemma forms remain part of the Old English evidence, but the Old English form here of this entry is the oblique cell.

Development to Old English From **fúrxtīnaz* ‘fright’, the oblique in-stem develops through the loss and weakening of the final ending, yielding *fyrhte* ‘fright’. The form compared here therefore follows the ordinary Old English reduction of the abstract ending in this paradigm.

The later nominative forms with *-u* or *-o* belong to a subsequent analogical reshaping of the paradigm. The Old English form here is earlier in that sense: it is the attested OE cell in which the inherited in-stem development remains most transparent.

Paradigm comparison The comparison below sets the relevant forms side by side. It separates the attested oblique form from the later remodeled nominative line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation in-stem headword	**furxtīn*	broader noun-class label	wider family context	useful lexeme label, but not the cell compared here
remodeled nominative line	nominative in-stem forms	*fyrhtu* ‘fright’ / *fyrhto* ‘fright’ type lemma forms	*fyrhtu* ‘fright’ / *fyrhto* ‘fright’	genuine OE evidence, but later remodeled
selected oblique singular	**fúrxtīnaz*	regular output: *fyrhte* ‘fright’	*fyrhte* ‘fright’	exact match between input, output, and attested cell

The oblique in-stem form is the relevant comparison form. It yields attested *fyrhte* ‘fright’ directly, while the more familiar nominative forms belong to a later analogical layer.

hammer — OE *hameres*

Derivation: citation reconstruction **xámaraz*; form followed here **xámaras* > *hameres* (late analogy).

Derivation trace Proto input: **xámaras*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**xámœræs*
[no change]	OE Unstressed AE Merger	**xámeres*
Early Anglo-Frisian		
[no change]		

Old English form: *hameres*

Reconstruction and comparative evidence The inherited noun is the masculine a-stem **xámaraz* ‘hammer’, reflected in Old English citation forms such as *hamor* ‘hammer’ and *hamer* ‘hammer’ (Kroonen 2013, 206; Orel 2003, 197; Clark Hall 1960, 160). The derivational input **xámaras* ‘hammer’ is the genitive singular of that same noun rather than a different lexeme.

The citation tradition is already mixed in its unstressed vowel, while the genitive singular gives a closer comparison form. This is a cell choice within one paradigm, not a change of stem class.

Old English evidence Bosworth-Toller directly records *hameres* ‘hammer’ in an Old English genitival phrase (Bosworth and Toller 1898, 78). Clark Hall preserves the simplex headword as *hamer* ‘hammer’ / *hamor* ‘hammer’ (Clark Hall 1960, 160).

Sievers-Brunner gives a paradigm line *hamor* ‘hammer’ — *hamores* ‘hammer’, which shows that the oblique tradition itself was not entirely uniform (Brunner 1965, sect. 245). The relevant comparison form here is the attested genitive singular *hameres* ‘hammer’.

Development to Old English From **xámaras* ‘hammer’, Anglo-Frisian brightening and the subsequent merger of unstressed *æ* with *e* yield *hameres* ‘hammer’. The derivation of that oblique form is straightforward once the genitive singular cell is selected.

The noun as a whole retains a mixed citation tradition in *hamor* ‘hammer’ and *hamer* ‘hammer’, and the selected oblique cell avoids making that variation carry the argument of the entry.

Paradigm comparison The comparison below sets the relevant forms side by side. It separates the attested genitive singular from the less stable citation tradition.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**xámaraz*	regular citation form *hamer* / *hamor*	*hamor* / *hamer*	good lexical background, but not the Old English form here
genitive singular	**xámaras*	regular output: *hameres*	*hameres*	exact match between input, output, and attested cell
later oblique tradition	oblique a-stem forms	*hamores* type evidence	*hamores*	attested background variant, but not the chosen comparison form

have — OE *hæfeþ*

Derivation: citation reconstruction **xabēną*; form followed here **xábēθi* > *hæfeþ* (late analogy).

Derivation trace Proto input: **xábēθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**xæbēθ*
PWGmc Early I Apocope	**xábēθ*	OE Velar Fricative Palatalization	**çæbēθ*
Early Anglo-Frisian		PGmc B Allophony	**çæβēθ*
		OE Unstressed Long Vowel Shortening	**çæβeθ*
[no change]			

Old English form: *hæfeþ*

Reconstruction and comparative evidence The verb belongs to the inherited class-III weak paradigm usually cited under **xabēną* ‘have’ and Old English *habban* ‘have’ (Kroonen 2013, 237; Ringe and Taylor 2014, 93). Within that paradigm, however, the infinitive and the singular present indicative do not continue the same stem. Ringe and Taylor distinguish a *-ja-* stem in the infinitive from a non-geminating -ai- / *-ē-* stem in the 2sg and 3sg present forms (Ringe and Taylor 2014, 93).

The derivational input **xábēθi* ‘have’ is therefore the 3sg present cell rather than a rephrasing of the infinitive. For the present analysis, that finite cell is the cleaner comparator for the inherited non-geminating stem.

Old English evidence The ordinary Old English headword is *habban* ‘have’ (Clark Hall 1960, 157). Campbell’s Anglian paradigm includes unsyncopated 3sg forms of the *hæfed* ‘has’ type, and the present paradigm therefore shows forms of the *hæf-* type that support the normalized target *hæfeþ* ‘has’ (Campbell 1959, sect. 762).

The target form is therefore a normalized finite cell rather than the ordinary dictionary lemma. It represents the inherited non-geminating present stem more directly than *habban* ‘have’ does.

Development to Old English From **xábēθi* ‘have’, the finite form yields *hæfeþ* ‘has’ regularly. Ringe and Taylor discuss this non-geminating present stem under *habban* ‘have’ (Ringe and Taylor 2014, 364). Campbell’s Anglian paradigms include unsyncopated 3sg forms of the *hæfeþ* ‘has’ / *hæfed* ‘has’ type (Campbell 1959, sect. 762).

The wider lexeme is less straightforward only because the infinitive *habban* ‘have’ shows later leveling. The 3sg present cell is therefore the closer comparison form.

Paradigm comparison The comparison below sets the relevant forms side by side. It separates the analogically leveled citation form from the regular 3sg present line.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive	*-ja-* stem of **xabēną*	citation form *habban*	*habban*	important headword, but shaped by later leveling
3sg present	**xábēθi*	regular output: *hæfeþ*	*hæfeþ*	exact match between input, output, and finite form compared here
syncopated finite tradition	same present stem	*hæfþ* type evidence	*hæfþ*	genuine later OE finite form, but not the normalized target used here

knee — OE *cneowe*

Derivation: citation reconstruction **knéwą*; form followed here **knéwai* > *cneowe* (late analogy).

Derivation trace Proto input: **knéwai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**knéowē*
PNWGmc Unstressed Ai Monophthongization	**knéwē*	OE Unstressed Long Vowel Shortening	**knéowe*
Early Anglo-Frisian			
[no change]			

Old English form: *cneowe*

Reconstruction and selection of the dative The citation reconstruction is the neuter noun **knéwą* ‘knee’. Ringe and Taylor expressly give the singular **knewą* ‘knee’ and distinguish its endingless development from the inflected stem (Ringe and Taylor 2014, 387, §7.2.4). The acute in the selected notation marks root stress; it does not make the vowel long.

The derivational input **knéwai* ‘knee (dat.sg.)’ belongs to another cell of the same inherited paradigm. It combines the stem with the conventional dative ending. Fulk regards Proto-Germanic *-ai* as the most likely source of the West Germanic dative singular, while considering the competing *-ē* analysis and the uncertainties of the wider Germanic comparison (Fulk 2018, 147, §7.8). The whole-word input is therefore an explicit paradigm construction under that defended working account, not a verbatim reconstruction quoted from Hogg or Campbell. It is neither a later Old English transponent nor a nominative from which the dative has been generated by an invented sound law.

The distinction between citation reconstruction and derivational input is essential. Selecting the dative does not deny the existence of the familiar nominative *cnēow* ‘knee’, nor replace the inherited neuter reconstruction with a different lexeme. It selects the cell in which the regular pre-ending development can be compared without importing the nominative’s analogically restored consonant.

Old English quantity and paradigm evidence The citation and dative are different cells, not rival reconstructions of one cell. The source comparison accordingly adds an uncertainty about the inherited ending, not another dispute over the already documented short dative. Fulk’s preferred *-ai* account licenses the selected paradigm construction without establishing a uniquely recoverable complete ancestor (Fulk 2018, 147). The detailed quantity and levelling evidence below remains independently necessary.

Hogg explicitly uses cneowe ‘knee (dat.sg.)’ as an example of short *eo*. His adjacent long-diphthong example, *cnēowe* ‘know (past subjunctive)’, is a verb form, not an alternative knee dative. In the discussion of breaking he again names the knee dative as a regular example before *w*, while assigning the final *w* of nominative *cnēo(w)* to analogy [Hogg (2011), pp. 21–22, §2.33; p. 86, §5.22 and note 7]. The quantity contrast is an explicit grammatical analysis, not an inference from an unmarked manuscript digraph.

Campbell’s regular paradigm similarly has genitive *cneowes* ‘knee (gen.sg.)’ and dative *cneowe* beside long endingless *cnēo(w)*. He then allows the long diphthong to extend into the inflected forms, giving the secondary *cnēowes* type (Campbell 1959, 232–33, §584). Luick independently gives *cneowe(s)* as the genitive and dative products of short-vowel diphthongization before *w*, with the following-*i* exclusion (Luick 1914--1940, 139, §134). These accounts support the selected short dative. They do not establish that every occurrence of an inflected knee form necessarily retained the original quantity.

Ringe and Taylor give the same basic opposition: long endingless *cnēo* and short *cneow-* before syllabic endings. They cautiously consider subsequent extension of long quantity into knee’s

inflected stem, noting the difficulty of finding decisive verse evidence [Ringe and Taylor (2014), pp. 187–188, §6.2.4; p. 387, §7.2.4]. Fulk describes the wider two-way leveling pattern of this stem class: the endingless diphthong spreads into inflected forms, and their *w* spreads into uninflected forms. His explicit illustrative example is the servant paradigm, not an independent metrical attestation for knee (Fulk 2018, 154, §7.12).

The regular selected development Before a retained syllabic ending, the root vowel remains short. The singleton *w* conditions short *e* to *eo*, unless followed by a high front vocalic. Hogg's knee dative and Luick's corresponding examples belong to this regular pre-ending domain (Hogg 2011, 86; Luick 1914--1940, 139; Ringe and Taylor 2014, 187–88). The weakening of the selected dative ending leaves Old English *-e*; the medial *w* has not become a word-final offglide and does not need to be restored.

The modeled short diphthong is supplied by SC044 OEBreaking, not by assigning long quantity at SC033 OEEwLongDiphthong. The latter now excludes this singleton path and retains only the separately disclosed j-created representation discussed with *hīew* 'hue'. The inherited-ending and short-diphthong developments are independently motivated; arriving at the right final spelling is not itself their historical justification.

Why the nominative is a comparison, not the selected target The endingless development follows a different route. Ringe and Taylor derive long *cnēo* from the inherited singular through an offglide-bearing intermediate, and explain the *w* in *cnēow* as transfer from the inflected forms (Ringe and Taylor 2014, 174, 387). Campbell likewise distinguishes endingless diphthong formation from the later addition of *w* from the obliques (Campbell 1959, 46–47, 232–33). Luick's older account distinguishes the endingless knee/tree *eo* and *eu* sources and their subsequent leveling (Luick 1914--1940, 118, §101). Agreement on the short/long paradigm opposition does not make all these detailed early reconstructions identical.

The nominative is thus not an appropriate obligation for a purely phonological derivation that retains *w* and lengthens the root before its ending disappears. Such a derivation reaches a real Old English form by conflating two histories. The selected dative avoids that conflation without a knee-specific lengthening rule, an analogical sound change, or grammatical conditioning of a phonetic law. It also avoids choosing a long oblique which itself reflects the quantity leveling under investigation.

Paradigm comparison and confidence

Cell or interpretation	Reconstruction / form	Historical role
PGmc citation singular	*knéwą	Preserved lexeme-level reconstruction
Selected PGmc dative singular	*knéwai	Explicit inherited-cell construction under the preferred *-ai analysis
OE selected dative singular	cneowe	Regular short pre-ending diphthong
OE regular genitive singular	cneowes	Independent short-stem comparison
OE endingless singular	cnēo	Regular long offglide-derived form
OE familiar nominative	cnēow	Endingless form with restored w
Later long obliques	cnēowe, cnēowes	Possible quantity leveling; not the selected regular comparison

Confidence is strong in the regular paradigm distinction: Campbell, Luick, Hogg and Ringe–Taylor independently identify the short pre-ending stem. Confidence is more qualified in the ex-

act inherited dative-ending reconstruction and in how widely later long quantity spread through knee's attested obliques. Neither uncertainty authorizes an unrestricted short-*ew* to long-*ēow* sound law.

The authors' exact inputs are not silently harmonized. Hogg's breaking footnote prints **knewaz* 'knee', whereas Ringe and Taylor give the neuter *knewą* (Hogg 2011, 86, note 7; Ringe and Taylor 2014, 387). The latter supplies the retained citation reconstruction. This discrepancy in the quoted ending is separate from their agreement on short pre-ending quantity; it is not evidence for replacing the selected dative with a nominative or for claiming identical whole-word inputs.

This entry belongs to late analogy and paradigm-cell selection because the conventional citation paradigm has undergone later reshaping. That placement does not mean that the selected dative itself requires an analogical operation. It follows the same principle as the regular selected obliques used for cow, night and meed: distinguish the history of the lexeme's paradigm from the sound-law path of the chosen cell.

learn (3sg) — OE *leornaþ*

Derivation: citation reconstruction **liznōjaną*; form followed here **líznōθi* > *leornaþ* (late analogy).

Derivation trace Proto input: **líznōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**léornōθ*
PWGmc Early I Apocope	**líznōθ*	OE Late O Shortening	**léornoθ*
PNWGmc I Lowering	**léznōθ*	OE Final Unstressed O Lowering	**léornaθ*
EAF Rhotacism	**lérnōθ*		
Early Anglo-Frisian			
[no change]			

Old English form: *leornaþ*

A textually attested third singular The selected Old English leornaþ 'learns' corresponds to Miller's printed *leornað* 'learns'. In the Bede passage, *swa hwylc swa* 'whoever' governs coordinated *rædeð* 'reads' and *leornað*. The singular indefinite-relative subject establishes the person and number; the ending alone would not suffice. Final eth is represented by thorn without changing the vowel or cell (Miller 1891, 380, ll. 26–27).

The input **líznōθi* 'learns' is a finite third-singular selection, not the infinitive citation label. Its inherited **lizn-* root remains distinct from the later lowered intermediate. The wider family has a class III prehistory and subsequent conjugational remodelling; the English class II form does not recover that complete earlier paradigm (Kroonen 2013, 340; Orel 2003, 247; Jasanoff 1973, 865, 868).

SC015 PNWGmcILowering lowers the selected root before **zn*. Rhotacism, English breaking, ending reduction and dental spelling then give *leornaþ*. This explanation leaves the inherited input high-vowelled. It does not select an **e*-grade just because the Old English word contains *eo* (Hill 2017, 153; Ringe and Taylor 2014, 38, 84).

The high-vowel *liorn- tradition remains important, but the infinitive's spelling cannot establish the prehistory of this specific cell. Jasanoff's proposed lost **liznią* and Hill's later-mutation account identify possible morphological sources of high vowels; neither yields a fully certain inherited paradigm. The family classification accordingly retains conjugational remodelling without claiming that the selected vowel itself requires analogical replacement (Jasanoff 1973, 865, 868; Hill 2017, 153, n. 25; Tarsi and Laker 2026, 42–43).

learn (iptv.2sg) — OE *leorna*

Derivation: citation reconstruction **liznōjanǫ*; form followed here **líznô* > *leorna* (late analogy).

Derivation trace Proto input: **líznô*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**léornô*
PNWGmc I Lowering	**léznô*	OE Unstressed Long Vowel Shortening	**léorna*
EAF Rhotacism	**lérnô*		
Early Anglo-Frisian			
[no change]			

Old English form: *leorna*

The imperative singular Old English leorna ‘learn!’ is the second-person singular imperative, not an abbreviated infinitive. Deuteronomy14:23 prints the command followed by *þæt þu ondræde Drihten* ‘that you fear the Lord’. The addressee is singular, and the form is read directly in the edited biblical text (Crawford 1922, 351).

The derivational input **líznô* ‘learn!’ is the imperative cell, distinct from the lexeme-level infinitive label. The inherited **lizn-* family is supported by the comparative dictionaries; its historical weak class III formation and later remodelling remain uncertain (Kroonen 2013, 340; Orel 2003, 247; Jasanoff 1973, 865, 868).

Under SC015 PNWGmcILowering, lowering before preconsonantal **z* produces **léznô*; rhotacism gives **lérnô*, and English breaking and ending reduction lead to *leorna*. The companion infinitive *leornian* and present third singular *leornaþ* are different cells of the same family. Their agreement does not multiply the independent etymological evidence (Hill 2017, 153; Crawford 1922, 71, 351; Miller 1891, 380).

The attested alternative *liorna* ‘learn!’ remains relevant to the *liorn-/leorn- distribution. Mutation in an earlier paradigm, later levelling and dialect history must be considered rather than identifying every high vowel as unchanged inheritance. The retained family classification marks conjugational remodelling, not finite-cell selection or an extra replacement needed for this imperative (Campbell 1959, 48, 59, 125; Jasanoff 1973, 865, 868).

lick (3sg) — OE *liccaþ*

Derivation: citation reconstruction **likkōnǫ*; form followed here **líkkōθi* > *liccaþ* (late analogy).

Derivation trace Proto input: **líkkōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Late O Shortening	**líkkoθ*
PWGmc Early I Apocope	**líkkōθ*	OE Final Unstressed O Lowering	**líkkaθ*
Early Anglo-Frisian			
[no change]			

Old English form: *liccaþ*

Reconstruction and comparative evidence Ringe and Taylor give PWGmc **ekkōn* ‘lick (3sg.)’ continuing as Old English *liccian* ‘lick’, Old Saxon *likkon* ‘lick’, and Old High German *lecchon* ‘lick’ (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* ‘lick’ with the same Old English continuation (Orel 2003, 285).

The form compared here in this entry is the non-*j* present third singular **líkkōθi* ‘licks’, not the remodeled infinitive. Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80).

Old English evidence Bosworth-Toller lemmatizes the verb as *liccian* ‘lick’ (Bosworth and Toller 1898, 614). The same consonantal frame appears in Campbell’s and Brunner’s grammatical citations of *liccian* ‘lick’ (Campbell 1959, sect. 398.1; Brunner 1965, sect. 45 Anm. 3). The Old English headword is therefore clear even though the entry here is not about the citation form.

The form treated here is the present third singular *liccaþ* ‘licks’. It is a selected paradigm form beside the lemma *liccian* ‘lick’ and the related imperative *licca* ‘lick’, not a separately lemmatized headword.

Development to Old English **líkkōθi* ‘licks’ first loses final *-i*, giving **líkkōθ* ‘licks’. Campbell’s class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support the regular 3sg outcome *-aþ* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). Because this ending never contains *-j-*, the form does not pass through an i-umlauted *-eþ* stage.

Paradigm comparison The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian* ‘lick’	*liccian* ‘lick’	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**líkkô*	regular output *licca* ‘lick’	*licca* ‘lick’	useful family control, but not the target of this entry
present third singular	**líkkōθi*	regular output *liccaþ* ‘licks’	*liccaþ* ‘licks’	exact match between the derivational input and the Old English form here

lick (iptv.2sg) — OE *licca*

Derivation: citation reconstruction **likkōną*; form followed here **líkkô* > *licca* (late analogy).

Derivation trace Proto input: **líkkô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**líkka*
[no change]		
Early Anglo-Frisian		
[no change]		

Old English form: *licca*

Reconstruction and comparative evidence Ringe and Taylor give PWGmc **ekkōn* ‘lick (3sg.)’ continuing as Old English *liccian* ‘lick’, Old Saxon *likkon* ‘lick’, and Old High German *lecchon* ‘lick’ (Ringe and Taylor 2014, 50). Orel gives the fuller weak-verb reconstruction **likkōjanan* ‘lick’ with the same Old English continuation (Orel 2003, 285).

Campbell’s weak class-II discussion gives present forms such as lufas, *-aþ* (< *-ōsi*, *-ōþi)* (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)* and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The form treated here is therefore not that remodeled infinitive but a finite cell in bare trimoric **-ō*.

Old English evidence Bosworth-Toller lemmatizes the verb as *liccian* ‘lick’ (Bosworth and Toller 1898, 614). Campbell cites *liccian* ‘lick’ among Old English forms with preserved geminate *cc* (Campbell 1959, sect. 398.1). Brunner likewise cites *liccian* ‘lick’ (Brunner 1965, sect. 45 Anm. 3). The Old English evidence therefore establishes the verbal headword and its consonantal frame securely.

The Old English form here in this entry is the imperative singular *licca* ‘lick’. It is a paradigm form chosen beside the headword *liccian* ‘lick’ and the related present *liccaþ* ‘licks’, not a separately lemmatized citation word.

Development to Old English With the stem *licc-* established, the remaining development is brief. Campbell’s class-II present endings lufas, *-aþ* (< *-ōsi*, *-ōþi)* support late *-a* in this finite cell (Campbell 1959, sect. 356.4). Ringe and Taylor likewise note stable *a* in the class-II 2sg and 3sg (Ringe and Taylor 2014, 80). The same stem consonantism that appears in *liccian* ‘lick’ is preserved here, giving *cc* throughout the finite form.

Paradigm comparison The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**líkkōjaną*	regular output *liccian* ‘lick’	*liccian* ‘lick’	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**líkkô*	regular output *licca* ‘lick’	*licca* ‘lick’	exact match between the derivational input and the Old English form here
present third singular companion	**líkkōθi*	regular output *liccaþ*	*liccaþ*	useful family control, but not the target of this entry

live — OE *lifeþ*

Derivation: citation reconstruction **libēną*; form followed here **líbēθi* > *lifeþ* (late analogy).

Derivation trace Proto input: **líbēθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PGmc B Allophony	**líβēθ*
PWGmc Early I Apocope	**líbēθ*	OE Unstressed Long Vowel Shortening	**líβeθ*
Early Anglo-Frisian			
[no change]			

Old English form: *lifeþ*

Reconstruction and comparative evidence The inherited verb belongs to the class-III weak family cited by Kroonen under **libēn-*, reflected in Old English *libban* ‘live’ (Kroonen 2013, 336). Ringe and Taylor show that the paradigm also contained a separate 3sg present stem, continued in late Northumbrian *lifed* ‘lives’, which they treat as an archaism (Ringe and Taylor 2014, 364).

The derivational input **líbēθi* ‘live’ therefore represents a finite present cell rather than the citation infinitive. The later lemma tradition also includes remodeled forms such as *lifian* ‘live’.

Old English evidence The ordinary lemma tradition centers on *libban* ‘live’ and, in later remodeling, *lifian* ‘live’. For this entry, however, the relevant comparison form is the archaic 3sg present attested as *lifed* ‘lives’, here normalized as *lifeþ* ‘lives’ (Ringe and Taylor 2014, 364; Campbell 1959, sect. 762).

The target is thus a normalized finite form, not the ordinary dictionary lemma. It preserves the older present-stem history more clearly than the remodeled lemma tradition does.

Development to Old English From **líbēθi* ‘live’, regular reduction of the final syllable and later weakening of the unstressed vowel yield *lifeþ* ‘lives’. The attested spelling *lifed* ‘lives’ belongs to the same finite form in late Northumbrian orthography (Campbell 1959, sect. 762; Ringe and Taylor 2014, 364).

Paradigm comparison The comparison below sets the relevant forms side by side. It separates the archaic finite cell from the ordinary infinitival and later remodeled lemma lines.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation infinitive line	**libēną*	OE *libban* headword tradition	*libban*	establishes the lexeme, but not the Old English form here
3sg present	**líbēθi*	regular output: *lifeþ*; attested *lifed*	*lifed*, normalized here as *lifeþ*	selected archaic finite cell
later remodeled present tradition	later class-II-type forms	*lifian* and related finite remodeling	*lifian*	genuine OE development, but secondary to the cell compared here

make (3sg) — OE *macaþ*

Derivation: citation reconstruction **makōną*; form followed here **mákōθi* > *macaþ* (late analogy).

Derivation trace Proto input: **mákōθi*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening	**mækōθ*
PWGmc Early I Apocope	**mákōθ*	OE A Restoration	**makōθ*
Early Anglo-Frisian		OE Late O Shortening	**makoθ*
		OE Final Unstressed O Lowering	**makaθ*
[no change]			

Old English form: *macaþ*

Reconstruction and comparative evidence Kroonen derives the Old English verb from **makōjan-* on the make-family base **maka-* (Kroonen 2013, 350). Ringe and Taylor likewise derive Old English macian ‘make’ from PWGmc **makon* ‘make’ through **mekojan* ‘make’ (Ringe and Taylor 2014, 191).

The derivational input **mákōθi* ‘makes’ is therefore a finite 3sg cell of the same family, not the citation form of the verb.

Old English evidence Clark Hall lemmatizes the verb as macian ‘make’ (Clark Hall 1960, 193). The relevant comparison form here is normalized present-third-singular macaþ ‘makes’, set beside the dictionary headword and the related imperative singular *maca* ‘make’.

Campbell’s class-II paradigm gives ordinary 3sg ending *-aþ*, while his dialect survey allows secondary *-e-* spellings in some traditions (Campbell 1959, sect. 356.4; 1959, sect. 757). *Macaþ* ‘makes’ is thus the regular comparison form for the non-*j* 3sg cell.

Development to Old English After early loss of final *-i*, **mákōθi* ‘makes’ → **mákōθ* ‘makes’ after loss of final -i. Anglo-Frisian brightening gives **mækōθ* ‘makes’, but Campbell lists *macian* ‘make’ among class-II verbs with restored *a*, so the stem returns to *mak-* before the ending is reduced (Campbell 1959, sect. 159).

The ending then follows ordinary class-II 3sg development: **makōθ* ‘makes’ > **makaθ* ‘makes’ > macaþ_. Campbell’s_lufas, -aþ (< -ōsi, -ōþi)_and Ringe and Taylor’s discussion of stable class-II_a_in finite non-_j‘ cells support this sequence (Campbell 1959, sect. 356.4; Ringe and Taylor 2014, 80).

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the selected 3sg cell from the make-family lemma and from the companion imperative form.

PGmc cell / interpretation	Candidate input	OE outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**makōjaną*	dictionary headword macian ‘make’	macian ‘make’	family background, but not the cell compared here
3sg present	**mákōθi*	regular output macaþ ‘makes’	macaþ ‘makes’	exact match
imperative singular companion	**mákô*	related finite form *maca* ‘make’	*maca* ‘make’	useful control, but not the target

make (iptv.2sg) — OE *maca*

Derivation: citation reconstruction **makōną*; form followed here **mákô* > *maca* (late analogy).

Derivation trace Proto input: **mákô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**mækô*
[no change]	OE A Restoration	**makô*
	OE Unstressed Long Vowel Shortening	**maka*
Early Anglo-Frisian		
[no change]		

Old English form: *maca*

Reconstruction and comparative evidence The make-family belongs to the Old English class-II weak verbs. Campbell cites *lapian, macian* ‘make’ among verbs with restored *a* (Campbell 1959, sect. 159). Ringe and Taylor place the Germanic verb in the same class, comparing West Germanic continuants such as Old Frisian *makia* ‘make’, Old Saxon *makon* ‘make’, and Old High German *mahhon* ‘make’ (Ringe and Taylor 2014, 191).

The derivational input **mákô* ‘make (iptv.2sg)’ is not the citation form of the lexeme but a finite paradigm cell. Ringe and Taylor give the class-II weak imperative singular as -a < **-ō*, which makes this cell the relevant comparison point for the Old English form treated here (Ringe and Taylor 2014, 314).

Old English evidence The dictionary headword is *macian* ‘make’ (Clark Hall 1960, 193). The form compared here in this entry is therefore not the lemma but the imperative singular *maca* ‘make’, chosen as a paradigm form beside the headword *macian* ‘make’ and the related finite form *macaþ* ‘makes’.

The lexical history is still that of *macian* ‘make’, but the finite cell isolates the regular outcome of trimoric **ō* more cleanly than the citation form does.

Development to Old English From **mákô* ‘make (iptv.2sg)’, Anglo-Frisian brightening first gives **mækô* ‘make (iptv.2sg)’. Campbell cites *macian* ‘make’ among the class-II verbs with restored *a* (Campbell 1959, sect. 159). Ringe and Taylor’s class-II weak imperative singular -a < **-ō* then supports the later finite ending (Ringe and Taylor 2014, 314).

The same development explains why earlier fronted forms of the *mæċa* ‘make’ type do not control the entry. Once trimoric **ô* is treated as a back-vocalic trigger for restoration, the imperative singular falls into line with the broader *macian* ‘make’ family.

Paradigm comparison A paradigm comparison identifies which finite cell of the make-family matches the Old English form chosen here. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**mákōjaną*	comparative continuation *macian* ‘make’	*macian* ‘make’	ordinary headword of the verb, but not the finite form compared here
imperative singular	**mákô*	regular output: *maca* ‘make’	*maca* ‘make’	exact match between input, output, and selected paradigm form

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
present third singular companion	**mákōθi*	comparative companion *macaþ*	*macaþ*	useful family control, but not the target of this entry

man — OE *mannes*

Derivation: citation reconstruction **mánnaz*; form followed here **mánnas* > *mannes* (late analogy).

Derivation trace Proto input: **mánnas*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening Unstressed	**mánnæs*
[no change]	OE Unstressed AE Merger	**mánnes*
Early Anglo-Frisian		
[no change]		

Old English form: *mannes*

Reconstruction and comparative evidence The lexeme-level reconstruction is not uniform. Kroonen cites **mannan-*, and Orel has **mannz* 'man' (Kroonen 2013, 354; Orel 2003, 299). The derivational input **mánnas* 'man' belongs to a different level: it is the genitive-singular cell chosen for the Old English comparison.

The target is not the ordinary citation form. Its medial geminate precedes the genitive ending.

Old English evidence Campbell gives the paradigm *mann* 'man', *man* 'man' / *mannes* 'man's / *menn* 'men' (Campbell 1959, sect. 621). Sievers-Brunner likewise cites *man* 'man' *mannes* 'man's (Brunner 1965, sect. 226). He also explains that word-final simplification underlies forms such as *man* 'man' beside inflected *monnes* 'man's (Brunner 1965, sect. 231). Clark Hall keeps the dictionary headword under *mann* (Clark Hall 1960, 197).

The relevant comparison form is therefore the attested genitive singular *mannes* 'man's, not the citation lemma *mann* 'man'.

Development to Old English Campbell's paradigm *mann*, *man* / *mannes* / *menn* 'men' confirms the selected genitive singular *mannes* (Campbell 1959, sect. 621). In the present analysis, **mánnas* 'man' develops through Anglo-Frisian brightening and later unstressed merger to *mannes*. In this cell the geminate remains medial before *-es*. The citation form behaves differently because word-final gemination was simplified in Old English (Brunner 1965, sect. 231).

Paradigm comparison The comparison below sets the relevant forms side by side. It separates the citation-form line from the genitive singular.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mannăz*	expected citation-form outcome *man*	*mann* / *monn*	establishes the lexeme, but not the Old English form here
accusative singular	**mannǫ*	expected *man*	*man*	same word-final simplification as the nominative
dative singular	**mannăi*	expected *manne*	*manne*	preserves medial *nn*, but not the chosen cell
genitive singular	**mánnas*	regular output: *mannes*	*mannes*	exact match between input, output, and attested comparator

meed — OE *meorde*

Derivation: citation reconstruction **mizdō*; form followed here **mízdai* > *meorde* (late analogy).

Derivation trace Proto input: **mízdai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Breaking	**méordē*
PNWGmc Unstressed Ai Monophthongization	**mízdē*	OE Unstressed Long Vowel Shortening	**méorde*
PNWGmc I Lowering	**mézdē*		
EAF Rhotacism	**mérdē*		
Early Anglo-Frisian			
[no change]			

Old English form: *meorde*

Reconstruction and comparative evidence The lexeme-level reconstruction is **mizdō* ‘meed’, but the derivational input **mízdai* ‘meed’ is a dative-singular cell rather than the citation form. The Old English evidence for the *meord* ‘meed, reward’ side is oblique.

The wider history of competing *mēd* ‘meed, reward’ remains disputed. Kroonen and Fulk explain it through some form of *z*-loss and compensatory lengthening (Kroonen 2013, 370; Fulk 2018, 52), while Orel keeps a doublet analysis (Orel 2003, 272). The comparison here concerns the attested oblique line *meorde*.

Old English evidence The directly attested forms are obliques: *meorde* ‘meed’ as a dative singular and *meorda* ‘meed’ as a genitive plural (Bright 1971, 328; Bosworth and Toller 1898, 647). Lexicographers reconstruct a bare nominative *meord* ‘meed’ from those obliques, while West Saxon prose more commonly shows the competing doublet *mēd* ‘meed’ (Clark Hall 1960, 214; Bosworth and Toller 1898, 647).

The target of this entry is therefore the attested oblique *meorde* ‘meed’, not the reconstructed lemma *meord* ‘meed’ and not the better-known West Saxon citation form *mēd* ‘meed’.

Development to Old English Ringe and Taylor give the broader noun history as **mizdo* ‘reward’ > **mizdu* ‘reward’ > OE meord ~ méd (Ringe and Taylor 2014, 84). The fuller oblique-cell path modeled here spells out the intermediate monophthongization, lowering, rhotacism, breaking, and unstressed-shortening steps needed for the selected dative-singular comparison.

This entry therefore follows the attested oblique line within the broader development. It does not depend on a full decision about the history of the competing *mēd* ‘meed, reward’ tradition.

The consonantal distinction is substantial. Retained **z* can become a rhotic, producing the consonantal group that conditions English breaking in *meord-*. *Loss of the sibilant with compensatory lengthening instead leads toward the long-vowel* mēd* tradition. These are alternative histories of the consonantal sequence, not two interchangeable spellings of the selected dative (Kroonen 2013, 370; Crist 2002, 3–6).

SC015 PNWGmcILowering places lowering before either continuation: its condition is the original preconsonantal **z*, not the later rhotic or a particular following vowel. The selected **mízdai* therefore supplies a positive application with a historically relevant consonant group. It does not independently establish the entire history of the rival nominative or the absolute date of every lowering environment (Hill 2017, 153; Crist 2002, 3–6).

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the attested oblique target from the broader lemma history.

PGmc cell / interpretation	Candidate input	Expected or documented OE outcome	OE comparison form	Result
citation nominative singular	**mizdō*	inferred lemma outcome *meord* ‘meed’	*meord* ‘meed’	useful background, but the bare lemma is reconstructed rather than directly attested
selected dative singular	**mízdai*	regular output: *meorde* ‘meed’	*meorde* ‘meed’	exact match between derivational input and attested target
genitive singular	**mizdōz*	regular output: *meorde* ‘meed’	*meorde* ‘meed’	converges on the same attested string, but the dat.sg. has the clearest direct support
genitive plural control	plural oblique line	attested *meorda* ‘meed’	*meorda* ‘meed’	confirms the broader oblique tradition, but not the chosen singular target

night — OE *niht*

Derivation: citation reconstruction **náxtz*; form followed here **náxti* > *niht* (late analogy).

Derivation trace Proto input: **náxti*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**nœxti*
[no change]	OE Breaking	**neaxti*
	OE I Umlaut	**niexti*
Early Anglo-Frisian	OE Ws Palatal Umlaut	**nixti*
[no change]	OE High Vowel Apocope	**nixt*

Old English form: *niht*

Reconstruction and comparative evidence Ringe and Taylor cite gen.sg. **nahtiz* ‘night’, dat.sg. **nahti* ‘night’, and nom.pl. **nahtiz* ‘night’ for the high-vowel side of the paradigm, and derive West Saxon *niht* ‘night’ from that side (Ringe and Taylor 2014, 240). The citation reconstruction **náxtz* ‘night’ therefore belongs to the nominative-like headword, while the derivational input **náxti* ‘night’ represents the dative-singular cell.

The word later became the model for endingless datives. Ringe and Taylor explicitly explain forms such as *dæg* ‘day’ by analogy with dat.sg. niht < **nahti* ‘night’ (Ringe and Taylor 2014, 380).

Old English evidence Clark Hall lemmatizes *niht* ‘night’ and cross-references forms such as *neaht* ‘night’, *neht* ‘night’, and *nieht* ‘night’ (Clark Hall 1960, 215). Campbell likewise preserves the fluctuation between *neaht* ‘night’ and *niht* ‘night’, giving genitive *nihte, nihtes*, dative *niht, nihte*, nominative plural *niht* ‘night’, and the contrasting plural-side forms represented by *neahtas* ‘night’ (Campbell 1959, sect. 628.3).

The comparison form used here is therefore an attested Old English *niht* ‘night’, not a reconstructed substitute. The broader lexical record still preserves the non-umlauted side of the paradigm in *neaht* ‘night’-type forms.

Development to Old English Ringe and Taylor derive West Saxon *niht* ‘night’ from **nahti* ‘night’ via **nehti* ‘night’ and **neahti* ‘night’ (Ringe and Taylor 2014, 240). Campbell and Brunner preserve the contrasting non-umlauted *neaht* ‘night’-type forms elsewhere in the paradigm (Campbell 1959, sect. 628.3; Brunner 1965, sect. 284).

The modeled path is therefore **náxti* ‘night’ > **nœxti* ‘night’ > **neaxti* ‘night’ > **niexti* ‘night’ > **nixti* ‘night’ > niht.

Paradigm comparison The comparison below sets the relevant forms side by side. It identifies the inherited cell that matches the attested Old English form.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**náxtz*	expected non-umlauted outcome *neaht*	*neaht*	useful background, but not the comparison used for *niht*
selected dative singular	**náxti*	regular output: *niht* ‘night’	*niht* ‘night’	exact match between input, output, and paradigm cell

rest — OE *ræste*

Derivation: citation reconstruction **rastō*; form followed here **rástōz* > *ræste* (late analogy).

Derivation trace Proto input: **rástōz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Surviving Bimoric O Unrounding	**rástā*
EAF Final Z Deletion	**rástō*	EAF Brightening	**ræstā*
Early Anglo-Frisian		EAF Brightening Long Final	**ræstǣ*
		OE Unstressed Long Vowel Shortening	**ræstæ*
[no change]		OE Unstressed AE Merger	**ræste*

Old English form: *ræste*

Reconstruction and comparative evidence Kroonen treats the noun as a feminine ō-stem **rastō-*, continued by Old English *ræst* ‘rest’ (Kroonen 2013, 445). The derivational input **rástōz* ‘rest’ therefore does not replace the lexeme-level headword. It identifies one oblique singular cell on the side of the paradigm that yields *ræste* ‘rest’.

The source tradition used here labels that cell specifically as genitive singular, but the broader local synthesis of the ō-stem paradigm shows that the oblique singulars converge on the same front-vocalic *ræste* ‘rest’ side, in contrast to a nominative singular that would remain *rast* ‘rest’.

Old English evidence The ordinary Old English citation form is *ræst* ‘rest’ (Kroonen 2013, 445). Clark Hall likewise gives *ræst* ‘rest’ (Clark Hall 1960, 239). Bosworth-Toller also preserves oblique uses of *ræste* ‘rest’, including prepositional examples such as *on ræste* ‘at rest’ and *tó ræste* ‘to rest’ (Bosworth and Toller 1898, 121).

The comparison form used here is therefore an attested oblique *ræste* ‘rest’, not a reconstructed surrogate. The dictionary headword *ræst* ‘rest’ remains an equally real part of the Old English record.

Development to Old English Once final **z* is lost, the derivational input moves through the front-vocalic oblique side of the paradigm rather than the back-vocalic nominative side. In the modeled derivation, the surviving final long vowel is first exposed, unrounded, fronted, shortened, and then reduced to the final *-e* of *ræste* ‘rest’.

The key point is the paradigm split. Nominative **rastō* ‘rest’ yields a regular *rast* ‘rest’, whereas the selected oblique input yields *ræste* ‘rest’. The later citation form *ræst* ‘rest’ is best explained as leveling from that oblique *ræst-* stem.

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the nominative citation form from the oblique singular chosen here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**rastō*	expected regular outcome *rast*	*ræst* ‘rest’	useful background, but not the cell that matches attested oblique *ræste*
selected oblique singular	**rástōz*	regular output: *ræste* ‘rest’	*ræste* ‘rest’	exact match between derivational input and attested OE oblique form

shoulder — OE *sċuldrum*

Derivation: citation reconstruction **skuldrō*; form followed here **skúldramiz* > *sċuldrum* (late analogy).

Derivation trace Proto input: **skúldramiz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Sk Palatalization	**fúldrum*
PNWGmc A To U Before M	**skúldrumiz*		
PWGmc Early I Apocope	**skúldrumz*		
EAF Final Z Deletion	**skúldrum*		
Early Anglo-Frisian			
[no change]			

Old English form: *sċuldrum*

Reconstruction and comparative evidence The handbooks do not agree on the reconstruction of the Germanic word. Orel gives **skuldr(j)ō* as a feminine ō-/jō-stem and explicitly notes that Old English *sculdor* ‘shoulder’ is masculine beside OFrisian *skulder* ‘shoulder’, Middle Low German schulder ‘shoulder’, and Old High German *scultra* ‘shoulder’, *scultirra* ‘shoulder’ (Orel 2003, 345). Kroonen reconstructs **skuldra-*, a masculine a-stem, and derives the Old High German feminine forms from **skuldrjōn-* (Kroonen 2013, 478). Ringe and Taylor cite PWGmc **skuldru* ‘shoulder’ for the Old English branch (Ringe and Taylor 2014, 142).

These forms imply different stem classes and different expectations for the Old English inflection. The question is which inflectional cell best aligns with the Old English evidence.

A dative/instrumental plural form **skúldramiz* ‘shoulder’ aligns with the inherited plural ending that later yields Old English *-um*, and it corresponds directly to the attested dative plural discussed below.

Old English evidence The ordinary Old English headword is *sculdor* ‘shoulder’. Clark Hall lemmatizes *sculdor* ‘shoulder’ as the normal dictionary form (Clark Hall 1960, 257). Bosworth-Toller also preserves the dative plural *sculdrum* ‘shoulder’ (Bosworth and Toller 1898, 85).

Bosworth-Toller’s Supplement records a weak-feminine *sculdra* ‘shoulder’ (Bosworth and Toller 1898, 699), so *sculdra* ‘shoulder’ belongs to the Old English record beside the stronger masculine paradigm headed by *sculdor* ‘shoulder’. Brunner and Luick also record later spellings such as *sceoldor* ‘shoulder’ and the i-mutated dative plural *scyldrum* ‘shoulder’, which reflect secondary phonological and analogical reshaping within Old English (Brunner 1965, sect. 92.2.a; Luick 1914--1940, 230).

The singular and plural evidence point to different parts of the paradigm. The relevant comparison form here is the attested dative plural *sċuldrum* ‘shoulder (dat.pl.)’. The spelling with *sċ-* is a normalized representation of the same Old English initial cluster.

Development to Old English Proto-Germanic **skúldramiz* ‘shoulder’ can be interpreted as a dative/instrumental plural form. In this environment the post-tonic *a* before *m* is raised to *u*, giving a form of the **skúldrumiz* ‘shoulder’ type. Unstressed *u* is regularly preserved before *m*, especially in the dative plural ending *-um*: Campbell states this explicitly, and Hogg formulates the same condition for the dative plural inflexion (Campbell 1959, sect. 373; Hogg 1992b, sect. 3.3.1.3). Brunner points in the same direction by excluding *m* from the environments in which medial *o* became general in West Saxon (Brunner 1965, sect. 44 Anm. 7).

Subsequent reduction of the ending removes the final **i* and **z*, so that the inflectional ending appears in Old English as *-um*. The initial cluster is written here as *sċ-*, and the development is **skúldramiz* ‘shoulder’ > **skúldrumiz* ‘shoulder’ > **skúldrum* ‘shoulder’ > *sċuldrum* ‘shoulder’.

Paradigm comparison A paradigm comparison identifies the Proto-Germanic inflectional cell that corresponds to an established Old English paradigm form. The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
singular-oriented citation input	**skúldrō*	probe output: *sċoldor*	*sculdor* ‘shoulder’	fails: the singular output has root *o*, not the attested *u*
serious plural-based singular alternative	**skúldru*	probe output: *sċuldor*	*sculdor* ‘shoulder’	close formally, but it compares a plural-stage input with a singular form
dat./inst.pl. input	**skúldramiz* ‘shoulder’	regular output: *sċuldrum* ‘shoulder (dat.pl.)’	*sculdrum* ‘shoulder’	matches both the output and the dative plural comparison form
later weak-feminine singular	—	OE *sculdra* ‘shoulder’	*sculdra* ‘shoulder’	secondary doublet, useful as a control rather than the inherited target

The dative plural line is decisive because it matches both the output and the paradigm cell of Old English *sculdrum* ‘shoulder’. Singular-oriented candidates either lower the root vowel or compare unlike cells.

shove — OE *sċēaf*

Derivation: citation reconstruction **skéubaną*; form followed here **skáub* > *sċēaf* (late analogy).

Derivation trace Proto input: **skáub*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Au Brightening	**skáeub*
[no change]	OE Diphthong Leveling	**skēab*
	PGmc B Allophony	**skēaβ*
Early Anglo-Frisian	OE Sk Palatalization	**ʃēaβ*
[no change]		

Old English form: *sċēaf*

Reconstruction and comparative evidence Kroonen reconstructs the strong verb as **skeuban- ~ *skūban-* and cites Old English present forms *scēofan* ‘shove’, *scūfan* ‘shove’ (Kroonen 2013, 444). Those present-system forms belong to the same verb family, but the comparison here uses the singular preterite **skáub* ‘shove’, not the infinitive.

Old English evidence The ordinary dictionary verb is *scūfan* ‘shove’/*scēofan* ‘shove’, but the preterite itself is well attested. Bright gives the principal parts *scufan* ‘shove’, *sceaf* ‘shove’, *scufon* ‘shove’, *scofen* ‘shove’ (Bright 1971, 347). Sweet gives the same paradigm (Sweet 1953, 29). The normalized form here is sċēaf ‘sheaf’, regularizing the attested spellings *sceaf* ‘shove’ and prefixed *āsceaf* ‘shove’.

Development to Old English From **skáub* ‘shove’, the development is straightforward. **au* fronts and levels to *ēa*, final **b* becomes a fricative and is written *f*, and initial **sk-* undergoes the usual Old English palatalized spelling in this environment. The derivation therefore gives **skáub* ‘shove’ > **skáeub* ‘shove’ > **skēab* ‘shove’ > **skēaβ* ‘shove’ > *sċēaf* ‘sheaf’.

Paradigm comparison A paradigm comparison is needed here because the ordinary citation verb and *sċēaf* ‘sheaf’ belong to different cells of the same strong paradigm. The comparison below is manual.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation infinitive	**skéubaną*	inherited infinitive line *sċēofan*; present system also leveled *scūfan* ‘shove’	*scēofan* ‘shove’ / *scūfan* ‘shove’	necessary background, but not the comparison used for *sċēaf* ‘sheaf’
1/3 sg. preterite	**skáub*	documented regular output: sċēaf ‘sheaf’	sċēaf ‘sheaf’	direct match for the singular preterite
preterite plural	**skúbun*	later leveled plural *scufon* ‘shove’ beside expected *sċufun* under the corrected cascade	*scufon* ‘shove’	poorer comparison for the singular-preterite target
past participle	**skúbanaz*	attested participial line *scofen* ‘shove’	*scofen* ‘shove’	valid alternative cell, but not the form compared here

show (3sg) — OE *sċēawaþ*

Derivation: citation reconstruction **skawōną*; form followed here **skáwōθi* > *sċēawaþ* (late analogy).

Derivation trace Proto input: **skáwōθi*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Aw Long Diphthong	**skɛ́awōθ*
PWGmc Early I Apocope **skáwōθ*	OE Sk Palatalization	**ʃɛ́awōθ*
Early Anglo-Frisian	OE Late O Shortening	**ʃɛ́awoθ*
[no change]	OE Final Unstressed O Lowering	**ʃɛ́awaθ*

Old English form: *sċēawaþ*

Reconstruction and comparative evidence Orel reconstructs the verb as **skawōjanan* ‘show’ and cites Old English *sceáwian* ‘show’ beside Old Frisian *skawia* ‘show’, Old Saxon *skawōn* ‘show’, and Old High German *scouwōn* ‘show’ (Orel 2003, 337). This is infinitive-family evidence. CAPR assembles the present third singular **skáwōθi* ‘show (3sg)’, a finite class-II cell with stable *a* in the 3sg ending (Ringe and Taylor 2014, 80).

Campbell states the class-II present endings as lufas, *-aþ* (< *-ōsi*, *-ōþi)* in his late-shortening discussion (Campbell 1959, 148). Ringe and Taylor likewise note that class-II weak present 2sg. *-as(t)*

and 3sg. *-aþ* have stable *a* (Ringe and Taylor 2014, 80). The relevant comparison is therefore the 3sg cell itself, not an i-umlauted alternative.

Old English evidence The held fourth edition of Bright's reader contains the simplex infinitive *scēawian* 'show' and imperative *scēawa* 'show' in the Blickling homily (Bright 1917, 71). The Wanderer contains prefixed *geondscēawað* 'surveys' (Bright 1917, 161). This supports the finite ending pattern, not a separately quoted unprefixed third singular or a passage in the separate Cassidy/Ringler edition.

The form written here as *sċēawaþ* 'shows' is the normalized simplex comparison form for that weak class-II pattern. It is therefore not a dictionary headword but a finite comparison form aligned with the attested *scēaw-* evidence and the directly cited prefixed *geondscēawað* 'surveys' ending pattern.

Development to Old English The infinitive entry and SC034 OEAwLongDiphthong explain the dictionary-singleton versus handbook-geminate root premise. The same computational proxy remains here without claiming that output fit resolves it (Orel 2003, 337; Kroonen 2013, 442; Campbell 1959, 45; Brunner 1965, 49).

The complete trace of **skáwōθi* 'show (3sg)' has already lost final *i* before that proxy: the actual checkpoint is **s*k*á*w*ō*θ*. Thus absence of *j* alone is not the explanation of the compiled lack of mutation. Campbell's late-shortening account and Ringe and Taylor's stable-*a* discussion independently support *-aþ*, not *-eþ* (Campbell 1959, 148; Ringe and Taylor 2014, 80).

Paradigm comparison The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjanǫ*	regular output *sċēawian*	*scēawian* 'show'	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular companion	**skáwô*	regular output *sċēawa* 'show'	*scēawa* 'show'	useful family control, but not the target of this entry
present third singular	**skáwōθi*	regular output *sċēawaþ* 'shows'	normalized *sċēawaþ* 'shows'; source-side pattern *sceawað*	exact match for the finite form compared here

show (iptv.2sg) — OE *sċēawa*

Derivation: citation reconstruction **skawōnǫ*; form followed here **skáwô* > *sċēawa* (late analogy).

Derivation trace Proto input: **skáwô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Aw Long Diphthong	**skéawô*
[no change]	OE Sk Palatalization	**ʃéawô*
	OE Unstressed Long Vowel Shortening	**ʃéawa*
Early Anglo-Frisian		
[no change]		

Old English form: *sċēawa*

Reconstruction and comparative evidence Orel reconstructs the verb as **skawōjanan* ‘show’ and cites Old English *sceáwian* ‘show’ beside Old Frisian *skawia* ‘show’, Old Saxon *skawōn* ‘show’, and Old High German *scouwōn* ‘show’ (Orel 2003, 337). This is infinitive-family evidence, not a quotation of the selected imperative. CAPR assembles imperative singular **skáwô* ‘show (iptv.2sg)’, using the finite class-II cell with imperative -a < **-ō* (Ringe and Taylor 2014, 314).

The imperative singular provides the direct comparison with the Old English form. The lexical history still belongs to the *sceáwian* ‘show’ verb, but the cell compared here isolates the finite *-a* outcome more clearly than the citation form does.

Old English evidence The held fourth edition of Bright’s reader contains *scēawian* ‘show’ and the imperative singular *scēawa* ‘show’ in the Blickling homily (Bright 1917, 71). The form treated here is therefore an attested finite paradigm form, not a reconstructed convenience form; it is not attributed to an unverified passage in the separate Cassidy/Ringler edition.

The spelling used in this entry is normalized *sċēawa* ‘show’, while Bright’s cited text gives source spelling *scēawa* ‘show’. The ordinary Old English headword remains *scēawian* ‘show’; *sċēawa* ‘show’ is the imperative singular chosen beside it.

Development to Old English The infinitive entry and SC034 OEAwLongDiphthong distinguish the dictionary singleton stem from Campbell and Brunner’s inherited-geminate analysis. The same computational proxy is retained here; successful finite output does not resolve that prehistoric root history (Orel 2003, 337; Campbell 1959, 45; Brunner 1965, 49).

Ringe and Taylor’s class-II weak imperative singular -a < **-ō* supports the finite ending that yields *sċēawa* ‘show’ (Ringe and Taylor 2014, 314). The ending comparison is therefore justified independently of the unresolved root quantity history. The inherited *late_analogy* corpus classification is not changed by this root-only adjudication; it does not make the proxy a demonstrated analogical sound law.

Paradigm comparison The comparison below sets the relevant forms side by side.

PGmc cell / interpretation	Candidate input	Old English outcome or comparison	OE comparison form	Result
lexeme-level infinitive	**skáwōjanǫ*	regular output *sċēawian*	*scēawian* ‘show’	ordinary dictionary headword of the verb, but not the finite form compared here
imperative singular	**skáwô*	regular output *sċēawa* ‘show’	*scēawa* ‘show’ / normalized *sċēawa* ‘show’	exact match between the derivational input and the Old English form here
present third singular companion	**skáwōθi*	regular output *sċēawaþ* ‘show’	*sċēawaþ* ‘show’	useful family control, but not the target of this entry

span — OE *spanne*

Derivation: citation reconstruction **spannō*; form followed here **spánnai* > *spanne* (late analogy).

Derivation trace Proto input: **spánnai*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Unstressed Long Vowel Shortening	**spánne*
PNWGmc Unstressed Ai Monophthongization	**spánnē*		
Early Anglo-Frisian			
[no change]			

Old English form: *spanne*

Reconstruction and comparative evidence Seebold gives Old English *spann* ‘span’ under this noun family (Seebold 1970, 450). The form followed here, **spánnai* ‘span’, is therefore not a rival headword, but the specific dative singular form compared on the model of the feminine ō-stem paradigm (Brunner 1965, sect. 252; 1965, sect. 255.2).

Old English evidence The reviewed lexicographic evidence more directly supports the citation noun *spann* ‘span’ than the exact form *spanne* ‘span’. Clark Hall gives *spann* ‘span’ (Clark Hall 1960, 286), and *spanne* ‘span’ is accordingly treated as the regular dative singular form compared here rather than as a dictionary headword.

Development to Old English Citation **spannō* ‘span’ yields *span* ‘span’. The oblique cell **spánnai* ‘span’ therefore supplies the conservative comparison form: it preserves the medial geminate and yields *spanne* ‘span’, while citation **spannō* gives the nominative background form.

Paradigm comparison The comparison below sets the citation form beside the dative singular form compared here.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**spannō*	regular output: *span* ‘span’	*spann* ‘span’	useful citation-form background, but not the form compared here
dative singular compared here	**spánnai*	regular output: *spanne* ‘span’	*spanne* ‘span’	exact match for that conservative form

straw — OE *strēaw*

Derivation: **stráwą* yields regular **strēa*; the Old English form here is *strēaw* (late analogy).

Derivation trace Proto input: **stráwą*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**stráeu*
PWGmc Endingless Aw	**stráu*	OE Diphthong Leveling	**strēa*
Early Anglo-Frisian			
[no change]			

Regular outcome: **strēa*

Old English form: *strēaw*

Reconstruction and comparative evidence Kroonen cites the neuter stem **strawa-* 'straw' and Old English *strēa(w)* 'straw' (Kroonen 2013, 484). The selected **stráwą* 'straw' is CAPR's accented nominative/accusative singular, not Kroonen's verbatim full-word quotation. His derivation of the noun from the strew formation concerns genealogical direction; it does not explain the later restoration of the glide (Kroonen 2013, 484).

Old English evidence The comparison remains attested *strēaw* 'straw', not reconstructed **strēa* 'straw'. Campbell discusses restoration of the glide and the compound *strēaberige* 'strawberry' (Campbell 1959, 46–47, 232–33). Brunner explicitly treats the regular free simplex and short obliques as unattested; his paradigm note limits the West Saxon straw evidence to nominative/accusative singular and a doubtful plural. The compound preserves an endingless pattern, not an independently attested free noun or a complete surviving regular paradigm (Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 173, 387).

Development to Old English The regular endingless branch loses the final vowel and vocalizes final **w*: **stráwą* 'straw' > **stráu* 'straw'. The narrow fragment SC111 PWGmcEndinglessAw combines those operations; its computational holding site does not establish their order against unrelated changes. Its product feeds SC030 OEAuBrightening, giving **stráeu* 'straw', and SC032 OEDiphthongLeveling, giving regular **strēa* 'straw' (Campbell 1959, 46–47; Ringe and Taylor 2014, 172–73).

Attested *strēaw* 'straw' combines that long diphthong with a glide restored from inflected forms. Reciprocally, the long diphthong spreads into the formerly short oblique stem. This mutual levelling explains the long-plus-glide form without inventing a sound law that preserves final singleton **w* in the regular endingless branch (Campbell 1959, 232–33; Brunner 1965, 204–5; Ringe and Taylor 2014, 387). The cascade stops before that levelling: its reconstructed output remains an explicit mismatch, while the attested comparison and derivational input are unchanged.

Paradigm comparison The table distinguishes source-supported historical comparisons from the one selected compiled derivation; it is not a set of newly compiled oblique inputs.

Cell or evidence	Regular development	Attested comparison and qualification
selected nominative/accusative singular	**stráwą* > **strēa*	*strēaw* 'straw': restored glide; regular free simplex unattested
short oblique stem, exemplified by genitive singular	**strawes* 'of straw'; corresponding short dative also reconstructed	Brunner's regular short obliques are unattested; long quantity is extended from the endingless cells, not demonstrated by a surviving short straw paradigm

Cell or evidence	Regular development	Attested comparison and qualification
remodelled singular and limited plural evidence	long diphthong plus restored glide	West Saxon *strēaw* ‘straw’; Brunner records plural *strewu* ‘straws’ with a question mark, not as a secure complete paradigm
compound witness	endingless pattern	*strēaberige* ‘strawberry’, not an attested free **strēa* ‘straw’

These qualifications follow Brunner’s paradigm and attestation discussion and Ringe and Taylor’s explicit mutual levelling account (Brunner 1965, 49, 110, 204–5; Ringe and Taylor 2014, 173, 387). The noun/verb genealogical direction remains a separate qualified question; this entry adopts neither a new ancestry nor an analogical transducer rule.

thistle — OE *þistles*

Derivation: citation reconstruction **θéstilaz*; form followed here **θístilas* > *þistles* (late analogy).

Derivation trace Proto input: **θístilas*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		EAF Brightening Unstressed	**θéstilæs*
PNWGmc I Lowering	**θéstilas*	OE I Umlaut	**θistilæs*
Early Anglo-Frisian		OE L Adjacent Syncope	**θistlæs*
		OE Unstressed AE Merger	**θistles*
[no change]			

Old English form: *þistles*

Reconstruction and comparative evidence Orel prints **þe(x)stilaz* ‘thistle’ for the lexeme (Orel 2003, 458). The comparative label **θéstilaz* ‘thistle’ therefore remains in view as the lexeme-level headword, while the derivational input **θístilas* ‘thistle’ is a specific genitive singular cell.

Old English evidence The ordinary simplex headword tradition is broken *þistel* ‘thistle’ / *ðistel* ‘thistle’. Clark Hall gives *ðistel* as the noun headword (Clark Hall 1960, 326). The Old English form here here is the genitive singular *þistles* ‘thistle’, which preserves the same stem in an oblique form where the cluster is medial.

Development to Old English Campbell’s discussion of cluster nouns shows the contrast clearly. Simplex forms often develop a parasite vowel in word-final obstruent + sonorant clusters, while comparable medial clusters remain unbroken; his examples include *hrefn*, *tacn* ‘token, sign’, *wépn* ‘weapon’, and *botm* ‘bottom’ beside forms with parasitic vowels elsewhere in the same lexical class (Campbell 1959, 151). The genitive singular **θístilas* ‘thistle’ therefore supplies the conservative comparison form: the cluster is medial and the regular development yields *þistles* ‘thistle’, while the simplex nominative belongs to the broken headword tradition *þistel* ‘thistle’.

Paradigm comparison The comparison below sets the relevant forms side by side. It shows the contrast between the citation form and the genitive singular cell.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
citation nominative singular	**θéstilaz*	computed output: *þistl*	*þistel* ‘thistle’	useful citation-form background, but not the Old English form here
genitive singular	**θístilas*	computed output: *þistles* ‘thistle’	*þistles* ‘thistle’	exact match for the conservative cell

Part V. Reconstructed Old English comparators

Direct attestation does not supply the required comparator in these entries. The target is an explicitly reconstructed Old English form and carries the corresponding evidential burden.

knob — OE **cnobba*

Derivation: citation reconstruction **knúppaz*; form followed here **knúbbô* > **cnobba* (reconstructed Old English comparator).

Derivation trace Proto input: **knúbbô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE Unstressed Long Vowel Shortening	**knóbba*
PNWGmc U Lowering **knóbbô*		
Early Anglo-Frisian		
[no change]		

Old English form: **cnobba*

Reconstruction and comparative evidence The wider knob-family is not uniform. Kroonen's discussion of the Germanic n-stems points to related voiced and voiceless branches within this group (Kroonen 2011, 297). The citation reconstruction **knúppaz* 'knob' represents the broader cognate-set headword, while the form followed here, **knúbbô* 'knob', represents the voiced weak-noun branch treated here.

The Old English record is uneven. The better attested OE material belongs to the voiceless branch, but the present entry represents the reconstructed OE form that would continue the voiced branch behind later English knob.

Old English evidence Clark Hall preserves Old English evidence of the *cnoppa* 'knob' type (Clark Hall 1960, 79). Those forms are genuine Old English evidence, but they belong to the voiceless branch of the family.

The target **cnobba* 'knob' is a reconstructed Old English form, not a directly attested one. I use it for the voiced branch because attested *cnoppa* 'knob' continues the voiceless branch and therefore represents a different prehistory.

Development to Old English From the weak-noun form followed here, **knúbbô* 'knob', the regular Old English outcome is **cnobba* 'knob', with Proto-Germanic *kn-* represented in Old English as *cn-* and with the expected weak-noun ending.

The entry therefore does not claim that **cnobba* 'knob' is attested. Its claim is different: if the voiced weak-noun branch is the one to be represented, then **cnobba* is the regular Old English form corresponding to that branch.

Reconstruction status

Form	Status	Relevance to this entry
**knúbbô* > **cnobba*	reconstructed OE form; regular derivation	reconstructed Old English form compared here
cnopp / *cnoppa*	attested OE branch	important control form, but belongs to the voiceless branch
cnæp	attested OE form from another family	not part of the present lexeme line

This remains the most review-sensitive item here, because the choice between reconstructed **cnobba* ‘knob’ and attested *cnoppa* ‘knob’ is still a comparator-policy question rather than a settled point of OE attestation.

reek — OE **rēac*

Derivation: citation reconstruction **ráukiz*; form followed here **ráukaz* > **rēac* (reconstructed Old English comparator).

Derivation trace Proto input: **ráukaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Au Brightening	**ráeuka*
EAF Final Z Deletion	**ráuka*	OE Diphthong Leveling	**rēaka*
Early Anglo-Frisian		PWGmc Final Bare A Loss	**rēak*
[no change]			

Old English form: **rēac*

Reconstruction and comparative evidence The wider noun family is represented by **ráukiz* ‘reek’ / **rauki-*, with Old English *rēc* ‘reek’ as the attested noun reflex in the comparative dictionaries (Kroonen 2013, 446; Orel 2003, 338). The derivational input **ráukaz* ‘reek’ is therefore not the lexeme-level headword, but the form used here for the Old English derivation.

Old English evidence The attested noun is *rēc* ‘reek’, not **rēac* ‘reek’. Clark Hall records *rēc* ‘reek’ as the noun and also preserves related forms such as *rēcels* ‘incense’; Kroonen likewise gives OE *rēc* ‘reek’ under the noun family (Clark Hall 1960, 255; Kroonen 2013, 446). Clark Hall and Seebold also record verbal *rēac* ‘reek’ as the preterite of *rēocan* ‘reek’, but that verbal form is separate from the noun treated here (Clark Hall 1960, 254; Seebold 1970, 380).

The Old English form here **rēac* ‘reek’ is therefore a reconstructed West Saxon noun form, not a directly attested manuscript headword.

Development to Old English From **ráukaz* ‘reek’, the regular West Saxon development gives **rēac* ‘reek’. The attested noun *rēc* ‘reek’ belongs to the same lexical family, but reflects a later smoothed surface form rather than the regular noun target represented here.

Form note The distinction here is between an attested noun headword *rēc* ‘reek’ and a reconstructed regular West Saxon target **rēac* ‘reek’. The latter is treated as the modelling target, while the former remains philological background.

strew — OE **strīeġan*

Derivation: **stráwjaną* > **strīeġan* (reconstructed Old English comparator).

Derivation trace Proto input: **stráwjaną*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE Awwj Resolution	**stráujaną*
PWGmc J Gemination	**stráwwjaną*	OE Au Brightening	**stráeujaną*
Early Anglo-Frisian		OE Diphthong Leveling	**strēajaną*
		OE Heavy Syllable Nasal Apocope	**strēajan*
[no change]		OE Secondary Nasalization	**strēająn*
		OE I Umlaut	**strījąn*
		OE Weak Tail Reduction	**strījan*
		OE J Strengthening After Front Diphthong	**strīʒan*

Old English form: **strīeġan*

Reconstruction and comparative evidence Kroonen cites the inherited weak verb as **straujan-* and gives Old English *streowian* ‘strew’ as its dictionary continuation (Kroonen 2013, 483). Ringe and Taylor separate the inherited class-I branch from the remodelled class-II branch: Anglian *strēgan* ‘strew’ continues the inherited verb, while West Saxon *streowian* ‘strew’ is remodelled (Ringe and Taylor 2014, sect. 6.1 n. 27).

Luick groups **strauwjan* ‘strew’ with the same set as **hauwja-* and **kauwjan* ‘chew’, yielding Anglian *strēzan* ‘strew’ beside West Saxon forms of the *hīez* ‘hew’, *ciezan* ‘choose’ type (Luick 1914--1940, sect. 98). Fulk likewise allows an early West Saxon **striegan* ‘strew’ directly from Proto-Germanic **straujana* ‘strew’ (Fulk 2018, sect. 4.10 n. 1).

Old English evidence The attested inherited Old English form is *strēgan* ‘strew’ in Anglian. The attested West Saxon citation forms are *strewian* ‘strew’, *streowian* ‘strew’, and *strēawian* ‘strew’, which belong to the remodelled class-II branch (Ringe and Taylor 2014, sect. 6.1 n. 27; Campbell 1959, sect. 753.7).

The target **strīeġan* ‘strew’ is therefore a reconstructed Old English form, not an attested manuscript lemma. It represents the inferred West Saxon reflex of the inherited class-I branch.

Development to Old English From **stráwjaną* ‘strew’, the inherited West Saxon line passes through **straujaną* ‘strew’, then a fronted **strēajan-* stage, then **strīejan* ‘strew’ by i-umlaut, with retained/strengthened glide after the front diphthong to yield reconstructed **strīeġan* ‘strew’.

This differs from Anglian *strēgan* ‘strew’, where smoothing removes the diphthongal sequence, and from West Saxon *strewian* ‘strew’ / *streowian* ‘strew’ / *strēawian* ‘strew’, where the verb has already been remodelled into class II (Fulk 2018, sect. 4.10 n. 1; Campbell 1959, sect. 753.7).

Reconstruction status

Form or branch	Status	Relevance to this entry
strēgan ‘strew’	attested Anglian inherited class-I form	proves that the inherited verb survived into Old English
**strīeġan* ‘strew’	reconstructed West Saxon inherited class-I form; trace-supported	Old English form here
strewian / *streowian* ‘strew’ / *strēawian*	attested remodelled West Saxon class-II forms	genuine OE evidence, but not the inherited branch modeled here

Part VI. Known but unmodelled developments

The historical development is sufficiently supported by comparative and philological evidence, but the current transducer cascade does not yet implement the phonological or morphological process required to derive the target. Naming the gap explains the status without pretending that no historical account exists.

fire — OE *fȳre*

Derivation: **fūri* yields regular **fȳr*; the Old English form here is *fȳre* (known but unmodelled development).

Derivation trace Proto input: **fūri*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	OE I Umlaut	**fȳri*
[no change]	OE High Vowel Apocope	**fȳr*
Early Anglo-Frisian		
[no change]		

Regular outcome: **fȳr*

Old English form: *fȳre*

Reconstruction and comparative evidence Kroonen places the lexeme in a heteroclitic family **fōr* ‘fire’ ~ **fun-* and explains the front-mutated West Germanic forms from an oblique form of the **fu(w)eri* type (Kroonen 2013, 151). The derivational input **fūri* ‘fire’ is CAPR’s oblique-input proxy supplying the *i* needed for i-umlaut. The account does not independently certify that exact complete word, its long *ū*, or its cell.

The Old English target combines a regular inherited form *fȳr* ‘fire’ with an attested oblique surface form *fȳre* ‘fire’, interpreted by CAPR as analogically restored.

Competing ancestry and the legitimacy of the input Kroonen distinguishes the dative **fu(w)eni*, its *r*-remodelled **fu(w)eri*, and metathesized **feuri*. His preferred pre-Germanic nominative is **péh₂-ur*; the **ph₂-uōr* proposal reported from Simms is rejected, not endorsed (Kroonen 2013, 151). Ringe instead supplies a reaccented zero-grade collective, printed **ph2uōŕ*, and separately considers post-Proto-Germanic oblique dissimilation and *n* replacement by *r* (Ringe 2017, 147, 309). These are consequential formation and accent alternatives, not interchangeable quoted dative inputs.

Ringe and Taylor reconstruct apparently disyllabic West Germanic **fuir* with *r* levelled into the oblique stem, but doubt that the endingless dative evidence establishes inherited **fuiri*. A front-rounded intermediate in the history of *fȳr* is plausible; the intervening stages remain uncertain (Ringe and Taylor 2014, 119, 225). Neither passage endorses CAPR’s exact long-*ū* proxy. Equal citation and input fields select the Proto-Germanic stage by project convention, not by a new scholarly quotation. Quantity, cell and historical input legitimacy require separate scientific assessment; no input migration is made here.

Old English evidence Toller’s Supplement records *fyr* ‘fire’ as a noun in its *fyr. Add* entry and separately preserves oblique *fyre* ‘fire’, including *in fyre* (Toller 1921, 276). These attestations establish the forms, not the date or mechanism of restoration. The first is the regular output of the phonological development from the selected input; the second is interpreted by CAPR as restoration of final *-e* within the paradigm, rather than demonstrated as such by the dictionary.

The entry therefore distinguishes a regular path from an adopted input proxy, the comparative legitimacy of that proxy, and an attested surface form whose additional inflection CAPR treats as morphological remodeling.

Development to Old English From **fūri* ‘fire’, i-umlaut changes *ū* to *ȳ* (Hogg 1992b, sect. 3.3.3.1). Subsequent loss of the final high vowel after a heavy syllable yields *fȳr* ‘fire’ (Campbell 1959, sect. 345). The selected phonological path is complete at that point.

CAPR interprets the final *-e* of *fȳre* ‘fire’ as analogical restoration in the Old English paradigm rather than retention of the derivational input’s ending. The form therefore remains a known but unmodelled remodelling: the deterministic phonology is regular, but the attested surface form includes later morphological rebuilding.

Paradigm comparison The comparison below sets the relevant forms side by side. It distinguishes the lexeme-level etymological background from the inherited cell that actually produces the front-mutated form and from the later analogical surface result.

PGmc cell / interpretation	Candidate input	OE output or comparison	OE comparison form	Result
lexeme-level heteroclitic headword	**fōr* ‘fire’ ~ **fun-*	comparative background only	fire family	explains the wider lexeme, but not the selected oblique input
selected oblique-input proxy	**fūri*	regular output: *fȳr* ‘fire’	*fȳr* ‘fire’	regular output; exact ancestral cell remains qualified
later analogical surface form	—	attested *fȳre* ‘fire’ with restored *-e*	*fȳre* ‘fire’	genuine OE target, but not the direct phonological output

tap — OE *tæppa*

Derivation: **táppô* yields regular **tappa*; the Old English form here is *tæppa* (known but unmodelled development).

Derivation trace Proto input: **táppô*

Earlier Germanic changes	Old English changes	
Northwest and West Germanic	EAF Brightening	**tæppô*
[no change]	OE A Restoration	**tappô*
	OE Unstressed Long Vowel Shortening	**tappa*
Early Anglo-Frisian		
[no change]		

Regular outcome: **tappa*

Old English form: *tæppa*

Reconstruction and comparative evidence Orel gives the noun under **tappòn* ‘tap’ and already connects it with Old English *tæppa* ‘tap’ (Orel 2003, 402). The derivational input **táppô* ‘tap’ is therefore the inherited noun itself; the entry does not depend on a different lexeme-level proto or a different inherited noun cell.

Old English evidence The Old English noun family is well attested. Orel gives *tæppa* ‘tap’, and Clark Hall records *tæppa* ‘tap’ together with derivatives *tæppere* ‘tapster’ and *tæppestre* ‘tapster’ (Orel 2003, 402; Clark Hall 1960, 305). The target is therefore a real Old English noun form, not a reconstructed convenience spelling.

Development to Old English From **táppô* ‘tap’, the regular inherited noun path gives *tappa* ‘tap’. The attested target *tæppa* ‘tap’ therefore stands outside that regular phonological development.

The mismatch is historically intelligible, but it is not solved here by a new inherited input. A related j-verb pathway would give *teppan* ‘tap’, not the noun target *tæppa* ‘tap’. The entry accordingly remains a known but unmodelled case.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**táppô*	regular output: *tappa* ‘tap’	regular output, but not the target
attested OE target	—	*tæppa* ‘tap’	genuine target form, but analogically remodelled in the present classification
related j-verb background	**táppjanǫ*	*teppan* ‘tap’	related formation, but not the noun target

Part VII. Unexplained or deliberately unmodelled exceptions

No sufficiently supported account yet reconciles the regular output with the Old English form. An ad hoc sound law would conceal rather than explain the mismatch.

buck — OE *bucc*

Derivation: **búkkaz* yields regular **bocc*; the Old English form here is *bucc* (unexplained exception).

Derivation trace Proto input: **búkkaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**bókk*
PNWGmc U Lowering	**bókkaz*		
EAF Final Z Deletion	**bókka*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **bocc*

Old English form: *bucc*

Reconstruction and comparative evidence Kroonen and Orel both reconstruct the word with a geminate stop, **bukkaz* 'buck' (Kroonen 2013, 121; Orel 2003, 61). Orel also preserves parallel n-stem material behind Old English *bucca* 'buck' (Orel 2003, 62). The derivational input therefore remains identical with the lexeme label: no alternative inherited cell accounts for the form.

Old English evidence Old English preserves a mixed lexical picture. Campbell cites *bucca* 'buck' in the exception set for this phonological environment (Campbell 1959, sect. 115). Clark Hall and Bosworth-Toller show that Old English has both *bucca* 'buck' and *bucc* 'buck' (Clark Hall 1960, 53; Bosworth and Toller 1898, 122). The a-stem citation form *bucc* 'buck' is the target treated here; *bucca* 'buck' supplies genuine philological background from the same lexical family.

Development to Old English From **búkkaz* 'buck', the regular inherited path gives *bocc* 'buck'. That is the form expected under the ordinary lowering pattern in this environment. *bucc* 'buck' therefore remains outside the deterministic phonology.

No accepted inherited cell repairs the mismatch. A high-vowel alternative would introduce i-umlaut and produce a *byċċ* 'buck'-type form rather than the target. *bucc* 'buck' is therefore best treated as a documented exception, not as a regular paradigm-cell survival.

Form comparison

Form type	Input or form	OE output or comparison	Result
regular inherited noun path	**búkkaz*	regular output: *bocc* 'buck'	regular output, but not the target
attested OE target	—	*bucc* 'buck'	genuine target form, but unexplained in the present classification
parallel OE lexical background	—	*bucca* 'buck'	related n-stem form, not the present target

fowl — OE *fugol*

Derivation: **fúglaz* yields regular **fogol*; the Old English form here is *fugol* (unexplained exception).

Derivation trace Proto input: **fúglaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**fógl*
PNWGmc U Lowering	**fóglaz*	OE Epenthetic Vowel	**fógol*
EAF Final Z Deletion	**fógla*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **fogol*

Old English form: *fugol*

Reconstruction and comparative evidence The noun is the ordinary Germanic a-stem **fúglaz* ‘fowl’, continued by forms such as Old Norse *fugl* and Old High German *fogal* ‘bird, fowl’ (Kroonen 2013, 197; Orel 2003, 155). There is no stem-class or paradigm-cell dispute behind this entry. The comparative headword and the derivational input are the same.

The difficulty lies instead in the stressed root vowel. Under the regular West Germanic and Old English development, that *u* should lower before the following non-high vowel, yielding an *o*-vocalism (Ringe and Taylor 2014, 42–43; Campbell 1959, 43).

Old English evidence Old English dictionaries record the noun as *fugol* ‘fowl’, with variant spelling *fugel* ‘fowl’ (Bosworth and Toller 1898, 282; Clark Hall 1960, 138). The target is therefore an attested ordinary Old English noun, not a reconstructed or selectively chosen paradigm form.

The attested word already contains the crucial problem. Its medial *-o-* is the ordinary parasite vowel of Old English cluster phonology, but the root *fu-* retains *u* where the regular history predicts *fo-* (Campbell 1959, 150).

Development to Old English From **fúglaz* ‘fowl’, the regular cascade yields *fogol* ‘fowl’: the root vowel lowers before the following non-high vowel (Ringe and Taylor 2014, 42–43; Campbell 1959, 43), final *z* is lost, and the cluster is resolved by the usual medial vowel (Ringe and Taylor 2014, 345; Campbell 1959, 150). That is the expected inherited outcome.

The attested Old English noun is *fugol* ‘fowl’, not *fogol* ‘fowl’. Luick and later handbooks treat this preservation of *u* as a small inherited residue, not as a categorical sound law (Luick 1914--1940, 148; Ringe and Taylor 2014, 47). The item therefore remains a genuine lexical exception rather than the output of a recoverable regular mechanism.

Expected and attested forms The comparison below sets the relevant forms side by side. It distinguishes the regular prediction from the attested Old English noun.

Form	Status	Relevance to this entry
fogol	computed regular output from **fúglaz*	establishes the expected inherited outcome
fugol	attested Old English form	Old English form here; preserves unexplained root *u*
fugel	attested variant spelling	secondary spelling variant of the attested noun

The unresolved point lies only in the root vowel. The medial *-o-* is regular, but no accepted lautgesetzlich pathway has been found from **fúglaz* ‘fowl’ to attested *fugol* ‘fowl’.

rust — OE *rust*

Derivation: **rústaz* yields regular **rost*; the Old English form here is *rust* (unexplained exception).

Derivation trace Proto input: **rústaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**róst*
PNWGmc U Lowering	**róstaz*		
EAF Final Z Deletion	**rósta*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **rost*

Old English form: *rust*

Reconstruction and comparative evidence Orel gives both **rustō* ‘rust’ and **rustaz* ‘rust’, with masculine/feminine alternatives and Old English rust ‘rust’, Old Saxon rost ‘rust’ and Old High German rost ‘rust’ (Orel 2003, 309). The conventional feminine *ō*-stem and masculine *az*-stem are distinct morphological alternatives within his account. Ringe independently reconstructs **rustaz* ‘rust’; his discussion of alternative suffixal and cluster histories for *-st-* concerns the formation, not the explanation of retained *u* (Ringe 2017, 247–48, especially p. 248).

The masculine Old English noun favors the masculine citation reconstruction, selected here as **rústaz* ‘rust’ with the project’s acute stress notation. This is a morphological and comparative choice, not an exclusive disagreement between Orel and Ringe or a claim that Orel’s *ō*-stem is unsupported. The comparative lowered forms provide context, but output agreement does not choose a stem class: both formations regularly predict **rost* ‘rust’, not the attested Old English noun.

Old English evidence The noun is attested, not reconstructed. Clark Hall gives rūst ‘rust’ (m.) (Clark Hall 1960, 245), and Toller’s supplement to Bosworth-Toller records rúst ‘rust’, beside *rust*, with a qualification about quantity (Toller 1921, 691). Their lexicographical signs are retained as source spellings; the selected comparator remains normalized *rust*, without a newly adopted quantity interpretation. The evidence establishes the noun and masculine gender, not an independent explanation of its retained vowel.

Development to Old English SC017 PNWGmcULowering regularly lowers the root *u* of **rústaz* ‘rust’ before the following non-high vowel, giving **róstaz* ‘rust’. Final-*z* loss and subsequent regular developments complete the predicted **rost* ‘rust’, not the attested comparator *rust*. Campbell describes the lowering and its unresolved retained-*u* residues (Campbell 1959, 43, §115); Ringe and Taylor discuss the remaining exceptions without establishing a general consonantal exemption (Ringe and Taylor 2014, 27–34).

The control **frústą* ‘frost’ regularly gives forst ‘frost’, so *-st-* cannot be exempted generally. Selecting the masculine stem is not a phonological repair: *rust* remains an unexplained lexical retention and the mismatch total does not fall. No exception is inserted into the sound law.

Paradigm comparison Fresh production and isolated-wrapper diagnostics give the following results. The oblique-type constructions test conditioning; they are not newly attested Proto-Germanic paradigm cells.

Form	Role	Result
**rústaz*	Selected masculine citation form	**rost*, mismatch
**rústō*	Orel's alternative *ō*-stem	**rost*, mismatch
**rústas*	Genitive-singular-type probe	**rostes*
**rústai*	Dative-singular-type probe	**roste*
**rústi*	High-front-vowel diagnostic	**ryst*
**rústu*	Marginal high-vowel diagnostic construction	**rust*

The hypothetical instrumental-type **rústu* 'rust' is a diagnostic construction, not an attested Proto-Germanic citation form or an adopted historical solution. Its matching output does not establish a source for the ordinary citation noun.

wolf — OE *wulf*

Derivation: **wúlfaz* yields regular **wolf*; the Old English form here is *wulf* (unexplained exception).

Derivation trace Proto input: **wúlfaz*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		PWGmc Final Bare A Loss	**wólf*
PNWGmc U Lowering	**wólfaz*		
EAF Final Z Deletion	**wólfa*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **wolf*

Old English form: *wulf*

Reconstruction and comparative evidence The inherited noun is an a-stem: Kroonen gives **wulfa-*, and Ringe and Taylor list PGmc **wulfaz* 'wolf' among the inherited words that preserve *u* in Old English beside Old High German *wolf* 'wolf' (Kroonen 2013, 638; Ringe and Taylor 2014, 47). Campbell accordingly names Old English *wulf* as an exception to the regular lowering of stressed *u* before a following non-high vowel (Campbell 1959, sect. 115).

The older literature often notices that the exceptional words cluster near labials. Bülbring lists *full*, *wulle*, and *wulf* together, but he also says that the ordinary rule still gives *o* in comparable forms such as *folc* and *bolt* (Bülbring 1902, sect. 116). Luick rejects a categorical labial blocker on the same grounds and prefers a lexical or analogical account instead (Luick 1914--1940, 148).

Old English evidence Campbell treats *wulf* 'wolf' as part of the exceptional *u* set (Campbell 1959, sect. 115). Sievers-Brunner notes that oblique *wulfe* 'wolf (dat.)' continues *wulfi* 'wolf (dat.)' or older *wulfai* 'wolf (dat.)' (Brunner 1965, sect. 160).

The surviving oblique forms do not supply a regular route back to bare *wulf* 'wolf'. They belong to the same lexeme, but they do not remove the explanatory problem presented by the citation form.

Development to Old English Under the ordinary Northwest Germanic lowering of stressed *u* before a following non-high vowel, the citation-form input would point toward *o*-vocalism (Ringe and Taylor 2014, 42–44). The compact trace shows exactly that path: **wúlfaz* 'wolf' > **wólfaz* 'wolf' > **wólfa* 'wolf' > wolf.

A high-vowel oblique input would behave differently. There the following high vowel would block the lowering of *u*, but the same environment would also trigger i-umlaut, so the regular control result would be *wylf* ‘wolf’ or *wylfe*, not bare *wulf*. The attested noun therefore remains unexplained at the citation-form level.

Expected and attested forms The comparison below sets the regular inherited outcomes beside the attested Old English noun.

Form / interpretation	Status	Relevance to this entry
**wúlfaz* > *wolf*	computed regular output from the citation form	shows the regular development expected from the inherited a-stem
OHG wolf	comparative regular cognate	confirms that the *o*-vocalism is the ordinary outcome
**wúlfi* / **wúlfis* > *wylf* / *wylfe*	expected high-vowel control forms	shows why oblique high-vowel cells do not solve the noun’s vowel history
wulf	attested Old English noun	attested Old English form; the preservation of *u* remains unexplained

wool — OE *wull*

Derivation: **wúllō* yields regular **woll*; the Old English form here is *wull* (unexplained exception).

Derivation trace Proto input: **wúllō*

Earlier Germanic changes		Old English changes	
Northwest and West Germanic		OE High Vowel Apocope	**wóll*
PNWGmc U Lowering	**wóllō*		
PNWGmc Final Long O Raising	**wóllu*		
Early Anglo-Frisian			
[no change]			

Regular outcome: **woll*

Old English form: *wull*

The inherited paradigm The selected **wúllō* ‘wool’ represents the feminine nominative singular. It must be distinguished from accusative **wúllǭ*, whose final vowel was nasalized. Ringe explicitly reconstructs both cells; Orel and Kroonen discuss the inherited formation and its comparative history (Ringe 2017, 171–73, 241–44; Orel 2003, 474; Kroonen 2013, 598–99). Attested *wull* ‘wool’ and *wulle* ‘wool’ preserve a high root vowel. The latter is not, merely by its shape, evidence that the whole noun was originally a weak n-stem (Campbell 1959, 43, §115; Bülbring 1902, 45–46, §116).

Regular development and the alternative explanation CAPR provisionally follows Fulk’s lowering-before-final-raising chronology (Fulk 2018, 56). Both oral **ō* and nasal **ǭ* condition root lowering before their distinct subsequent developments:

Inherited cell	Preferred chronology	Raising before lowering
nominative **wúllō*	**wóllō* → **wóllu* → regular **woll*	**wúllu* → regular **wull*

Inherited cell	Preferred chronology	Raising before lowering
accusative **wúllǭ*	**wóllǭ* → regular **wolle*	**wóllǭ* → regular **wolle*

These are sound-law predictions, not additional attestations. Earlier raising supplies a genuine nominative **wullu* with high root vowel and an independently conditioned low-root oblique. Generalization of the nominative vowel could then produce the attested high-root paradigm: this is Ringe and Taylor's plausible but unproved explanation, not an impossible escape cell (Ringe and Taylor 2014, 32). Old Frisian *wolle* 'wool' and Old High German *wolla* 'wool' support the comparative reality of a lowered root, but do not attest that particular early English accusative (Kroonen 2013, 598–99).

The main comparison of the two chronologies, the evidence from other ō-stems and the costs of the proposed analogies is in SC019 PNWGmcFinalLongORaising. The preferred ordering does not regularly derive the attested high root here. The derivational input, target and classification therefore remain unchanged: regular **woll* 'wool', not attested *wull*, is what the current sound laws produce. No lexical exception or analogical levelling has been inserted into those laws.

Luick discusses the alternations in §§77–79, especially §78 (printed pp. 105–106), not §§105–106. He rejects a proposed labial protection rule using regular low-root counterexamples (Luick 1914--1940, 104–6, §§77–79). The separate trimoric genitive **-ôz* must not be substituted for either cell above: its analysis differs from the bimoric genitive and its complete computational pathway remains unmodeled (Fulk 2018, 155; Ringe and Taylor 2014, 59, 299).

References

Bammesberger, Alfred. 1990. *Die Morphologie Des Urgermanischen Nomens*. Vol. 2. Untersuchungen Zur Vergleichenden Grammatik Der Germanischen Sprachen. Carl Winter.

Bammesberger, Alfred. 1997. 'Die Vorform von Altenglisch hærfest'. *Anglia: Zeitschrift für Englische Philologie* 115 (2): 223–30. https://doi.org/10.1515/angl.1997.115.2.223.

Bosworth, Joseph, and T. Northcote Toller. 1898. *An Anglo-Saxon Dictionary*. Clarendon Press.

Bright, James W. 1917. *An Anglo-Saxon Reader*. 4th edn. Henry Holt; Company.

Bright, James W. 1971. *Bright's Old English Grammar and Reader*. 3rd edn. Edited by Frederic G. Cassidy and Richard N. Ringler. Holt, Rinehart; Winston.

Brunner, Karl. 1965. *Altenglische Grammatik nach der angelsächsischen Grammatik von Eduard Sievers*. 3rd edn. Max Niemeyer.

Bülbring, Karl D. 1902. *Altenglisches Elementarbuch. 1: Lautlehre*. Carl Winter.

Campbell, Alistair. 1959. *Old English Grammar*. Clarendon Press.

Čermák, Jan. 2005. 'Notes on Obscured Compounds in Old English'. In *Patterns: A Festschrift for Libuše Dušková*, edited by Jan Čermák, Aleš Klégr, Markéta Malá, and Pavlína Šaldová. Karolinum.

Ciszek-Kiliszewska, Ewa. 2015. 'Old English Suffix -Cund and Its Fate'. *Kwartalnik Neofilologiczny* 62 (3): 487–504.

Clark Hall, J. R. 1960. *A Concise Anglo-Saxon Dictionary*. 4th edn. Cambridge University Press.

Colman, Fran. 2011. 'On the Moneyers' Names Buga and Boia on Anglo-Saxon Coins'. *Nomina* 34: 91–120. https://www.snsbi.org.uk/Nomina_articles/Nomina_34_Colman.pdf.

Crawford, S. J., ed. 1922. *The Old English Version of the Heptateuch, Ælfric's Treatise on the Old and New Testament and His Preface to Genesis*. Humphrey Milford, Oxford University Press, for the Early English Text Society.

Crist, Sean J. 2002. *An Analysis of *z Loss in West Germanic*. Paper presented at the Linguistic Society of America Annual Meeting.

Fulk, R. D. 2018. *A Comparative Grammar of the Early Germanic Languages*. Vol. 3. Studies in Germanic Linguistics. John Benjamins.

Goering, Nelson. 2023. 'The Hēafudu-Problem: Early Old English Foot Structure'. In *Prosody in Medieval English and Norse*. British Academy. https://doi.org/10.5871/bacad/9780197267462.003.0004.

Hill, Eugen. 2017. 'West Germanic Monosyllabic Lengthening and Gothic Breaking as Partially Proto-Germanic Developments: The Evidence of Pronominal Place Adverbs "Here", "Where" and "There"'. *NOWELE. North-Western European Language Evolution* 70 (2): 135–70. https://doi.org/10.1075/nowele.70.2.02hil.

Hogg, Richard M. 1992a. *A Grammar of Old English. Volume 1: Phonology*. Blackwell.

Hogg, Richard M. 1992b. 'Phonology and Morphology'. In *The Cambridge History of the English Language. Volume I: The Beginnings to 1066*, edited by Richard M. Hogg. Cambridge University Press.

Hogg, Richard M. 1996. ‘Tertiary Stress in Old English: Some Reflections on Explanatory Inadequacy’. In *English Historical Linguistics 1994*, edited by Derek Britton, vol. 135. Current Issues in Linguistic Theory. John Benjamins. https://doi.org/10.1075/cilt.135.03hog.

Hogg, Richard M. 2011. *A Grammar of Old English. Volume 1: Phonology*. Wiley-Blackwell.

Jasanoff, Jay H. 1973. ‘The Germanic Third Weak Class’. *Language* 49 (4): 850–70. https://doi.org/10.2307/412065.

Kastovsky, Dieter. 1992. ‘Semantics and Vocabulary’. In *The Cambridge History of the English Language. Volume I: The Beginnings to 1066*, edited by Richard M. Hogg. Cambridge University Press.

Kluge, Friedrich. 2011. *Etymologisches Wörterbuch der deutschen Sprache*. 25th edn. Edited by Elmar Seebold. De Gruyter.

Kroonen, Guus. 2011. *The Proto-Germanic n-Stems: A Study in Diachronic Morphophonology*. Vol. 18. Leiden Studies in Indo-European. Rodopi.

Kroonen, Guus. 2013. *Etymological Dictionary of Proto-Germanic*. Vol. 11. Leiden Indo-European Etymological Dictionary Series. Brill.

Lloyd, Albert L. 1966. ‘Is There an a-Umlaut of i in Germanic?’ *Language* 42 (4): 738–45. https://doi.org/10.2307/411829.

Luick, Karl. 1914--1940. *Historische Grammatik der englischen Sprache*. Tauchnitz.

McCully, C. B., and R. M. Hogg. 1990. ‘An Account of Old English Stress’. *Journal of Linguistics* 26 (2): 315–39. https://doi.org/10.1017/S0022226700014699.

Miller, Thomas, ed. 1891. *The Old English Version of Bede’s Ecclesiastical History of the English People*. I, second half. For the Early English Text Society.

Minkova, Donka. 1996. ‘Verse Structure as Evidence for Prosodic Reconstruction in Old English’. In *English Historical Linguistics 1994*, edited by Derek Britton, vol. 135. Current Issues in Linguistic Theory. John Benjamins. https://doi.org/10.1075/cilt.135.04min.

Minkova, Donka, and Z. L. Zhou. 2023. ‘Nominal Compounds in Old English Meter and Prosody’. *Journal of Germanic Linguistics* 35 (1): 69–95. https://doi.org/10.1017/S1470542722000083.

Orel, Vladimir. 2003. *A Handbook of Germanic Etymology*. Brill.

Ringe, Don. 2017. *From Proto-Indo-European to Proto-Germanic*. 2nd edn. Vol. 1. A Linguistic History of English. Oxford University Press.

Ringe, Don, and Ann Taylor. 2014. *The Development of Old English*. Vol. 2. A Linguistic History of English. Oxford University Press.

Sauer, Hans. 1992. *Nominalkomposita Im Frühmittelenglischen: Mit Ausblicken Auf Die Geschichte Der Englischen Nominalkomposition*. Vol. 30. Buchreihe Der Anglia. Max Niemeyer.

Seebold, Elmar. 1970. *Vergleichendes und etymologisches Wörterbuch der germanischen starken Verben*. Mouton.

Stanley, Eric. 2022. *Metricalities: Studies in Old English Verse*. Edited by Andy Orchard. CLASP Ancillary Publications 2. CLASP. https://clasp.ell.ox.ac.uk/metricalities.pdf.

Stiles, Patrick V. 1985. ‘The Fate of the Numeral “4” in Germanic (1)’. *NOWELE: North-Western European Language Evolution* 6: 81–104. https://doi.org/10.1075/nowele.6.06sti.

Stiles, Patrick V. 2012. ‘Older Runic Evidence for North-West Germanic a-Umlaut of u (and “the Converse of Polivanov’s Law”)’. In *Laws and Rules in Indo-European*, edited by Philomen Probert and Andreas Willi. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199609925.003.0004.

Sweet, Henry. 1953. *An Anglo-Saxon Primer*. 9th edn. Clarendon Press.

Tarsi, Matteo, and Stephen Laker. 2026. ‘“To Know”, “to Teach”, and “to Learn” in Germanic’. *Journal of Germanic Linguistics* 38: 40–58. https://doi.org/10.1017/S1470542726100178.

Toller, T. Northcote. 1921. *An Anglo-Saxon Dictionary: Supplement*. Oxford University Press.

Versloot, Arjen P. 2026. ‘Reconstructing the Historical Phonology of Old English’. *English Language and Linguistics* 30 (1): 103–41. https://doi.org/10.1017/S1360674324000674.

Viredaz, Rémy. 2025. *Germanic, Slavic and Baltic ‘Thousand’ Once More*. Unpublished manuscript (unfinished), Geneva, 20 October 2025. https://www.academia.edu/144462167/2025_Germanic_Slavic_and_Baltic_thousand_once_more_full_text_unfinished_.